

```
Private Sub Worksheet_Change(ByVal target As Range)
    ' 只有當 (2,2) 被修改時才觸發
    If target.row = 2 And target.Column = 2 Then
        If IsNumeric(target.Value) And target.Value > 0 Then
            Application.EnableEvents = False ' 防止遞迴觸發
            Call BuildCorrelationMatrix
            Application.EnableEvents = True
        End If
    End If
End Sub
```

```
Private Sub Worksheet_Change(ByVal target As Range)
```

```
    ' 1. 安全機制：確認只更改了一個儲存格，避免您一次選取/貼上大範圍資料時引發錯誤
```

```
    If target.CountLarge > 1 Then Exit Sub
```

```
    ' 2. 目標鎖定：確認被更改的儲存格 (Target) 是否與 B1 重疊 (Intersect)
```

```
    If Not Intersect(target, Me.Range("B1")) Is Nothing Then
```

```
        ' 3. 條件檢查：確保 B1 裡面確實有輸入文字，而不是被按 Delete 清空
```

```
        If Trim(target.Value) <> "" Then
```

```
            ' 4. 暫停事件監聽：因為 FetchSingleReport 執行時會大幅修改工作表內容(ws.cells.Clear)，
```

```
            ' 這會引發無數次的 Change 事件導致無限迴圈當機。因此必須先暫停監聽。
```

```
            Application.EnableEvents = False
```

```
            On Error GoTo ErrorHandler ' 加上錯誤捕捉，確保發生錯誤時能重新開啟事件監聽
```

```
            ' 5. 呼叫您的主程式
```

```
            Call FetchSingleReport
```

```
ErrorHandler:
```

```
    ' 6. 恢復事件監聽：無論程式是正常跑完還是中途出錯，都必須把監聽器打開
```

```
    Application.EnableEvents = True
```

```
    ' 如果真的有錯誤，顯示錯誤訊息
```

```
    If Err.Number <> 0 Then
```

```
        MsgBox "執行時發生錯誤：" & Err.Description, vbCritical
```

```
    End If
```

```
End If
```

```
End If
```

```
End Sub
```



```
Private Sub Worksheet_Change(ByVal target As Range)
    ' 只關心 B1 (ticker) 與 H5 (price target) 兩格
    Dim hitB1 As Boolean
    Dim hitH5 As Boolean
    hitB1 = Not Intersect(target, Me.Range("B1")) Is Nothing
    hitH5 = Not Intersect(target, Me.Range("H5")) Is Nothing
    If Not (hitB1 Or hitH5) Then Exit Sub

    ' 防呆：避免一次選多格
    If target.count > 1 Then Exit Sub

    ' 防呆：B1 被清空時不執行 (避免 Delete 觸發報錯)
    If hitB1 And Trim(Me.Range("B1").Value) = "" Then Exit Sub

    ' 暫停事件監聽，避免巨集寫入時遞迴觸發
    Application.EnableEvents = False
    On Error GoTo ErrorHandler

    ' 派發到正確的 refresh
    If hitB1 Then
        Call RefreshTickerInsight(CStr(Me.Range("B1").Value))
    ElseIf hitH5 Then
        Call RefreshTickerProjection
    End If

ErrorHandler:
    Application.EnableEvents = True
    If Err.Number <> 0 Then
        MsgBox "Ticker Insight 執行錯誤: " & Err.Description, _
            vbCritical, "錯誤代碼 " & Err.Number
    End If
End Sub
```

```
Private Sub CmdBtn0_Click()  
    Call ClearAllData  
End Sub  
  
Private Sub CmdBtn1_Click()  
    Call OpenDeleteForm  
End Sub  
  
Private Sub CmdBtn2_Click()  
    Call RebuildPortfolioDashboard  
End Sub  
  
Private Sub CmdBtn3_Click()  
    Call OpenTransactionForm  
End Sub  
Private Sub CmdBtn6_Click()  
    Call StartAutoRefresh  
End Sub  
  
Private Sub CmdBtn7_Click()  
    Call StopAutoRefresh  
End Sub  
Private Sub CmdBtn4_Click()  
    Call BuildHoldingsCorrelation  
End Sub  
Private Sub CmdBtn5_Click()  
    Call BuildDrawdownShadow  
End Sub  
  
Private Sub CmdBtn8_Click()  
    Call RunSystemDebug  
End Sub
```

```
Private Sub Worksheet_Change(ByVal target As Range)
    ' 1. 檢查變動的儲存格是否包含 B2
    ' Intersect 函數用來判斷兩個範圍是否有交集，若有交集代表 B2 被更動了
    If Not Intersect(target, Me.Range("B2")) Is Nothing Then

        ' 2. 防呆機制：避免使用者一次選取多個儲存格（例如按下 Delete 刪除大範圍）導致程式錯誤
        If target.count > 1 Then Exit Sub

        ' 3. 防呆機制：如果 B2 被清空，則不執行程式
        If Trim(Me.Range("B2").Value) = "" Then Exit Sub

        ' 4. 安全機制：暫時關閉事件監聽
        ' 因為接下來呼叫的巨集會自動寫入 A 到 F 欄的資料，
        ' 為了避免巨集寫入資料時再次觸發 Worksheet_Change 造成無窮迴圈或效能拖累，必須先關閉。
        Application.EnableEvents = False

        ' 5. 錯誤處理宣告，確保即使主程式出錯，也能把事件監聽重新打開
        On Error GoTo ErrorHandler

        ' 6. 呼叫您的主程式進行計算
        Call UpdateVolatilityAnalysis_Pro

ErrorHandler:
        ' 7. 復原機制：重新開啟事件監聽
        Application.EnableEvents = True

        ' 若有捕捉到錯誤，顯示錯誤訊息
        If Err.Number <> 0 Then
            MsgBox "自動執行時發生錯誤：" & Err.Description, vbCritical, "錯誤代碼 " & Err.Number
        End If

    End If
End Sub
```

```
Private Sub Workbook_Open()  
    ' 關閉螢幕更新，讓切換過程更順暢，不會有畫面閃爍  
    Application.ScreenUpdating = False  
  
    On Error Resume Next ' 防呆：如果該工作表被重新命名或刪除，避免跳出偵錯錯誤  
  
    ' 指定每次開啟時要強制切換到的工作表名稱  
    ThisWorkbook.Sheets("Tickers volatility").Activate  
  
    ' (可選) 如果您希望開啟時，游標固定停在特定的儲存格 (例如 B2 讓您直接輸入)  
    ThisWorkbook.Sheets("Tickers volatility").Range("B2").Select  
  
    On Error GoTo 0 ' 恢復正常的錯誤提示  
  
    ' 重新開啟螢幕更新  
    Application.ScreenUpdating = True  
End Sub
```

frmDeleteTransaction - 1

Option Explicit

'--- 初始化：開啟表單時載入資料 ---

```
Private Sub UserForm_Initialize()  
    Call LoadTransactions  
End Sub
```

'--- 核心功能：載入交易紀錄到 ListBox ---

```
Private Sub LoadTransactions()  
    Dim wsTrans As Worksheet  
    Dim lastRow As Long, i As Long  
  
    Set wsTrans = ThisWorkbook.Sheets("Transactions")  
    lastRow = wsTrans.Cells(wsTrans.Rows.Count, "A").End(xlUp).row
```

```
    ' 清空列表  
    lstTransactions.Clear
```

```
    ' 設定標題列 (非必要，但若 ListBox 有開 ColumnHeads 則需要)  
    ' 這裡我們直接讀取資料
```

```
    If lastRow < 2 Then  
        lstTransactions.AddItem "無交易紀錄"  
        Exit Sub  
    End If
```

' 迴圈讀取資料 (從第 2 列開始)

' 修正後的假設：

' A欄(1)=ID, B欄(2)=日期, C欄(3)=代號, D欄(4)=動作, E欄(5)=股數, I欄(9)=總金額

```
For i = 2 To lastRow  
    With lstTransactions  
        ' .AddItem 新增第一欄 (ListBox Column 0) -> 放日期  
        .AddItem wsTrans.Cells(i, 2).Value ' 原本是 1，改為 2 (B欄: 日期)  
  
        ' 依序填入後續欄位  
        .List(.ListCount - 1, 1) = wsTrans.Cells(i, 3).Value ' 原本是 2，改為 3 (C欄: 代號)  
        .List(.ListCount - 1, 2) = wsTrans.Cells(i, 4).Value ' 原本是 3，改為 4 (D欄: 動作)  
        .List(.ListCount - 1, 3) = wsTrans.Cells(i, 5).Value ' 原本是 4，改為 5 (E欄: 股數)  
  
        ' 金額原本在 H欄(8)，因為前面多一欄 ID，推測現在移到 I欄(9)  
        ' 如果您的金額還是在 H欄，請改回 8  
        .List(.ListCount - 1, 4) = wsTrans.Cells(i, 9).Value  
  
        ' 隱藏的列號 (用於刪除功能)，保持 i 不變  
        .List(.ListCount - 1, 5) = i  
    End With  
Next i  
End Sub
```

'--- 按鈕動作：刪除選取項目 ---

```
Private Sub btnDelete_Click()  
    Dim wsTrans As Worksheet  
    Dim i As Long  
    Dim rowToDelete As Long  
    Dim selectedIndex As Long  
    Dim response As VbMsgBoxResult  
    Dim tickerInfo As String  
  
    selectedIndex = lstTransactions.ListIndex  
  
    ' 1. 防呆：檢查有無選取  
    If selectedIndex = -1 Then  
        MsgBox "請先點選一筆要刪除的交易！", vbExclamation  
        Exit Sub  
    End If
```

```
    ' 讀取隱藏的列號 (第6欄，索引為5)  
    rowToDelete = lstTransactions.List(selectedIndex, 5)
```

' 讀取資訊做確認提示

```
tickerInfo = lstTransactions.list(selectedIndex, 0) & " - " & _  
            lstTransactions.list(selectedIndex, 1) & " " & _  
            lstTransactions.list(selectedIndex, 2)  
  
' 2. 二次確認  
response = MsgBox("確定要刪除這筆紀錄嗎?" & vbCrLf & vbCrLf & _  
                  "內容: " & tickerInfo & vbCrLf & _  
                  "Excel 列號: " & rowToDelete & vbCrLf & vbCrLf & _  
                  "此動作無法復原!", vbYesNo + vbCritical, "刪除確認")  
  
If response = vbNo Then Exit Sub  
  
' 3. 執行刪除  
Set wsTrans = ThisWorkbook.Sheets("Transactions")  
wsTrans.Rows(rowToDelete).Delete  
  
' 4. 刪除後動作  
MsgBox "刪除成功! 系統將自動重新計算 Dashboard。", vbInformation  
  
' 重新載入列表 (因為列號變了, 必須重抓)  
Call LoadTransactions  
  
' 呼叫主程式更新 Dashboard (確保 Portfolio 準確)  
Call RebuildPortfolioDashboard  
End Sub
```

Option Explicit

' 初始化：預設選台股

Private Sub UserForm_Initialize()

optTW.Value = True

Call PopulateSectors("TW")

End Sub

' 切換到台股

Private Sub optTW_Click()

Call PopulateSectors("TW")

End Sub

' 切換到美股

Private Sub optUS_Click()

Call PopulateSectors("US")

End Sub

'--- 更新版：根據市場填充板塊清單（各 30 個細分類） ---

Private Sub PopulateSectors(market As String)

cmbSector.Clear

If market = "TW" Then

' === 台股 30 大細產業分類 ===

cmbSector.AddItem "01. 半導體 - 晶圓代工 (Foundry)"

cmbSector.AddItem "02. 半導體 - IC設計 (IC Design)"

cmbSector.AddItem "03. 半導體 - 封測 (Packaging)"

cmbSector.AddItem "04. 半導體 - IP/ASIC (Silicon IP)"

cmbSector.AddItem "05. 電腦週邊 - AI 伺服器 (AI Server)"

cmbSector.AddItem "06. 電腦週邊 - 品牌與代工 (OEM/ODM)"

cmbSector.AddItem "07. 電腦週邊 - 工業電腦 (IPC)"

cmbSector.AddItem "08. 通信網路 - 網通設備 (Networking)"

cmbSector.AddItem "09. 光電業 - 面板與光學 (Panel & Optics)"

cmbSector.AddItem "10. 電子零組件 - PCB (印刷電路板)"

cmbSector.AddItem "11. 電子零組件 - 被動元件 (Passive)"

cmbSector.AddItem "12. 電子零組件 - 連接器 (Connectors)"

cmbSector.AddItem "13. 電子零組件 - 電源供應 (Power Supply)"

cmbSector.AddItem "14. 金融 - 金控 (FHC)"

cmbSector.AddItem "15. 金融 - 銀行與證券 (Bank/Securities)"

cmbSector.AddItem "16. 航運 (Container Shipping)"

cmbSector.AddItem "17. 傳產 - 散裝與航空 (Bulk & Aviation)"

cmbSector.AddItem "18. 鋼鐵 (Steel)"

cmbSector.AddItem "19. 塑膠 (Plastics)"

cmbSector.AddItem "20. 紡織 (Textiles)"

cmbSector.AddItem "21. 水泥 (Cement)"

cmbSector.AddItem "22. 電機機械"

cmbSector.AddItem "23. 電器電纜"

cmbSector.AddItem "24. 生技醫療 (Biotech)"

cmbSector.AddItem "25. 化學工業 (Chemicals)"

cmbSector.AddItem "26. 汽車工業 (Auto)"

cmbSector.AddItem "27. 建材營造 (Construction)"

cmbSector.AddItem "28. 食品工業 (Food)"

cmbSector.AddItem "29. 觀光餐旅 (Tourism)"

cmbSector.AddItem "30. 綠能環保 (Green Energy)"

Else

' === 美股 30 大 Finviz 風格分類 ===

cmbSector.AddItem "01. Tech - Mega Cap (七巨頭)"

cmbSector.AddItem "02. Tech - Software Infrastructure (軟體基建)"

cmbSector.AddItem "03. Tech - Software Application (應用軟體)"

cmbSector.AddItem "04. Tech - Semiconductors (半導體)"

cmbSector.AddItem "05. Tech - Consumer Electronics (消費電子)"

cmbSector.AddItem "06. Tech - Info Services (資訊服務)"

cmbSector.AddItem "07. Financial - Banks Diversified (綜合銀行)"

cmbSector.AddItem "08. Financial - Credit Services (信貸服務)"

cmbSector.AddItem "09. Financial - Asset Management (資產管理)"

cmbSector.AddItem "10. Financial - Insurance (保險)"

cmbSector.AddItem "11. Healthcare - Drug Manufacturers (製藥)"

cmbSector.AddItem "12. Healthcare - Biotechnology (生技)"

cmbSector.AddItem "13. Healthcare - Medical Devices (醫材)"

cmbSector.AddItem "14. Healthcare - Plans (健保)"

cmbSector.AddItem "15. Cyclical - Auto Manufacturers (汽車)"

cmbSector.AddItem "16. Cyclical - Internet Retail (電商)"

```
        cmbSector.AddItem "17. Cyclical - Restaurants (餐飲)"
        cmbSector.AddItem "18. Cyclical - Travel & Leisure (旅遊)"
        cmbSector.AddItem "19. Defensive - Discount Stores (零售)"
        cmbSector.AddItem "20. Defensive - Beverages (飲料)"
        cmbSector.AddItem "21. Defensive - Household (家用品)"
        cmbSector.AddItem "22. Comm - Internet Content (網路內容)"
        cmbSector.AddItem "23. Comm - Telecom (電信)"
        cmbSector.AddItem "24. Comm - Entertainment (娛樂)"
        cmbSector.AddItem "25. Industrials - Aerospace (航太軍工)"
        cmbSector.AddItem "26. Industrials - Logistics (物流)"
        cmbSector.AddItem "27. Industrials - Farm & Heavy (重工)"
        cmbSector.AddItem "28. Energy - Oil & Gas (石油天然氣)"
        cmbSector.AddItem "29. Basic Materials (原物料)"
        cmbSector.AddItem "30. Real Estate (REITs)"
        cmbSector.AddItem "31. ETFs - Major Indices (指數與區域)"
    End If
    cmbSector.ListIndex = 0 ' 預設選第一個
End Sub
```

' 按下掃描按鈕

```
Private Sub btnRunScan_Click()
```

```
    Dim market As String
```

```
    Dim Sector As String
```

```
    If optTW.Value Then market = "TW" Else market = "US"
```

```
    Sector = cmbSector.Value
```

' 隱藏表單並執行主程式

```
Me.Hide
```

```
Call RunCompanyResearch(market, Sector)
```

```
Unload Me
```

```
End Sub
```

Option Explicit

' 表單初始化：設定下拉選單與預設值

Private Sub UserForm_Initialize()

Me.txtDate.Value = Format(Date, "yyyy/mm/dd")

With Me.cmbAction

.AddItem "Buy"

.AddItem "Sell"

.AddItem "UpdateInfo"

.ListIndex = 0

End With

With Me.cmbSector

.AddItem "資訊科技 (Information Technology)"

.AddItem "金融 (Financials)"

.AddItem "醫療保健 (Health Care)"

.AddItem "非必需消費 (Consumer Discretionary)"

.AddItem "必需消費 (Consumer Staples)"

.AddItem "工業 (Industrials)"

.AddItem "能源 (Energy)"

.AddItem "公用事業 (Utilities)"

.AddItem "房地產 (Real Estate)"

.AddItem "原物料 (Materials)"

.AddItem "通訊服務 (Communication Services)"

.AddItem "ETF / 指數"

End With

With Me.cmbStrategy

.AddItem "EMA6 ADD - EMA 13 CUT"

.AddItem "EMA13 ADD - EMA 30 CUT"

.AddItem "EMA30 ADD - EMA 30 CUT"

.AddItem "SMA60 ADD Mid-term Holdings"

.AddItem "Long-term Holdings"

End With

With Me.cmbCurrency

.Clear

.AddItem "TWD"

.AddItem "USD"

.ListIndex = 0

End With

' ★ 新增：券商選單

With Me.cmbBroker

.AddItem "國泰世華"

.AddItem "元大證券"

.ListIndex = 0

End With

Me.txtShares.Value = "0"

Me.txtPrice.Value = "0"

Me.txtFee.Value = "0"

Me.txtTax.Value = "0"

Me.txtTarget.Value = "0"

Me.txtBeta.Value = "0"

End Sub

' [按鈕] 計算 Beta

Private Sub btnCalcBeta_Click()

Dim ticker As String

Dim Sector As String

Dim benchmark As String

Dim betaVal As Double

ticker = Trim(Me.txtTicker.Value)

Sector = Me.cmbSector.Value

```

If ticker = "" Then
    MsgBox "請先輸入股票代號 (Ticker)!", vbExclamation
    Exit Sub
End If

Me.btnCalcBeta.Caption = "計算中..."
DoEvents

benchmark = GetBenchmarkTicker(Sector, ticker)
betaVal = CalculateBeta30D(ticker, benchmark)

If betaVal = -999 Then
    MsgBox "Beta 計算失敗 (可能資料不足或代號錯誤)", vbExclamation
    Me.txtBeta.Value = "0"
Else
    Me.txtBeta.Value = Format(betaVal, "0.00")
End If

Me.btnCalcBeta.Caption = "CalcBeta"
End Sub

' =====
' [按鈕] 儲存交易
' =====

Private Sub btnSave_Click()
    Dim wsLog As Worksheet
    Dim NextRow As Long
    Dim tDate As Date
    Dim tShares As Double, tPrice As Double
    Dim tFee As Double, tTax As Double, tNetAmount As Double
    Dim estSellFee As Double
    Dim estSellTax As Double
    Dim estSellAmount As Double

    If Me.txtTicker.Value = "" Then
        MsgBox "請輸入股票代號 (Ticker)", vbExclamation
        Me.txtTicker.SetFocus
        Exit Sub
    End If

    If Not IsDate(Me.txtDate.Value) Then
        MsgBox "日期格式錯誤，請輸入 yyyy/mm/dd", vbExclamation
        Exit Sub
    End If

    Set wsLog = ThisWorkbook.Sheets("Transactions")
    NextRow = wsLog.Cells(wsLog.Rows.Count, "A").End(xlUp).row + 1

    tDate = CDate(Me.txtDate.Value)
    tShares = val(Me.txtShares.Value)
    tPrice = val(Me.txtPrice.Value)
    tFee = val(Me.txtFee.Value)
    tTax = val(Me.txtTax.Value)

    If Me.cmbAction.Value = "Buy" Then
        estSellAmount = tShares * tPrice

        If Me.cmbCurrency.Value = "USD" Then
            If InStr(1, Me.cmbSector.Value, "ETF", vbTextCompare) > 0 Then
                estSellFee = 3
            Else
                estSellFee = Round(estSellAmount * 0.001, 2)
                ' ★ 元大美股個股低消 $1 USD
                If Me.cmbBroker.Value = "元大證券" Then
                    If estSellFee < 1 Then estSellFee = 1
                End If
            End If
            estSellTax = 0
        Else
            ' ★ 台股：依券商折扣率計算預估賣出手續費

```

```

        Dim dBrokerRateSave As Double
        If Me.cmbBroker.Value = "元大證券" Then
            dBrokerRateSave = 0.1
        Else
            dBrokerRateSave = 0.08
        End If
        estSellFee = Int(estSellAmount * 0.001425 * dBrokerRateSave)
        If estSellFee < 1 Then estSellFee = 1

        If InStr(1, Me.cmbSector.Value, "ETF", vbTextCompare) > 0 Then
            estSellTax = Int(estSellAmount * 0.001)
        Else
            estSellTax = Int(estSellAmount * 0.003)
        End If
    End If

    tNetAmount = (tShares * tPrice) + tFee + tTax + estSellFee + estSellTax

```

```

ElseIf Me.cmbAction.Value = "Sell" Then
    tNetAmount = (tShares * tPrice) - tFee - tTax
Else
    tNetAmount = 0
End If

```

```

With wsLog
    .cells(NextRow, "A").Value = "T-" & Format(Now, "yyyymmdd-HHmss")
    .cells(NextRow, "B").Value = tDate
    .cells(NextRow, "C").Value = UCase(Trim(Me.txtTicker.Value))
    .cells(NextRow, "D").Value = Me.cmbAction.Value
    .cells(NextRow, "E").Value = tShares
    .cells(NextRow, "F").Value = tPrice
    .cells(NextRow, "G").Value = tFee
    .cells(NextRow, "H").Value = tTax
    .cells(NextRow, "I").Value = tNetAmount
    .cells(NextRow, "J").Value = Me.cmbSector.Value
    .cells(NextRow, "K").Value = val(Me.txtTarget.Value)
    .cells(NextRow, "L").Value = Me.cmbStrategy.Value
    .cells(NextRow, "M").Value = val(Me.txtBeta.Value)
    .cells(NextRow, "N").Value = Me.cmbBroker.Value ' ★ 新增券商欄
End With

```

```
MsgBox "交易儲存成功！(買入成本已包含預估賣出費稅)", vbInformation
```

```

Me.txtTicker.Value = ""
Me.txtShares.Value = "0"
Me.txtPrice.Value = "0"
Me.txtFee.Value = "0"
Me.txtTax.Value = "0"
Me.txtTarget.Value = "0"
Me.txtBeta.Value = "0"
Me.cmbBroker.ListIndex = 0 ' ★ 重設為國泰世華
Me.txtTicker.SetFocus
End Sub

```

```
End Sub
```

```

' =====
' [按鈕] 關閉視窗
' =====

```

```

Private Sub btnClose_Click()
    Unload Me
End Sub

```

```

' =====
' [核心功能] 自動計算手續費與證交稅
' =====

```

```

Private Sub AutoCalculateCosts()
    On Error Resume Next

```

```

    Dim dShares As Double
    Dim dPrice As Double
    Dim dAmount As Double
    Dim dFee As Double

```

```

Dim dTax As Double
Dim sCurrency As String
Dim sAction As String
Dim sSector As String

```

```

dShares = val(Me.txtShares.Value)
dPrice = val(Me.txtPrice.Value)
sCurrency = Me.cmbCurrency.Value
sAction = Me.cmbAction.Value
sSector = Me.cmbSector.Value

```

```
dAmount = dShares * dPrice
```

```

If dAmount <= 0 Then
    Me.txtFee.Value = "0"
    Me.txtTax.Value = "0"
    Exit Sub
End If

```

```

If sCurrency = "TWD" Then
    ' ★ 依券商套用對應折扣率
    Dim dBrokerRate As Double
    If Me.cmbBroker.Value = "元大證券" Then
        dBrokerRate = 0.1
    Else
        dBrokerRate = 0.08 ' 國泰世華
    End If
    dFee = Int(dAmount * 0.001425 * dBrokerRate)
    If dFee < 1 Then dFee = 1

    If sAction = "Sell" Then
        If InStr(1, sSector, "ETF", vbTextCompare) > 0 Then
            dTax = Int(dAmount * 0.001)
        Else
            dTax = Int(dAmount * 0.003)
        End If
    Else
        dTax = 0
    End If

```

```

ElseIf sCurrency = "USD" Then
    If InStr(1, sSector, "ETF", vbTextCompare) > 0 Then
        dFee = 3
    Else
        dFee = Round(dAmount * 0.001, 2)
        ' ★ 元大美股個股低消 $1 USD
        If Me.cmbBroker.Value = "元大證券" Then
            If dFee < 1 Then dFee = 1
        End If
    End If
    dTax = 0
End If

```

```

Me.txtFee.Value = CStr(dFee)
Me.txtTax.Value = CStr(dTax)

```

```
End Sub
```

```

' [事件觸發] 當數值改變時自動計算

```

```

Private Sub txtShares_Change()
    Call AutoCalculateCosts
End Sub

```

```

Private Sub txtPrice_Change()
    Call AutoCalculateCosts
End Sub

```

```

Private Sub cmbAction_Change()
    Call AutoCalculateCosts
End Sub

```

```
Private Sub cmbSector_Change()  
    Call AutoCalculateCosts  
End Sub
```

```
Private Sub cmbCurrency_Change()  
    Call AutoCalculateCosts  
End Sub
```

' ★ 新增：券商變更時重新計算

```
Private Sub cmbBroker_Change()  
    Call AutoCalculateCosts  
End Sub
```

Option Explicit

```

' =====
' SHARED FUNCTIONS MODULE
' Used by: PortfolioDashboard_v3, UserForms, Correlation sheet
' =====

```

```

Public Type TechIndicators
    price           As Double
    ChangePercent   As Double
    Change180D      As Double
    DistFromHigh    As Double
    Bias5           As Double
    Bias20          As Double
    Bias60          As Double
    Bias120         As Double
    Bias240         As Double
    PERatio         As Double
    ForwardPE       As Double
    PEGRatio        As Double
    EPSNextQ        As Double
    EPSNextY        As Double
    Success         As Boolean
End Type

```

```

' =====
' PRICE CACHE (in-memory cache to avoid duplicate API calls)
' =====
Public g_PriceCache As Object
Public g_CacheTime As Date

```

```

Function GetCompanyName(ticker As String) As String
    Dim url As String, http As Object, response As String
    Dim nameStr As String, startPos As Long, endPos As Long

    ticker = UCase(Trim(ticker))
    If InStr(ticker, ".TW") = 0 And InStr(ticker, ".TWO") = 0 And IsNumeric(ticker) Then
        ticker = ticker & ".TW"
    End If

    url = "https://queryl.finance.yahoo.com/v8/finance/chart/" & ticker & "?interval=1d&range=1d"

    On Error Resume Next
    Set http = CreateObject("WinHttp.WinHttpRequest.5.1")
    If http Is Nothing Then Set http = CreateObject("MSXML2.XMLHTTP")
    On Error GoTo 0

    With http
        .Open "GET", url, False
        .setRequestHeader "User-Agent", "Mozilla/5.0"
        .send
        response = .responseText
    End With

    Dim searchTag As String
    searchTag = """"longName""": """"
    startPos = InStr(response, searchTag)

    If startPos > 0 Then
        startPos = startPos + Len(searchTag)
        endPos = InStr(startPos, response, """"")
        GetCompanyName = Mid(response, startPos, endPos - startPos)
    Else
        searchTag = """"shortName""": """"
        startPos = InStr(response, searchTag)
        If startPos > 0 Then
            startPos = startPos + Len(searchTag)
            endPos = InStr(startPos, response, """"")
            GetCompanyName = Mid(response, startPos, endPos - startPos)
        End If
    End If
End Function

```

```

Else
    GetCompanyName = ticker
End If
End If
Set http = Nothing
End Function

```

```

Function GetCurrencyType(ticker As String) As String
    ticker = UCase(Trim(ticker))
    If InStr(ticker, ".TW") > 0 Or InStr(ticker, ".TWO") > 0 Then
        GetCurrencyType = "TWD"
    Else
        GetCurrencyType = "USD"
    End If
End Function

```

```

' =====
' Technical Analysis
' =====

```

```

Function GetTechData(ticker As String) As TechIndicators
    Dim http As Object, response As String
    Dim result As TechIndicators
    Dim closeArr() As Double
    Dim i As Long, count As Long
    Dim currentPrice As Double, maxHigh As Double
    Dim rawTicker As String

    rawTicker = UCase(Trim(ticker))
    result.Success = False
    ticker = rawTicker

    If InStr(ticker, ".TW") = 0 And InStr(ticker, ".TWO") = 0 And IsNumeric(ticker) Then
        ticker = ticker & ".TW"
    End If

    On Error Resume Next
    Set http = CreateObject("WinHttp.WinHttpRequest.5.1")
    If http Is Nothing Then Set http = CreateObject("MSXML2.XMLHTTP")
    On Error GoTo 0

    response = FetchYahooData(http, ticker, "chart")

    If InStr(response, """"timestamp""") = 0 And IsNumeric(rawTicker) Then
        ticker = rawTicker & ".TWO"
        response = FetchYahooData(http, ticker, "chart")
    End If

    Dim closeStart As Long, closeEnd As Long, closeStr As String, rawArr() As String
    Dim qPos As Long
    qPos = InStr(response, """"quote"":[")
    If qPos = 0 Then qPos = 1

    closeStart = InStr(qPos, response, """"close"":[") + 9
    closeEnd = InStr(closeStart, response, "]")

    If closeStart > 8 And closeEnd > closeStart Then
        closeStr = Mid(response, closeStart, closeEnd - closeStart)
        rawArr = Split(closeStr, ",")
        ReDim closeArr(0 To UBound(rawArr))
        count = 0: maxHigh = 0

        For i = 0 To UBound(rawArr)
            If IsNumeric(rawArr(i)) Then
                closeArr(count) = val(rawArr(i))
                If closeArr(count) > maxHigh Then maxHigh = closeArr(count)
                count = count + 1
            End If
        Next i

        If count > 0 Then

```

```

ReDim Preserve closeArr(0 To count - 1)
currentPrice = closeArr(count - 1)

' -----
Dim metaPos As Long: metaPos = InStr(response, ""regularMarketPrice"":")
If metaPos > 0 Then
    Dim vals As Long: vals = metaPos + 21
    Dim valE As Long: valE = InStr(vals, response, ",")
    If valE = 0 Then valE = InStr(vals, response, "}")
    If valE > vals Then
        Dim metaVal As String: metaVal = Trim(Mid(response, vals, valE - vals))
        If IsNumeric(metaVal) And val(metaVal) > 0 Then
            Dim metaPx As Double: metaPx = val(metaVal)
            ' -----
            If metaPx <> currentPrice Then
                currentPrice = metaPx
                closeArr(count - 1) = currentPrice
                If currentPrice > maxHigh Then maxHigh = currentPrice
            End If
        End If
    End If
End If
' -----

result.price = currentPrice

If count >= 2 Then
    Dim prev As Double: prev = closeArr(count - 2)
    If prev > 0 Then result.ChangePercent = (currentPrice - prev) / prev
End If

Dim idx180 As Long: idx180 = count - 126
If idx180 >= LBound(closeArr) And idx180 <= UBound(closeArr) Then
    If closeArr(idx180) > 0 Then
        result.Change180D = (currentPrice - closeArr(idx180)) / closeArr(idx180)
    End If
End If

If maxHigh > 0 Then result.DistFromHigh = (currentPrice - maxHigh) / maxHigh

result.Bias5 = CalculateBias(closeArr, 5, currentPrice)
result.Bias20 = CalculateBias(closeArr, 20, currentPrice)
result.Bias60 = CalculateBias(closeArr, 60, currentPrice)
result.Bias120 = CalculateBias(closeArr, 120, currentPrice)
result.Bias240 = CalculateBias(closeArr, 240, currentPrice)
result.Success = True

```

```

Call GetFundamentalData(http, ticker, result)

```

```

End If

```

```

End If

```

```

GetTechData = result

```

```

End Function

```

```

' =====
' GET STOCK PRICE (with cache)
' =====

```

```

Public Function GetStockPrice(ticker As String) As Double
    ticker = UCase(Trim(ticker))

```

```

If g_PriceCache Is Nothing Then

```

```

    Set g_PriceCache = CreateObject("Scripting.Dictionary")

```

```

    g_CacheTime = Now

```

```

End If

```

```

If Now - g_CacheTime > TimeSerial(0, 5, 0) Then

```

```

    g_PriceCache.RemoveAll

```

```

    g_CacheTime = Now

```

```

End If

```

```

If g_PriceCache.Exists(ticker) Then

```

```

    GetStockPrice = g_PriceCache(ticker)

```

```

Exit Function

```

```

End If

Dim fullTicker As String: fullTicker = ticker
If InStr(fullTicker, ".TW") = 0 And InStr(fullTicker, ".TWO") = 0 And IsNumeric(fullTicker) Then
    fullTicker = fullTicker & ".TW"
End If

Dim http As Object
On Error Resume Next
Set http = CreateObject("WinHttp.WinHttpRequest.5.1")
If http Is Nothing Then Set http = CreateObject("MSXML2.XMLHTTP")
On Error GoTo 0

Dim safeTicker As String: safeTicker = Replace(fullTicker, "^", "%5E")
Dim url As String
Dim resp As String: resp = ""
url = "https://query1.finance.yahoo.com/v8/finance/chart/" & safeTicker & "?interval=1d&range=5d"

On Error Resume Next
With http
    .Open "GET", url, False
    .setRequestHeader "User-Agent", "Mozilla/5.0"
    .setRequestHeader "Accept", "application/json"
    .send
    resp = .responseText
End With
On Error GoTo 0

Dim px As Double: px = 0
px = ParseClosePrice(resp)

' .TWO fallback
If px = 0 And IsNumeric(ticker) And InStr(fullTicker, ".TWO") = 0 Then
    url = "https://query1.finance.yahoo.com/v8/finance/chart/" & ticker & ".TWO?interval=1d&range=5d"
    On Error Resume Next
    With http
        .Open "GET", url, False
        .setRequestHeader "User-Agent", "Mozilla/5.0"
        .send
        resp = .responseText
    End With
    On Error GoTo 0
    px = ParseClosePrice(resp)
End If

If px > 0 Then g_PriceCache(ticker) = px
GetStockPrice = px
Set http = Nothing
End Function

```

```

' -----
Private Function ParseClosePrice(resp As String) As Double
    ParseClosePrice = 0

```

```

' -----
Dim metaPos As Long: metaPos = InStr(resp, """"regularMarketPrice"":")
If metaPos > 0 Then
    Dim vals As Long: vals = metaPos + 21 ' -- W Len("""regularMarketPrice"":")
    Dim valE As Long: valE = InStr(vals, resp, ",")
    If valE = 0 Then valE = InStr(vals, resp, "}") ' -- 内 -- b

    If valE > vals Then
        Dim metaVal As String: metaVal = Trim(Mid(resp, vals, valE - vals))
        If IsNumeric(metaVal) And val(metaVal) > 0 Then
            ParseClosePrice = val(metaVal)
            Exit Function ' -- \ -- o -- 抑s -- A -- _ -- }
        End If
    End If
End If
End If

```

```

' -----

```

```

Dim csPos As Long: csPos = InStr(resp, """"close"":[")
If csPos = 0 Then Exit Function
Dim csS As Long: csS = csPos + 9
Dim csE As Long: csE = InStr(csS, resp, "]")
If csE <= csS Then Exit Function

Dim parts() As String: parts = Split(Mid(resp, csS, csE - csS), ",")
Dim i As Long
For i = UBound(parts) To 0 Step -1
    Dim pt As String: pt = Trim(parts(i))
    If IsNumeric(pt) Then
        If CDBl(pt) > 0 Then
            ParseClosePrice = CDBl(pt)
            Exit Function
        End If
    End If
Next i
End Function

Sub GetFundamentalData(http As Object, ticker As String, ByRef res As TechIndicators)
    Dim response As String
    response = FetchYahooData(http, ticker, "quoteSummary")
    If Len(response) < 100 Then Exit Sub

    response = Replace(Replace(Replace(Replace(response, " ", ""), vbCr, ""), vbLf, ""), vbTab, "")

    res.PERatio = JsonExtract(response, "trailingPE")
    res.ForwardPE = JsonExtract(response, "forwardPE")
    res.PEGRatio = JsonExtract(response, "pegRatio")
    res.EPSNextQ = JsonExtractTrend(response, 1)
    res.EPSNextY = JsonExtractTrend(response, 3)

    Dim apiPrice As Double: apiPrice = JsonExtract(response, "regularMarketPrice")
    Dim apiPrev As Double: apiPrev = JsonExtract(response, "regularMarketPreviousClose")
    If apiPrice > 0 And apiPrev > 0 Then
        res.ChangePercent = (apiPrice - apiPrev) / apiPrev
    End If
End Sub

Public Function FetchYahooData(http As Object, ticker As String, mode As String) As String
    Dim url As String
    If mode = "chart" Then
        url = "https://query1.finance.yahoo.com/v8/finance/chart/" & ticker & "?interval=1d&range=2y"
    Else
        url = "https://query1.finance.yahoo.com/v10/finance/quoteSummary/" & ticker & _
            "?modules=summaryDetail,defaultKeyStatistics,earningsTrend,price"
    End If
    On Error Resume Next
    With http
        .Open "GET", url, False
        .setRequestHeader "User-Agent", "Mozilla/5.0"
        .send
        FetchYahooData = .responseText
    End With
    On Error GoTo 0
End Function

Function CalculateBias(priceArr() As Double, period As Long, currentPrice As Double) As Double
    If UBound(priceArr) + 1 < period Then CalculateBias = 0: Exit Function
    Dim k As Double: k = 2 / (period + 1)
    Dim ema As Double: ema = priceArr(0)
    Dim i As Long
    For i = 1 To UBound(priceArr)
        ema = (priceArr(i) - ema) * k + ema
    Next i
    If ema > 0 Then CalculateBias = (currentPrice - ema) / ema Else CalculateBias = 0
End Function

Function JsonExtract(json As String, key As String) As Double
    Dim searchStr As String: searchStr = """" & key & """:{""raw"":""
    Dim pos As Long: pos = InStr(json, searchStr)
    If pos = 0 Then JsonExtract = 0: Exit Function

```

```

Dim valStart As Long: valStart = pos + Len(searchStr)
Dim valEnd As Long: valEnd = InStr(valStart, json, ",")
If valEnd = 0 Then valEnd = InStr(valStart, json, "}")
If valEnd > valStart Then
    Dim tmp As String: tmp = Mid(json, valStart, valEnd - valStart)
    If IsNumeric(tmp) Then JsonExtract = val(tmp)
End If
End Function

```

```

Function JsonExtractTrend(json As String, blockIndex As Long) As Double
    Dim pos As Long: pos = 1
    Dim i As Long
    For i = 0 To blockIndex
        pos = InStr(pos, json, ""earningsEstimate":")
        If pos = 0 Then JsonExtractTrend = 0: Exit Function
        pos = pos + Len(""earningsEstimate":")
    Next i
    Dim searchStr As String: searchStr = ""avg"":{"raw":
    pos = InStr(pos, json, searchStr)
    If pos > 0 Then
        Dim valStart As Long: valStart = pos + Len(searchStr)
        Dim valEnd As Long: valEnd = InStr(valStart, json, ",")
        If valEnd = 0 Then valEnd = InStr(valStart, json, "}")
        If valEnd > valStart Then
            Dim tmp As String: tmp = Mid(json, valStart, valEnd - valStart)
            If IsNumeric(tmp) Then JsonExtractTrend = val(tmp)
        End If
    End If
End Function

```

```

' =====
' Beta Calculation
' =====

```

```

Function GetBenchmarkTicker(Sector As String, stockTicker As String) As String
    If InStr(UCase(stockTicker), ".TW") > 0 Or InStr(UCase(stockTicker), ".TWO") > 0 Then
        GetBenchmarkTicker = "^TWII": Exit Function
    End If
    Select Case Sector
        Case ChrW(&H8CC7) & ChrW(&H8A0A) & ChrW(&H79D1) & ChrW(&H6280) & " (Information Technology)": GetBenchmarkTicker = "^IXIC"
        Case ChrW(&H5DE5) & ChrW(&H696D) & " (Industrials)": GetBenchmarkTicker = "^DJI"
        Case ChrW(&H91D1) & ChrW(&H878D) & " (Financials)": GetBenchmarkTicker = "XLF"
        Case ChrW(&H91AB) & ChrW(&H7642) & ChrW(&H4FDD) & ChrW(&H5065) & " (Health Care)": GetBenchmarkTicker = "XLV"
        Case ChrW(&H80FD) & ChrW(&H6E90) & " (Energy)": GetBenchmarkTicker = "XLE"
        Case Else: GetBenchmarkTicker = "^GSPC"
    End Select
End Function

```

```

Function CalculateBeta30D(stockTicker As String, marketTicker As String) As Double
    Dim stockPrices As Object, marketPrices As Object
    Set stockPrices = GetHistoryPrices(stockTicker)
    Set marketPrices = GetHistoryPrices(marketTicker)

    If stockPrices.count < 5 Or marketPrices.count < 5 Then
        CalculateBeta30D = -999: Exit Function
    End If

```

```

' Only use the last 30 trading days

```

```

Dim keys As Variant: keys = stockPrices.keys
Dim totalKeys As Long: totalKeys = UBound(keys) + 1
Dim startIdx As Long: startIdx = IIf(totalKeys > 30, totalKeys - 30, 0)

```

```

Dim arrStock() As Double
Dim arrMarket() As Double
ReDim arrStock(1 To totalKeys)
ReDim arrMarket(1 To totalKeys)

```

```

Dim validPairs As Long
Dim prevStock As Double, prevMarket As Double
Dim i As Long, dateKey As Variant

```

```

For i = startIdx To UBound(keys)
    dateKey = keys(i)
    If marketPrices.Exists(dateKey) Then
        Dim currStock As Double: currStock = stockPrices(dateKey)
        Dim currMarket As Double: currMarket = marketPrices(dateKey)
        If prevStock > 0 Then
            validPairs = validPairs + 1
            arrStock(validPairs) = (currStock - prevStock) / prevStock
            arrMarket(validPairs) = (currMarket - prevMarket) / prevMarket
        End If
        prevStock = currStock: prevMarket = currMarket
    End If
Next i

If validPairs >= 10 Then
    Dim cleanS() As Double, cleanM() As Double
    ReDim cleanS(1 To validPairs): ReDim cleanM(1 To validPairs)
    For i = 1 To validPairs
        cleanS(i) = arrStock(i): cleanM(i) = arrMarket(i)
    Next i
    On Error Resume Next
    CalculateBeta30D = Application.WorksheetFunction.Slope(cleanS, cleanM)
    If Err.Number <> 0 Then CalculateBeta30D = -999
    On Error GoTo 0
Else
    CalculateBeta30D = -999
End If
End Function

Function GetHistoryPrices(ticker As String) As Object
    Dim dict As Object: Set dict = CreateObject("Scripting.Dictionary")
    ticker = UCase(Trim(ticker))
    If InStr(ticker, ".TW") = 0 And InStr(ticker, ".TWO") = 0 And IsNumeric(ticker) Then
        ticker = ticker & ".TW"
    End If

    Dim http As Object, response As String
    On Error Resume Next
    Set http = CreateObject("WinHttp.WinHttpRequest.5.1")
    If http Is Nothing Then Set http = CreateObject("MSXML2.XMLHTTP")
    With http
        .Open "GET", "https://query1.finance.yahoo.com/v8/finance/chart/" & ticker & "?range=1y&interval=1d", False
        .setRequestHeader "User-Agent", "Mozilla/5.0"
        .send
        response = .responseText
    End With
    On Error GoTo 0

    Dim tsStart As Long: tsStart = InStr(response, ""timestamp"":[") + 13
    Dim tsEnd As Long: tsEnd = InStr(tsStart, response, "]")
    Dim csStart As Long: csStart = InStr(response, ""adjclose"":[") + 12
    Dim csEnd As Long: csEnd = InStr(csStart, response, "]")

    If tsStart > 12 And tsEnd > tsStart And csStart > 11 And csEnd > csStart Then
        Dim tsArr() As String: tsArr = Split(Mid(response, tsStart, tsEnd - tsStart), ",")
        Dim closeArr() As String: closeArr = Split(Mid(response, csStart, csEnd - csStart), ",")
        If UBound(tsArr) = UBound(closeArr) Then
            Dim i As Long
            For i = 0 To UBound(tsArr)
                If IsNumeric(tsArr(i)) And IsNumeric(closeArr(i)) Then
                    Dim dv As Date: dv = Int(CLng(tsArr(i)) / 86400 + 25569)
                    If Not dict.Exists(dv) Then dict.Add dv, CDbl(closeArr(i))
                End If
            Next i
        End If
    End If
    Set GetHistoryPrices = dict
End Function

Sub CalculateRealizedPnL()
    Dim wsTrans As Worksheet, wsReal As Worksheet, wsPort As Worksheet
    Set wsTrans = ThisWorkbook.Sheets("Transactions")
    Set wsReal = ThisWorkbook.Sheets("Realized")

```

```

Set wsPort = ThisWorkbook.Sheets("RR4")
Application.ScreenUpdating = False
wsReal.Range("A2:I10000").ClearContents

Dim hdrs As Variant
hdrs = Array("TICKER", "PNL", "SHARES", "AVG COST", "NET AMT", "STRATEGY", "PNL(TWD)", "DATE", "BROKER")
Dim hc As Integer
For hc = 0 To 8
    wsReal.Cells(1, hc + 1).Value = hdrs(hc)
    wsReal.Cells(1, hc + 1).Font.Bold = True
Next hc

' ===== FIFO lot queue per posKey =====
Dim dictLots As Object: Set dictLots = CreateObject("Scripting.Dictionary")

Dim exRate As Double
On Error Resume Next: exRate = val(wsPort.Range("B2").Value): On Error GoTo 0
If exRate <= 0 Then exRate = 32

Dim lastRow As Long: lastRow = wsTrans.Cells(wsTrans.Rows.Count, "A").End(xlUp).row
Dim writeRow As Long: writeRow = 2
Dim rr As Long

For rr = 2 To lastRow
    Dim tDate As Variant: tDate = wsTrans.Cells(rr, 2).Value
    Dim ticker As String: ticker = UCase(Trim(wsTrans.Cells(rr, 3).Value))
    Dim action As String: action = UCase(Trim(wsTrans.Cells(rr, 4).Value))
    Dim shares As Double: shares = val(wsTrans.Cells(rr, 5).Value)
    Dim NetAmount As Double: NetAmount = val(wsTrans.Cells(rr, 9).Value)
    Dim strat As String: strat = CStr(wsTrans.Cells(rr, 12).Value)
    Dim broker As String: broker = Trim(CStr(wsTrans.Cells(rr, 14).Value))
    If broker = "" Then broker = "Default"

    Dim currType As String
    If InStr(ticker, ".TW") > 0 Or InStr(ticker, ".TWO") > 0 Then
        currType = "TWD"
    Else
        currType = "USD"
    End If

    If ticker <> "" And shares > 0 Then
        Dim posKey As String: posKey = ticker & "!" & broker

        If Not dictLots.Exists(posKey) Then
            Dim newCol As Collection
            Set newCol = New Collection
            dictLots.Add posKey, newCol
        End If

        Select Case action
            Case "BUY"
                Dim CostPerShare As Double
                CostPerShare = NetAmount / shares
                dictLots(posKey).Add Array(shares, CostPerShare)

            Case "ADJUSTCOST"
                Dim totalSh As Double: totalSh = 0
                Dim ll As Long
                For ll = 1 To dictLots(posKey).count
                    totalSh = totalSh + dictLots(posKey)(ll)(0)
                Next ll
                If totalSh > 0 Then
                    Dim adjPerSh As Double: adjPerSh = NetAmount / totalSh
                    For ll = 1 To dictLots(posKey).count
                        Dim lotArr As Variant: lotArr = dictLots(posKey)(ll)
                        lotArr(1) = lotArr(1) + adjPerSh
                        If ll < dictLots(posKey).count Then
                            dictLots(posKey).Add lotArr, Before:=ll + 1
                            dictLots(posKey).Remove ll
                        Else
                            dictLots(posKey).Add lotArr
                        End If
                    Next ll
                End If
            End Select
        End If
    End If
Next rr

```

```

        dictLots(posKey).Remove 11
    End If
Next 11
End If

Case "SELL"
    Dim remaining As Double: remaining = shares
    Dim totalCostBasis As Double: totalCostBasis = 0

    Do While remaining > 0.00001 And dictLots(posKey).count > 0
        Dim frontLot As Variant: frontLot = dictLots(posKey)(1)
        Dim lotSh As Double: lotSh = frontLot(0)
        Dim lotCps As Double: lotCps = frontLot(1)

        If lotSh <= remaining + 0.00001 Then
            totalCostBasis = totalCostBasis + lotSh * lotCps
            remaining = remaining - lotSh
            dictLots(posKey).Remove 1
        Else
            totalCostBasis = totalCostBasis + remaining * lotCps
            Dim updatedLot As Variant
            updatedLot = Array(lotSh - remaining, lotCps)
            If dictLots(posKey).count > 1 Then
                dictLots(posKey).Add updatedLot, Before:=2
            Else
                dictLots(posKey).Add updatedLot
            End If
            dictLots(posKey).Remove 1
            remaining = 0
        End If
    Loop

    Dim fifoAvgC As Double
    If shares > 0 Then fifoAvgC = totalCostBasis / shares Else fifoAvgC = 0
    Dim r1Pnl As Double: r1Pnl = NetAmount - totalCostBasis
    Dim r1TWD As Double: r1TWD = IIf(currType = "USD", r1Pnl * exRate, r1Pnl)

    Dim finalDate As Date
    If IsDate(tDate) Then finalDate = CDate(tDate) Else finalDate = Date

    With wsReal
        .cells(writeRow, 1) = ticker
        .cells(writeRow, 2) = r1Pnl: .cells(writeRow, 2).NumberFormat = "#,##0.00"
        .cells(writeRow, 3) = shares
        .cells(writeRow, 4) = fifoAvgC
        .cells(writeRow, 5) = NetAmount
        .cells(writeRow, 6) = strat
        .cells(writeRow, 7) = r1TWD: .cells(writeRow, 7).NumberFormat = "#,##0"
        .cells(writeRow, 8) = finalDate: .cells(writeRow, 8).NumberFormat = "yyyy/m/d"
        .cells(writeRow, 9) = broker
        If r1TWD > 0 Then
            .cells(writeRow, 2).Font.Color = RGB(255, 100, 100)
            .cells(writeRow, 7).Font.Color = RGB(255, 100, 100)
        Else
            .cells(writeRow, 2).Font.Color = RGB(0, 255, 100)
            .cells(writeRow, 7).Font.Color = RGB(0, 255, 100)
        End If
    End With
    writeRow = writeRow + 1

End Select
End If
Next rr

' ===== Update RR4 Entry PX =====
Dim portLastRow As Long: portLastRow = wsPort.cells(wsPort.Rows.count, "A").End(xlUp).row
Dim pr As Long
For pr = 5 To portLastRow
    Dim pTicker As String: pTicker = UCase(Trim(wsPort.cells(pr, 1).Value))
    Dim pBroker As String: pBroker = Trim(CStr(wsPort.cells(pr, 13).Value))
    If pBroker = "" Then pBroker = "Default"

```

```

Dim pKey As String: pKey = pTicker & "I" & pBroker

If dictLots.Exists(pKey) And pTicker <> "" Then
    Dim totSh As Double: totSh = 0
    Dim totCost As Double: totCost = 0
    For ll = 1 To dictLots(pKey).count
        totSh = totSh + dictLots(pKey)(ll)(0)
        totCost = totCost + dictLots(pKey)(ll)(0) * dictLots(pKey)(ll)(1)
    Next ll
    If totSh > 0.00001 Then
        wsPort.cells(pr, 9).Value = totCost / totSh
    End If
End If
Next pr

wsReal.Columns("A:I").AutoFit
Application.ScreenUpdating = True
End Sub

Sub ClearAllData()
    If MsgBox("Clear ALL data? This cannot be undone!", vbYesNo + vbExclamation + vbDefaultButton2) = vbNo Then Exit Sub

    Dim wsTrans As Worksheet, wsPort As Worksheet, wsReal As Worksheet, wsHist As Worksheet
    On Error Resume Next
    Set wsTrans = ThisWorkbook.Sheets("Transactions")
    Set wsPort = ThisWorkbook.Sheets("RR4")
    Set wsReal = ThisWorkbook.Sheets("Realized")
    Set wsHist = ThisWorkbook.Sheets("HistoryLog")
    On Error GoTo 0

    Dim LR As Long
    If Not wsTrans Is Nothing Then
        LR = wsTrans.Cells(wsTrans.Rows.count, "A").End(xlUp).row
        If LR > 1 Then wsTrans.Range("A2:N" & LR).ClearContents ' -- M -- N
    End If
    If Not wsReal Is Nothing Then
        LR = wsReal.Cells(wsReal.Rows.count, "A").End(xlUp).row
        If LR > 1 Then wsReal.Range("A2:I" & LR).ClearContents ' -- H -- I
    End If
    If Not wsHist Is Nothing Then
        LR = wsHist.Cells(wsHist.Rows.count, "A").End(xlUp).row
        If LR > 1 Then wsHist.Range("A2:G" & LR).ClearContents
    End If

    MsgBox "All data cleared.", vbInformation
End Sub

' =====
' Form Launchers
' =====

Sub OpenTransactionForm()
    frmTransaction.Show
End Sub

Sub OpenDeleteForm()
    frmDeleteTransaction.Show
End Sub

' =====
' Realized Sheet Filter UI
' =====

Sub InitFilterInterface()
    Dim ws As Worksheet: Set ws = ThisWorkbook.Sheets("Realized")
    On Error Resume Next: ws.Unprotect: On Error GoTo 0
    ws.AutoFilterMode = False
    With ws.Cells
        .Interior.Color = RGB(0, 0, 0)
        .Font.Color = RGB(255, 255, 255)
        .Font.Name = "Calibri"
        .Font.Size = 12
    End With
    With ws.Range("J1:L20")
        .UnMerge: .ClearContents: .ColumnWidth = 12
    End With

```

```

End With
With ws
    .Range("J1:L6").Interior.Color = RGB(30, 30, 30)
    .Range("J1:L1").Merge
    .Range("J1").HorizontalAlignment = xlCenter
    .Range("J1").Font.Bold = True
    .Range("J1").Font.Color = RGB(255, 215, 0)
    .Range("J2").Value = "Start Date:"
    .Range("J3").Value = "End Date:"
    .Range("K2").Value = DateSerial(Year(Date), 1, 1)
    .Range("K3").Value = Date
    With .Range("K2:K3")
        .NumberFormat = "yyyy/m/d"
        .Interior.Color = RGB(60, 60, 60)
        .Borders.LineStyle = xlContinuous
        .Font.Color = RGB(255, 255, 255)
        .Font.Bold = True
    End With
    .Range("J5").Value = "Period P&L:"
    .Range("J5").Font.Bold = True
    .Range("K5").NumberFormat = "$#,##0"
    .Range("K5").Font.Size = 14
    .Range("K5").Font.Bold = True
    .Range("L2").Value = "(e.g. 2025/1/1)"
    .Range("L2").Font.Color = RGB(150, 150, 150)
    .Range("L2").Font.Size = 10
End With
End Sub

Sub FilterRealizedData()
    Dim ws As Worksheet: Set ws = ThisWorkbook.Sheets("Realized")
    Dim LR As Long: LR = ws.Cells(ws.Rows.Count, "A").End(xlUp).row
    If LR < 2 Then MsgBox "No data to filter", vbInformation: Exit Sub

    Dim sd As Date, ed As Date
    On Error Resume Next
    If IsDate(ws.Range("K2").Value) Then sd = CDate(ws.Range("K2").Value)
    If IsDate(ws.Range("K3").Value) Then ed = CDate(ws.Range("K3").Value)
    On Error GoTo 0
    If sd = 0 Or ed = 0 Then MsgBox "Enter valid dates in K2 and K3", vbExclamation: Exit Sub

    ws.AutoFilterMode = False
    ws.Range("A1:I" & LR).AutoFilter Field:=8, Criteria1:=">=" & sd, Operator:=xlAnd, Criteria2:="<=" & ed

    Dim PnL As Double
    On Error Resume Next
    PnL = Application.WorksheetFunction.Subtotal(109, ws.Range("G2:G" & LR))
    On Error GoTo 0

    ws.Range("K5").Value = PnL
    ws.Range("K5").Font.Color = IIf(PnL >= 0, RGB(255, 100, 100), RGB(0, 255, 100))
    MsgBox "Filter applied!", vbInformation
End Sub

Sub ClearRealizedFilter()
    Dim ws As Worksheet: Set ws = ThisWorkbook.Sheets("Realized")
    ws.AutoFilterMode = False
    Dim LR As Long: LR = ws.Cells(ws.Rows.Count, "A").End(xlUp).row
    Dim total As Double
    If LR >= 2 Then total = Application.WorksheetFunction.Sum(ws.Range("G2:G" & LR))
    ws.Range("K5").Value = total
    ws.Range("K5").Font.Color = IIf(total >= 0, RGB(255, 100, 100), RGB(0, 255, 100))
End Sub

' =====
' GitHub Coverage Report (kept as-is)
' =====

Sub QueryCoverageReport()
    Dim ticker As String
    ticker = InputBox("Enter TW stock code (e.g. 2330):", "Coverage Report")
    ticker = Trim(ticker)

```

```

    If ticker = "" Then Exit Sub
    Dim pureNum As String
    pureNum = Replace(Replace(UCase(ticker), ".TWO", ""), ".TW", "")
    If Not IsNumeric(pureNum) Then MsgBox "Numeric code only (e.g. 2330)", vbExclamation: Exit Sub

    Application.StatusBar = "Searching " & pureNum & "..."
    Application.ScreenUpdating = False

    Dim filePath As String: filePath = SearchGitHubFile(pureNum)
    If filePath = "" Then
        Application.StatusBar = False: Application.ScreenUpdating = True
        MsgBox "Report not found for " & pureNum, vbExclamation: Exit Sub
    End If

    Dim content As String: content = FetchRawContent(filePath)
    If content = "" Then
        Application.StatusBar = False: Application.ScreenUpdating = True
        MsgBox "Failed to load report.", vbExclamation: Exit Sub
    End If

    Call RenderReportToSheet(content, pureNum)
    Application.StatusBar = pureNum & " loaded."
    Application.ScreenUpdating = True
End Sub

Function EncodeURL(text As String) As String
    Dim i As Long, result As String, c As String
    For i = 1 To Len(text)
        c = Mid(text, i, 1)
        Select Case c
            Case "A" To "Z", "a" To "z", "0" To "9", "-", "_", ".", "/", "~"
                result = result & c
            Case Else
                Dim bytes() As Byte: bytes = StrConv(c, vbFromUnicode)
                Dim b As Long
                For b = 0 To UBound(bytes)
                    result = result & "%" & UCase(Hex(bytes(b)))
                Next b
            End Select
        Next i
    EncodeURL = result
End Function

Function SearchGitHubFile(tickerNum As String) As String
    Dim http As Object, response As String
    Set http = CreateObject("WinHttp.WinHttpRequest.5.1")
    Dim ghToken As String: ghToken = GetGitHubToken()
    On Error Resume Next
    With http
        .Open "GET", "https://api.github.com/repos/Timeverse/My-TW-Coverage/git/trees/master?recursive=1", False
        .setRequestHeader "User-Agent", "VBA-Excel-Client"
        .setRequestHeader "Accept", "application/vnd.github.v3+json"
        If ghToken <> "" Then .setRequestHeader "Authorization", "token " & ghToken
        .send
        response = .responseText
    End With
    On Error GoTo 0
    If Len(response) < 100 Then SearchGitHubFile = "": Exit Function

    Dim pat As String: pat = """"path"":""Pilot_Reports"""
    Dim pos As Long: pos = 1
    Do
        pos = InStr(pos, response, pat)
        If pos = 0 Then SearchGitHubFile = "": Exit Function
        Dim vs As Long: vs = pos + 8
        Dim ve As Long: ve = InStr(vs, response, """"")
        If ve > vs Then
            Dim pv As String: pv = Mid(response, vs, ve - vs)
            If InStr(pv, "/" & tickerNum & "_") > 0 Then
                SearchGitHubFile = pv: Exit Function
            End If
        End If
    Loop

```

```

        End If
        pos = pos + Len(pat)
    Loop
End Function

Function GetGitHubToken() As String
    On Error Resume Next
    GetGitHubToken = ThisWorkbook.Sheets("HistoryLog").Range("Z3").Value
    On Error GoTo 0
End Function

Sub SetGitHubToken()
    Dim token As String
    token = InputBox("Paste GitHub Token (ghp_...):", "Set Token")
    If token = "" Then Exit Sub
    ThisWorkbook.Sheets("HistoryLog").Range("Z3").Value = token
    ThisWorkbook.Sheets("HistoryLog").Range("Z3").Font.Color = RGB(0, 0, 0)
    MsgBox "Token saved!", vbInformation
End Sub

Function FetchRawContent(filePath As String) As String
    Dim http As Object, ghToken As String
    Set http = CreateObject("WinHttp.WinHttpRequest.5.1")
    ghToken = GetGitHubToken()
    Dim rawUrl As String
    rawUrl = "https://raw.githubusercontent.com/Timeverse/My-TW-Coverage/master/" & EncodeURL(filePath)
    On Error Resume Next
    With http
        .Open "GET", rawUrl, False
        .setRequestHeader "User-Agent", "VBA-Excel-Client"
        If ghToken <> "" Then .setRequestHeader "Authorization", "token " & ghToken
        .send
        If .status = 200 Then FetchRawContent = .responseText Else FetchRawContent = ""
    End With
    On Error GoTo 0
End Function

Sub RenderReportToSheet(content As String, tickerNum As String)
    Dim ws As Worksheet
    On Error Resume Next: Set ws = ThisWorkbook.Sheets("Coverage"): On Error GoTo 0
    If ws Is Nothing Then
        Set ws = ThisWorkbook.Sheets.Add(After:=ThisWorkbook.Sheets(ThisWorkbook.Sheets.count))
        ws.Name = "Coverage"
    End If
    With ws
        .cells.ClearContents: .cells.ClearFormats
        .cells.Interior.Color = RGB(10, 12, 18)
        .cells.Font.Color = RGB(210, 210, 210)
        .cells.Font.Name = "Calibri"
        .cells.Font.Size = 11
        .Columns("A").ColumnWidth = 5
        .Columns("B").ColumnWidth = 100
        .Rows.RowHeight = 16
    End With

    Dim lines() As String: lines = Split(content, vbLf)
    Dim writeRow As Long: writeRow = 2
    Dim i As Long
    For i = 0 To UBound(lines)
        Dim lt As String: lt = Replace(lines(i), vbCr, "")
        If lt = "" Then writeRow = writeRow + 1: GoTo NL
        With ws.cells(writeRow, "B")
            Dim dt As String: dt = lt
            If Left(lt, 2) = "# " Then
                dt = CleanWikilinks(Mid(lt, 3))
                .Value = dt: .Font.Size = 18: .Font.Bold = True
                .Font.Color = RGB(255, 215, 0): .Interior.Color = RGB(25, 25, 35)
                writeRow = writeRow + 1
            ElseIf Left(lt, 3) = "## " Then
                dt = CleanWikilinks(Mid(lt, 4))
                .Value = "## " & dt: .Font.Size = 13: .Font.Bold = True
            End If
        End With
    Next i
    NL:

```

```

        .Font.Color = RGB(80, 180, 255): .Interior.Color = RGB(20, 22, 32)
    ElseIf Left(lt, 4) = "### " Then
        dt = CleanWikilinks(Mid(lt, 5))
        .Value = " ### " & dt: .Font.Size = 12: .Font.Bold = True
        .Font.Color = RGB(120, 230, 160)
    ElseIf Left(lt, 2) = "- " Or Left(lt, 2) = "* " Then
        .Value = " * " & CleanWikilinks(Mid(lt, 3))
        .Font.Color = RGB(200, 200, 200)
    ElseIf Left(lt, 1) = "|" Then
        .Value = CleanWikilinks(lt): .Font.Name = "Courier New"
        .Font.Size = 10: .Font.Color = RGB(180, 220, 180)
        .Interior.Color = RGB(18, 22, 28)
    ElseIf Left(lt, 3) = "```" Then
        GoTo NL
    Else
        .Value = RemoveBoldMarkers(CleanWikilinks(lt))
        .Font.Color = RGB(200, 200, 200)
    End If
End With
writeRow = writeRow + 1
NL:
Next i
ws.Activate: ws.cells(1, 1).Select: ActiveWindow.ScrollRow = 1
End Sub

```

```

Function CleanWikilinks(text As String) As String
    Dim r As String: r = text
    Do While InStr(r, "[[") > 0
        Dim s As Long: s = InStr(r, "[[")
        Dim e As Long: e = InStr(s, r, "]]")
        If e > s Then
            r = Left(r, s - 1) & Mid(r, s + 2, e - s - 2) & Mid(r, e + 2)
        Else
            Exit Do
        End If
    Loop
    CleanWikilinks = r
End Function

```

```

Function RemoveBoldMarkers(text As String) As String
    RemoveBoldMarkers = Replace(text, "***", "")
End Function

```

```

' =====
' Array Helpers
' =====

```

```

Function CollectionToArray(col As Collection) As Variant
    If col.count = 0 Then CollectionToArray = Empty: Exit Function
    Dim arr() As Variant: ReDim arr(0 To col.count - 1)
    Dim i As Long
    For i = 0 To col.count - 1: arr(i) = col(i + 1): Next i
    CollectionToArray = arr
End Function

```

```

Sub SortArray(ByRef arr As Variant)
    If IsEmpty(arr) Or Not IsArray(arr) Then Exit Sub
    If UBound(arr) - LBound(arr) < 1 Then Exit Sub
    Dim i As Long, j As Long, tmp As Variant
    For i = LBound(arr) To UBound(arr) - 1
        For j = i + 1 To UBound(arr)
            If arr(i) > arr(j) Then tmp = arr(i): arr(i) = arr(j): arr(j) = tmp
        Next j
    Next i
End Sub

```

```

' =====
' DEBUG SYSTEM -- -- g -- J "DebugLog" -- u -- @ --
' =====

```

```

Sub RunSystemDebug()
    Application.ScreenUpdating = False

```

```

' -----
Dim wsDB As Worksheet
On Error Resume Next: Set wsDB = ThisWorkbook.Sheets("DebugLog"): On Error GoTo 0
If wsDB Is Nothing Then
    Set wsDB = ThisWorkbook.Sheets.Add(After:=ThisWorkbook.Sheets(ThisWorkbook.Sheets.count))
    wsDB.Name = "DebugLog"
End If

wsDB.cells.Clear
With wsDB.cells
    .Interior.Color = RGB(0, 0, 0)
    .Font.Color = RGB(200, 200, 200)
    .Font.Name = "Consolas"
    .Font.Size = 9
End With
wsDB.Activate
ActiveWindow.DisplayGridlines = False
wsDB.Columns("A").ColumnWidth = 22
wsDB.Columns("B").ColumnWidth = 45
wsDB.Columns("C").ColumnWidth = 18
wsDB.Columns("D").ColumnWidth = 30

' -----
With wsDB.cells(1, 1)
    .Value = "SYSTEM DEBUG LOG | " & Format(Now, "yyyy/m/d hh:mm:ss")
    .Font.Color = RGB(255, 192, 0)
    .Font.Bold = True
    .Font.Size = 13
End With
wsDB.Rows(1).RowHeight = 26
With wsDB.Range(wsDB.cells(2, 1), wsDB.cells(2, 4))
    .Interior.Color = RGB(255, 192, 0)
    .RowHeight = 3
End With

Dim wr As Long: wr = 4 ' -- g -- J -- C --

' -----
Call DB_Header(wsDB, wr, "CHECK", "ITEM", "STATUS", "DETAIL")
wr = wr + 1

' =====
' 1. -- u -- @ -- s -- b --
' =====

Call DB_Section(wsDB, wr, "WORKSHEET EXISTENCE")
wr = wr + 1

Dim sheetNames As Variant
sheetNames = Array("RR4", "Transactions", "Realized", "HistoryLog", _
    "Analysis", "HoldingsCorr", "Correlation", _
    "DrawdownChart", "DebugLog")

Dim sn As Variant
For Each sn In sheetNames
    Dim wsCheck As Worksheet
    On Error Resume Next: Set wsCheck = ThisWorkbook.Sheets(CStr(sn)): On Error GoTo 0
    If wsCheck Is Nothing Then
        Call DB_Row(wsDB, wr, "Sheet", CStr(sn), "MISSING", "Sheet not found", RGB(255, 80, 80))
    Else
        Call DB_Row(wsDB, wr, "Sheet", CStr(sn), "OK", "Exists", RGB(0, 210, 100))
    End If
    wr = wr + 1
    Set wsCheck = Nothing
Next sn

' =====
' 2. Transactions data check
' =====

Call DB_Section(wsDB, wr, "TRANSACTIONS DATA")
wr = wr + 1

```

```

Dim wsTr As Worksheet
On Error Resume Next: Set wsTr = ThisWorkbook.Sheets("Transactions"): On Error GoTo 0
If Not wsTr Is Nothing Then
    Dim trLR As Long: trLR = wsTr.Cells(wsTr.Rows.Count, "A").End(xlUp).row
    Dim trCount As Long: trCount = trLR - 1

    Call DB_Row(wsDB, wr, "Transactions", "Total rows", "INFO", trCount & " records", RGB(255, 192, 0))
    wr = wr + 1

    ' -----
    Dim blankDate As Long, blankTicker As Long, blankAction As Long, blankAmt As Long
    Dim tr As Long
    For tr = 2 To trLR
        If Trim(CStr(wsTr.Cells(tr, 2).Value)) = "" Then blankDate = blankDate + 1
        If Trim(CStr(wsTr.Cells(tr, 3).Value)) = "" Then blankTicker = blankTicker + 1
        If Trim(CStr(wsTr.Cells(tr, 4).Value)) = "" Then blankAction = blankAction + 1
        If Not IsNumeric(wsTr.Cells(tr, 9).Value) Then blankAmt = blankAmt + 1
    Next tr

    Call DB_Row(wsDB, wr, "Transactions", "Blank dates (Col B)", _
        IIf(blankDate = 0, "OK", "WARN"), blankDate & " rows", _
        IIf(blankDate = 0, RGB(0, 210, 100), RGB(255, 140, 0)))
    wr = wr + 1
    Call DB_Row(wsDB, wr, "Transactions", "Blank Ticker (Col C)", _
        IIf(blankTicker = 0, "OK", "WARN"), blankTicker & " rows", _
        IIf(blankTicker = 0, RGB(0, 210, 100), RGB(255, 140, 0)))
    wr = wr + 1
    Call DB_Row(wsDB, wr, "Transactions", "Blank Action (Col D)", _
        IIf(blankAction = 0, "OK", "WARN"), blankAction & " rows", _
        IIf(blankAction = 0, RGB(0, 210, 100), RGB(255, 140, 0)))
    wr = wr + 1
    Call DB_Row(wsDB, wr, "Transactions", "Non-numeric NetAmt (Col I)", _
        IIf(blankAmt = 0, "OK", "WARN"), blankAmt & " rows", _
        IIf(blankAmt = 0, RGB(0, 210, 100), RGB(255, 140, 0)))
    wr = wr + 1

    ' -----
    Dim invalidAction As Long
    For tr = 2 To trLR
        Dim act As String: act = UCase(Trim(CStr(wsTr.Cells(tr, 4).Value)))
        If act <> "BUY" And act <> "SELL" And act <> "ADJUSTCOST" And act <> "" Then
            invalidAction = invalidAction + 1
        End If
    Next tr
    Call DB_Row(wsDB, wr, "Transactions", "Invalid Action", _
        IIf(invalidAction = 0, "OK", "ERROR"), invalidAction & " rows (must be BUY/SELL/ADJUSTCOST)", _
        IIf(invalidAction = 0, RGB(0, 210, 100), RGB(255, 80, 80)))
    wr = wr + 1
End If

' =====
' 3. HistoryLog update check
' =====

Call DB_Section(wsDB, wr, "HISTORYLOG UPDATE CHECK")
wr = wr + 1

Dim wsH As Worksheet
On Error Resume Next: Set wsH = ThisWorkbook.Sheets("HistoryLog"): On Error GoTo 0
If Not wsH Is Nothing Then
    Dim h1LR As Long: h1LR = wsH.Cells(wsH.Rows.Count, "A").End(xlUp).row
    Dim h1Count As Long: h1Count = h1LR - 1
    Call DB_Row(wsDB, wr, "HistoryLog", "Total rows", "INFO", h1Count & " records", RGB(255, 192, 0))
    wr = wr + 1

    If h1LR >= 2 Then
        Dim lastLogTime As Variant: lastLogTime = wsH.Cells(h1LR, "A").Value
        Dim lastLogMkt As Variant: lastLogMkt = wsH.Cells(h1LR, "B").Value
        Dim lastLogPnL As Variant: lastLogPnL = wsH.Cells(h1LR, "C").Value

        ' -----
        Dim timeDiff As Double

```

```

If IsDate(lastLogTime) Then
    timeDiff = Now - CDate(lastLogTime)
    Dim diffMin As Long: diffMin = CLng(timeDiff * 1440)
    Dim timeStatus As String
    Dim timeColor As Long
    If diffMin < 5 Then
        timeStatus = "FRESH": timeColor = RGB(0, 210, 100)
    ElseIf diffMin < 60 Then
        timeStatus = "OK": timeColor = RGB(0, 210, 100)
    ElseIf diffMin < 480 Then
        timeStatus = "STALE": timeColor = RGB(255, 140, 0)
    Else
        timeStatus = "OLD": timeColor = RGB(255, 80, 80)
    End If
    Call DB_Row(wsDB, wr, "HistoryLog", "Last update time", timeStatus, _
        Format(CDate(lastLogTime), "m/d hh:mm") & " (" & diffMin & " min ago)", timeColor)
Else
    Call DB_Row(wsDB, wr, "HistoryLog", "Last update time", "ERROR", "Cannot parse date", RGB(255, 80, 80))
End If
wr = wr + 1

' -----
If IsNumeric(lastLogMkt) And CDBl(lastLogMkt) > 0 Then
    Call DB_Row(wsDB, wr, "HistoryLog", "Last market value (Col B)", "OK", _
        Format(CDBl(lastLogMkt), "#,##0") & " TWD", RGB(0, 210, 100))
Else
    Call DB_Row(wsDB, wr, "HistoryLog", "Last market value (Col B)", "ERROR", "Zero or non-numeric", RGB(255, 80, 80))
End If
wr = wr + 1

' M -- JPort Ret Chg% -- ^ -- 0 -- _ --
Dim mLr As Long: mLr = wsH.Cells(wsH.Rows.count, "M").End(xlUp).row
Dim mCount As Long: mCount = mLr - 1
Call DB_Row(wsDB, wr, "HistoryLog", "M-col data count (Port Ret)", _
    IIf(mCount > 0, "OK", "ERROR"), mCount & " rows", _
    IIf(mCount > 0, RGB(0, 210, 100), RGB(255, 80, 80)))
wr = wr + 1

' -----
Dim todayLogged As Boolean: todayLogged = False
If IsDate(lastLogTime) Then
    If Int(CDate(lastLogTime)) = Date Then todayLogged = True
End If
Call DB_Row(wsDB, wr, "HistoryLog", "Logged today?", _
    IIf(todayLogged, "OK", "WARN"), _
    IIf(todayLogged, "Logged today", "Not yet logged today"), _
    IIf(todayLogged, RGB(0, 210, 100), RGB(255, 140, 0)))
wr = wr + 1
End If
End If

' =====
' 4. RR4 -- D -- 0
' =====

Call DB_Section(wsDB, wr, "RR4 DASHBOARD CHECK")
wr = wr + 1

Dim wsP As Worksheet
On Error Resume Next: Set wsP = ThisWorkbook.Sheets("RR4"): On Error GoTo 0
If Not wsP Is Nothing Then
    ' FX rate
    Dim exRate As Double: exRate = 0
    On Error Resume Next: exRate = CDBl(wsP.Range("C22").Value): On Error GoTo 0
    Call DB_Row(wsDB, wr, "RR4", "FX rate C22 (USD/TWD)", _
        IIf(exRate > 20 And exRate < 50, "OK", "WARN"), _
        IIf(exRate > 0, Format(exRate, "0.00"), "No value"), _
        IIf(exRate > 20 And exRate < 50, RGB(0, 210, 100), RGB(255, 140, 0)))
    wr = wr + 1

    ' -- \ --
    Dim totalMkt As Double: totalMkt = 0

```

```

On Error Resume Next: totalMkt = Cdbl(wsP.Cells(1, 1).Value): On Error GoTo 0
Call DB_Row(wsDB, wr, "RR4", "Total market value (Cell A1)", _
    IIf(totalMkt > 0, "OK", "ERROR"), _
    IIf(totalMkt > 0, Format(totalMkt, "#,##0") & " TWD", "No value or zero"), _
    IIf(totalMkt > 0, RGB(0, 210, 100), RGB(255, 80, 80)))
wr = wr + 1

' Positions
Dim posLR As Long: posLR = wsP.Cells(wsP.Rows.Count, 1).End(xlUp).row
Dim posCount As Long: posCount = posLR - 4
Call DB_Row(wsDB, wr, "RR4", "Position count (Row 5+)", _
    IIf(posCount > 0, "OK", "WARN"), posCount & " positions", _
    IIf(posCount > 0, RGB(0, 210, 100), RGB(255, 140, 0)))
wr = wr + 1

' InceptionDate / StartingCapital -- R -- W -- d --
' -----
Dim incDate As Date
Dim stCap As Double
On Error Resume Next
incDate = CDate(wsP.Range("C2").Value) ' -- A -- 培s -- 顯漲 -- m
stCap = Cdbl(wsP.Range("E2").Value) ' -- A -- 培s -- 顯漲 -- m
On Error GoTo 0
Call DB_Row(wsDB, wr, "RR4", "InceptionDate (named range)", _
    IIf(incDate > DateSerial(2000, 1, 1), "OK", "ERROR"), _
    IIf(incDate > DateSerial(2000, 1, 1), Format(incDate, "yyyy/m/d"), "Not set"), _
    IIf(incDate > DateSerial(2000, 1, 1), RGB(0, 210, 100), RGB(255, 80, 80)))
wr = wr + 1
Call DB_Row(wsDB, wr, "RR4", "StartingCapital (named range)", _
    IIf(stCap > 0, "OK", "ERROR"), _
    IIf(stCap > 0, Format(stCap, "#,##0") & " TWD", "Not set"), _
    IIf(stCap > 0, RGB(0, 210, 100), RGB(255, 80, 80)))
wr = wr + 1
End If

' =====
' 5. Yahoo Finance API -- s -- u --
' =====

Call DB_Section(wsDB, wr, "YAHOO FINANCE API TEST")
wr = wr + 1

Dim testTickers As Variant
testTickers = Array("SPY", "QQQ", "2330.TW", "^TWII")
Dim tt As Variant
For Each tt In testTickers
    Dim http As Object
    On Error Resume Next
    Set http = CreateObject("WinHttp.WinHttpRequest.5.1")
    On Error GoTo 0

    Dim apiURL As String
    apiURL = "https://query1.finance.yahoo.com/v8/finance/chart/" & _
        CStr(tt) & "?interval=1d&range=5d"

    Dim apiResp As String: apiResp = ""
    On Error Resume Next
    With http
        .Open "GET", apiURL, False
        .setRequestHeader "User-Agent", "Mozilla/5.0"
        .setRequestHeader "Accept", "application/json"
        .send
        apiResp = .responseText
    End With
    On Error GoTo 0

    ' -----
    Dim apiPx As Double: apiPx = 0
    Dim csPos As Long
    csPos = InStr(apiResp, """"close"":["")
    If csPos > 0 Then
        Dim csS As Long: csS = csPos + 9

```

```

Dim csE As Long: csE = InStr(csS, apiResp, "]")
If csE > csS Then
    Dim csStr As String: csStr = Mid(apiResp, csS, csE - csS)
    Dim csParts() As String: csParts = Split(csStr, ",")
    Dim cpIdx As Long
    ' -----
    For cpIdx = UBound(csParts) To 0 Step -1
        Dim cpTrim As String: cpTrim = Trim(csParts(cpIdx))
        If IsNumeric(cpTrim) And CDBl(cpTrim) > 0 Then
            apiPx = CDBl(cpTrim)
            Exit For
        End If
    Next cpIdx
End If

' -----
Dim httpStatus As Long: httpStatus = 0
On Error Resume Next: httpStatus = http.status: On Error GoTo 0

Dim detailMsg As String
If apiPx > 0 Then
    detailMsg = "Price = " & Format(apiPx, "#,##0.00") & _
        " (HTTP " & httpStatus & ")"
ElseIf httpStatus = 429 Then
    detailMsg = "HTTP 429 - Too Many Requests (rate limited)"
ElseIf httpStatus = 401 Or httpStatus = 403 Then
    detailMsg = "HTTP " & httpStatus & " - Auth required / forbidden"
ElseIf httpStatus = 0 Then
    detailMsg = "No response (network or firewall)"
Else
    detailMsg = "HTTP " & httpStatus & " - Cannot parse response"
End If

Call DB_Row(wsDB, wr, "API Test", CStr(tt), _
    IIf(apiPx > 0, "OK", "FAIL"), detailMsg, _
    IIf(apiPx > 0, RGB(0, 210, 100), RGB(255, 80, 80)))
wr = wr + 1
Next tt

' =====
' 6. Per-module function existence check
' =====

Call DB_Section(wsDB, wr, "MODULE FUNCTION CHECK")
wr = wr + 1

Dim funcNames As Variant
funcNames = Array("GetStockPrice", "GetCompanyName", "GetCurrencyType", _
    "GetHistoryPrices", "GetTechData", "CorrPearson", _
    "CorrGetCommonDates", "BuildPositions", "CalculateRealizedPnL")

Dim fn As Variant
For Each fn In funcNames
    Dim fnExists As Boolean: fnExists = False
    On Error Resume Next
    Dim testCall As Variant
    testCall = Application.Run(CStr(fn))
    If Err.Number = 0 Or Err.Number = 1004 Or Err.Number = 450 Or Err.Number = 13 Then
        fnExists = True ' -- s -- b -- 桴 -- ] -- ` -- ^
    ElseIf Err.Number = 1000 Or Err.Number = 35 Then
        fnExists = False ' -- 鈔?--
    Else
        fnExists = True ' -- L -- ~ -- s -- b
    End If
    Err.Clear
    On Error GoTo 0
    Call DB_Row(wsDB, wr, "Function", CStr(fn), _
        IIf(fnExists, "OK", "MISSING"), _
        IIf(fnExists, "Callable", "Function not found, check module"), _
        IIf(fnExists, RGB(0, 210, 100), RGB(255, 80, 80)))
    wr = wr + 1
Next fn

```

```

' =====
' 7. System health summary
' =====

wr = wr + 1
Call DB_Section(wsDB, wr, "HEALTH SUMMARY")
wr = wr + 1

' -----
Dim okCount As Long, warnCount As Long, errCount As Long
Dim scanR As Long
For scanR = 4 To wr
    Dim statusVal As String: statusVal = CStr(wsDB.cells(scanR, 3).Value)
    Select Case statusVal
        Case "OK", "FRESH": okCount = okCount + 1
        Case "WARN", "STALE": warnCount = warnCount + 1
        Case "ERROR", "FAIL", "MISSING", "OLD": errCount = errCount + 1
    End Select
Next scanR

Dim healthScore As Long
If okCount + warnCount + errCount > 0 Then
    healthScore = CLng(okCount / (okCount + warnCount + errCount) * 100)
End If

Dim scoreColor As Long
If healthScore >= 90 Then
    scoreColor = RGB(0, 210, 100)
ElseIf healthScore >= 70 Then
    scoreColor = RGB(255, 192, 0)
Else
    scoreColor = RGB(255, 80, 80)
End If

With wsDB.cells(wr, 1)
    .Value = "HEALTH SCORE"
    .Font.Color = RGB(255, 192, 0)
    .Font.Bold = True
End With
With wsDB.cells(wr, 2)
    .Value = healthScore & " / 100"
    .Font.Color = scoreColor
    .Font.Bold = True
    .Font.Size = 14
End With
With wsDB.cells(wr, 3)
    .Value = "OK:" & okCount & " WARN:" & warnCount & " ERR:" & errCount
    .Font.Color = RGB(200, 200, 200)
End With

wsDB.Columns("A:D").AutoFit
wsDB.Activate
wsDB.cells(1, 1).Select

Application.ScreenUpdating = True
Application.StatusBar = "Debug complete - Health: " & healthScore & "/100"
MsgBox "Debug complete. Health score: " & healthScore & "/100" & vbCrLf & _
    "OK: " & okCount & " WARN: " & warnCount & " ERROR: " & errCount, vbInformation
End Sub

' -----
Private Sub DB_Section(ws As Worksheet, r As Long, title As String)
    With ws.Range(ws.cells(r, 1), ws.cells(r, 4))
        .Interior.Color = RGB(20, 20, 40)
        .Font.Color = RGB(100, 160, 255)
        .Font.Bold = True
    End With
    ws.cells(r, 1).Value = "### " & title
    ws.Rows(r).RowHeight = 18
End Sub

```

```

-----
Private Sub DB_Header(ws As Worksheet, r As Long, _
                    c1 As String, c2 As String, c3 As String, c4 As String)
    With ws.Range(ws.Cells(r, 1), ws.Cells(r, 4))
        .Interior.Color = RGB(10, 10, 10)
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
    End With
    ws.Cells(r, 1).Value = c1
    ws.Cells(r, 2).Value = c2
    ws.Cells(r, 3).Value = c3
    ws.Cells(r, 4).Value = c4
    ws.Rows(r).RowHeight = 16
End Sub

```

```

-----
Private Sub DB_Row(ws As Worksheet, r As Long, _
                 category As String, item As String, _
                 status As String, detail As String, statusColor As Long)
    ws.Rows(r).RowHeight = 16
    With ws.Cells(r, 1)
        .Value = category
        .Font.Color = RGB(150, 150, 150)
        .Interior.Color = IIf(r Mod 2 = 0, RGB(8, 8, 8), RGB(12, 12, 12))
    End With
    With ws.Cells(r, 2)
        .Value = item
        .Font.Color = RGB(200, 200, 200)
        .Interior.Color = IIf(r Mod 2 = 0, RGB(8, 8, 8), RGB(12, 12, 12))
    End With
    With ws.Cells(r, 3)
        .Value = status
        .Font.Color = statusColor
        .Font.Bold = True
        .HorizontalAlignment = xlCenter
        .Interior.Color = IIf(r Mod 2 = 0, RGB(8, 8, 8), RGB(12, 12, 12))
    End With
    With ws.Cells(r, 4)
        .Value = detail
        .Font.Color = RGB(180, 180, 180)
        .Interior.Color = IIf(r Mod 2 = 0, RGB(8, 8, 8), RGB(12, 12, 12))
    End With
End Sub

```

Option Explicit

' --- 自訂型別 ---

Private Type TradeRecord

TransID As String
 TradeDate As Long
 ticker As String
 action As String
 shares As Double
 price As Double
 Fee As Double
 Tax As Double
 NetAmount As Double
 Sector As String
 target As Double
 Strategy As String
 Beta As Double

End Type

Private Type RealizedRecord

ticker As String
 PnL As Double
 shares As Double
 AvgCost As Double
 SellAmount As Double
 Strategy As String
 pnlTWD As Double
 SellDate As Long
 Review As String

End Type

Private Type CampaignResult

ticker As String
 Legs As Long
 FirstEntry As Long
 LastExit As Long
 holdDays As Long
 totalCost As Double
 totalPnL As Double
 ReturnPct As Double
 Win As Long

End Type

' =====
' 主程式
' =====

Public Sub RebuildCampaigns()

Application.ScreenUpdating = False
 Application.Calculation = xlCalculationManual
 On Error GoTo ErrHandler

Dim trades() As TradeRecord
 Dim realized() As RealizedRecord
 trades = ReadTransactions()
 realized = ReadRealized()

Dim campaigns() As CampaignResult
 campaigns = BuildCampaigns(trades, realized)
 SortCampaignsByPnL campaigns
 WriteCampaignsSheet campaigns

Application.Calculation = xlCalculationAutomatic
 Application.ScreenUpdating = True
 MsgBox "完成! 共 " & (UBound(campaigns) - LBound(campaigns) + 1) & " 個 Campaigns", vbInformation
 Exit Sub

ErrHandler:

Application.Calculation = xlCalculationAutomatic
 Application.ScreenUpdating = True
 MsgBox "錯誤: " & Err.Description, vbCritical

End Sub

' 讀取 Transactions

```
Private Function ReadTransactions() As TradeRecord()
    Dim ws As Worksheet
    Set ws = ThisWorkbook.Worksheets("Transactions")
    Dim lastRow As Long
    lastRow = ws.Cells(ws.Rows.Count, "A").End(xlUp).row
    If lastRow < 2 Then Err.Raise 1001, , "Transactions 沒有資料"

    Dim arr() As TradeRecord
    ReDim arr(1 To lastRow - 1)
    Dim data As Variant
    data = ws.Range("A2:M" & lastRow).Value

    Dim i As Long
    For i = 1 To UBound(data, 1)
        arr(i).TransID = CStr(data(i, 1))
        arr(i).TradeDate = CLng(data(i, 2))
        arr(i).ticker = Trim(CStr(data(i, 3)))
        arr(i).action = Trim(CStr(data(i, 4)))
        arr(i).shares = CDBl(data(i, 5))
        arr(i).price = CDBl(data(i, 6))
        arr(i).Fee = CDBl(data(i, 7))
        arr(i).Tax = CDBl(data(i, 8))
        arr(i).NetAmount = CDBl(data(i, 9))
        arr(i).Sector = CStr(data(i, 10))
        If IsNumeric(data(i, 11)) Then arr(i).target = CDBl(data(i, 11))
        arr(i).Strategy = CStr(data(i, 12))
        If IsNumeric(data(i, 13)) Then arr(i).Beta = CDBl(data(i, 13))
    Next i
    ReadTransactions = arr
End Function
```

' 讀取 Realized

```
Private Function ReadRealized() As RealizedRecord()
    Dim ws As Worksheet
    Set ws = ThisWorkbook.Worksheets("Realized")
    Dim lastRow As Long
    lastRow = ws.Cells(ws.Rows.Count, "A").End(xlUp).row
    If lastRow < 2 Then Err.Raise 1002, , "Realized 沒有資料"

    Dim arr() As RealizedRecord
    ReDim arr(1 To lastRow - 1)
    Dim data As Variant
    data = ws.Range("A2:I" & lastRow).Value

    Dim i As Long
    For i = 1 To UBound(data, 1)
        arr(i).ticker = Trim(CStr(data(i, 1)))
        arr(i).PnL = CDBl(data(i, 2))
        arr(i).shares = CDBl(data(i, 3))
        arr(i).AvgCost = CDBl(data(i, 4))
        arr(i).SellAmount = CDBl(data(i, 5))
        arr(i).Strategy = CStr(data(i, 6))
        arr(i).pnlTWD = CDBl(data(i, 7))
        arr(i).SellDate = CLng(data(i, 8))
        If Not IsEmpty(data(i, 9)) Then arr(i).Review = CStr(data(i, 9))
    Next i
    ReadRealized = arr
End Function
```

' 建構 Campaigns (同 Ticker gap ? 7 天為同一 Campaign)

```
Private Function BuildCampaigns(trades() As TradeRecord, _
                                realized() As RealizedRecord) As CampaignResult()
    Dim dictDates As Object
    Set dictDates = CreateObject("Scripting.Dictionary")
```

```
dictDates.CompareMode = vbTextCompare
```

```
Dim i As Long, tk As String
```

```
For i = LBound(trades) To UBound(trades)
```

```
    tk = trades(i).ticker
```

```
    If Not dictDates.Exists(tk) Then Set dictDates(tk) = CreateObject("System.Collections.ArrayList")
```

```
    Dim al As Object: Set al = dictDates(tk)
```

```
    If Not al.Contains(trades(i).TradeDate) Then al.Add trades(i).TradeDate
```

```
Next i
```

```
For i = LBound(realized) To UBound(realized)
```

```
    tk = realized(i).ticker
```

```
    If Not dictDates.Exists(tk) Then Set dictDates(tk) = CreateObject("System.Collections.ArrayList")
```

```
    Set al = dictDates(tk)
```

```
    If Not al.Contains(realized(i).SellDate) Then al.Add realized(i).SellDate
```

```
Next i
```

```
Dim campaignList As Object
```

```
Set campaignList = CreateObject("System.Collections.ArrayList")
```

```
Dim keys As Variant: keys = dictDates.keys
```

```
Dim k As Long
```

```
For k = 0 To UBound(keys)
```

```
    tk = keys(k)
```

```
    Set al = dictDates(tk)
```

```
    al.Sort
```

```
Dim groups As Object: Set groups = CreateObject("System.Collections.ArrayList")
```

```
Dim curGroup As Object: Set curGroup = CreateObject("System.Collections.ArrayList")
```

```
curGroup.Add CLng(al(0))
```

```
Dim j As Long
```

```
For j = 1 To al.count - 1
```

```
    If CLng(al(j)) - CLng(al(j - 1)) > 7 Then
```

```
        groups.Add curGroup
```

```
        Set curGroup = CreateObject("System.Collections.ArrayList")
```

```
    End If
```

```
    curGroup.Add CLng(al(j))
```

```
Next j
```

```
groups.Add curGroup
```

```
Dim g As Long
```

```
For g = 0 To groups.count - 1
```

```
    Dim grp As Object: Set grp = groups(g)
```

```
    Dim minDate As Long: minDate = CLng(grp(0))
```

```
    Dim maxDate As Long: maxDate = CLng(grp(grp.count - 1))
```

```
Dim legCount As Long, totalCost As Double
```

```
legCount = 0: totalCost = 0
```

```
For i = LBound(trades) To UBound(trades)
```

```
    If StrComp(trades(i).ticker, tk, vbTextCompare) = 0 Then
```

```
        If trades(i).TradeDate >= minDate And trades(i).TradeDate <= maxDate Then
```

```
            Select Case trades(i).action
```

```
                Case "Buy": legCount = legCount + 1: totalCost = totalCost + trades(i).NetAmount
```

```
                Case "AdjustCost": totalCost = totalCost + trades(i).NetAmount
```

```
            End Select
```

```
        End If
```

```
    End If
```

```
Next i
```

```
Dim totalPnL As Double, realizedCount As Long
```

```
totalPnL = 0: realizedCount = 0
```

```
For i = LBound(realized) To UBound(realized)
```

```
    If StrComp(realized(i).ticker, tk, vbTextCompare) = 0 Then
```

```
        If realized(i).SellDate >= minDate And realized(i).SellDate <= maxDate Then
```

```
            totalPnL = totalPnL + realized(i).pnlTWD
```

```
            realizedCount = realizedCount + 1
```

```
        End If
```

```
    End If
```

```
Next i
```

```
If legCount = 0 And realizedCount = 0 Then GoTo NextGroup
```

```
Dim holdDays As Long: holdDays = IIf(maxDate > minDate, maxDate - minDate, 1)
```

```
Dim retPct As Double
```

```
If totalCost > 0 Then
```

```
    retPct = totalPnL / totalCost
```

```
ElseIf totalCost < 0 Then
```

```
    retPct = totalPnL / Abs(totalCost)
```

```
Else
```

```
    retPct = 0
```

```
End If
```

```
campaignList.Add tk & "|" & IIf(legCount > 0, legCount, realizedCount) & "|" & _  
    minDate & "|" & maxDate & "|" & holdDays & "|" & _  
    totalCost & "|" & totalPnL & "|" & retPct & "|" & _  
    IIf(totalPnL > 0, 1, 0)
```

```
NextGroup:
```

```
    Next g
```

```
Next k
```

```
If campaignList.count = 0 Then Err.Raise 1003, , "没有找到任何 Campaign"
```

```
Dim result() As CampaignResult
```

```
ReDim result(1 To campaignList.count)
```

```
For i = 0 To campaignList.count - 1
```

```
    Dim parts As Variant: parts = Split(CStr(campaignList(i)), "|")
```

```
    result(i + 1).ticker = parts(0)
```

```
    result(i + 1).Legs = CLng(parts(1))
```

```
    result(i + 1).FirstEntry = CLng(parts(2))
```

```
    result(i + 1).LastExit = CLng(parts(3))
```

```
    result(i + 1).holdDays = CLng(parts(4))
```

```
    result(i + 1).totalCost = CDBl(parts(5))
```

```
    result(i + 1).totalPnL = CDBl(parts(6))
```

```
    result(i + 1).ReturnPct = CDBl(parts(7))
```

```
    result(i + 1).Win = CLng(parts(8))
```

```
Next i
```

```
BuildCampaigns = result
```

```
End Function
```

```
'=====
```

```
' 按 P&L 降序排序
```

```
'=====
```

```
Private Sub SortCampaignsByPnL(ByRef arr() As CampaignResult)
```

```
    Dim i As Long, j As Long, tmp As CampaignResult
```

```
    For i = LBound(arr) To UBound(arr) - 1
```

```
        For j = i + 1 To UBound(arr)
```

```
            If arr(j).totalPnL > arr(i).totalPnL Then
```

```
                tmp = arr(i): arr(i) = arr(j): arr(j) = tmp
```

```
            End If
```

```
        Next j
```

```
    Next i
```

```
End Sub
```

```
Private Sub WriteCampaignsSheet(campaigns() As CampaignResult)
```

```
    Dim ws As Worksheet
```

```
On Error Resume Next
```

```
Set ws = ThisWorkbook.Worksheets("Campaigns")
```

```
If Not ws Is Nothing Then
```

```
    Application.DisplayAlerts = False
```

```
    ws.Delete
```

```
    Application.DisplayAlerts = True
```

```
End If
```

```
On Error GoTo 0
```

```
Set ws = ThisWorkbook.Worksheets.Add(After:=ThisWorkbook.Worksheets(ThisWorkbook.Worksheets.count))
```

```
ws.Name = "Campaigns"
```

```
ws.Cells.Interior.Color = RGB(0, 0, 0)
```

```
ws.Cells.Font.Color = RGB(200, 200, 200)
```

```
ws.Cells.Font.Name = "Consolas"
```

```
ws.Tab.Color = RGB(255, 140, 0)
```

```
ws.Activate
ActiveWindow.DisplayGridlines = False
```

```
Dim n As Long: n = UBound(campaigns) - LBound(campaigns) + 1
```

```
' ----- 全域列高 & 欄距 -----
ws.Rows(1).RowHeight = 36      ' 主標題列
ws.Rows(2).RowHeight = 26      ' 欄位標頭列
Dim rowIdx As Long
For rowIdx = 3 To n + 10
    ws.Rows(rowIdx).RowHeight = 20 ' 資料列統一 20
Next rowIdx
ws.Columns("D").ColumnWidth = 3  ' Section B / C 隔間
ws.Columns("J").ColumnWidth = 3  ' Section C / D 隔間
ws.Columns("T").ColumnWidth = 3  ' Section D / E 隔間
ws.Columns("W").ColumnWidth = 3  ' Section E / F 隔間
```

```
' =====
' Section A: 標題
' =====
```

```
With ws.Range("A1")
    .Value = "Roll & Scale-up"
    .Font.Bold = True
    .Font.Size = 14
    .Font.Color = RGB(255, 140, 0)
    .IndentLevel = 1
End With
```

```
' =====
' Section B: 關鍵指標統計
' =====
```

```
Dim totalLegs As Long, wins As Long, losses As Long
Dim grossWin As Double, grossLoss As Double, sumPnL As Double, sumHold As Long
Dim maxDD As Double, runningPeak As Double, runningPnL As Double
Dim maxConsecWin As Long, maxConsecLoss As Long, curConsecWin As Long, curConsecLoss As Long
```

```
Dim sorted() As CampaignResult: sorted = campaigns
Dim ii As Long, jj As Long, tmpC As CampaignResult
For ii = LBound(sorted) To UBound(sorted) - 1
    For jj = ii + 1 To UBound(sorted)
        If sorted(jj).LastExit < sorted(ii).LastExit Then
            tmpC = sorted(ii): sorted(ii) = sorted(jj): sorted(jj) = tmpC
        End If
    Next jj
Next ii
```

```
For ii = LBound(sorted) To UBound(sorted)
    totalLegs = totalLegs + sorted(ii).Legs
    sumHold = sumHold + sorted(ii).holdDays
    If sorted(ii).Win = 1 Then
        wins = wins + 1: grossWin = grossWin + sorted(ii).totalPnL
        curConsecWin = curConsecWin + 1: curConsecLoss = 0
        If curConsecWin > maxConsecWin Then maxConsecWin = curConsecWin
    Else
        losses = losses + 1: grossLoss = grossLoss + Abs(sorted(ii).totalPnL)
        curConsecLoss = curConsecLoss + 1: curConsecWin = 0
        If curConsecLoss > maxConsecLoss Then maxConsecLoss = curConsecLoss
    End If
    runningPnL = runningPnL + sorted(ii).totalPnL
    If runningPnL > runningPeak Then runningPeak = runningPnL
    Dim dd As Double: dd = runningPeak - runningPnL
    If dd > maxDD Then maxDD = dd
Next ii
```

```
sumPnL = grossWin - grossLoss
Dim avgLegs As Double: avgLegs = totalLegs / n
Dim winRate As Double: winRate = wins / n
Dim profitFactor As Double: profitFactor = IIf(grossLoss > 0, grossWin / grossLoss, 0)
Dim expectancy As Double: expectancy = sumPnL / n
Dim avgWin As Double: avgWin = IIf(wins > 0, grossWin / wins, 0)
Dim avgLoss As Double: avgLoss = IIf(losses > 0, grossLoss / losses, 0)
```

```

Dim rrr As Double:      rrr = IIf(avgLoss > 0, avgWin / avgLoss, 0)
Dim avgHold As Double:  avgHold = sumHold / n
Dim marRatio As Double: marRatio = IIf(maxDD > 0, sumPnL / maxDD, 0)

```

```
' 欄位標頭
```

```

ws.Range("A2").Value = "關鍵指標"
ws.Range("B2").Value = "值"
ws.Range("C2").Value = "說明"
With ws.Range("A2:C2")
    .Font.Bold = True
    .Font.Color = RGB(200, 200, 200)
    .IndentLevel = 1
End With

```

```
Dim labels As Variant, vals As Variant, desc As Variant
```

```

labels = Array("Total Campaigns", "Total Trades (legs)", "Avg Legs / Campaign", _
    "Win Rate (Campaign)", "Profit Factor", "Expectancy / Campaign", _
    "Avg Win", "Avg Loss", "賞盈比 (RRR)", "Avg Hold Days (Campaign)", _
    "Max Drawdown (TWD)", "MAR Ratio", "Net P&L (TWD)", _
    "Max Consec. Wins", "Max Consec. Losses", "Gross Win (TWD)")
vals = Array(n, totalLegs, Round(avgLegs, 2), Round(winRate, 4), Round(profitFactor, 2), _
    Round(expectancy, 2), Round(avgWin, 2), Round(avgLoss, 2), Round(rrr, 2), _
    Round(avgHold, 1), Round(maxDD, 2), Round(marRatio, 4), Round(sumPnL, 2), _
    maxConsecWin, maxConsecLoss, Round(grossWin, 2))
descs = Array("把同 ticker 連續 trades (gap<=7天) 合併", "原始交易筆數", _
    "平均每個 campaign 加碼次數", "Campaign 維度勝率", "毛利 / 毛損 — > 2.0 為健康", _
    "平均每 campaign 期望值 (TWD)", "平均獲利 campaign P&L", "平均虧損 campaign P&L", _
    "賞盈比", "Campaign 平均持有天數", "累計 P&L 曲線最大回撤", "Net P&L / Max DD", _
    "淨損益合計", "最大連勝次數", "最大連敗次數", "毛利合計")

```

```

For ii = 0 To UBound(labels)
    Dim r As Long: r = 3 + ii
    With ws.cells(r, 1)
        .Value = labels(ii)
        .Font.Bold = True
        .Font.Color = RGB(255, 255, 255) ' ← 白色
        .IndentLevel = 1
    End With
    With ws.cells(r, 2)
        .Value = vals(ii)
        .IndentLevel = 1
    End With
    With ws.cells(r, 3)
        .Value = desc(ii)
        .Font.Color = RGB(180, 180, 180) ' ← 淡灰說明
        .IndentLevel = 1
    End With
Next ii

```

```

ws.Range("B6").NumberFormat = "0.00%"
ws.Columns("A").ColumnWidth = 26
ws.Columns("B").ColumnWidth = 18
ws.Columns("C").ColumnWidth = 44

```

```

ApplyThreshold ws.cells(6, 2), winRate, 0.5
ApplyThreshold ws.cells(7, 2), profitFactor, 2
ApplyPnLColor ws.cells(8, 2), expectancy
ws.cells(9, 2).Font.Color = RGB(255, 140, 0)
ws.cells(10, 2).Font.Color = RGB(110, 110, 110)
ApplyThreshold ws.cells(11, 2), rrr, 1.5
ws.cells(13, 2).Font.Color = RGB(110, 110, 110)
ApplyThreshold ws.cells(14, 2), marRatio, 1
ApplyPnLColor ws.cells(15, 2), sumPnL
ws.cells(16, 2).Font.Color = RGB(255, 140, 0)
ws.cells(17, 2).Font.Color = RGB(110, 110, 110)
ws.cells(18, 2).Font.Color = RGB(255, 140, 0)

```

```

' =====
' Section C: Leg Analysis
' =====

```

```
With ws.Range("E1")
```

```

        .Value = "加碼次序 vs 表現 (Leg Position Analysis)"
        .Font.Bold = True: .Font.Color = RGB(255, 140, 0): .IndentLevel = 1
    End With
    With ws.Range("E2:I2")
        .Font.Bold = True: .Font.Color = RGB(200, 200, 200): .IndentLevel = 1
    End With
    ws.Range("E2").Value = "Leg #": ws.Range("F2").Value = "次數"
    ws.Range("G2").Value = "勝率": ws.Range("H2").Value = "平均報酬%"
    ws.Range("I2").Value = "Total P&L"

    Dim maxLeg As Long
    For ii = LBound(campaigns) To UBound(campaigns)
        If campaigns(ii).Legs > maxLeg Then maxLeg = campaigns(ii).Legs
    Next ii
    If maxLeg > 7 Then maxLeg = 7

    Dim legRow As Long: legRow = 3
    Dim leg As Long
    For leg = 1 To maxLeg
        Dim legCnt As Long, legWin As Long, legPnL As Double, legRetSum As Double
        legCnt = 0: legWin = 0: legPnL = 0: legRetSum = 0
        For ii = LBound(campaigns) To UBound(campaigns)
            If campaigns(ii).Legs = leg Then
                legCnt = legCnt + 1: legPnL = legPnL + campaigns(ii).totalPnL
                legRetSum = legRetSum + campaigns(ii).ReturnPct
                If campaigns(ii).Win = 1 Then legWin = legWin + 1
            End If
        Next ii
        If legCnt > 0 Then
            ws.cells(legRow, 5).Value = leg: ws.cells(legRow, 5).IndentLevel = 1
            ws.cells(legRow, 6).Value = legCnt: ws.cells(legRow, 6).IndentLevel = 1

            Dim wr As Double: wr = legWin / legCnt
            ws.cells(legRow, 7).Value = Round(wr, 4)
            ws.cells(legRow, 7).NumberFormat = "0.00%"
            ApplyThreshold ws.cells(legRow, 7), wr, 0.5

            Dim avgRet As Double: avgRet = legRetSum / legCnt
            ws.cells(legRow, 8).Value = Round(avgRet, 4)
            ws.cells(legRow, 8).NumberFormat = "0.00%"
            ApplyPnLColor ws.cells(legRow, 8), avgRet

            ws.cells(legRow, 9).Value = Round(legPnL, 2)
            ws.cells(legRow, 9).NumberFormat = "#,##0"
            ApplyPnLColor ws.cells(legRow, 9), legPnL

            legRow = legRow + 1
        End If
    Next leg

    ws.Columns("E").ColumnWidth = 8
    ws.Columns("F").ColumnWidth = 8
    ws.Columns("G").ColumnWidth = 10
    ws.Columns("H").ColumnWidth = 12
    ws.Columns("I").ColumnWidth = 14

```

```

' =====
' Section D: Campaign 明細表
' =====

```

```

With ws.Range("K1")
    .Value = "全部 Campaigns 明細 (依 P&L 排序)"
    .Font.Bold = True: .Font.Color = RGB(255, 140, 0): .IndentLevel = 1
End With
With ws.Range("K2:S2")
    .Font.Bold = True: .Font.Color = RGB(200, 200, 200): .IndentLevel = 1
End With
ws.Range("K2").Value = "Ticker": ws.Range("L2").Value = "Legs"
ws.Range("M2").Value = "First Entry": ws.Range("N2").Value = "Last Exit"
ws.Range("O2").Value = "True Hold Days": ws.Range("P2").Value = "Total Cost"
ws.Range("Q2").Value = "Total P&L": ws.Range("R2").Value = "Return %"
ws.Range("S2").Value = "Win"

```

```

For ii = LBound(campaigns) To UBound(campaigns)
    r = 2 + ii
    Dim c As Long
    For c = 11 To 19: ws.cells(r, c).IndentLevel = 1: Next c

    ws.cells(r, 11).Value = campaigns(ii).ticker
    ws.cells(r, 11).Font.Color = RGB(255, 255, 255)
    ws.cells(r, 12).Value = campaigns(ii).Legs
    ws.cells(r, 13).Value = CDate(campaigns(ii).FirstEntry)
    ws.cells(r, 14).Value = CDate(campaigns(ii).LastExit)
    ws.cells(r, 15).Value = campaigns(ii).holdDays
    ws.cells(r, 16).Value = Round(campaigns(ii).totalCost, 2)

    ws.cells(r, 17).Value = Round(campaigns(ii).totalPnL, 2)
    ApplyPnLColor ws.cells(r, 17), campaigns(ii).totalPnL

    ws.cells(r, 18).Value = campaigns(ii).ReturnPct
    ApplyPnLColor ws.cells(r, 18), campaigns(ii).ReturnPct

    ws.cells(r, 19).Value = campaigns(ii).Win
    If campaigns(ii).Win = 1 Then
        ws.cells(r, 19).Font.Color = RGB(255, 140, 0)
    Else
        ws.cells(r, 19).Font.Color = RGB(110, 110, 110)
    End If
Next ii

ws.Range("M3:N" & (2 + n)).NumberFormat = "yyyy-mm-dd"
ws.Range("P3:Q" & (2 + n)).NumberFormat = "#,##0"
ws.Range("R3:R" & (2 + n)).NumberFormat = "0.00%"

ws.Columns("K").ColumnWidth = 14: ws.Columns("L").ColumnWidth = 7
ws.Columns("M").ColumnWidth = 13: ws.Columns("N").ColumnWidth = 13
ws.Columns("O").ColumnWidth = 15: ws.Columns("P").ColumnWidth = 14
ws.Columns("Q").ColumnWidth = 14: ws.Columns("R").ColumnWidth = 11
ws.Columns("S").ColumnWidth = 6

' =====
' Section E: 並行時序
' =====

With ws.Range("U1")
    .Value = "並行 Campaign 數時序"
    .Font.Bold = True: .Font.Color = RGB(255, 140, 0): .IndentLevel = 1
End With
With ws.Range("U2:V2")
    .Font.Bold = True: .Font.Color = RGB(200, 200, 200): .IndentLevel = 1
End With
ws.Range("U2").Value = "Date": ws.Range("V2").Value = "Concurrent"

Dim dictAllDates As Object: Set dictAllDates = CreateObject("Scripting.Dictionary")
For ii = LBound(campaigns) To UBound(campaigns)
    Dim d As Long
    For d = campaigns(ii).FirstEntry To campaigns(ii).LastExit
        If Not dictAllDates.Exists(d) Then dictAllDates(d) = 0
    Next d
Next ii

Dim dateKeys As Variant: dateKeys = dictAllDates.keys
Dim dk As Long, dl As Long, tmpD As Variant
For dk = 0 To UBound(dateKeys) - 1
    For dl = dk + 1 To UBound(dateKeys)
        If dateKeys(dl) < dateKeys(dk) Then
            tmpD = dateKeys(dk): dateKeys(dk) = dateKeys(dl): dateKeys(dl) = tmpD
        End If
    Next dl
Next dk

Dim outRow As Long: outRow = 3
For dk = 0 To UBound(dateKeys)
    Dim concurrent As Long: concurrent = 0

```

```

For ii = LBound(campaigns) To UBound(campaigns)
    If dateKeys(dk) >= campaigns(ii).FirstEntry And _
        dateKeys(dk) <= campaigns(ii).LastExit Then concurrent = concurrent + 1
Next ii
ws.cells(outRow, 21).Value = CDate(CLng(dateKeys(dk))): ws.cells(outRow, 21).IndentLevel = 1
ws.cells(outRow, 22).Value = concurrent: ws.cells(outRow, 22).IndentLevel = 1
outRow = outRow + 1
Next dk

ws.Range("U3:U" & (outRow - 1)).NumberFormat = "yyyy-mm-dd"
ws.Columns("U").ColumnWidth = 14: ws.Columns("V").ColumnWidth = 13

' =====
' Section F: Leg# vs TotalPnL
' =====

With ws.Range("X1")
    .Value = "Leg# vs TotalPnL"
    .Font.Bold = True: .Font.Color = RGB(255, 140, 0): .IndentLevel = 1
End With
With ws.Range("X2:Y2")
    .Font.Bold = True: .Font.Color = RGB(200, 200, 200): .IndentLevel = 1
End With
ws.Range("X2").Value = "Leg#": ws.Range("Y2").Value = "TotalPnL"

legRow = 3
For leg = 1 To maxLeg
    legPnL = 0
    For ii = LBound(campaigns) To UBound(campaigns)
        If campaigns(ii).Legs = leg Then legPnL = legPnL + campaigns(ii).totalPnL
    Next ii
    ws.cells(legRow, 24).Value = leg: ws.cells(legRow, 24).IndentLevel = 1
    ws.cells(legRow, 25).Value = Round(legPnL, 2): ws.cells(legRow, 25).IndentLevel = 1
    ws.cells(legRow, 25).NumberFormat = "#,##0"
    ApplyPnLColor ws.cells(legRow, 25), legPnL
    legRow = legRow + 1
Next leg

ws.Columns("X").ColumnWidth = 8: ws.Columns("Y").ColumnWidth = 14

```

End Sub

```

' =====
' 正值橘 / 負値暗灰
' =====

```

```
Private Sub ApplyPnLColor(cell As Range, Value As Double)
```

```
    If Value > 0 Then
```

```
        cell.Font.Color = RGB(255, 140, 0)
```

```
    ElseIf Value < 0 Then
```

```
        cell.Font.Color = RGB(110, 110, 110)
```

```
    End If
```

End Sub

```

' =====
' 達標橘 / 未達標暗灰
' =====

```

```
Private Sub ApplyThreshold(cell As Range, Value As Double, threshold As Double)
```

```
    If Value >= threshold Then
```

```
        cell.Font.Color = RGB(255, 140, 0)
```

```
    Else
```

```
        cell.Font.Color = RGB(110, 110, 110)
```

```
    End If
```

End Sub

Option Explicit

```

' =====
' COMMODITY TERM STRUCTURE — Bloomberg-Style Dashboard
' v3.0 | Dark Theme | Section 1: Curve Signal | Section 2: Spread Matrix
' =====

' — Sheet / layout constants —
Private Const SH_MAIN As String = "BBG_COMM"
Private Const SH_TICK As String = "TickerMap"
Private Const SH_CFG As String = "_Config"
Private Const MAX_RETRY As Integer = 3

' — Dashboard layout columns (BBG_COMM) —
Private Const COL_FIRST As Long = 1
Private Const COL_LAST As Long = 13 ' STRUCTURE column
Private Const COL_PREVCACHE As Long = 14 ' hidden cache for prior-session M1
Private Const COL_UNUSED_GAP As Long = 12 ' visually reserved blank column

' — Theme tokens (pure-black canvas, monospace) —
Private Const THEME_FONT As String = "Consolas"
Private Const THEME_FONT_SIZE As Long = 10
Private Const C_BG As Long = 0 ' RGB(0,0,0) - pure black canvas

' =====
' ENTRY POINT - refresh full dashboard
' - TickerMap layout (V3): A=Instrument | B=Base | C=M1 | D=M2 | E=M3 | F=M6 | G=M12
' - Computes % change vs previous close (cached in hidden column 14)
' - Calls RefreshSpreadMatrix and writes timestamp
' - Per-row error guard: a single failed fetch does not abort the run
' =====

Sub RefreshDashboard()
    Dim wsM As Worksheet
    Dim wsTick As Worksheet
    Dim wsCfg As Worksheet
    Dim sec1 As Long, instCnt As Long, sec2Data As Long
    Dim i As Long
    Const PREV_COL As Integer = 14 ' hidden cache for prior-session M1 close

    Set wsM = ThisWorkbook.Sheets(SH_MAIN)
    Set wsTick = ThisWorkbook.Sheets(SH_TICK)
    Set wsCfg = ThisWorkbook.Sheets(SH_CFG)

    sec1 = wsCfg.Range("B1").Value
    instCnt = wsCfg.Range("B2").Value
    sec2Data = wsCfg.Range("B3").Value

    Application.ScreenUpdating = False
    Application.StatusBar = "Refreshing Commodity Term Structure..."
    wsM.Columns(PREV_COL).Hidden = True

    Dim okCount As Long, errCount As Long
    okCount = 0: errCount = 0

    For i = 1 To instCnt
        Dim dataRow As Long
        dataRow = sec1 + i - 1

        Application.StatusBar = "Fetching [" & i & "/" & instCnt & "] " & _
            wsM.Cells(dataRow, 2).Value

        ' Per-row error guard so a single failed fetch cannot abort the whole run.
        On Error GoTo RowFail

        ' V3 TickerMap layout: C..G = M1, M2, M3, M6, M12.
        Dim tM1 As String, tM2 As String, tM3 As String
        Dim tM6 As String, tM12 As String
        tM1 = Trim(wsTick.Cells(i + 1, 3).Value)
        tM2 = Trim(wsTick.Cells(i + 1, 4).Value)
        tM3 = Trim(wsTick.Cells(i + 1, 5).Value)
        tM6 = Trim(wsTick.Cells(i + 1, 6).Value)

```

```

tM12 = Trim(wsTick.cells(i + 1, 7).Value)

Dim baseSym As String
baseSym = Trim(wsTick.cells(i + 1, 2).Value)

Dim pM1 As Double, pM2 As Double, pM3 As Double
Dim pM6 As Double, pM12 As Double
pM1 = FetchPriceWithFallback(tM1, baseSym)
pM2 = FetchPriceWithFallback(tM2, baseSym)
pM3 = FetchPriceWithFallback(tM3, baseSym)
pM6 = FetchPriceWithFallback(tM6, baseSym)
pM12 = FetchPriceWithFallback(tM12, baseSym)

If pM1 > 0 Then
    ' % change vs cached prior M1 close.
    Dim prevClose As Double: prevClose = 0
    If IsNumeric(wsM.cells(dataRow, PREV_COL).Value) Then
        prevClose = CDBl(wsM.cells(dataRow, PREV_COL).Value)
    End If
    With wsM.cells(dataRow, 3)
        .Font.Name = THEME_FONT
        .Font.Size = THEME_FONT_SIZE
        .Interior.Color = C_BG
        .HorizontalAlignment = xlCenter
        .VerticalAlignment = xlCenter
        If prevClose > 0 Then
            Dim chg As Double: chg = (pM1 - prevClose) / prevClose
            .Value = chg
            .NumberFormat = "+0.00%;-0.00%;0.00%"
            .Font.Bold = True
            If chg > 0 Then
                .Font.Color = RGB(0, 255, 136)
            ElseIf chg < 0 Then
                .Font.Color = RGB(255, 51, 51)
            Else
                .Font.Color = RGB(255, 255, 0)
            End If
        Else
            .Value = "new"
            .Font.Color = RGB(136, 136, 136)
        End If
    End With
    wsM.cells(dataRow, PREV_COL).Value = pM1

    ' Prices
    WritePrice wsM.cells(dataRow, 4), pM1
    WritePrice wsM.cells(dataRow, 5), pM2
    WritePrice wsM.cells(dataRow, 6), pM3
    WritePrice wsM.cells(dataRow, 7), pM6
    WritePrice wsM.cells(dataRow, 8), pM12

    ' Spreads (guard against zero far-month price -> blank, not bogus spread)
    If pM2 > 0 Then WriteSpread wsM.cells(dataRow, 9), pM1 - pM2 _
        Else wsM.cells(dataRow, 9).Value = "-"
    If pM6 > 0 Then WriteSpread wsM.cells(dataRow, 10), pM1 - pM6 _
        Else wsM.cells(dataRow, 10).Value = "-"
    If pM12 > 0 Then WriteSpread wsM.cells(dataRow, 11), pM1 - pM12 _
        Else wsM.cells(dataRow, 11).Value = "-"

    ' Structure: prefer M12, fall back to M6 then M2 when far months unavailable.
    Dim farPrice As Double
    If pM12 > 0 Then
        farPrice = pM12
    ElseIf pM6 > 0 Then
        farPrice = pM6
    Else
        farPrice = pM2
    End If
    If farPrice > 0 Then ApplyStructure wsM.cells(dataRow, 13), pM1, farPrice

    okCount = okCount + 1

```

```

Else
    MarkError wsM, dataRow
    errCount = errCount + 1
End If

On Error GoTo 0
GoTo RowDone

RowFail:
    MarkError wsM, dataRow
    errCount = errCount + 1
    On Error GoTo 0

RowDone:
    Next i

' Wipe stray rows between the last Section-1 instrument and the
' Section-2 title band. Stop at sec2Data - 3 so we never touch:
'   sec2Data - 2 (Section 2 title row, merged A:M)
'   sec2Data - 1 (Section 2 header row)
' Unmerge each row first so the wipe cannot trip "Cannot change
' part of a merged cell" (Excel error 1004).
Dim wipeRow As Long, cc As Integer
For wipeRow = sec1 + instCnt To sec2Data - 3
    On Error Resume Next
    wsM.Range(wsM.cells(wipeRow, 1), wsM.cells(wipeRow, COL_LAST)).UnMerge
    On Error GoTo 0
    For cc = 1 To 13
        With wsM.cells(wipeRow, cc)
            .ClearContents
            .Interior.Color = C_BG
            .Font.Color = C_BG
            .Font.Name = THEME_FONT
            .Font.Size = THEME_FONT_SIZE
        End With
    Next cc
Next wipeRow

' Section 2: Calendar Spread Matrix (reads selected row from _Config!B5).
Call RefreshSpreadMatrix(wsM, wsTick, wsCfg, sec1, sec2Data)

' Timestamp: spread matrix occupies 6 rows -> first free row is sec2Data + 6.
Dim tsRow As Long
tsRow = sec2Data + 6
With wsM.cells(tsRow, 2)
    .Value = "Last updated: " & Format(Now(), "yyyy/mm/dd hh:mm:ss")
    .Font.Name = THEME_FONT
    .Font.Size = 8
    .Font.Color = RGB(100, 100, 100)
    .Interior.Color = C_BG
    .HorizontalAlignment = xlLeft
End With

Application.ScreenUpdating = True
Application.StatusBar = "Dashboard refreshed - " & okCount & " OK / " & errCount & " errors"
End Sub

' Backward-compat alias for any toolbar button still pointing at V2.
Sub RefreshDashboardV2()
    Call RefreshDashboard
End Sub

Private Sub RefreshSpreadMatrix(wsM As Worksheet, wsTick As Worksheet, _
                                wsCfg As Worksheet, sec1 As Long, sec2Data As Long)
    ' Resolve which Section-1 instrument feeds the spread matrix.
    ' _Config!B5 stores a 1-based offset within Section 1 (1 = first instrument).
    ' Defaults to 1 if missing / out of range / non-numeric.
    Dim instCnt As Long
    instCnt = CLng(wsCfg.Range("B2").Value)
    Dim selOffset As Long: selOffset = 1
    If IsNumeric(wsCfg.Range("B5").Value) Then
        selOffset = CLng(wsCfg.Range("B5").Value)
        If selOffset < 1 Or selOffset > instCnt Then selOffset = 1
    End If

```

```

End If
Dim selRow As Long: selRow = secl + selOffset - 1

' Robustness: if the user-selected row has no usable M1 price, scan
' forward from secl for the first row that does. Avoids the matrix
' silently displaying all-zero spreads when _Config is stale.
If Not RowHasValidPrices(wsM, selRow) Then
    Dim probe As Long
    For probe = secl To secl + instCnt - 1
        If RowHasValidPrices(wsM, probe) Then
            selRow = probe
            Exit For
        End If
    Next probe
End If

' Echo the selected instrument name into the section header so the user
' can see which curve is being decomposed.
On Error Resume Next
With wsM.cells(sec2Data - 1, 7)
    .Value = "Selected: " & wsM.cells(selRow, 2).Value
    .Font.Name = THEME_FONT
    .Font.Size = THEME_FONT_SIZE - 1
    .Font.Color = RGB(180, 180, 180)
    .Font.Italic = True
    .HorizontalAlignment = xlCenter
End With
On Error GoTo 0

' Read M1..M12 prices (BBG_COMM columns 4..8).
Dim prices(1 To 5) As Double
Dim c As Integer
For c = 1 To 5
    Dim v As Variant: v = wsM.cells(selRow, 3 + c).Value
    If IsNumeric(v) Then
        If CDbl(v) > 0 Then prices(c) = CDbl(v) Else prices(c) = 0
    Else
        prices(c) = 0
    End If
Next c

' Populate the TICKERS column (col 3 in the matrix) from the
' TickerMap row that corresponds to selRow.
Dim tickRow As Long: tickRow = (selRow - secl) + 2 ' +2: header row at 1, data starts at 2
Dim labels As Variant
labels = Array("M1-M2", "M1-M3", "M1-M6", "M1-M12", "M2-M3", "M3-M6")
Dim tickCols As Variant
' Column indexes in TickerMap (V3 layout): C=M1, D=M2, E=M3, F=M6, G=M12
tickCols = Array(Array(3, 4), Array(3, 5), Array(3, 6), Array(3, 7), Array(4, 5), Array(5, 6))
Dim jj As Integer
For jj = 0 To 5
    Dim ta As String, tb As String
    On Error Resume Next
    ta = CStr(wsTick.cells(tickRow, tickCols(jj)(0)).Value)
    tb = CStr(wsTick.cells(tickRow, tickCols(jj)(1)).Value)
    On Error GoTo 0
    With wsM.cells(sec2Data + jj, 3)
        .Value = ta & " / " & tb
        .Font.Name = THEME_FONT
        .Font.Size = 8
        .Font.Color = RGB(140, 140, 140)
        .Interior.Color = C_BG
        .HorizontalAlignment = xlCenter
        .VerticalAlignment = xlCenter
    End With
Next jj

Dim spreads(1 To 6) As Double
spreads(1) = prices(1) - prices(2)
spreads(2) = prices(1) - prices(3)
spreads(3) = prices(1) - prices(4)

```



```

        .Value = spd / prices(1)
        If spd >= 0 Then
            .Font.Color = RGB(0, 255, 136)
        Else
            .Font.Color = RGB(255, 51, 51)
        End If
    Else
        .Value = "-"
        .Font.Color = RGB(136, 136, 136)
    End If
End With

' Signal Flag (col 13)
With wsM.cells(r, 13)
    ApplyMatrixCellTheme wsM.cells(r, 13)
    .Font.Bold = True
    If Abs(spd) > 5 Then
        .Value = ">> WATCH": .Font.Color = RGB(255, 51, 51)
    ElseIf Abs(spd) > 1 Then
        .Value = "!! SHARP MOVE": .Font.Color = RGB(255, 192, 0)
    Else
        .Value = "OK STABLE": .Font.Color = RGB(136, 136, 136)
    End If
End With

' Reserved gap col 12 stays clean.
With wsM.cells(r, COL_UNUSED_GAP)
    .ClearContents
    .Interior.Color = C_BG
End With
Next j
End Sub

' Shared per-cell theme for Section-2 matrix cells (Consolas / centered / black bg).
Private Sub ApplyMatrixCellTheme(cell As Range)
    cell.Font.Name = THEME_FONT
    cell.Font.Size = THEME_FONT_SIZE
    cell.Interior.Color = C_BG
    cell.HorizontalAlignment = xlCenter
    cell.VerticalAlignment = xlCenter
End Sub

' =====
' ApplyDarkTheme
' One-shot UI sweep for BBG_COMM:
'   - Paint the entire used range pure black
'   - Set every cell font to Consolas
'   - Re-render Section 1 / Section 2 title bands (merged across A:M)
'   - Re-render legend band at top (rows 2-3)
'   - Center the instrument column (B) left, headers center
'   - Hide the prev-close cache column (14)
' Run this once after any layout change, or simply re-run; it is idempotent.
' =====
Sub ApplyDarkTheme()
    Dim wsM As Worksheet: Set wsM = ThisWorkbook.Sheets(SH_MAIN)
    Dim wsCfg As Worksheet: Set wsCfg = ThisWorkbook.Sheets(SH_CFG)

    Dim sec1 As Long: sec1 = CLng(wsCfg.Range("B1").Value)
    Dim instCnt As Long: instCnt = CLng(wsCfg.Range("B2").Value)
    Dim sec2Data As Long: sec2Data = CLng(wsCfg.Range("B3").Value)

    Application.ScreenUpdating = False

    ' Kill default Excel gridlines on this sheet so the pure-black canvas
    ' has no light-grey artefact lines bleeding through.
    On Error Resume Next
    wsM.Activate
    ActiveWindow.DisplayGridlines = False
    ActiveWindow.DisplayHeadings = True
    On Error GoTo 0

```

```

' --- 1. Paint everything pure black + Consolas across the dashboard band.
Dim lastRow As Long: lastRow = sec2Data + 8
Dim canvas As Range
Set canvas = wsM.Range(wsM.cells(1, 1), wsM.cells(lastRow, COL_LAST))
With canvas
    .Interior.Color = C_BG
    .Font.Name = THEME_FONT
    .Font.Size = THEME_FONT_SIZE
    .Font.Color = RGB(220, 220, 220)
    .VerticalAlignment = xlCenter
    ' Strip any leftover borders so frame color is fully managed by us.
    .Borders.LineStyle = xlNone
End With

' Whole-sheet sweep so neighbouring cells (units, helper text) also get Consolas.
With wsM.cells
    .Font.Name = THEME_FONT
End With

' --- 2. Legend band (rows 2-3). Merge B:M (column A stays as gutter).
UnmergeBand wsM, 2, 3
With wsM.Range(wsM.cells(2, 2), wsM.cells(2, COL_LAST))
    .Merge
    .Value = "BACKWARDATION -> Supply tightness. No storage incentive. Physical scarcity drives spot premium over forward."
    .Font.Name = THEME_FONT
    .Font.Color = RGB(0, 255, 136)
    .Font.Italic = True
    .Font.Size = THEME_FONT_SIZE
    .HorizontalAlignment = xlLeft
    .VerticalAlignment = xlCenter
    .IndentLevel = 1
    .Borders.LineStyle = xlNone
End With
With wsM.Range(wsM.cells(3, 2), wsM.cells(3, COL_LAST))
    .Merge
    .Value = "CONTANGO -> Oversupply. Storage economical - buy spot cheap, sell forward at premium. Physical demand weak."
    .Font.Name = THEME_FONT
    .Font.Color = RGB(255, 51, 51)
    .Font.Italic = True
    .Font.Size = THEME_FONT_SIZE
    .HorizontalAlignment = xlLeft
    .VerticalAlignment = xlCenter
    .IndentLevel = 1
    .Borders.LineStyle = xlNone
End With

' --- 3. Section 1 title band (row sec1 - 2) and header row (sec1 - 1).
' Title merges B:M (column A stays as gutter).
Dim sec1Title As Long: sec1Title = sec1 - 2
Dim sec1Head As Long: sec1Head = sec1 - 1
If sec1Title >= 4 Then
    UnmergeBand wsM, sec1Title, sec1Title
    With wsM.Range(wsM.cells(sec1Title, 2), wsM.cells(sec1Title, COL_LAST))
        .Merge
        .Value = "< SECTION 1 > CURVE STRUCTURE SIGNAL - FRONT MONTH vs DEFERRED MONTHS"
        .Font.Name = THEME_FONT
        .Font.Color = RGB(220, 220, 220)
        .Font.Bold = True
        .Font.Size = THEME_FONT_SIZE + 1
        .HorizontalAlignment = xlCenter
        .VerticalAlignment = xlCenter
        .Interior.Color = RGB(20, 20, 20)
        .Borders.LineStyle = xlNone
        .Borders(xlEdgeBottom).LineStyle = xlContinuous
        .Borders(xlEdgeBottom).Color = RGB(60, 60, 60)
        .Borders(xlEdgeBottom).Weight = xlThin
    End With
End If
PaintHeaderRow wsM, sec1Head, _
Array("", "INSTRUMENT", "% Change", "M1", "M2", "M3", "M6", "M12", _
    "M1-M2 SPRD", "M1-M6 SPRD", "M1-M12 SPRD", "", "STRUCTURE")

```

```
' --- 4. Section 1 instrument column (B) left-align bold, dim units.
```

```
Dim i As Long
```

```
For i = 0 To instCnt - 1
```

```
    With wsM.cells(sec1 + i, 2)
```

```
        .Font.Name = THEME_FONT
```

```
        .Font.Size = THEME_FONT_SIZE
```

```
        .Font.Bold = True
```

```
        .Font.Color = RGB(220, 220, 220)
```

```
        .HorizontalAlignment = xlLeft
```

```
        .VerticalAlignment = xlCenter
```

```
        .IndentLevel = 1
```

```
    End With
```

```
Next i
```

```
' --- 5. Section 2 title + header. Title merges B:M (column A stays gutter).
```

```
Dim sec2Title As Long: sec2Title = sec2Data - 2
```

```
Dim sec2Head As Long: sec2Head = sec2Data - 1
```

```
If sec2Title >= sec1 + instCnt Then
```

```
    UnmergeBand wsM, sec2Title, sec2Title
```

```
    With wsM.Range(wsM.cells(sec2Title, 2), wsM.cells(sec2Title, COL_LAST))
```

```
        .Merge
```

```
        .Value = "< SECTION 2 >  CALENDAR SPREAD MATRIX  -  SELECT INSTRUMENT VIA _Config!B5"
```

```
        .Font.Name = THEME_FONT
```

```
        .Font.Color = RGB(220, 220, 220)
```

```
        .Font.Bold = True
```

```
        .Font.Size = THEME_FONT_SIZE + 1
```

```
        .HorizontalAlignment = xlCenter
```

```
        .VerticalAlignment = xlCenter
```

```
        .Interior.Color = RGB(20, 20, 20)
```

```
        .Borders.LineStyle = xlNone
```

```
        .Borders(xlEdgeBottom).LineStyle = xlContinuous
```

```
        .Borders(xlEdgeBottom).Color = RGB(60, 60, 60)
```

```
        .Borders(xlEdgeBottom).Weight = xlThin
```

```
    End With
```

```
End If
```

```
PaintHeaderRow wsM, sec2Head, _
```

```
    Array("", "SPREAD PERIOD", "TICKERS", "VALUE", "DIRECTION", "STRENGTH", _  
        "INTERPRETATION", "ROLL YIELD %", "PRIOR WEEK", "WoW CHANGE", "MOMENTUM", "", "SIGNAL FLAG")
```

```
' --- 6. Section 2 first column (SPREAD PERIOD labels).
```

```
Dim sLabels As Variant
```

```
sLabels = Array("M1 - M2  [ 1mo ]", "M1 - M3  [ 2mo ]", _  
                "M1 - M6   [ 5mo ]", "M1 - M12 [11mo ]", _  
                "M2 - M3   [ roll ]", "M3 - M6   [ roll ]")
```

```
Dim k As Long
```

```
For k = 0 To 5
```

```
    With wsM.cells(sec2Data + k, 2)
```

```
        .Value = sLabels(k)
```

```
        .Font.Name = THEME_FONT
```

```
        .Font.Size = THEME_FONT_SIZE
```

```
        .Font.Bold = True
```

```
        .Font.Color = RGB(220, 220, 220)
```

```
        .HorizontalAlignment = xlLeft
```

```
        .IndentLevel = 1
```

```
    End With
```

```
Next k
```

```
' --- 7. Unified column widths sized for the LONGEST content per column.
```

```
' A : 3  - left gutter (visible margin so row-2/3/5 merges read as "from col B")
```

```
' B : 26 - INSTRUMENT names / SPREAD PERIOD labels
```

```
' C : 22 - TICKERS pair "CLM26.NYM / CLM26.NYM" (Section 2) / % Change (Section 1)
```

```
' D : 12 - M1 / VALUE
```

```
' E : 13 - M2 / DIRECTION ("[+] BKWD")
```

```
' F : 11 - M3 / STRENGTH ("MODERATE")
```

```
' G : 36 - M6 / INTERPRETATION ("TIGHT SUPPLY / STRONG SPOT DEMAND" = 33)
```

```
' H : 13 - M12 / ROLL YIELD %
```

```
' I : 13 - M1-M2 SPRD / PRIOR WEEK
```

```
' J : 13 - M1-M6 SPRD / WoW CHANGE
```

```
' K : 13 - M1-M12 SPRD / MOMENTUM
```

```
' L : 3  - reserved gap
```

```

' M : 22 - STRUCTURE ("BACKWARDATION") / SIGNAL FLAG ("!! SHARP MOVE")
wsM.Columns(1).ColumnWidth = 3
wsM.Columns(2).ColumnWidth = 26
wsM.Columns(3).ColumnWidth = 22
wsM.Columns(4).ColumnWidth = 12
wsM.Columns(5).ColumnWidth = 13
wsM.Columns(6).ColumnWidth = 11
wsM.Columns(7).ColumnWidth = 36
wsM.Columns(8).ColumnWidth = 13
wsM.Columns(9).ColumnWidth = 13
wsM.Columns(10).ColumnWidth = 13
wsM.Columns(11).ColumnWidth = 13
wsM.Columns(COL_UNUSED_GAP).ColumnWidth = 3
wsM.Columns(COL_LAST).ColumnWidth = 22
wsM.Columns(COL_PREVCACHE).Hidden = True

' --- 7b. Uniform row heights. Rows 21-23 came out short because
' SyncLayoutFromTickerMap appended new instrument rows without resetting
' the row height. Normalise every data row and the title bands.
Dim rh As Long
For rh = sec1 To sec1 + instCnt - 1
    wsM.Rows(rh).RowHeight = 20
Next rh
For rh = sec2Data To sec2Data + 5
    wsM.Rows(rh).RowHeight = 20
Next rh
wsM.Rows(2).RowHeight = 18
wsM.Rows(3).RowHeight = 18
If sec1Title >= 4 Then wsM.Rows(sec1Title).RowHeight = 24
If sec2Title >= sec1 + instCnt Then wsM.Rows(sec2Title).RowHeight = 24
wsM.Rows(sec1Head).RowHeight = 22
wsM.Rows(sec2Head).RowHeight = 22

' --- 8. Freeze panes at first data row of Section 1 so headers stay visible.
On Error Resume Next
wsM.Activate
wsM.cells(sec1, 1).Select
ActiveWindow.FreezePanes = False
ActiveWindow.FreezePanes = True
On Error GoTo 0

' --- 9. Sheet tab color: Bloomberg-style amber so BBG_COMM stands out
' in the workbook tab bar.
On Error Resume Next
wsM.Tab.Color = RGB(230, 170, 40)
On Error GoTo 0

Application.ScreenUpdating = True
MsgBox "Dark theme applied. Run RefreshDashboard to repopulate prices.", vbInformation
End Sub

' Helper - unmerge a horizontal band before re-merging (idempotent).
' Sweeps a wide range so legacy merges that include column A or extend past
' COL_LAST are also broken before the new merge is applied.
Private Sub UnmergeBand(wsM As Worksheet, rTop As Long, rBot As Long)
    On Error Resume Next
    wsM.Range(wsM.cells(rTop, 1), wsM.cells(rBot, COL_LAST + 4)).UnMerge
    On Error GoTo 0
End Sub

' Helper - paint a header row from an array of labels (COL_LAST cells wide).
Private Sub PaintHeaderRow(wsM As Worksheet, rowIdx As Long, labels As Variant)
    Dim c As Long
    For c = 1 To COL_LAST
        With wsM.cells(rowIdx, c)
            .Value = labels(c - 1)
            .Font.Name = THEME_FONT
            .Font.Size = THEME_FONT_SIZE
            .Font.Bold = True
            .Font.Color = RGB(180, 180, 180)
            .Interior.Color = RGB(25, 25, 25)
        End With
    Next c
End Sub

```

```

        .HorizontalAlignment = xlCenter
        .VerticalAlignment = xlCenter
        .Borders(xlEdgeBottom).LineStyle = xlContinuous
        .Borders(xlEdgeBottom).Color = RGB(80, 80, 80)
        .Borders(xlEdgeBottom).Weight = xlThin
    End With
Next c
End Sub
' =====
' Price fetcher with dual-endpoint fallback + retry
' =====
Private Function FetchPrice(symbol As String) As Double
    On Error GoTo Bail
    If symbol = "" Then FetchPrice = 0: Exit Function
    Dim attempt As Integer, result As Double
    For attempt = 1 To MAX_RETRY
        result = FetchYahoo(symbol)
        If result > 0 Then FetchPrice = result: Exit Function
        Call PauseMS(600 * attempt)
    Next attempt
    FetchPrice = 0
    Exit Function
Bail:
    FetchPrice = 0
End Function

' Tries the per-month ticker first, then falls back to the front-month
' continuous contract for this base if the monthly symbol returns nothing.
' Yahoo's coverage of CME monthly tickers (e.g. CLM26.NYM, ESH26.CME) is
' inconsistent, so this fallback keeps the dashboard populated rather than
' showing ERR for every far-month contract.
'
' Equity index futures (ES/NQ/YM/RTY/VX) are short-circuited: Yahoo only
' carries the continuous symbol for these, so we skip the monthly lookup
' entirely to avoid wasting 3 retries per cell on a guaranteed 404.
Private Function FetchPriceWithFallback(symbol As String, baseSym As String) As Double
    If IsIndexFutureBase(baseSym) Then
        FetchPriceWithFallback = FetchPrice(FrontMonthYahooSymbol(baseSym))
        Exit Function
    End If

    Dim p As Double
    p = FetchPrice(symbol)
    If p > 0 Then FetchPriceWithFallback = p: Exit Function

    If LenB(baseSym) > 0 Then
        Dim alt As String: alt = FrontMonthYahooSymbol(baseSym)
        ' Avoid an infinite loop if caller already passed the same form.
        If StrComp(alt, symbol, vbTextCompare) <> 0 Then
            p = FetchPrice(alt)
            If p > 0 Then FetchPriceWithFallback = p: Exit Function
        End If
    End If
    FetchPriceWithFallback = 0
End Function

' Whitelist of equity index futures whose individual-month contracts are
' NOT carried by Yahoo Finance. Treat as front-month-only.
Private Function IsIndexFutureBase(baseSym As String) As Boolean
    Select Case UCase(Trim(baseSym))
        Case "ES", "NQ", "YM", "RTY", "VX"
            IsIndexFutureBase = True
        Case Else
            IsIndexFutureBase = False
    End Select
End Function

' Resolves the Yahoo symbol that actually serves data for this base's
' front-month / spot quote. Most futures use <base>=F, but a handful of
' instruments need a special route:
'   VX -> ^VIX (Yahoo has no VX=F; ^VIX is the VIX spot index, which is

```

```
Sub FixAndRebuild()  
    '?w?w ?B?Jl?G???? _Config ??? ?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?  
    Dim wsCfg As Worksheet  
    On Error Resume Next  
    Set wsCfg = ThisWorkbook.Sheets("_Config")  
    On Error GoTo 0  
    If wsCfg Is Nothing Then  
        Set wsCfg = ThisWorkbook.Sheets.Add  
        wsCfg.Name = "_Config"  
        wsCfg.Visible = xlSheetHidden  
    End If  
  
    ' ????? BBG_COMM ?u?@????????c  
    Dim wsM As Worksheet  
    Set wsM = ThisWorkbook.Sheets("BBG_COMM")  
  
    ' ???y?? Section1 ?_?l?C?]??uINSTRUMENT?v???D?^  
    Dim sec1 As Long, i As Long  
    For i = 1 To 50  
        If InStr(wsM.cells(i, 2).Value, "INSTRUMENT") > 0 Then  
            sec1 = i + 1      ' ???q???D?U?@?C?}?l  
            Exit For  
        End If  
    Next i  
    If sec1 = 0 Then sec1 = 7      ' fallback ?w]???  
  
    ' ???y?? Section2 ?_?l?C?]??uSPREAD PERIOD?v???D?^  
    Dim sec2Hdr As Long, sec2Data As Long  
    For i = sec1 To 100  
        If InStr(wsM.cells(i, 2).Value, "SPREAD PERIOD") > 0 Then  
            sec2Hdr = i  
            sec2Data = i + 1  
            Exit For  
        End If  
    Next i  
    If sec2Data = 0 Then sec2Data = sec1 + 20      ' fallback  
  
    ' ?p???~??q?}sec1 ?? sec2Hdr ?????????e???C?^  
    Dim instCnt As Long  
    For i = sec1 To sec2Hdr - 1  
        If Trim(wsM.cells(i, 2).Value) <> "" Then instCnt = instCnt + 1  
    Next i  
    If instCnt = 0 Then instCnt = 14      ' fallback  
  
    '?g?J _Config  
    wsCfg.Range("A1") = "SEC1_START": wsCfg.Range("B1") = sec1  
    wsCfg.Range("A2") = "INST_COUNT": wsCfg.Range("B2") = instCnt  
    wsCfg.Range("A3") = "SEC2_DATA_START": wsCfg.Range("B3") = sec2Data  
    ' Spread-matrix instrument selector (1-based offset within Section 1).
```

```

If Not IsNumeric(wsCfg.Range("B5").Value) Then
    wsCfg.Range("A5") = "SpreadMatrix_Offset": wsCfg.Range("B5") = 1
End If

MsgBox "? _Config ?w??_?I" & vbCr & vbCr & _
    "sec1      = " & sec1 & vbCr & _
    "instCnt   = " & instCnt & vbCr & _
    "sec2Data  = " & sec2Data & vbCr & vbCr & _
    "?{?b?i?H??? RefreshDashboard", vbInformation
End Sub

Private Function FetchYahoo(symbol As String) As Double
    ' Full error trap: any HTTP / parse failure returns 0 instead of bubbling
    ' up to the caller. A fresh MSXML2.XMLHTTP is created per endpoint so a
    ' stuck state on attempt #1 cannot poison attempt #2.
    On Error GoTo Bail

    ' URL-encode the few characters that appear in Yahoo symbols but break
    ' raw URL parsing in MSXML. Most important: ^VIX -> %5EVIX.
    Dim safeSym As String: safeSym = symbol
    safeSym = Replace(safeSym, "^", "%5E")

    Dim eps(1) As String
    eps(0) = "https://query1.finance.yahoo.com/v8/finance/chart/" & safeSym & "?interval=1d&range=1d"
    eps(1) = "https://query2.finance.yahoo.com/v8/finance/chart/" & safeSym & "?interval=1d&range=1d"

    Dim ep As Integer
    For ep = 0 To 1
        Dim http As Object
        Set http = Nothing
        Set http = CreateObject("MSXML2.XMLHTTP")

        Dim sent As Boolean: sent = False
        On Error Resume Next
        http.Open "GET", eps(ep), False
        http.setRequestHeader "User-Agent", "Mozilla/5.0 (Windows NT 10.0)"
        http.send
        sent = (Err.Number = 0)
        On Error GoTo Bail

        If sent Then
            Dim status As Long: status = 0
            On Error Resume Next
            status = http.status
            On Error GoTo Bail

            If status = 200 Then
                Dim response As String: response = ""
                On Error Resume Next
                response = http.responseText
                On Error GoTo Bail

                Dim parsed As Double
                parsed = ParseYahooPrice(response)
                If parsed > 0 Then FetchYahoo = parsed: Exit Function
            End If
        End If
        Set http = Nothing
    Next ep

    FetchYahoo = 0
    Exit Function
Bail:
    FetchYahoo = 0
End Function

' Pulls regularMarketPrice from the Yahoo chart JSON, with a fallback to
' the last numeric value in the close[] array.
Private Function ParseYahooPrice(response As String) As Double
    On Error GoTo Bail
    If LenB(response) = 0 Then ParseYahooPrice = 0: Exit Function

```

```

Dim pos As Long
pos = InStr(response, """"regularMarketPrice"":")
If pos > 0 Then
    Dim tmp As String
    tmp = Mid(response, pos + 22, 20)
    tmp = Split(tmp, ",")(0)
    tmp = Split(tmp, "}")(0)
    tmp = Trim(tmp)
    If IsNumeric(tmp) Then
        If Cdbl(tmp) > 0 Then ParseYahooPrice = Cdbl(tmp): Exit Function
    End If
End If

pos = InStr(response, """"close"":[")
If pos > 0 Then
    Dim arr As String
    arr = Mid(response, pos + 9, 300)
    arr = Split(arr, "]")(0)
    Dim parts() As String: parts = Split(arr, ",")
    ' Walk from the tail backwards, skipping nulls that Yahoo sometimes
    ' emits for the most recent (still-open) bar.
    Dim k As Long
    For k = UBound(parts) To LBound(parts) Step -1
        Dim cand As String: cand = Trim(parts(k))
        If IsNumeric(cand) Then
            If Cdbl(cand) > 0 Then ParseYahooPrice = Cdbl(cand): Exit Function
        End If
    Next k
End If

ParseYahooPrice = 0
Exit Function
Bail:
ParseYahooPrice = 0
End Function

```

```

Sub FixTickerMap()
    Dim wsTick As Worksheet
    On Error Resume Next
    Set wsTick = ThisWorkbook.Sheets("TickerMap")
    On Error GoTo 0
    If wsTick Is Nothing Then
        Set wsTick = ThisWorkbook.Sheets.Add
        wsTick.Name = "TickerMap"
        wsTick.Visible = xlSheetHidden
    End If

    wsTick.cells.Clear

    ' ???D
    Dim h As Variant
    Dim headers As Variant
    headers = Array("Instrument", "M1", "M2", "M3", "M6", "M12")
    Dim ci As Integer
    For ci = 0 To 5
        wsTick.cells(1, ci + 1).Value = headers(ci)
    Next ci

```

```

' ?w?w Yahoo Finance ?s??X??u?? =F?]??^?A
'      ????? +1/+2/+5/+11 ??????N?X
'      ?榛?GCLK25.NYM, CLM25.NYM... ???W Yahoo ???
'      ??i?a???k?G?????ΦP?@?? =F ?N?X?A?? VBA ?????
'      ?????? Quandl/CME ?榛?A?X?? M2 spread ?????

```

```

Dim data As Variant
data = Array( _
    Array("WTI CRUDE OIL", "CL=F", "CL=F", "CL=F", "CL=F", "CL=F"), _
    Array("BRENT CRUDE", "BZ=F", "BZ=F", "BZ=F", "BZ=F", "BZ=F"), _
    Array("RBOB GASOLINE", "RB=F", "RB=F", "RB=F", "RB=F", "RB=F"), _
    Array("ULSD DIESEL", "HO=F", "HO=F", "HO=F", "HO=F", "HO=F"), _
    Array("NATURAL GAS", "NG=F", "NG=F", "NG=F", "NG=F", "NG=F"), _

```

```
Array("GOLD", "GC=F", "GC=F", "GC=F", "GC=F", "GC=F"), _  
    Array("COPPER", "HG=F", "HG=F", "HG=F", "HG=F", "HG=F"), _  
    Array("SILVER", "SI=F", "SI=F", "SI=F", "SI=F", "SI=F"), _  
    Array("CORN", "ZC=F", "ZC=F", "ZC=F", "ZC=F", "ZC=F"), _  
    Array("SOYBEANS", "ZS=F", "ZS=F", "ZS=F", "ZS=F", "ZS=F"), _  
    Array("WHEAT", "ZW=F", "ZW=F", "ZW=F", "ZW=F", "ZW=F"), _  
    Array("ALUMINUM", "ALI=F", "ALI=F", "ALI=F", "ALI=F", "ALI=F"), _  
    Array("LUMBER", "LBS=F", "LBS=F", "LBS=F", "LBS=F", "LBS=F"), _  
    Array("STEEL HRC", "HR=F", "HR=F", "HR=F", "HR=F", "HR=F"), _  
    Array("S&P 500 E-MINI", "ES=F", "ES=F", "ES=F", "ES=F", "ES=F"), _  
    Array("NASDAQ-100 EMINI", "NQ=F", "NQ=F", "NQ=F", "NQ=F", "NQ=F"), _  
    Array("DOW E-MINI", "YM=F", "YM=F", "YM=F", "YM=F", "YM=F"), _  
    Array("RUSSELL 2000", "RTY=F", "RTY=F", "RTY=F", "RTY=F", "RTY=F"), _  
    Array("VIX", "VX=F", "VX=F", "VX=F", "VX=F", "VX=F") _  
  
)  
  
Dim r As Integer, col As Integer  
For r = 0 To UBound(data)  
    For col = 0 To 5  
        wsTick.cells(r + 2, col + 1).Value = data(r)(col)  
    Next col  
Next r  
  
MsgBox "? TickerMap ?w??s?I?@ " & UBound(data) + 1 & " ???~?C" & vbCr &  
    "{?b???? RefreshDashboard", vbInformation
```

End Sub

' =====
' ??A?p??U??X???N?X & % Change ???
' ???N???? FixTickerMap ?M RefreshDashboard ????????q??
' =====

' ?w?w ????N?X???? ?w?
' Yahoo Finance ???f???GF=1?? G=2 H=3 J=4 K=5 M=6
' N=7?? Q=8 U=9 V=10 X=11 Z=12

Function MonthCode(m As Integer) As String
 Dim codes As String: codes = "FGHIJKMNQUVYZ"
 MonthCode = Mid(codes, m, 1)

End Function

' ?w?w ?p??? n ?????X???N?X ?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?w?

Function FuturesTicker(baseSymbol As String, exchange As String, _
 monthsAhead As Long) As String

 Dim targetDate As Date
 targetDate = DateAdd("m", monthsAhead, Date)

 Dim y As Integer: y = Year(targetDate)
 Dim m As Integer: m = Month(targetDate)

 '?榛?GCLK26.NYM
 FuturesTicker = baseSymbol & MonthCode(m) & Right(CStr(y), 2) & exchange

End Function

' ?w?w ?U??~????????? ?w?
' .NYM = NYMEX .CMX = COMEX .CBT = CBOT

Function GetExchange(commodity As String) As String
 Select Case commodity
 Case "CL", "RB", "HO", "NG": GetExchange = ".NYM"
 Case "BZ": GetExchange = ".NYM" ' Brent ?P NYMEX
 Case "GC", "SI", "HG", "ALI": GetExchange = ".CMX"
 Case "HR": GetExchange = ".CMX"
 Case "ZC", "ZS", "ZW": GetExchange = ".CBT"
 ' --- Equity index futures ---
 Case "ES", "NQ", "RTY": GetExchange = ".CME" ' E-mini S&P / Nasdaq / Russell
 Case "YM": GetExchange = ".CBT" ' E-mini Dow on CBOT
 Case "VX": GetExchange = ".CFE" ' VIX on CFE
 Case Else: GetExchange = ".CMX"

End Select

End Function

' Index futures only roll on quarterly cycle (H/M/U/Z = Mar/Jun/Sep/Dec)

```
Function QuarterlyMonthCode(m As Integer) As String
```

```
    Select Case m
```

```
        Case 1, 2, 3:    QuarterlyMonthCode = "H"    ' March
        Case 4, 5, 6:    QuarterlyMonthCode = "M"    ' June
        Case 7, 8, 9:    QuarterlyMonthCode = "U"    ' September
        Case 10, 11, 12: QuarterlyMonthCode = "Z"    ' December
```

```
    End Select
```

```
End Function
```

```
' Snap requested month-ahead forward to the NEXT quarterly contract.
```

```
Function FuturesTickerQuarterly(baseSymbol As String, exchange As String, _  
                                monthsAhead As Long) As String
```

```
    Dim targetDate As Date
```

```
    targetDate = DateAdd("m", monthsAhead, Date)
```

```
    Dim y As Integer: y = Year(targetDate)
```

```
    Dim m As Integer: m = Month(targetDate)
```

```
    FuturesTickerQuarterly = baseSymbol & QuarterlyMonthCode(m) & Right(CStr(y), 2) & exchange
```

```
End Function
```

```
' Returns True for symbols that only trade on the H/M/U/Z quarterly cycle.
```

```
Function IsQuarterlyOnly(baseSymbol As String) As Boolean
```

```
    Select Case baseSymbol
```

```
        Case "ES", "NQ", "YM", "RTY"  
            IsQuarterlyOnly = True
```

```
        Case Else
```

```
            IsQuarterlyOnly = False
```

```
    End Select
```

```
End Function
```

```
'  
=====
```

```
' ??? TickerMap ?X ???A???N?X
```

```
Sub RebuildTickerMapDynamic()
```

```
    Dim wsTick As Worksheet
```

```
    On Error Resume Next
```

```
    Set wsTick = ThisWorkbook.Sheets("TickerMap")
```

```
    On Error GoTo 0
```

```
    If wsTick Is Nothing Then
```

```
        Set wsTick = ThisWorkbook.Sheets.Add
```

```
        wsTick.Name = "TickerMap"
```

```
        wsTick.Visible = xlSheetHidden
```

```
    End If
```

```
    wsTick.cells.Clear
```

```
' ???D
```

```
wsTick.cells(1, 1) = "Instrument"
```

```
wsTick.cells(1, 2) = "Base"
```

```
wsTick.cells(1, 3) = "M1"
```

```
wsTick.cells(1, 4) = "M2"
```

```
wsTick.cells(1, 5) = "M3"
```

```
wsTick.cells(1, 6) = "M6"
```

```
wsTick.cells(1, 7) = "M12"
```

```
' [??~?W??, ??!N?X]
```

```
Dim items As Variant
```

```
items = Array( _
```

```
    Array("WTI CRUDE OIL", "CL"), _
```

```
    Array("BRENT CRUDE", "BZ"), _
```

```
    Array("RBOB GASOLINE", "RB"), _
```

```
    Array("ULSD DIESEL", "HO"), _
```

```
    Array("NATURAL GAS", "NG"), _
```

```
    Array("GOLD", "GC"), _
```

```
    Array("COPPER", "HG"), _
```

```
    Array("SILVER", "SI"), _
```

```
    Array("CORN", "ZC"), _
```

```
    Array("SOYBEANS", "ZS"), _
```

```
    Array("WHEAT", "ZW"), _
```

```
    Array("ALUMINUM", "ALI"), _
```

```
    Array("STEEL HRC", "HR") _
```

```
)
```

```

Dim offsets As Variant
offsets = Array(1, 2, 3, 6, 12) ' M1, M2, M3, M6, M12

Dim i As Integer, j As Integer
For i = 0 To UBound(items)
    Dim row As Integer: row = i + 2
    Dim base As String: base = items(i)(1)
    Dim exch As String: exch = GetExchange(base)

    wsTick.cells(row, 1) = items(i)(0)
    wsTick.cells(row, 2) = base

    For j = 0 To 4
        wsTick.cells(row, j + 3) = FuturesTicker(base, exch, CInt(offsets(j)))
    Next j
Next i

' ???G?w??
Dim preview As String
preview = "??A?????X??N?X?]H WTI ???^?G" & vbCr & vbCr
For j = 0 To 4
    Dim mo As Variant: mo = Array("M1", "M2", "M3", "M6", "M12")
    preview = preview & mo(j) & " = " & wsTick.cells(2, j + 3).Value & vbCr
Next j

MsgBox preview & vbCr & "?@ " & UBound(items) + 1 & " ???~?w??s?C", vbInformation
End Sub

```

```

' V3 TickerMap builder
' - Uses Yahoo-style per-month tickers (<base><MonthCode><YY><.EXCH>)
' - Honors quarterly-only contracts for equity index futures (ES/NQ/YM/RTY)
' - Adds equity index futures (S&P 500, Nasdaq-100, Dow, Russell 2000, VIX)
'   forward curve alongside the commodity universe.
' Layout (same as RebuildTickerMapDynamic):
'   A=Instrument | B=Base | C=M1 | D=M2 | E=M3 | F=M6 | G=M12
' Use with RefreshDashboardV2.

```

```

Sub RebuildTickerMapV3()
    Dim wsTick As Worksheet
    On Error Resume Next
    Set wsTick = ThisWorkbook.Sheets("TickerMap")
    On Error GoTo 0
    If wsTick Is Nothing Then
        Set wsTick = ThisWorkbook.Sheets.Add
        wsTick.Name = "TickerMap"
        wsTick.Visible = xlSheetHidden
    End If
    wsTick.cells.Clear

    wsTick.cells(1, 1) = "Instrument"
    wsTick.cells(1, 2) = "Base"
    wsTick.cells(1, 3) = "M1"
    wsTick.cells(1, 4) = "M2"
    wsTick.cells(1, 5) = "M3"
    wsTick.cells(1, 6) = "M6"
    wsTick.cells(1, 7) = "M12"

```

```

' [Instrument display name, Base symbol]
Dim items As Variant
items = Array( _
    Array("WTI CRUDE OIL", "CL"), _
    Array("BRENT CRUDE", "BZ"), _
    Array("RBOB GASOLINE", "RB"), _
    Array("ULSD DIESEL", "HO"), _
    Array("NATURAL GAS", "NG"), _
    Array("GOLD", "GC"), _
    Array("COPPER", "HG"), _
    Array("SILVER", "SI"), _
    Array("CORN", "ZC"), _
    Array("SOYBEANS", "ZS"), _

```

```

    Array("WHEAT", "ZW"), _
    Array("ALUMINUM", "ALI"), _
    Array("STEEL HRC", "HR"), _
    Array("S&P 500 E-MINI", "ES"), _
    Array("NASDAQ-100 EMINI", "NQ"), _
    Array("DOW E-MINI", "YM"), _
    Array("RUSSELL 2000", "RTY"), _
    Array("VIX", "VX") _
)

Dim offsets As Variant
offsets = Array(1, 2, 3, 6, 12) ' M1, M2, M3, M6, M12

Dim i As Integer, j As Integer
For i = 0 To UBound(items)
    Dim r As Integer: r = i + 2
    Dim base As String: base = items(i)(1)
    Dim exch As String: exch = GetExchange(base)

    wsTick.cells(r, 1) = items(i)(0)
    wsTick.cells(r, 2) = base

    For j = 0 To 4
        If IsQuarterlyOnly(base) Then
            wsTick.cells(r, j + 3) = FuturesTickerQuarterly(base, exch, CLng(offsets(j)))
        Else
            wsTick.cells(r, j + 3) = FuturesTicker(base, exch, CLng(offsets(j)))
        End If
    Next j
Next i

MsgBox "TickerMap V3 rebuilt - " & UBound(items) + 1 & " instruments " & _
    "(commodities + equity index futures forward curve)." & vbCr & vbCr & _
    "WTI M1 = " & wsTick.cells(2, 3).Value & vbCr & _
    "S&P M1 = " & wsTick.cells(15, 3).Value & vbCr & _
    "S&P M6 = " & wsTick.cells(15, 6).Value & vbCr & vbCr & _
    "Run SyncLayoutFromTickerMap then RefreshDashboard.", vbInformation
End Sub

```

```

' SyncLayoutFromTickerMap
' - Reads every Instrument row from TickerMap and pushes it into
'   BBG_COMM Section 1 column B, then updates _Config!B2 (INST_COUNT).
' - This is what makes newly added index futures actually appear in
'   the dashboard. Run once after RebuildTickerMapV3.
' - If Section 1 is about to overflow into Section 2, shifts the
'   Section 2 anchor (_Config!B3) downward so they do not collide.

```

```

Sub SyncLayoutFromTickerMap()
    Dim wsM As Worksheet: Set wsM = ThisWorkbook.Sheets(SH_MAIN)
    Dim wsTick As Worksheet: Set wsTick = ThisWorkbook.Sheets(SH_TICK)
    Dim wsCfg As Worksheet: Set wsCfg = ThisWorkbook.Sheets(SH_CFG)

    Dim sec1 As Long: sec1 = CLng(wsCfg.Range("B1").Value)
    Dim sec2Data As Long: sec2Data = CLng(wsCfg.Range("B3").Value)
    Dim oldCnt As Long: oldCnt = CLng(wsCfg.Range("B2").Value)

    ' Count TickerMap data rows (skip header at row 1, stop on blank Instrument).
    Dim newCnt As Long: newCnt = 0
    Dim i As Long
    For i = 2 To 200
        If Trim(CStr(wsTick.cells(i, 1).Value)) = "" Then Exit For
        newCnt = newCnt + 1
    Next i
    If newCnt = 0 Then
        MsgBox "TickerMap is empty - run RebuildTickerMapV3 first.", vbExclamation
        Exit Sub
    End If

    ' If the new instrument list runs past the existing Section 2 anchor,
    ' push Section 2 down (plus 1 blank gap row).

```

```

If sec1 + newCnt > sec2Data Then
    Dim shift As Long: shift = (sec1 + newCnt + 1) - sec2Data
    sec2Data = sec2Data + shift
    wsCfg.Range("B3").Value = sec2Data
End If

' Clear any rows beyond the new count (was 18 -> now 14 case).
' Unmerge each row first to survive previously merged title bands.
If newCnt < oldCnt Then
    Dim wr As Long, cc As Integer
    For wr = sec1 + newCnt To sec1 + oldCnt - 1
        On Error Resume Next
        wsM.Range(wsM.cells(wr, 1), wsM.cells(wr, COL_LAST)).UnMerge
        On Error GoTo 0
        For cc = 1 To 13
            With wsM.cells(wr, cc)
                .ClearContents
                .Interior.Color = C_BG
                .Font.Color = C_BG
                .Font.Name = THEME_FONT
                .Font.Size = THEME_FONT_SIZE
            End With
        Next cc
    Next wr
End If

' Write instrument names into BBG_COMM column B.
For i = 1 To newCnt
    Dim r As Long: r = sec1 + i - 1
    Dim instName As String: instName = CStr(wsTick.cells(i + 1, 1).Value)
    Dim baseSym As String: baseSym = CStr(wsTick.cells(i + 1, 2).Value)
    With wsM.cells(r, 2)
        .Value = instName & " [" & baseSym & "]"
        .Font.Name = THEME_FONT
        .Font.Size = THEME_FONT_SIZE
        .Font.Bold = True
        .Font.Color = RGB(220, 220, 220)
        .Interior.Color = C_BG
        .HorizontalAlignment = xlLeft
        .VerticalAlignment = xlCenter
        .IndentLevel = 1
    End With
Next i

wsCfg.Range("B2").Value = newCnt

MsgBox "Layout synced: " & newCnt & " instruments (was " & oldCnt & ")." & vbCr & _
"Section 2 anchor: row " & sec2Data & vbCr & vbCr & _
"Now run RefreshDashboard.", vbInformation
End Sub

```

```

' =====
' Cell-writing helpers
' =====

```

```

Private Sub WritePrice(cell As Range, price As Double)
    cell.Interior.Color = C_BG
    cell.Font.Name = THEME_FONT
    cell.Font.Size = THEME_FONT_SIZE
    cell.Font.Bold = False
    cell.HorizontalAlignment = xlRight
    cell.VerticalAlignment = xlCenter
    If price > 0 Then
        cell.NumberFormat = "#,##0.00"
        cell.Value = price
        cell.Font.Color = RGB(220, 220, 220)
    Else
        ' No usable quote for this contract month - show a soft dash so
        ' the user can distinguish "not available" from a real zero price.
        cell.NumberFormat = "@"
        cell.Value = "-"
        cell.Font.Color = RGB(100, 100, 100)
    End If
End Sub

```

```

        cell.HorizontalAlignment = xlCenter
    End If
End Sub

Private Sub WriteSpread(cell As Range, spd As Double)
    ' Show "0.00" for a genuine zero so e.g. equity-index rows (where M1..M12
    ' all equal the front-month spot) read as "calculated and equal to zero"
    ' rather than the dash format which implies missing data.
    cell.NumberFormat = "#,##0.00;(#,##0.00);0.00"
    cell.Value = spd
    cell.Font.Name = THEME_FONT
    cell.Font.Size = THEME_FONT_SIZE
    cell.Font.Bold = True
    cell.HorizontalAlignment = xlCenter
    cell.VerticalAlignment = xlCenter
    If spd > 0.005 Then
        cell.Font.Color = RGB(0, 255, 136)
        cell.Interior.Color = RGB(0, 40, 20)
    ElseIf spd < -0.005 Then
        cell.Font.Color = RGB(255, 51, 51)
        cell.Interior.Color = RGB(40, 0, 0)
    Else
        cell.Font.Color = RGB(255, 255, 0)
        cell.Interior.Color = RGB(40, 40, 0)
    End If
End Sub

```

```

Private Sub ApplyStructure(cell As Range, pM1 As Double, pM12 As Double)
    Dim spd As Double: spd = pM1 - pM12
    cell.Font.Name = THEME_FONT
    cell.Font.Size = THEME_FONT_SIZE
    cell.Font.Bold = True
    cell.HorizontalAlignment = xlCenter
    cell.VerticalAlignment = xlCenter
    If spd > 0.005 Then
        cell.Value = "BACKWARDATION"
        cell.Font.Color = RGB(0, 255, 136)
        cell.Interior.Color = RGB(0, 51, 25)
    ElseIf spd < -0.005 Then
        cell.Value = "CONTANGO"
        cell.Font.Color = RGB(255, 51, 51)
        cell.Interior.Color = RGB(51, 0, 0)
    Else
        cell.Value = "MIXED / FLAT"
        cell.Font.Color = RGB(255, 255, 0)
        cell.Interior.Color = RGB(40, 40, 0)
    End If
End Sub

```

' Only marks a row as ERR when *every* price column is empty/zero.
 ' If at least one month came back valid, the prior WritePrice calls already
 ' rendered the cells correctly and we should not overwrite them.
 ' Skips COL_UNUSED_GAP (col 12) which is reserved as a visual divider.

```

Private Sub MarkError(ws As Worksheet, r As Long)
    Dim c As Integer
    For c = 4 To COL_LAST
        If c = COL_UNUSED_GAP Then GoTo NextCol
        Dim v As Variant: v = ws.Cells(r, c).Value
        Dim hasData As Boolean: hasData = False
        If IsNumeric(v) Then
            If CDbl(v) > 0 Then hasData = True
        End If
        If Not hasData Then
            With ws.Cells(r, c)
                .NumberFormat = "@"
                .Value = "ERR"
                .Font.Name = THEME_FONT
                .Font.Size = THEME_FONT_SIZE
                .Font.Bold = True
                .Font.Color = RGB(255, 102, 0)
                .Interior.Color = C_BG
            End With
        End If
    Next c
Next r

```

```

        .HorizontalAlignment = xlCenter
        .VerticalAlignment = xlCenter
    End With
End If
NextCol:
Next c
' Make sure the gap column stays clean.
With ws.cells(r, COL_UNUSED_GAP)
    .ClearContents
    .Interior.Color = C_BG
End With
End Sub

Private Function GetCellPrice(cell As Range) As Double
    If IsNumeric(cell.Value) Then GetCellPrice = CDBl(cell.Value) Else GetCellPrice = 0
End Function

```

```

' =====
' Utilities
' =====

```

```

Private Sub PauseMS(ms As Long)
    Dim t As Single: t = Timer + ms / 1000
    Do While Timer < t: DoEvents: Loop
End Sub

```

```

Sub ExportSnapshot()
    Dim ws As Worksheet: Set ws = ThisWorkbook.Sheets(SH_MAIN)
    Dim wsCfg As Worksheet: Set wsCfg = ThisWorkbook.Sheets(SH_CFG)

    ' Resolve a writable output directory.
    ' ThisWorkbook.Path is empty when the workbook has never been saved -
    ' that produced the Open-failure ("TermStructure_*.csv" has no drive).
    Dim outDir As String
    If LenB(ThisWorkbook.path) > 0 Then
        outDir = ThisWorkbook.path
    Else
        outDir = Environ$("USERPROFILE") & "\Desktop"
        If Len(Dir(outDir, vbDirectory)) = 0 Then outDir = Environ$("TEMP")
    End If

    Dim outPath As String
    outPath = outDir & "\TermStructure_" & Format(Now(), "yyyymmdd_hhmmss") & ".csv"

    ' Derive the export row range from _Config instead of hard-coding 60.
    Dim sec1 As Long: sec1 = CLng(wsCfg.Range("B1").Value)
    Dim instCnt As Long: instCnt = CLng(wsCfg.Range("B2").Value)
    Dim sec2Data As Long: sec2Data = CLng(wsCfg.Range("B3").Value)
    Dim lastRow As Long: lastRow = sec2Data + 6 ' include timestamp row

    Dim fNum As Integer: fNum = FreeFile

    On Error GoTo BadOpen
    Open outPath For Output As #fNum
    On Error GoTo 0

    Dim r As Long, c As Integer, line As String
    For r = 1 To lastRow
        line = ""
        For c = 2 To COL_LAST
            Dim v As String: v = CStr(ws.cells(r, c).Value)
            ' RFC 4180: wrap in quotes when the cell contains a comma,
            ' embedded quote, or newline; escape internal quotes by doubling.
            If InStr(v, ",") > 0 Or InStr(v, "\"") > 0 Or InStr(v, vbLf) > 0 Then
                v = "\"" & Replace(v, "\"", "\"\"") & "\""
            End If
            line = line & IIf(c > 2, ",", "") & v
        Next c
        Print #fNum, line
    Next r
    Close #fNum

```

```
MsgBox "Exported:" & vbCr & outPath, vbInformation  
Exit Sub
```

```
BadOpen:
```

```
Dim errDesc As String: errDesc = Err.Description  
On Error Resume Next  
Close #fNum  
On Error GoTo 0  
MsgBox "Cannot write CSV:" & vbCr & outPath & vbCr & vbCr & _  
    "Error " & Err.Number & ": " & errDesc, vbCritical
```

```
End Sub
```

```

Sub BuildCorrelationMatrix()
    Dim wsC As Worksheet
    On Error Resume Next
    Set wsC = ThisWorkbook.Sheets("Correlation")
    On Error GoTo 0
    If wsC Is Nothing Then
        Set wsC = ThisWorkbook.Sheets.Add(After:=ThisWorkbook.Sheets(ThisWorkbook.Sheets.count))
        wsC.Name = "Correlation"
    End If

    ' ----- Layout constants -----
    Dim MATRIX_ROW As Long: MATRIX_ROW = 2
    Dim MATRIX_COL As Long: MATRIX_COL = 4

    ' ----- Read days BEFORE clearing (preserve user input) -----
    Dim nDays As Long: nDays = 60
    If IsNumeric(wsC.cells(2, 2).Value) And wsC.cells(2, 2).Value > 0 Then
        nDays = CLng(wsC.cells(2, 2).Value)
    End If

    ' ----- Base style -----
    wsC.cells.Clear
    With wsC.cells
        .Interior.Color = RGB(0, 0, 0)
        .Font.Color = RGB(200, 200, 200)
        .Font.Name = "Consolas"
        .Font.Size = 8
    End With
    wsC.Activate
    ActiveWindow.DisplayGridlines = False

    ' ----- Title -----
    With wsC.cells(1, 1)
        .Value = "US SECTOR CORRELATION MATRIX"
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
        .Font.Size = 13
    End With

    ' ----- Days label + input -----
    wsC.cells(2, 1).Value = "DAYS:"
    wsC.cells(2, 1).Font.Color = RGB(150, 150, 150)
    wsC.cells(2, 1).Font.Bold = True
    With wsC.cells(2, 2)
        .Value = nDays
        .Interior.Color = RGB(30, 30, 0)
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
        .NumberFormat = "0"
        .HorizontalAlignment = xlCenter
    End With

    ' ----- Sector ETF list -----
    Dim tickers As Variant
    Dim labels As Variant
    tickers = Array( _
        "XLK", "XLF", "XLV", "XLE", "XLY", "XLP", "XLI", "XLU", "XLRE", "XLB", "XLC", _
        "IGV", "KBE", "XBI", "JETS", "XHB", "XRT", "XME", "XOP", "KIE", _
        "SMH", "SOXX", "ARKK", "ARKG", "KWEB", "GDX", "SLX", "MOO", "PHO", "ICLN")
    labels = Array( _
        "Tech", "Finance", "Health", "Energy", "ConDisc", "ConStap", "Industrl", "Utility", "RealEst", "Material", "CommSvc", _
        "SoftETF", "Banks", "BioTech", "Airlines", "HomeBld", "Retail", "Metals", "OilGas", "Insure", _
        "SemiETF", "SOXX", "ARKK", "ARKG", "China", "Gold", "Steel", "Agri", "Water", "CleanE")

    Dim nSec As Integer: nSec = UBound(tickers) + 1

    ' ----- Fetch return data -----
    Application.StatusBar = "Fetching price history for " & nSec & " sectors..."
    Dim retData() As Double
    ReDim retData(0 To nSec - 1, 0 To nDays - 2)

```

```

Dim si As Integer
For si = 0 To nSec - 1
    Application.StatusBar = "Fetching [" & si + 1 & "/" & nSec & "] " & tickers(si)
    DoEvents
    Dim prices() As Double
    prices = FetchPriceArray(CStr(tickers(si)), nDays)
    Dim j As Long
    For j = 0 To nDays - 2
        If j + 1 <= UBound(prices) And prices(j) > 0 Then
            retData(si, j) = (prices(j + 1) - prices(j)) / prices(j)
        Else
            retData(si, j) = 0
        End If
    Next j
Next si

```

```

' ----- Column widths -----
wsC.Columns(1).ColumnWidth = 36
wsC.Columns(2).ColumnWidth = 12
wsC.Columns(3).Hidden = True
Dim ci As Integer
For ci = 0 To nSec - 1
    wsC.Columns(MATRIX_COL + ci).ColumnWidth = 6
Next ci

```

```

' ----- Row heights -----
wsC.Rows(1).RowHeight = 22
wsC.Rows(MATRIX_ROW).RowHeight = 18
Dim ri As Integer
For ri = 0 To nSec - 1
    wsC.Rows(MATRIX_ROW + 1 + ri).RowHeight = 15
Next ri

```

```

' ----- Column headers (rotated) -----
For ci = 0 To nSec - 2
    With wsC.cells(MATRIX_ROW, MATRIX_COL + ci)
        .Value = labels(ci)
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
        .Font.Size = 8
        .Interior.Color = RGB(10, 10, 10)
        .HorizontalAlignment = xlCenter
        .VerticalAlignment = xlBottom
        .Orientation = 0
        .WrapText = False
    End With
Next ci

```

```

' ----- Matrix -----
For ri = 0 To nSec - 1
    Dim dataRow As Long: dataRow = MATRIX_ROW + 1 + ri

    With wsC.cells(dataRow, 2)
        .Value = labels(ri)
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
        .Font.Size = 8
        .HorizontalAlignment = xlRight
        .VerticalAlignment = xlCenter
    End With

    For ci = 0 To nSec - 1
        Dim corr As Double
        If ri = ci Then
            corr = 1
        Else
            corr = CalcCorrelation(retData, ri, ci, nDays - 1)
        End If

        With wsC.cells(dataRow, MATRIX_COL + ci)
            .Value = corr

```

```
.NumberFormat = "0.00"
.HorizontalAlignment = xlCenter
.VerticalAlignment = xlCenter
.Font.Size = 8
.Font.Bold = (ri = ci)
```

```
Select Case True
    Case corr >= 0.7
        .Interior.Color = RGB(120, 0, 0)
        .Font.Color = RGB(255, 180, 180)
    Case corr >= 0.4
        .Interior.Color = RGB(70, 15, 15)
        .Font.Color = RGB(230, 130, 130)
    Case corr >= 0.1
        .Interior.Color = RGB(25, 25, 25)
        .Font.Color = RGB(190, 190, 190)
    Case corr >= -0.1
        .Interior.Color = RGB(10, 10, 10)
        .Font.Color = RGB(120, 120, 120)
    Case corr >= -0.4
        .Interior.Color = RGB(10, 35, 10)
        .Font.Color = RGB(100, 200, 100)
    Case Else
        .Interior.Color = RGB(0, 60, 0)
        .Font.Color = RGB(80, 220, 80)
```

```
End Select
```

```
If ri = ci Then
    .Interior.Color = RGB(35, 25, 0)
    .Font.Color = RGB(255, 192, 0)
```

```
End If
```

```
' Black border
```

```
With .Borders
```

```
.LineStyle = xlContinuous
```

```
.Color = RGB(0, 0, 0)
```

```
.Weight = xlThin
```

```
End With
```

```
End With
```

```
Next ci
```

```
Next ri
```

```
' ——— Gold separator ———
```

```
With wsC.Range(wsC.Cells(MATRIX_ROW, 1), _
    wsC.Cells(MATRIX_ROW, MATRIX_COL + nSec - 1)).Borders(xlEdgeBottom)
```

```
.LineStyle = xlContinuous
```

```
.Color = RGB(255, 192, 0)
```

```
.Weight = xlThin
```

```
End With
```

```
' ——— Legend ———
```

```
Dim legRow As Long: legRow = MATRIX_ROW + nSec + 2
```

```
wsC.Cells(legRow, 1).Value = "COLOUR LEGEND:"
```

```
wsC.Cells(legRow, 1).Font.Color = RGB(150, 150, 150)
```

```
wsC.Cells(legRow, 1).Font.Bold = True
```

```
Dim legItems As Variant
```

```
legItems = Array( _
    Array(">= 0.7 High +corr", RGB(120, 0, 0), RGB(255, 180, 180)), _
    Array("0.4~0.7 Mid +corr", RGB(70, 15, 15), RGB(230, 130, 130)), _
    Array("-0.1~0.4 Neutral", RGB(25, 25, 25), RGB(190, 190, 190)), _
    Array("-0.4~-0.1 Mild -corr", RGB(10, 35, 10), RGB(100, 200, 100)), _
    Array("<= -0.4 High -corr", RGB(0, 60, 0), RGB(80, 220, 80)))
```

```
Dim li As Integer
```

```
For li = 0 To UBound(legItems)
```

```
    With wsC.Cells(legRow + 1 + li, 1)
```

```
        .Value = legItems(li)(0)
```

```
        .Interior.Color = legItems(li)(1)
```

```
        .Font.Color = legItems(li)(2)
```

```
        .Font.Size = 9
```

```
        .Font.Bold = True
```

```

    End With
Next li

' ——— Freeze panes ———
wsC.cells(MATRIX_ROW + 1, MATRIX_COL).Select
ActiveWindow.FreezePanes = False
ActiveWindow.FreezePanes = True
wsC.cells(1, 1).Select

Application.StatusBar = "Correlation matrix done — " & Format(Now, "hh:mm:ss")
MsgBox "Done! (" & nSec & " sectors, " & nDays & " days)", vbInformation
End Sub

' ——— Fetch closing price array (nDays entries) ———
Private Function FetchPriceArray(ticker As String, nDays As Long) As Double()
    Dim result() As Double
    ReDim result(0 To nDays - 1)

    Dim http As Object
    Set http = CreateObject("MSXML2.XMLHTTP")
    Dim url As String
    url = "https://query1.finance.yahoo.com/v8/finance/chart/" & ticker & _
        "?interval=1d&range=" & CStr(CLng(nDays * 1.5 / 365 * 12)) & "mo"
    If CLng(nDays * 1.5) > 365 Then url = Replace(url, "range=", "range=2y&fake=")
    url = "https://query1.finance.yahoo.com/v8/finance/chart/" & ticker & "?interval=1d&range=1y"

    On Error Resume Next
    http.Open "GET", url, False
    http.setRequestHeader "User-Agent", "Mozilla/5.0"
    http.send
    On Error GoTo 0

    If http.status <> 200 Then FetchPriceArray = result: Exit Function

    Dim resp As String: resp = http.responseText
    Dim pos As Long: pos = InStr(resp, """"close"":["")
    If pos = 0 Then FetchPriceArray = result: Exit Function

    Dim arrStr As String
    arrStr = Mid(resp, pos + 9, 5000)
    arrStr = Split(arrStr, "]")(0)
    Dim parts() As String: parts = Split(arrStr, ",")

    ' Take last nDays values
    Dim total As Long: total = UBound(parts) + 1
    Dim startIdx As Long: startIdx = IIf(total > nDays, total - nDays, 0)
    Dim idx As Long: idx = 0
    Dim k As Long
    For k = startIdx To UBound(parts)
        If IsNumeric(Trim(parts(k))) And idx < nDays Then
            result(idx) = CDbl(Trim(parts(k)))
            idx = idx + 1
        End If
    Next k

    FetchPriceArray = result
End Function

' ——— Pearson correlation between two return series ———
Private Function CalcCorrelation(retData() As Double, r1 As Integer, r2 As Integer, n As Long) As Double
    Dim sumX As Double, sumY As Double, sumXY As Double
    Dim sumX2 As Double, sumY2 As Double
    Dim i As Long, cnt As Long

    For i = 0 To n - 1
        Dim x As Double: x = retData(r1, i)
        Dim y As Double: y = retData(r2, i)
        If x <> 0 Or y <> 0 Then
            sumX = sumX + x
            sumY = sumY + y
            sumXY = sumXY + x * y
        End If
    Next i

    cnt = n
    CalcCorrelation = (sumXY * cnt - sumX * sumY) / Sqr((sumX2 * cnt - sumX * sumX) * (sumY2 * cnt - sumY * sumY))
End Function

```

CorrelationMatrix - 5

```
        sumX2 = sumX2 + x * x
        sumY2 = sumY2 + y * y
        cnt = cnt + 1
    End If
Next i

If cnt < 5 Then CalcCorrelation = 0: Exit Function

Dim num As Double: num = cnt * sumXY - sumX * sumY
Dim den As Double: den = Sqr((cnt * sumX2 - sumX ^ 2) * (cnt * sumY2 - sumY ^ 2))
If den = 0 Then CalcCorrelation = 0 Else CalcCorrelation = num / den
End Function
```

```
' —— Helper: safely name a cell (optional) ——
Private Sub Name_Safe(ws As Worksheet)
    ' placeholder, not used
End Sub
```

```
' —— Refresh with new days value ——
Sub RefreshCorrelation()
    Call BuildCorrelationMatrix
End Sub
```

Option Explicit

```

' =====
' DRAWDOWN SHADOW v2.0
' Builds drawdown shadow + pain-adjusted chart from HistoryLog!M
'
' v2 changes vs v1:
' 1. Detection rule:
'     v1: >= 2 consecutive negative-return days
'     v2: any 5-day rolling window with >= 4 negative days marks
'         all 5 days in that window as "in drawdown"
' 2. cumRet now SUMS ONLY THE NEGATIVE returns within a segment.
'     Positive bounce days inside the segment are zeroed out.
' 3. BUG FIX: the pain-adjusted shadowVal formula was missing a
'     '+' between the two terms (the '_' line-continuation in v1
'     silently dropped it). v2 restores it.
' =====

Private Const SH_HIST As String = "HistoryLog"
Private Const SH_OUT As String = "DrawdownChart"
Private Const DATA_ROW As Long = 3
Private Const PAIN_CAP As Double = 5
Private Const PAIN_OVERFLOW As Double = 9999

' Detection window parameters - change these to tune rule.
Private Const WIN_SIZE As Long = 5 ' rolling window length
Private Const WIN_MIN_NEG As Long = 4 ' min negative days within window

Public Sub BuildDrawdownShadow()
    Application.ScreenUpdating = False
    Application.StatusBar = "Building drawdown shadow v2..."
    On Error GoTo CleanFail

    ' --- Read HistoryLog!M into typed arrays ---
    Dim wsH As Worksheet
    On Error Resume Next: Set wsH = ThisWorkbook.Sheets(SH_HIST): On Error GoTo CleanFail
    If wsH Is Nothing Then MsgBox "HistoryLog sheet not found", vbExclamation: GoTo CleanExit

    Dim lastRow As Long
    lastRow = wsH.Cells(wsH.Rows.Count, "M").End(xlUp).Row
    If lastRow < 3 Then MsgBox "Not enough data in column M (need >= 5 rows for 5-day window)", vbInformation: GoTo CleanExit

    Dim n As Long: n = lastRow - 1
    Dim ret() As Double: ReDim ret(1 To n)
    Dim dateStr() As String: ReDim dateStr(1 To n)
    Dim valid() As Boolean: ReDim valid(1 To n)

    Dim r As Long, i As Long
    For r = 2 To lastRow
        i = r - 1
        Dim v As Variant: v = wsH.Cells(r, "M").Value
        If IsNumeric(v) Then
            ret(i) = CDbl(v)
            valid(i) = True
        Else
            ret(i) = 0
            valid(i) = False
        End If
        dateStr(i) = "R" & r
    On Error Resume Next
        Dim d0 As Date: d0 = CDate(wsH.Cells(r, "A").Value)
        If d0 > 0 Then dateStr(i) = Format(d0, "m/d/yy")
    On Error GoTo CleanFail
    Next r

    ' --- Mark DD days using 5-day rolling window ---
    Dim isDD() As Boolean: ReDim isDD(1 To n)
    Dim w As Long, negCount As Long, k As Long
    For w = 1 To n - WIN_SIZE + 1
        negCount = 0
        For k = w To w + WIN_SIZE - 1

```

```

        If valid(k) And ret(k) < 0 Then negCount = negCount + 1
    Next k
    If negCount >= WIN_MIN_NEG Then
        For k = w To w + WIN_SIZE - 1
            isDD(k) = True
        Next k
    End If
Next w

' --- Build segments from consecutive isDD-marked days ---
Dim segments As Collection: Set segments = New Collection
Dim segDates As Collection: Set segDates = New Collection

Dim s As Long, segStart As Long, segEnd As Long
s = 1
Do While s <= n
    If isDD(s) Then
        segStart = s
        Do While s <= n
            If Not isDD(s) Then Exit Do
            s = s + 1
        Loop
        segEnd = s - 1

        Dim segLen As Long: segLen = segEnd - segStart + 1
        Dim segArr() As Double: ReDim segArr(0 To segLen - 1)
        Dim p As Long
        For p = 0 To segLen - 1
            ' Ignore positive returns - only negatives accumulate
            If ret(segStart + p) < 0 Then
                segArr(p) = ret(segStart + p)
            Else
                segArr(p) = 0
            End If
        Next p

        segments.Add segArr
        segDates.Add dateStr(segStart)
    Else
        s = s + 1
    End If
Loop

If segments.count = 0 Then
    MsgBox "No drawdown segments found (no 5-day window has >= 4 negative days)", vbInformation
    GoTo CleanExit
End If

' --- Find max segment length ---
Dim maxDays As Long: maxDays = 0
For s = 1 To segments.count
    Dim a As Variant: a = segments(s)
    If UBound(a) + 1 > maxDays Then maxDays = UBound(a) + 1
Next s

' --- Create / clear DrawdownChart sheet ---
Dim wsD As Worksheet
On Error Resume Next: Set wsD = ThisWorkbook.Sheets(SH_OUT): On Error GoTo CleanFail
If wsD Is Nothing Then
    Set wsD = ThisWorkbook.Sheets.Add(After:=ThisWorkbook.Sheets(ThisWorkbook.Sheets.count))
    wsD.Name = SH_OUT
End If
wsD.Cells.Clear

Dim shp As Shape
For Each shp In wsD.Shapes
    shp.Delete
Next shp

With wsD.Cells
    .Interior.Color = RGB(0, 0, 0)

```

```

.Font.Color = RGB(200, 200, 200)
.Font.Name = "Consolas"
.Font.Size = 9
End With
wsD.Activate
ActiveWindow.DisplayGridlines = False

With wsD.cells(1, 1)
.Value = "DRAWDOWN SHADOW v2 (rule: " & WIN_MIN_NEG & "/" & WIN_SIZE & ") | " & _
    segments.count & " episodes | " & Format(Now, "hh:mm:ss")
.Font.Color = RGB(255, 192, 0)
.Font.Bold = True
.Font.Size = 12
End With
wsD.Rows(1).RowHeight = 24

wsD.cells(DATA_ROW - 1, 1).Value = "Day"
wsD.cells(DATA_ROW - 1, 1).Font.Color = RGB(150, 150, 150)

Dim d As Long
For d = 1 To maxDays
    wsD.cells(DATA_ROW + d - 1, 1).Value = d
Next d

' --- Compute shadow f(x) per segment per day ---
Dim minCum As Double: minCum = 0
Dim cumRet As Double, runMin As Double
Dim xVal As Double, fxVal As Double, cellVal As Double
Dim depth As Double, rVal As Long, colIdx As Long, segLenInner As Long

For s = 1 To segments.count
    Dim arr As Variant: arr = segments(s)
    segLenInner = UBound(arr) + 1
    colIdx = s + 1
    wsD.cells(DATA_ROW - 1, colIdx).Value = segDates(s)
    wsD.cells(DATA_ROW - 1, colIdx).Font.Color = RGB(180, 180, 180)
    wsD.cells(DATA_ROW - 1, colIdx).Font.Size = 8

    cumRet = 0
    runMin = 0
    For d = 0 To segLenInner - 1
        xVal = d + 1
        cumRet = cumRet + arr(d) ' arr(d) is 0 for positive days (ignored)
        fxVal = (Exp(xVal) / (xVal ^ 3 + WorksheetFunction.Pi() ^ 3) + Exp(Sqr(xVal))) * cumRet
        cellVal = fxVal
        wsD.cells(DATA_ROW + d, colIdx).Value = Round(cellVal, 4)
        wsD.cells(DATA_ROW + d, colIdx).NumberFormat = "0.0000"

        depth = Abs(cellVal)
        rVal = IIf(depth > 1000, 255, IIf(depth > 100, 230, IIf(depth > 10, 200, 170)))
        wsD.cells(DATA_ROW + d, colIdx).Font.Color = RGB(rVal, 60, 60)

        If cellVal < runMin Then runMin = cellVal
    Next d
    If runMin < minCum Then minCum = runMin
Next s

' --- Column widths / row heights ---
wsD.Columns(1).ColumnWidth = 6
Dim ci As Long
For ci = 2 To segments.count + 1
    wsD.Columns(ci).ColumnWidth = 9
Next ci
For d = 1 To maxDays
    wsD.Rows(DATA_ROW + d - 1).RowHeight = 15
Next d

' --- Build chart ---
Dim co As ChartObject
Set co = wsD.ChartObjects.Add(wsD.Columns("M").Left, wsD.Rows(1).Top, 700, 400)
With co.Chart

```

```

.ChartType = xlLine
.HasTitle = True
.ChartTitle.text = "Drawdown Shadow v2 (" & WIN_MIN_NEG & "/" & WIN_SIZE & " rule)"
.ChartTitle.Font.Color = RGB(255, 192, 0)
.ChartTitle.Font.Name = "Consolas"
.ChartTitle.Font.Bold = True

Do While .SeriesCollection.count > 0
    .SeriesCollection(1).Delete
Loop

Dim xVals() As Long: ReDim xVals(0 To maxDays - 1)
For d = 0 To maxDays - 1: xVals(d) = d + 1: Next d

Dim yVals() As Variant
Dim cum2 As Double, xv As Double
Dim ser As Series
Dim brightness As Long, lineClr As Long

For s = 1 To segments.count
    arr = segments(s)
    segLenInner = UBound(arr) + 1
    ReDim yVals(0 To maxDays - 1)
    cum2 = 0
    For d = 0 To maxDays - 1
        If d < segLenInner Then
            xv = d + 1
            cum2 = cum2 + arr(d)
            yVals(d) = Round((Exp(xv) / (xv ^ 3 + WorksheetFunction.Pi() ^ 3) + Exp(Sqr(xv))) * cum2, 4)
        Else
            yVals(d) = CVErr(xlErrNA)
        End If
    Next d
    Set ser = .SeriesCollection.NewSeries
    ser.Values = yVals
    ser.XValues = xVals
    ser.Name = CStr(segDates(s))

    brightness = 60 + CInt((s / segments.count) * 160)
    lineClr = RGB(brightness, 20, 20)
    With ser.Format.line
        .ForeColor.RGB = lineClr
        .Weight = IIf(s = segments.count, 2, 1)
        .Transparency = IIf(s = segments.count, 0, 0.4 - (s / segments.count) * 0.3)
    End With
    ser.MarkerStyle = xlMarkerStyleNone
Next s

.PlotArea.Interior.Color = RGB(8, 8, 8)
.PlotArea.Border.Color = RGB(40, 40, 40)
.ChartArea.Interior.Color = RGB(0, 0, 0)
.ChartArea.Border.Color = RGB(30, 30, 30)

With .Axes(xlCategory)
    .HasTitle = True
    .AxisTitle.text = "Days in Drawdown"
    .AxisTitle.Font.Color = RGB(255, 255, 255)
    .AxisTitle.Font.Name = "Consolas"
    .TickLabels.Font.Color = RGB(255, 255, 255)
    .TickLabels.Font.Name = "Consolas"
    .TickLabels.Font.Size = 8
    .MajorGridlines.Border.Color = RGB(35, 35, 35)
End With

With .Axes(xlValue)
    .HasTitle = True
    .AxisTitle.text = "f(x) = sum(neg) * (e^x/(x^3+pi^3) + e^sqrt(x))"
    .AxisTitle.Font.Color = RGB(255, 255, 255)
    .AxisTitle.Font.Name = "Consolas"
    .TickLabels.Font.Color = RGB(255, 255, 255)
    .TickLabels.Font.Name = "Consolas"
    .TickLabels.Font.Size = 8

```

```

        .TickLabels.NumberFormat = "0.00"
        .MajorGridlines.Border.Color = RGB(35, 35, 35)
        .MaximumScale = 0
        .CrossesAt = 0
    End With

    .HasLegend = True
    With .Legend
        .Interior.Color = RGB(10, 10, 10)
        .Border.Color = RGB(30, 30, 30)
        .Font.Color = RGB(255, 255, 255)
        .Font.Name = "Consolas"
        .Font.Size = 7
    End With
End With

' =====
' Pain-Adjusted Section (BUG FIX: missing '+' restored)
' =====

Dim painStartRow As Long: painStartRow = DATA_ROW + maxDays + 2
Dim painHeaderRow As Long: painHeaderRow = painStartRow + 1
Dim painDataRow As Long
Dim painVal As Double, shadowVal As Double
Dim minPain As Double: minPain = 0

With wsD.cells(painStartRow, 1)
    .Value = "PAIN-ADJUSTED DRAWDOWN |  $f(x) = \text{TAN}(\text{ASIN}(\text{shadow}/\text{cap})) * \text{cap}$ "
    .Font.Color = RGB(255, 99, 71)
    .Font.Bold = True
    .Font.Size = 10
End With

With wsD.cells(painStartRow, segments.count + 3)
    .Value = "PAIN CAP%"
    .Font.Color = RGB(255, 192, 0)
    .Font.Bold = True
End With

With wsD.cells(painStartRow, segments.count + 4)
    .Value = PAIN_CAP
    .Font.Color = RGB(0, 191, 255)
    .Font.Bold = True
End With

With wsD.cells(painHeaderRow, 1)
    .Value = "Day"
    .Font.Color = RGB(150, 150, 150)
End With

For d = 1 To maxDays
    With wsD.cells(painHeaderRow + d, 1)
        .Value = d
        .Font.Color = RGB(200, 200, 200)
    End With
Next d

For s = 1 To segments.count
    colIdx = s + 1
    With wsD.cells(painHeaderRow, colIdx)
        .Value = segDates(s)
        .Font.Color = RGB(180, 180, 180)
        .Font.Size = 8
    End With
Next s

' --- Compute pain values ---
For s = 1 To segments.count
    arr = segments(s)
    segLenInner = UBound(arr) + 1
    colIdx = s + 1
    cumRet = 0

    For d = 0 To segLenInner - 1

```

```

    xVal = d + 1
    cumRet = cumRet + arr(d)
    ' BUG FIX: previous version had '_' line continuation but missing '+'.
    ' Now the '+' is explicit so e^sqrt(x) is properly added.
    shadowVal = (Exp(xVal) / (xVal ^ 3 + WorksheetFunction.Pi() ^ 3) + _
        Exp(Sqr(xVal))) * cumRet

    If Abs(shadowVal) >= PAIN_CAP Then
        painVal = Sgn(shadowVal) * PAIN_OVERFLOW
    Else
        painVal = Tan(WorksheetFunction.Asin(shadowVal / PAIN_CAP)) * PAIN_CAP
    End If

    painDataRow = painHeaderRow + xVal
    With wsD.cells(painDataRow, colIdx)
        .Value = Round(painVal, 4)
        .NumberFormat = "0.0000"
        .Font.Size = 9
        depth = Abs(painVal)
        If depth >= PAIN_OVERFLOW Then
            .Font.Color = RGB(255, 0, 0): .Font.Bold = True
        ElseIf depth > 5 Then
            .Font.Color = RGB(255, 40, 20)
        ElseIf depth > 2 Then
            .Font.Color = RGB(240, 60, 30)
        ElseIf depth > 0.5 Then
            .Font.Color = RGB(210, 80, 50)
        Else
            .Font.Color = RGB(180, 100, 80)
        End If
    End With

    If painVal < minPain Then minPain = painVal
Next d
Next s

' --- Pain stats ---
Dim painStatCol As Long: painStatCol = segments.count + 3
With wsD.cells(painHeaderRow + 1, painStatCol)
    .Value = "PAIN STATS"
    .Font.Color = RGB(255, 99, 71)
    .Font.Bold = True
End With
With wsD.cells(painHeaderRow + 2, painStatCol)
    .Value = "Max Pain"
    .Font.Color = RGB(150, 150, 150)
End With
With wsD.cells(painHeaderRow + 2, painStatCol + 1)
    .Value = Round(minPain, 4)
    .NumberFormat = "0.0000"
    .Font.Color = RGB(255, 69, 0)
End With

If minCum <> 0 Then
    With wsD.cells(painHeaderRow + 3, painStatCol)
        .Value = "Pain/DD"
        .Font.Color = RGB(150, 150, 150)
    End With
    With wsD.cells(painHeaderRow + 3, painStatCol + 1)
        .Value = Format(minPain / minCum, "0.00") & "x"
        .Font.Color = RGB(255, 69, 0)
    End With
End If

' --- Summary panel ---
Dim sumRow As Long: sumRow = DATA_ROW - 1
Dim sumCol As Long: sumCol = segments.count + 3
With wsD.cells(sumRow, sumCol)
    .Value = "DRAWDOWN SUMMARY v2"
    .Font.Color = RGB(255, 192, 0)
    .Font.Bold = True

```

```

End With
wsD.cells(sumRow + 1, sumCol).Value = "Episodes"
wsD.cells(sumRow + 1, sumCol + 1).Value = segments.count
wsD.cells(sumRow + 2, sumCol).Value = "Max Days"
wsD.cells(sumRow + 2, sumCol + 1).Value = maxDays
wsD.cells(sumRow + 3, sumCol).Value = "Min f(x)"
wsD.cells(sumRow + 3, sumCol + 1).Value = Format(minCum, "0.0000")
wsD.cells(sumRow + 4, sumCol).Value = "Rule"
wsD.cells(sumRow + 4, sumCol + 1).Value = WIN_MIN_NEG & "/" & WIN_SIZE & " neg"

wsD.cells(sumRow + 6, sumCol).Value = "Start"
wsD.cells(sumRow + 6, sumCol + 1).Value = "Days"
wsD.cells(sumRow + 6, sumCol + 2).Value = "Sum Neg%"
wsD.cells(sumRow + 6, sumCol).Font.Color = RGB(255, 192, 0)
wsD.cells(sumRow + 6, sumCol + 1).Font.Color = RGB(255, 192, 0)
wsD.cells(sumRow + 6, sumCol + 2).Font.Color = RGB(255, 192, 0)

Dim sumNeg As Double
For s = 1 To segments.count
    arr = segments(s)
    sumNeg = 0
    For k = 0 To UBound(arr)
        sumNeg = sumNeg + arr(k) ' arr already zero-filled for positives
    Next k
    wsD.cells(sumRow + 6 + s, sumCol).Value = segDates(s)
    wsD.cells(sumRow + 6 + s, sumCol + 1).Value = UBound(arr) + 1
    wsD.cells(sumRow + 6 + s, sumCol + 2).Value = Format(sumNeg * 100, "0.00") & "%"
    wsD.cells(sumRow + 6 + s, sumCol + 2).Font.Color = RGB(200, 80, 80)
Next s

For ci = sumCol To sumCol + 2
    wsD.Columns(ci).ColumnWidth = 14
Next ci

CleanExit:
Application.ScreenUpdating = True
Application.StatusBar = False
If segments.count > 0 Then
    Application.StatusBar = "Drawdown Shadow v2 done: " & segments.count & " episodes"
    MsgBox "Done! Found " & segments.count & " drawdown segments." & vbCrLf & _
        "Rule: 5-day window with >= 4 negative days" & vbCrLf & _
        "Positive returns within segment are ignored in cumRet.", vbInformation
End If
Exit Sub

CleanFail:
Application.ScreenUpdating = True
Application.StatusBar = False
MsgBox "Error " & Err.Number & ": " & Err.Description, vbCritical
End Sub

```

Option Explicit

' 功能：個股波動率分佈與進階風險分析（支援多週期：100天，200天，1000天）
 ' 工作表：Tickers volatility
 ' 輸入：B2 (Ticker)
 ' 輸出：多週期報表（起始列分別為 4, 29, 54）

```
Sub UpdateVolatilityAnalysis_Pro()
    Dim ws As Worksheet
    Dim ticker As String
    Dim allPrices() As Double, allDates() As Date
    Dim totalDays As Long

    ' 1. 設定工作表與輸入
    Set ws = ThisWorkbook.Sheets("Tickers volatility")
    ticker = Trim(ws.Range("B2").Value)
    If ticker = "" Then MsgBox "請在 B2 輸入股票代碼", vbExclamation: Exit Sub

    Application.ScreenUpdating = False

    ' 2. 清除舊數據（範圍加大以容納三組分析表）
    ws.Range("A4:F100").ClearContents

    ' 3. 抓取歷史數據（改為抓取 10 年，確保有 1000 個交易日）
    If Not GetHistoricalData(ticker, allDates, allPrices) Then
        MsgBox "無法抓取數據，請檢查代碼或網絡。", vbCritical
        Application.ScreenUpdating = True
        Exit Sub
    End If

    totalDays = UBound(allPrices) + 1
    If totalDays < 20 Then
        MsgBox "數據筆數不足以進行進階分析。", vbExclamation
        Application.ScreenUpdating = True
        Exit Sub
    End If

    ' 4. 執行三個不同週期的分析（傳入：工作表，全部價格，欲分析天數，輸出起始列，標題）
    ' 1-a. 過去 100 天（輸出在 A4:F26）
    Call AnalyzeAndOutput(ws, allPrices, 100, 4, "Past 100 Days")

    ' 1-b. 過去 200 天（輸出在 A29:F51）
    Call AnalyzeAndOutput(ws, allPrices, 200, 29, "Past 200 Days")

    ' 1-c. 過去 1000 天（輸出在 A54:F76）
    Call AnalyzeAndOutput(ws, allPrices, 1000, 54, "Past 1000 Days")

    Application.ScreenUpdating = True
    MsgBox "100天、200天、1000天進階波動率分析更新完成！", vbInformation
End Sub
```

' 副程式：執行特定天數的計算與報表輸出

```
Sub AnalyzeAndOutput(ws As Worksheet, ByRef allPrices() As Double, lookbackDays As Long, startRow As Long, titleSuffix As String)
    Dim prices() As Double, returns() As Double
    Dim i As Long, j As Long, n As Long
    Dim totalData As Long, useDays As Long

    totalData = UBound(allPrices) + 1
    useDays = lookbackDays
    If totalData < useDays Then useDays = totalData ' 如果歷史數據不夠，則用最大可用的天數

    ' 1. 截取最近 useDays 的股價數據
    ReDim prices(0 To useDays - 1)
    For i = 0 To useDays - 1
```

```

    prices(i) = allPrices(UBound(allPrices) - useDays + 1 + i)
Next i

n = UBound(prices)
If n < 1 Then Exit Sub
ReDim returns(0 To n - 1)

' --- 統計變數宣告 ---
Dim mean As Double, stdDev As Double, maxRet As Double, minRet As Double
Dim annVol As Double, skew As Double, kurt As Double
Dim downDev As Double, mdd As Double
Dim currRollVol As Double, maxRollVol As Double
Dim upStreak As Long, downStreak As Long, maxUpStreak As Long, maxDownStreak As Long

' 2. 計算日漲跌幅 & 連續漲跌天數
maxUpStreak = 0: maxDownStreak = 0
upStreak = 0: downStreak = 0

For i = 1 To n
    If prices(i - 1) <> 0 Then
        returns(i - 1) = (prices(i) - prices(i - 1)) / prices(i - 1)
    Else
        returns(i - 1) = 0
    End If

    If returns(i - 1) > 0 Then
        upStreak = upStreak + 1
        downStreak = 0
        If upStreak > maxUpStreak Then maxUpStreak = upStreak
    ElseIf returns(i - 1) < 0 Then
        downStreak = downStreak + 1
        upStreak = 0
        If downStreak > maxDownStreak Then maxDownStreak = downStreak
    Else
        upStreak = 0
        downStreak = 0
    End If
Next i

' 3. 計算最大回撤 (MDD)
Dim peak As Double, DrawDown As Double
peak = prices(0)
mdd = 0
For i = 1 To n
    If prices(i) > peak Then peak = prices(i)
    DrawDown = (peak - prices(i)) / peak
    If DrawDown > mdd Then mdd = DrawDown
Next i

' 4. 計算下行標準差 (Downside Deviation)
Dim downReturns() As Double
Dim downCount As Long
downCount = 0
For i = LBound(returns) To UBound(returns)
    If returns(i) < 0 Then
        ReDim Preserve downReturns(0 To downCount)
        downReturns(downCount) = returns(i)
        downCount = downCount + 1
    End If
Next i

' 5. 計算 20日滾動波動率
Dim rollWindow(1 To 20) As Double
Dim rollVol As Double
maxRollVol = 0
If UBound(returns) >= 19 Then
    For i = 19 To UBound(returns)
        For j = 1 To 20
            rollWindow(j) = returns(i - 20 + j)
        Next j
        rollVol = Application.WorksheetFunction.StDev_S(rollWindow) * Sqr(252)
    Next i

```

```

        If rollVol > maxRollVol Then maxRollVol = rollVol
        If i = UBound(returns) Then currRollVol = rollVol
    Next i
End If

```

' 6. 基礎與進階統計指標

```

On Error Resume Next
mean = Application.WorksheetFunction.Average(returns)
stdDev = Application.WorksheetFunction.StDev_S(returns)
maxRet = Application.WorksheetFunction.Max(returns)
minRet = Application.WorksheetFunction.Min(returns)
annVol = stdDev * Sqr(252)
skew = Application.WorksheetFunction.skew(returns)
kurt = Application.WorksheetFunction.kurt(returns)
If downCount > 1 Then
    downDev = Application.WorksheetFunction.StDev_S(downReturns)
Else
    downDev = 0
End If
On Error GoTo 0

```

' 7. 填寫標題與數值 (結構化輸出)

```

Dim labelsArr As Variant, valsArr As Variant
labelsArr = Array("Mean (Daily)", "Std Dev (Daily)", "Annualized Volatility", _
    "Skewness (不對稱性)", "Kurtosis (肥尾效應)", _
    "Downside Deviation", "Max Drawdown (MDD)", _
    "Current 20D Volatility", "Max 20D Volatility", _
    "Max Up Streak (Days)", "Max Down Streak (Days)", _
    "Max Daily Chg%", "Min Daily Chg%")

valsArr = Array(mean, stdDev, annVol, skew, kurt, downDev, -mdd, _
    currRollVol, maxRollVol, maxUpStreak, maxDownStreak, maxRet, minRet)

```

' 寫入表頭

```

ws.cells(startRow, 1).Value = "Statistic (" & titleSuffix & ")"
ws.cells(startRow, 2).Value = "Value"

```

```

For i = LBound(labelsArr) To UBound(labelsArr)
    ws.cells(startRow + 1 + i, 1).Value = labelsArr(i)
    ws.cells(startRow + 1 + i, 2).Value = valsArr(i)
Next i

```

' 動態套用格式化

```

ws.Range("B" & startRow + 1 & ":" & B" & startRow + 6 & ":" & B" & startRow + 12 & ":" & startRow + 13).NumberFormat = "0.00%"
ws.Range("B" & startRow + 4 & ":" & B" & startRow + 5).NumberFormat = "0.00"
ws.Range("B" & startRow + 10 & ":" & B" & startRow + 11).NumberFormat = "0"

```

' 8. 建立分佈表

```

Dim bins(0 To 21) As Long, labels(0 To 21) As String
Dim retVal As Double, binIdx As Integer
Dim totalReturnsCount As Long

```

```

totalReturnsCount = UBound(returns) + 1
labels(0) = "< -10%"
labels(21) = "> 10%"

```

```

For i = 1 To 20
    Dim lowerLimit As Double
    lowerLimit = -0.1 + (i - 1) * 0.01
    labels(i) = Format(lowerLimit, "0%") & " ~ " & Format(lowerLimit + 0.01, "0%")
Next i

```

```

For i = LBound(returns) To UBound(returns)
    retVal = returns(i)
    If retVal < -0.1 Then
        bins(0) = bins(0) + 1
    ElseIf retVal >= 0.1 Then
        bins(21) = bins(21) + 1
    Else
        binIdx = Int((retVal + 0.1) / 0.01) + 1
    End If

```

```

        If binIdx >= 1 And binIdx <= 20 Then bins(binIdx) = bins(binIdx) + 1
    End If
Next i

```

```

' 填入分佈表格

```

```

ws.cells(startRow, 4).Value = "Range"
ws.cells(startRow, 5).Value = "Count"
ws.cells(startRow, 6).Value = "Freq %"

```

```

For i = 0 To 21
    ws.cells(startRow + 1 + i, 4).Value = "" & labels(i)
    ws.cells(startRow + 1 + i, 5).Value = bins(i)
    If totalReturnsCount > 0 Then
        ws.cells(startRow + 1 + i, 6).Value = bins(i) / totalReturnsCount
    Else
        ws.cells(startRow + 1 + i, 6).Value = 0
    End If
Next i
ws.Range("F" & startRow + 1 & ":F" & startRow + 22).NumberFormat = "0.00%"

```

```

End Sub

```

```

' 核心函數：抓取 10 年數據（供切割週期使用）

```

```

Function GetHistoricalData(ticker As String, ByRef outDates() As Date, ByRef outPrices() As Double) As Boolean

```

```

    Dim http As Object
    Dim url As String, response As String
    Dim tsStr As String, closeStr As String
    Dim tsArr() As String, closeArr() As String
    Dim i As Long

```

```

' 1. 處理 Ticker

```

```

If InStr(ticker, ".TW") = 0 And InStr(ticker, ".TWO") = 0 And IsNumeric(ticker) Then
    ticker = ticker & ".TW"
End If

```

```

' 2. 設定 URL (Range 改為 10y 確保涵蓋 1000 交易日)

```

```

url = "https://query1.finance.yahoo.com/v8/finance/chart/" & ticker & "?range=10y&interval=1d"

```

```

On Error Resume Next

```

```

Set http = CreateObject("WinHttp.WinHttpRequest.5.1")
If http Is Nothing Then Set http = CreateObject("MSXML2.XMLHTTP")

```

```

With http
    .Open "GET", url, False
    .setRequestHeader "User-Agent", "Mozilla/5.0"
    .send
    response = .responseText
End With

```

```

On Error GoTo 0

```

```

' 3. 解析 JSON

```

```

Dim tsStart As Long, tsEnd As Long
tsStart = InStr(response, """"timestamp"":["") + 12
tsEnd = InStr(tsStart, response, "]")
If tsStart > 12 And tsEnd > tsStart Then
    tsStr = Mid(response, tsStart, tsEnd - tsStart)
    tsArr = Split(tsStr, ",")

```

```

Else
    GetHistoricalData = False
    Exit Function
End If

```

```

Dim closeStart As Long, closeEnd As Long

```

```

closeStart = InStr(response, """"adjclose"":["""adjclose"":["") + 23
If closeStart < 24 Then closeStart = InStr(response, """"close"":["") + 8

```

```

closeEnd = InStr(closeStart, response, "]")
If closeStart > 20 And closeEnd > closeStart Then
    closeStr = Mid(response, closeStart, closeEnd - closeStart)
    closeArr = Split(closeStr, ",")

```

```
Else
    GetHistoricalData = False
    Exit Function
End If

Dim count As Long
count = UBound(tsArr)
If UBound(closeArr) < count Then count = UBound(closeArr)

ReDim outDates(0 To count)
ReDim outPrices(0 To count)

Dim validCount As Long
validCount = 0

For i = 0 To count
    If IsNumeric(tsArr(i)) And IsNumeric(closeArr(i)) Then
        outDates(validCount) = (Cdbl(tsArr(i)) / 86400) + 25569
        outPrices(validCount) = Cdbl(closeArr(i))
        validCount = validCount + 1
    End If
Next i

If validCount > 0 Then
    ReDim Preserve outDates(0 To validCount - 1)
    ReDim Preserve outPrices(0 To validCount - 1)
    GetHistoricalData = True
Else
    GetHistoricalData = False
End If
End Function
```

Option Explicit

```

' =====
' PORTFOLIO DASHBOARD v3.0 - Bloomberg Terminal Style
' =====

Private Const SH_PORT As String = "RR4"
Private Const SH_TRANS As String = "Transactions"
Private Const SH_REAL As String = "Realized"
Private Const SH_HIST As String = "HistoryLog"

Public g_AutoOn As Boolean
Public g_NextTime As Date

' =====
' MAIN ENTRY
' =====

Sub RebuildPortfolioDashboard()
    Dim wsP As Worksheet
    Dim wsTr As Worksheet
    Dim wsR As Worksheet
    Set wsP = ThisWorkbook.Sheets(SH_PORT)
    Set wsTr = ThisWorkbook.Sheets(SH_TRANS)
    Set wsR = ThisWorkbook.Sheets(SH_REAL)

    If Not g_PriceCache Is Nothing Then
        g_PriceCache.RemoveAll
        g_CacheTime = Now
    End If

    ' Backfill missing history rows

    Application.ScreenUpdating = False
    Dim swingRiskMap As Object
    Set swingRiskMap = CreateObject("Scripting.Dictionary")
    Dim srLastRow As Long
    srLastRow = wsP.Cells(wsP.Rows.Count, 1).End(xlUp).row
    Dim srRow As Long
    For srRow = 5 To srLastRow
        Dim srTicker As String: srTicker = CStr(wsP.Cells(srRow, 1).Value)
        Dim srVal As String: srVal = CStr(wsP.Cells(srRow, 17).Value)
        If srTicker <> "" And srVal <> "" Then
            swingRiskMap(srTicker) = srVal
        End If
    Next srRow
    Call ResetSheetStyle(wsP)

    Dim positions As Object
    Set positions = BuildPositions(wsTr)

    Dim exRate As Double
    exRate = GetExRate(wsP)

    Dim posData() As Variant
    Dim totalMktTWD As Double
    Dim totalCostTWD As Double
    Dim totalUnrlTWD As Double
    Dim posCount As Long

    Call CalcPositions(positions, exRate, posData, totalMktTWD, totalCostTWD, totalUnrlTWD, posCount)
    Call CalculateRealizedPnL

    Dim realPnL As Double
    On Error Resume Next
    realPnL = Application.WorksheetFunction.Sum(wsR.Columns("G"))
    On Error GoTo 0

    Dim portBeta As Double
    portBeta = CalcPortBeta(posData)

    Call DrawHeader(wsP, totalMktTWD, totalCostTWD, totalUnrlTWD, realPnL, portBeta, posCount, exRate)

```

```

Call DrawColumnHeaders(wsP)
Dim lastDataRow As Long
lastDataRow = WritePositionRows(wsP, posData, totalMktTWD, swingRiskMap)
Call DrawDisclaimer(wsP, lastDataRow)
Call LogHistory(totalMktTWD, totalUnrlTWD + realPnL, realPnL)

Application.ScreenUpdating = True
Call DrawDeepAnalysis(wsP, posData, totalMktTWD, totalCostTWD, totalUnrlTWD, realPnL)
Application.StatusBar = "Dashboard updated: " & Format(Now, "hh:mm:ss")
MsgBox "Dashboard updated!", vbInformation
End Sub

```

```

Private Sub LogHistory(totalMkt As Double, totalPnL As Double, realPnL As Double)
    Dim wsH As Worksheet
    On Error Resume Next
    Set wsH = ThisWorkbook.Sheets(SH_HIST)
    On Error GoTo 0
    If wsH Is Nothing Then Exit Sub

```

```

    Dim nr As Long: nr = wsH.Cells(wsH.Rows.Count, "A").End(xlUp).Row + 1
    If nr > 2 Then
        If Int(CDate(wsH.Cells(nr - 1, 1).Value)) = Date Then nr = nr - 1
    End If

```

```

    Dim priceSPY As Double, priceQQQ As Double, priceTWII As Double
    Dim priceSOX As Double, priceTWOII As Double

```

```

    priceSPY = GetStockPrice("SPY")
    priceQQQ = GetStockPrice("QQQ")
    priceTWII = GetStockPrice("^TWII")
    priceSOX = GetStockPrice("^SOX")
    priceTWOII = GetStockPrice("^TWOII")

```

```

With wsH

```

```

    .Cells(nr, "A").Value = Now
    .Cells(nr, "B").Value = totalMkt
    .Cells(nr, "C").Value = totalPnL
    .Cells(nr, "D").Value = priceSPY
    .Cells(nr, "E").Value = priceQQQ
    .Cells(nr, "F").Value = priceTWII
    .Cells(nr, "G").Value = realPnL
    .Cells(nr, "N").Value = priceSOX
    .Cells(nr, "O").Value = priceTWOII

```

```

    .Cells(nr, "P").Formula = "=(" & nr & "-$N$2)/$N$2"
    .Cells(nr, "Q").Formula = "=(" & nr & "-$O$2)/$O$2"

```

```

    .Cells(nr, "A").NumberFormat = "yyyy/m/d h:mm:ss"
    .Range("B" & nr & ":C" & nr).NumberFormat = "#,##0"
    .Range("D" & nr & ":F" & nr).NumberFormat = "#,##0.00"
    .Cells(nr, "G").NumberFormat = "#,##0"
    .Cells(nr, "N").NumberFormat = "#,##0.00"
    .Cells(nr, "O").NumberFormat = "#,##0.00"
    .Cells(nr, "P").NumberFormat = "0.00%"
    .Cells(nr, "Q").NumberFormat = "0.00%"

```

```

End With

```

```

End Sub

Function GetHistoricalPrice(ticker As String, targetDate As Date) As Double
    Dim http As Object
    Set http = CreateObject("MSXML2.XMLHTTP")

```

```

    ' Unix epoch is UTC; TW market is UTC+8 (8h ahead) - opening crosses a day boundary
    Dim p1 As Long, p2 As Long
    p1 = DateDiff("s", #1/1/1970#, targetDate) - 8 * 3600
    p2 = p1 + 86400

```

```

    ' ^ -- URL encode -- %5E
    Dim safeTicker As String
    safeTicker = Replace(ticker, "^", "%5E")

```

```

    Dim url As String

```

```
url = "https://query2.finance.yahoo.com/v8/finance/chart/" & safeTicker & _
      "?period1=" & p1 & "&period2=" & p2 & "&interval=1d"
```

```
On Error GoTo ErrHandler
http.Open "GET", url, False
http.setRequestHeader "User-Agent", "Mozilla/5.0"
http.send
```

```
Dim resp As String: resp = http.responseText
```

```
' -----
Dim pos As Long
pos = InStr(resp, """"close"":["")
If pos = 0 Then GoTo ErrHandler
pos = pos + 9
Dim endPos As Long: endPos = InStr(pos, resp, "]")
Dim closeVal As String: closeVal = Trim(Mid(resp, pos, endPos - pos))
If Len(closeVal) = 0 Or InStr(closeVal, "null") > 0 Then GoTo ErrHandler
```

```
GetHistoricalPrice = CDbl(Split(closeVal, ",")(0))
Exit Function
```

```
ErrHandler:
```

```
GetHistoricalPrice = 0
```

```
End Function
```

```
Sub BackfillHistory()
```

```
Dim wsH As Worksheet
```

```
On Error Resume Next
```

```
Set wsH = ThisWorkbook.Sheets(SH_HIST)
```

```
On Error GoTo 0
```

```
If wsH Is Nothing Then MsgBox "History sheet not found": Exit Sub
```

```
Dim lastRow As Long
```

```
lastRow = wsH.Cells(wsH.Rows.Count, "A").End(xlUp).Row
```

```
If lastRow < 2 Then MsgBox "No data to backfill": Exit Sub
```

```
Application.ScreenUpdating = False
```

```
Dim i As Long, countSOX As Long, countOTC As Long
```

```
For i = 2 To lastRow
```

```
    If wsH.Cells(i, "A").Value = "" Then GoTo NextRow
```

```
    ' H: refill SPY Ret%
```

```
    wsH.Cells(i, "H").Formula = "=(D" & i & " - $D$2)/$D$2"
```

```
    wsH.Cells(i, "H").NumberFormat = "0.00%"
```

```
    ' -----
```

```
    Dim targetDate As Date
```

```
    On Error Resume Next
```

```
    targetDate = CDate(Int(CDbl(wsH.Cells(i, "A").Value)))
```

```
    On Error GoTo 0
```

```
    If targetDate = 0 Then GoTo NextRow
```

```
    Dim price As Double
```

```
    ' N -- G -- ^ -- ^SOX -- v -- L --
```

```
    price = GetHistoricalPrice("^SOX", targetDate)
```

```
    If price > 0 Then
```

```
        wsH.Cells(i, "N").Value = price
```

```
        wsH.Cells(i, "N").NumberFormat = "#,##0.00"
```

```
        wsH.Cells(i, "P").Formula = "=(N" & i & " - $N$2)/$N$2"
```

```
        wsH.Cells(i, "P").NumberFormat = "0.00%"
```

```
        countSOX = countSOX + 1
```

```
    End If
```

```
    ' O -- G -- ^ -- ^TWOII -- v -- L --
```

```
    price = GetHistoricalPrice("^TWOII", targetDate)
```

```
    If price > 0 Then
```

```
        wsH.Cells(i, "O").Value = price
```

```
        wsH.Cells(i, "O").NumberFormat = "#,##0.00"
```

```
        wsH.Cells(i, "Q").Formula = "=(O" & i & " - $O$2)/$O$2"
```

```

        wsH.cells(i, "Q").NumberFormat = "0.00%"
        countOTC = countOTC + 1
    End If

    Application.StatusBar = "Backfilling row " & i & "/" & lastRow & _
        " SOX:" & countSOX & " OTC:" & countOTC
NextRow:
    Next i

    Application.ScreenUpdating = True
    Application.StatusBar = False
    MsgBox "Backfill complete!" & vbLf & _
        "H = SPY Ret%: refilled (total " & lastRow - 1 & " rows)" & vbLf & _
        "N = ^SOX: updated " & countSOX & " rows" & vbLf & _
        "O = ^TWII: updated " & countOTC & " rows" & vbLf & _
        "Existing / blank rows skipped automatically"
End Sub

' =====
' Sheet style reset
' =====

Private Sub ResetSheetStyle(ws As Worksheet)
    ws.cells.Clear
    With ws.cells
        .Interior.Color = RGB(0, 0, 0)
        .Font.Color = RGB(221, 221, 221)
        .Font.Name = "Consolas"
        .Font.Size = 10
        .Font.Bold = False
        .NumberFormat = "General"
        .HorizontalAlignment = xlLeft
        .VerticalAlignment = xlCenter
    End With
    ws.Activate
    ActiveWindow.DisplayGridlines = False

    Dim colWidths As Variant
    colWidths = Array(5, 22, 10, 8, 8, 10, 10, 8, 10, 10, 8, 11, 7, 8, 10, 10)
    ' Column widths (Excel units ? pixels / 7)
    ' A=240px?34 B=580px?83 F=580px?83 others=138px?20
    Dim i As Integer
    For i = 1 To 16
        Select Case i
            Case 1: ws.Columns(i).ColumnWidth = 17          ' A: 240px
            Case 2, 6: ws.Columns(i).ColumnWidth = 40       ' B, F: 580px
            Case Else: ws.Columns(i).ColumnWidth = 11       ' others: 138px
        End Select
    Next i

    Dim r As Long
    For r = 1 To 100
        ws.Rows(r).RowHeight = 18
    Next r
    ws.Rows(1).RowHeight = 28
    ws.Rows(4).RowHeight = 20
End Sub

' =====
' Header (rows 1-3)
' =====

Private Sub DrawHeader(ws As Worksheet, totalMkt As Double, totalCost As Double, _
    totalUnrl As Double, realPnL As Double, _
    portBeta As Double, posCount As Long, exRate As Double)

    Dim unrlPct As Double
    If totalCost > 0 Then unrlPct = totalUnrl / totalCost Else unrlPct = 0

    With ws.cells(1, 1)
        .Value = Format(totalMkt, "$#,##0")
        .Font.Color = RGB(255, 192, 0)
        .Font.Size = 16
    End With

```

```

        .Font.Bold = True
        .HorizontalAlignment = xlCenter
    End With

    ws.cells(1, 2).Value = " <- TOTAL MKT "
    ws.cells(1, 2).Font.Color = RGB(255, 192, 0)
    ws.cells(1, 2).Font.Bold = True

    ws.cells(1, 3).Value = "INCEPTION"
    ws.cells(1, 3).Font.Color = RGB(150, 150, 150)
    ws.cells(2, 3).Value = Format(GetInceptionDate(ws), "m/d/yyyy")
    ws.cells(2, 3).Font.Color = RGB(221, 221, 221)

    ws.cells(1, 4).Value = "DAYS"
    ws.cells(1, 4).Font.Color = RGB(150, 150, 150)
    ws.cells(2, 4).Value = Date - GetInceptionDate(ws)
    ws.cells(2, 4).Font.Color = RGB(221, 221, 221)

    ws.cells(1, 5).Value = "STARTING"
    ws.cells(1, 5).Font.Color = RGB(150, 150, 150)
    ws.cells(2, 5).Value = GetStartingCapital(ws)
    ws.cells(2, 5).Font.Color = RGB(221, 221, 221)
    ws.cells(2, 5).NumberFormat = "#,##0"

    ws.cells(1, 9).Value = "PORT.BETA"
    ws.cells(1, 9).Font.Color = RGB(150, 150, 150)
    ws.cells(1, 10).Value = portBeta
    ws.cells(1, 10).NumberFormat = "0.00"
    ws.cells(1, 10).Font.Color = RGB(255, 192, 0)
    ws.cells(1, 10).Font.Bold = True

    ws.cells(1, 11).Value = "POSITIONS"
    ws.cells(1, 11).Font.Color = RGB(150, 150, 150)
    ws.cells(1, 12).Value = posCount
    ws.cells(1, 12).Font.Color = RGB(221, 221, 221)

    ws.cells(2, 13).Value = "RL PNL"
    ws.cells(2, 13).Font.Color = RGB(150, 150, 150)
    ws.cells(2, 14).Value = realPnL
    ws.cells(2, 14).NumberFormat = "$#,##0"
    ws.cells(2, 14).Font.Color = PnLColor(realPnL)
    ws.cells(2, 14).Font.Bold = True

    Dim startCap As Double: startCap = GetStartingCapital(ws)
    Dim rlPct As Double
    If startCap > 0 Then rlPct = realPnL / startCap Else rlPct = 0
    ws.cells(2, 15).Value = "RL PNL%"
    ws.cells(2, 15).Font.Color = RGB(150, 150, 150)
    ws.cells(2, 16).Value = rlPct
    ws.cells(2, 16).NumberFormat = "0.00%"
    ws.cells(2, 16).Font.Color = PnLColor(rlPct)
    ws.cells(2, 16).Font.Bold = True

    ws.cells(2, 1).Value = "USD/TWD"
    ws.cells(2, 1).Font.Color = RGB(150, 150, 150)
    ws.cells(2, 2).Value = exRate
    ws.cells(2, 2).NumberFormat = "0.00"
    ws.cells(2, 2).Font.Color = RGB(221, 221, 221)

    ws.cells(2, 9).Value = "UNRL PNL%"
    ws.cells(2, 9).Font.Color = RGB(150, 150, 150)
    ws.cells(2, 10).Value = unrlPct
    ws.cells(2, 10).NumberFormat = "0.00%"
    ws.cells(2, 10).Font.Color = PnLColor(unrlPct)
    ws.cells(2, 10).Font.Bold = True

    ws.cells(2, 11).Value = "UNRL PNL"
    ws.cells(2, 11).Font.Color = RGB(150, 150, 150)
    ws.cells(2, 12).Value = totalUnrl
    ws.cells(2, 12).NumberFormat = "$#,##0"
    ws.cells(2, 12).Font.Color = PnLColor(totalUnrl)

```

```

ws.cells(2, 12).Font.Bold = True

With ws.Range(ws.cells(3, 1), ws.cells(3, 16))
    .Interior.Color = RGB(255, 192, 0)
    .RowHeight = 3
End With
End Sub

' =====
' DEEP ANALYSIS - Sector / Strategy / Currency / Technical
' Writes to a separate "Analysis" sheet, dark theme
' =====

Sub DrawDeepAnalysis(wsP As Worksheet, posData() As Variant, _
    totalMkt As Double, totalCost As Double, _
    totalUnrl As Double, realPnL As Double)

    ' --- Get or create Analysis sheet ---
    Dim wsA As Worksheet
    On Error Resume Next
    Set wsA = ThisWorkbook.Sheets("Analysis")
    On Error GoTo 0
    If wsA Is Nothing Then
        Set wsA = ThisWorkbook.Sheets.Add(After:=wsP)
        wsA.Name = "Analysis"
    End If

    wsA.cells.Clear
    With wsA.cells
        .Interior.Color = RGB(0, 0, 0)
        .Font.Color = RGB(221, 221, 221)
        .Font.Name = "Consolas"
        .Font.Size = 36
    End With
    wsA.Activate
    ActiveWindow.DisplayGridlines = False

    If Not IsArray(posData) Then Exit Sub
    Dim n As Long
    On Error Resume Next
    n = UBound(posData, 1)
    If Err.Number <> 0 Or n < 1 Then Exit Sub
    On Error GoTo 0

    ' -----
    With wsA.cells(1, 1)
        .Value = "DEEP ANALYSIS - " & Format(Now, "yyyy/mm/dd hh:mm")
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
        .Font.Size = 36
    End With
    With wsA.Range(wsA.cells(2, 1), wsA.cells(2, 20))
        .Interior.Color = RGB(255, 192, 0)
        .RowHeight = 3
    End With

    ' -----
    Dim totalPnL As Double: totalPnL = totalUnrl + realPnL
    Dim retPct As Double
    If totalCost > 0 Then retPct = totalPnL / totalCost

    Dim r As Long: r = 4
    wsA.cells(r, 1).Value = "RETURN SUMMARY"
    wsA.cells(r, 1).Font.Color = RGB(255, 192, 0)
    wsA.cells(r, 1).Font.Bold = True
    r = r + 1

    Call WriteKV(wsA, r, "Total Return %", retPct, "0.00%", True): r = r + 1
    Call WriteKV(wsA, r, "Total Cost", totalCost, "$#,##0", False): r = r + 1
    Call WriteKV(wsA, r, "Unrealized PnL", totalUnrl, "$#,##0", True): r = r + 1
    Call WriteKV(wsA, r, "Realized PnL", realPnL, "$#,##0", True): r = r + 1
    Call WriteKV(wsA, r, "Total PnL", totalPnL, "$#,##0", True): r = r + 1

```

```

' -----
r = r + 2
Dim sectorD As Object: Set sectorD = CreateObject("Scripting.Dictionary")
Dim stratD As Object: Set stratD = CreateObject("Scripting.Dictionary")
Dim currencyD As Object: Set currencyD = CreateObject("Scripting.Dictionary")

Dim i As Long
For i = 1 To n
    Dim sec As String: sec = CStr(posData(i, 7))
    Dim cur As String: cur = GetCurrencyType(CStr(posData(i, 2)))
    Dim mv As Double: mv = posData(i, 8)
    Dim stg As String
    ' strategy stored in posData(i,2) ticker -> need from positions
    ' use sector as proxy if strategy not in posData
    stg = CStr(posData(i, 7)) ' fallback: sector

    If sec <> "" Then If sectorD.Exists(sec) Then sectorD(sec) = sectorD(sec) + mv Else sectorD.Add sec, mv
    If cur <> "" Then If currencyD.Exists(cur) Then currencyD(cur) = currencyD(cur) + mv Else currencyD.Add cur, mv
Next i

r = DrawAllocTable(wsA, r, "SECTOR ALLOCATION", sectorD, totalMkt)
r = r + 2
r = DrawAllocTable(wsA, r, "CURRENCY ALLOCATION", currencyD, totalMkt)

' -----
r = r + 2
wsA.cells(r, 1).Value = "TECHNICAL & HOLDING ANALYSIS"
wsA.cells(r, 1).Font.Color = RGB(255, 192, 0)
wsA.cells(r, 1).Font.Bold = True
r = r + 1

' Header
Dim techHdrs As Variant
techHdrs = Array("TICKER", "NAME", "DAYS", "LAST", "HIGH DIST%", _
    "DAY CHG%", "BIAS5", "BIAS20", "BIAS60", "BIAS120", "BIAS240", "TREND")

Dim c As Integer
For c = 0 To UBound(techHdrs)
    With wsA.cells(r, c + 1)
        .Value = techHdrs(c)
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
        .Interior.Color = RGB(10, 10, 10)
        .HorizontalAlignment = xlCenter
    End With
Next c
r = r + 1

' Data rows
For i = 1 To n
    Dim tkr As String: tkr = CStr(posData(i, 2))
    Dim nm As String: nm = CStr(posData(i, 3))
    Dim ddays As Long: ddays = posData(i, 5)

    Dim rowBg As Long: rowBg = IIf(i Mod 2 = 1, RGB(15, 15, 15), RGB(22, 22, 22))
    With wsA.Range(wsA.cells(r, 1), wsA.cells(r, 12))
        .Interior.Color = rowBg
        .HorizontalAlignment = xlCenter
    End With

    ' Get tech data from existing function
    Dim tech As TechIndicators
    tech = GetTechData(tkr)

    wsA.cells(r, 1).Value = tkr
    wsA.cells(r, 2).Value = nm
    wsA.cells(r, 2).HorizontalAlignment = xlLeft
    wsA.cells(r, 3).Value = ddays
    wsA.cells(r, 4).Value = posData(i, 11)
    wsA.cells(r, 4).NumberFormat = "#,##0.00"

```

```

If tech.Success Then
    wsA.cells(r, 5).Value = tech.DistFromHigh
    wsA.cells(r, 6).Value = tech.ChangePercent
    wsA.cells(r, 7).Value = tech.Bias5
    wsA.cells(r, 8).Value = tech.Bias20
    wsA.cells(r, 9).Value = tech.Bias60
    wsA.cells(r, 10).Value = tech.Bias120
    wsA.cells(r, 11).Value = tech.Bias240

    ' Colour each metric cell
    Dim mc As Integer
    For mc = 5 To 11
        wsA.cells(r, mc).NumberFormat = "0.00%"
        wsA.cells(r, mc).Font.Color = PnLColor(wsA.cells(r, mc).Value)
    Next mc

    ' Trend
    If tech.Bias20 > 0 And tech.Bias60 > 0 Then
        wsA.cells(r, 12).Value = "BULLISH"
        wsA.cells(r, 12).Font.Color = RGB(255, 80, 80)
    ElseIf tech.Bias20 < 0 And tech.Bias60 < 0 Then
        wsA.cells(r, 12).Value = "BEARISH"
        wsA.cells(r, 12).Font.Color = RGB(0, 210, 100)
    Else
        wsA.cells(r, 12).Value = "NEUTRAL"
        wsA.cells(r, 12).Font.Color = RGB(200, 200, 200)
    End If
    wsA.cells(r, 12).Font.Bold = True
Else
    wsA.cells(r, 5).Value = "N/A"
End If
r = r + 1
Next i

wsA.Columns("A:L").AutoFit
wsP.Activate ' return to main sheet
End Sub

Sub RebuildActiveXButtons()
    Dim ws As Worksheet
    Set ws = ThisWorkbook.Sheets("RR4")

    ' ----- Unprotect sheet -----
    On Error Resume Next
    ws.Unprotect
    On Error GoTo 0

    ' ----- Remove existing shapes -----

    Dim shp As Shape
    Dim delList As New Collection
    For Each shp In ws.Shapes
        delList.Add shp.Name
    Next shp
    Dim nm As Variant
    For Each nm In delList
        On Error Resume Next
        ws.Shapes(CStr(nm)).Delete
        On Error GoTo 0
    Next nm

    ' ----- Probe ActiveX availability -----
    Dim testOle As OLEObject
    On Error Resume Next
    Set testOle = ws.OLEObjects.Add(ClassType:="Forms.CommandButton.1", _
        Left:=10, Top:=5, Width:=50, Height:=20)
    If Err.Number <> 0 Then
        MsgBox "ActiveX is unavailable. Reason: " & Err.Description & vbCr & vbCr & _
            "Please check:" & vbCr & _
            "1. File is saved as .xlsm" & vbCr & _
            "2. Trust Center > Enable ActiveX controls" & vbCr & _

```

```

        "3. Worksheet is not protected", vbExclamation
    Exit Sub
End If
testOle.Delete
On Error GoTo 0

' ----- Create new buttons -----
Dim topPos As Double: topPos = ws.Rows(35).Top

Dim btnDefs As Variant
btnDefs = Array( _
    Array("DELETE ALL", 10, topPos, 130, 20, RGB(30, 0, 0), RGB(255, 60, 60)), _
    Array("REGULATE", 150, topPos, 160, 20, RGB(10, 10, 30), RGB(100, 160, 255)), _
    Array("UPDATE", 320, topPos, 100, 20, RGB(0, 25, 0), RGB(0, 220, 100)), _
    Array("ADD TRANSACTION", 430, topPos, 160, 20, RGB(20, 15, 0), RGB(255, 192, 0)), _
    Array("HOLDINGS CORR", 600, topPos, 140, 20, RGB(15, 0, 30), RGB(180, 100, 255)), _
    Array("DRAWDOWN", 750, topPos, 110, 20, RGB(30, 5, 0), RGB(255, 120, 60)), _
    Array("AUTO ON", 870, topPos, 100, 20, RGB(0, 40, 0), RGB(0, 255, 136)), _
    Array("AUTO OFF", 980, topPos, 100, 20, RGB(40, 0, 0), RGB(255, 80, 80)), _
    Array("DEBUG", 1090, topPos, 80, 20, RGB(0, 10, 30), RGB(100, 160, 255)) _
)

Dim i As Integer
For i = 0 To UBound(btnDefs)
    Dim b As Variant: b = btnDefs(i)
    Dim ole As OLEObject
    Set ole = ws.OLEObjects.Add( _
        ClassType:="Forms.CommandButton.1", _
        Left:=b(1), Top:=b(2), Width:=b(3), Height:=b(4))
    ole.Name = "CmdBtn" & i
    With ole.Object
        .Caption = b(0)
        .Font.Name = "Consolas"
        .Font.Size = 10
        .Font.Bold = True
        .BackColor = b(5)
        .ForeColor = b(6)
        .BackStyle = 1
    End With
Next i

End Sub

' -----
Private Function DrawAllocTable(ws As Worksheet, startRow As Long, _
                                title As String, dict As Object, _
                                totalMkt As Double) As Long

    Dim r As Long: r = startRow
    ws.cells(r, 1).Value = title
    ws.cells(r, 1).Font.Color = RGB(255, 192, 0)
    ws.cells(r, 1).Font.Bold = True
    r = r + 1

    ' Header
    ws.cells(r, 1).Value = "CATEGORY"
    ws.cells(r, 2).Value = "VALUE"
    ws.cells(r, 3).Value = "WEIGHT"
    With ws.Range(ws.cells(r, 1), ws.cells(r, 3))
        .Font.Bold = True
        .Interior.Color = RGB(10, 10, 10)
        .HorizontalAlignment = xlCenter
        .Font.Color = RGB(255, 192, 0)
    End With
    r = r + 1

    Dim key As Variant
    Dim i As Long: i = 0
    For Each key In dict.keys
        Dim rowBg As Long: rowBg = IIf(i Mod 2 = 0, RGB(15, 15, 15), RGB(22, 22, 22))
        With ws.Range(ws.cells(r, 1), ws.cells(r, 3))
            .Interior.Color = rowBg

```

```

        .HorizontalAlignment = xlCenter
    End With
    ws.cells(r, 1).Value = key
    ws.cells(r, 1).HorizontalAlignment = xlLeft
    ws.cells(r, 2).Value = dict(key)
    ws.cells(r, 2).NumberFormat = "$#,##0"
    If totalMkt > 0 Then
        ws.cells(r, 3).Value = dict(key) / totalMkt
        ws.cells(r, 3).NumberFormat = "0.00%"
    End If
    r = r + 1: i = i + 1
Next key

DrawAllocTable = r
End Function
' =====
' Column header row (row 4)
' =====
Private Sub DrawColumnHeaders(ws As Worksheet)
    Dim headers As Variant
    headers = Array("ISIN", "NAME", "ENTRY DT", "DAYS", "BROKER", _
        "SECTOR", "NET EXPOS", "SHARES", "ENTRY PX", "LAST", _
        "% CHG", "UNRL PNL", "WT%", "W.BETA", "Beta 180D", "P.TARGET")

    Dim i As Integer
    For i = 0 To UBound(headers)
        With ws.cells(4, i + 1)
            .Value = headers(i)
            .Font.Color = RGB(255, 192, 0)
            .Font.Bold = True
            .Font.Size = 9
            .Font.Name = "Consolas"
            .Interior.Color = RGB(10, 10, 10)
            .HorizontalAlignment = xlCenter
        End With
    Next i

    ' SWING RISK header (col 17)
    With ws.cells(4, 17)
        .Value = "SWING RISK"
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
        .Font.Size = 9
        .Font.Name = "Consolas"
        .Interior.Color = RGB(10, 10, 10)
        .HorizontalAlignment = xlCenter
    End With

    ws.Columns(17).ColumnWidth = 20
    With ws.Range(ws.cells(4, 1), ws.cells(4, 17)).Borders(xlEdgeBottom)
        .LineStyle = xlContinuous
        .Color = RGB(255, 192, 0)
        .Weight = xlThin
    End With
End Sub

' -----
Private Function WritePositionRows(ws As Worksheet, posData() As Variant, _
    totalMkt As Double, ByVal swingRiskMap As Object) As Long

    WritePositionRows = 5
    If Not IsArray(posData) Then Exit Function
    Dim n As Long
    On Error Resume Next: n = UBound(posData, 1)
    If Err.Number <> 0 Or n < 1 Then Exit Function
    On Error GoTo 0

    Dim i As Long, r As Long: r = 5
    Dim currentBroker As String: currentBroker = ""
    Dim brokerMkt As Double: brokerMkt = 0
    Dim brokerUnrl As Double: brokerUnrl = 0

    For i = 1 To n

```

```

Dim thisBroker As String: thisBroker = CStr(posData(i, 6))

If thisBroker <> currentBroker Then
    If currentBroker <> "" Then
        Call DrawBrokerSubtotal(ws, r, currentBroker, brokerMkt, brokerUnrl)
        r = r + 1
    End If
    Call DrawBrokerHeader(ws, r, thisBroker)
    r = r + 1
    currentBroker = thisBroker
    brokerMkt = 0: brokerUnrl = 0
End If

Call WriteOnePositionRow(ws, r, i, posData, totalMkt, swingRiskMap)
brokerMkt = brokerMkt + posData(i, 8)
brokerUnrl = brokerUnrl + posData(i, 13)
r = r + 1
Next i

If currentBroker <> "" Then
    Call DrawBrokerSubtotal(ws, r, currentBroker, brokerMkt, brokerUnrl)
    r = r + 1
End If

```

```

WritePositionRows = r - 1
End Function

```

```

' -----
Private Sub DrawBrokerHeader(ws As Worksheet, r As Long, brokerName As String)
    ws.Rows(r).RowHeight = 22
    With ws.Range(ws.Cells(r, 1), ws.Cells(r, 17))
        .Interior.Color = RGB(35, 25, 0)
        .Font.Name = "Consolas"
        .Font.Size = 10
        .Font.Bold = True
        .Font.Color = RGB(255, 192, 0)
    End With
    ws.Cells(r, 1).Value = " = " & brokerName & " " & String(28, "=")
    ws.Cells(r, 1).HorizontalAlignment = xlLeft
End Sub

```

```

' -----
Private Sub DrawBrokerSubtotal(ws As Worksheet, r As Long, brokerName As String, _
    brokerMkt As Double, brokerUnrl As Double)
    ws.Rows(r).RowHeight = 18
    With ws.Range(ws.Cells(r, 1), ws.Cells(r, 17))
        .Interior.Color = RGB(25, 18, 0)
    End With
    With ws.Cells(r, 1)
        .Value = " Subtotal " & brokerName
        .Font.Color = RGB(200, 160, 0)
        .Font.Bold = True
        .Font.Name = "Consolas"
        .Font.Size = 9
        .HorizontalAlignment = xlLeft
    End With
    With ws.Cells(r, 7)
        .Value = brokerMkt
        .NumberFormat = "$#,##0"
        .Font.Color = RGB(200, 160, 0)
        .Font.Bold = True
    End With
    With ws.Cells(r, 12)
        .Value = brokerUnrl
        .NumberFormat = "$#,##0;-$,##0"
        .Font.Color = PnLColor(Round(brokerUnrl, 0))
        .Font.Bold = True
    End With
End Sub

```

```
Private Sub WriteOnePositionRow(ws As Worksheet, r As Long, i As Long, _
    posData() As Variant, totalMkt As Double, _
    swingRiskMap As Object)
```

```
Dim rowBg As Long
rowBg = IIf(i Mod 2 = 1, RGB(15, 15, 15), RGB(22, 22, 22))
```

```
With ws.Range(ws.Cells(r, 1), ws.Cells(r, 17))
```

```
.Interior.Color = rowBg
.Font.Name = "Consolas"
.Font.Size = 10
.Font.Color = RGB(210, 210, 210)
.HorizontalAlignment = xlCenter
```

```
End With
ws.Rows(r).RowHeight = 18
```

```
Dim tickerCode As String: tickerCode = CStr(posData(i, 1))
```

```
Dim nm As String: nm = posData(i, 3)
Dim entryDt As Date: entryDt = posData(i, 4)
Dim days As Long: days = posData(i, 5)
Dim broker As String: broker = posData(i, 6)
Dim Sector As String: Sector = posData(i, 7)
Dim netExpos As Double: netExpos = posData(i, 8)
Dim shares As Double: shares = posData(i, 9)
Dim entryPx As Double: entryPx = posData(i, 10)
Dim lastPx As Double: lastPx = posData(i, 11)
Dim chgPct As Double: chgPct = posData(i, 12)
Dim unrlPnl As Double: unrlPnl = posData(i, 13)
Dim Beta As Double: Beta = posData(i, 14)
Dim beta30d As Double: beta30d = posData(i, 15)
Dim pTgt As Double: pTgt = posData(i, 16)
```

```
Dim wtPct As Double: If totalMkt > 0 Then wtPct = netExpos / totalMkt
Dim wBeta As Double: wBeta = wtPct * Beta
```

```
ws.Cells(r, 1).Value = tickerCode
ws.Cells(r, 1).Font.Color = RGB(255, 192, 0)
ws.Cells(r, 1).Font.Bold = True
```

```
ws.Cells(r, 2).Value = nm
ws.Cells(r, 2).HorizontalAlignment = xlLeft
ws.Cells(r, 2).Font.Color = RGB(221, 221, 221)
```

```
ws.Cells(r, 3).Value = entryDt
ws.Cells(r, 3).NumberFormat = "m/d/yyyy"
```

```
ws.Cells(r, 4).Value = days
```

```
ws.Cells(r, 5).Value = broker
ws.Cells(r, 5).Font.Color = RGB(180, 180, 180)
```

```
ws.Cells(r, 6).Value = Sector
ws.Cells(r, 6).Font.Color = RGB(180, 180, 180)
```

```
ws.Cells(r, 7).Value = netExpos
ws.Cells(r, 7).NumberFormat = "$#,##0"
```

```
ws.Cells(r, 8).Value = shares
ws.Cells(r, 8).NumberFormat = "#,##0.##"
```

```
ws.Cells(r, 9).Value = entryPx
ws.Cells(r, 9).NumberFormat = "#,##0.00"
```

```
ws.Cells(r, 10).Value = lastPx
ws.Cells(r, 10).NumberFormat = "#,##0.00"
ws.Cells(r, 10).Font.Color = RGB(221, 221, 221)
```

```
ws.Cells(r, 11).Value = chgPct
ws.Cells(r, 11).NumberFormat = "0.00%"
ws.Cells(r, 11).Font.Bold = True
ws.Cells(r, 11).Font.Color = PnlColor(chgPct)
```

```

ws.cells(r, 12).Value = unrlPnl
ws.cells(r, 12).NumberFormat = "$#,##0;-$$#,##0"
ws.cells(r, 12).Font.Bold = True
ws.cells(r, 12).Font.Color = PnLColor(Round(unrlPnl, 0))

```

```

ws.cells(r, 13).Value = wtPct
ws.cells(r, 13).NumberFormat = "0.00%"

```

```

ws.cells(r, 14).Value = wBeta
ws.cells(r, 14).NumberFormat = "0.000"

```

```

ws.cells(r, 15).Value = beta30d
ws.cells(r, 15).NumberFormat = "0.000"
ws.cells(r, 15).Font.Color = RGB(180, 180, 255)

```

```

ws.cells(r, 16).Value = pTgt
ws.cells(r, 16).NumberFormat = "#,##0"
ws.cells(r, 16).Font.Color = RGB(255, 192, 0)

```

```

If pTgt > 0 And lastPx > pTgt Then
    ws.Range(ws.cells(r, 1), ws.cells(r, 17)).Interior.Color = RGB(40, 25, 0)
End If

```

```

If swingRiskMap.Exists(tickerCode) Then
    With ws.cells(r, 17)
        .Value = swingRiskMap(tickerCode)
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
        .HorizontalAlignment = xlCenter
    End With
End If
End Sub

```

Disclaimer (bottom rows)

```
Private Sub DrawDisclaimer(ws As Worksheet, startRow As Long)
```

```
    Dim r As Long: r = startRow + 3
```

```
    On Error Resume Next
```

```
    ws.Range(ws.cells(r, 1), ws.cells(r, 16)).UnMerge
```

```
    ws.Range(ws.cells(r, 1), ws.cells(r, 16)).Merge
```

```
    On Error GoTo 0
```

```
    ws.cells(r, 1).Value = "On the way in Medium-High Beta Between 2-2.5 , Remember always do the Eliminate underperformers"
```

```
    ws.cells(r, 1).Font.Color = RGB(255, 192, 0)
```

```
    ws.cells(r, 1).Font.Italic = True
```

```
    ws.cells(r, 1).HorizontalAlignment = xlLeft
```

```
    r = r + 1
```

```
    On Error Resume Next
```

```
    ws.Range(ws.cells(r, 1), ws.cells(r, 16)).UnMerge
```

```
    ws.Range(ws.cells(r, 1), ws.cells(r, 16)).Merge
```

```
    On Error GoTo 0
```

```
    ws.cells(r, 1).Value = "FOCUS ON WHAT MY ACTIONS ARE & DO YOUR OWN WORK ! => WHO SAYS YOU CAN'T FIND BETTER TRADE-IDEAS ? => Narrative + Money Flow + Technicals = Valuation Skyrocket"
```

```
    ws.cells(r, 1).Font.Color = RGB(255, 192, 0)
```

```
    ws.cells(r, 1).Font.Italic = True
```

```
    ws.cells(r, 1).HorizontalAlignment = xlLeft
```

```
    r = r + 2
```

```
    On Error Resume Next
```

```
    ws.Range(ws.cells(r, 1), ws.cells(r, 16)).UnMerge
```

```
    ws.Range(ws.cells(r, 1), ws.cells(r, 16)).Merge
```

```
    On Error GoTo 0
```

```
    ws.cells(r, 1).Value = "Expected to DO 1 => USA = Some tickers in hand can be replace => "
```

```
    ws.cells(r, 1).Font.Color = RGB(255, 255, 255)
```

```
    ws.cells(r, 1).HorizontalAlignment = xlLeft
```

```
    ws.cells(r, 1).Font.Bold = True
```

```
    r = r + 1
```

```
    On Error Resume Next
```

```

ws.Range(ws.cells(r, 1), ws.cells(r, 16)).UnMerge
ws.Range(ws.cells(r, 1), ws.cells(r, 16)).Merge
On Error GoTo 0
ws.cells(r, 1).Value = "Expected to DO 2 => TWII = Some tickers in hand can be replace => MLCC:2327國巨/3026禾伸堂/3090/6173 , ProbeCard: 2360/6223/6515 , 先進封裝: 3711/3481/6239 , "
ws.cells(r, 1).Font.Color = RGB(255, 255, 255)
ws.cells(r, 1).HorizontalAlignment = xlLeft
ws.cells(r, 1).Font.Bold = True

```

```

r = r + 1

```

```

On Error Resume Next

```

```

ws.Range(ws.cells(r, 1), ws.cells(r, 16)).UnMerge

```

```

ws.Range(ws.cells(r, 1), ws.cells(r, 16)).Merge

```

```

On Error GoTo 0

```

```

ws.cells(r, 1).Value = "Expected to DO 3 => ready to be assign"

```

```

ws.cells(r, 1).Font.Color = RGB(255, 255, 255)

```

```

ws.cells(r, 1).HorizontalAlignment = xlLeft

```

```

ws.cells(r, 1).Font.Bold = True

```

```

r = r + 2

```

```

On Error Resume Next

```

```

ws.Range(ws.cells(r, 1), ws.cells(r, 16)).UnMerge

```

```

ws.Range(ws.cells(r, 1), ws.cells(r, 16)).Merge

```

```

On Error GoTo 0

```

```

ws.cells(r, 1).Value = "Note => TWII focus on PCB/TGV/MLCC/Probe , USA focus on CHIP/SSD+HBM/"

```

```

ws.cells(r, 1).Font.Color = RGB(255, 255, 255)

```

```

ws.cells(r, 1).HorizontalAlignment = xlLeft

```

```

ws.cells(r, 1).Font.Bold = True

```

```

r = r + 1

```

```

On Error Resume Next

```

```

ws.Range(ws.cells(r, 1), ws.cells(r, 16)).UnMerge

```

```

ws.Range(ws.cells(r, 1), ws.cells(r, 16)).Merge

```

```

On Error GoTo 0

```

```

ws.cells(r, 1).Value = "Note => 3026 處置到 0702 , 7828 處置到 0702"

```

```

ws.cells(r, 1).Font.Color = RGB(255, 255, 255)

```

```

ws.cells(r, 1).HorizontalAlignment = xlLeft

```

```

ws.cells(r, 1).Font.Bold = True

```

```

End Sub

```

```

Private Sub CalcPositions(positions As Object, exRate As Double, _

```

```

    ByRef posData() As Variant, _

```

```

    ByRef totalMkt As Double, _

```

```

    ByRef totalCost As Double, _

```

```

    ByRef totalUnrl As Double, _

```

```

    ByRef posCount As Long)

```

```

totalMkt = 0: totalCost = 0: totalUnrl = 0: posCount = 0

```

```

Dim tkv As Variant

```

```

For Each tkv In positions.keys

```

```

    Dim d0 As Variant: d0 = positions(tkv)

```

```

    If d0(0) > 0 Then

```

```

        Dim pp0 As Long: pp0 = InStr(CStr(tkv), "I")

```

```

        Dim t0 As String: t0 = IIf(pp0 > 0, Left(CStr(tkv), pp0 - 1), CStr(tkv))

```

```

        Dim px0 As Double: px0 = GetStockPrice(t0)

```

```

        If px0 > 0 Then

```

```

            If GetCurrencyType(t0) = "USD" Then

```

```

                totalMkt = totalMkt + px0 * d0(0) * exRate

```

```

            Else

```

```

                totalMkt = totalMkt + px0 * d0(0)

```

```

            End If

```

```

        End If

```

```

    End If

```

```

Next tkv

```

```

Dim validCount As Long: validCount = 0

```

```

For Each tkv In positions.keys

```

```

    If positions(tkv)(0) > 0 Then validCount = validCount + 1

```

```

Next tkv

```

```
If validCount = 0 Then Exit Sub
```

```
' -----
```

```
Dim sortedKeys() As String
ReDim sortedKeys(0 To validCount - 1)
Dim ki As Long: ki = 0
```

```
Dim brokerOrder(0 To 1) As String
brokerOrder(0) = "Default"
brokerOrder(1) = ChrW(&H570B) & ChrW(&H6CF0) & ChrW(&H4E16) & ChrW(&H83EF) ' Cathay United
```

```
Dim bi As Long
For bi = 0 To 1
    Dim bName As String: bName = brokerOrder(bi)
    Dim twStart As Long: twStart = ki
```

```
' -- broker -- x --
```

```
For Each tkv In positions.keys
    If positions(tkv)(0) > 0 Then
        Dim ppl As Long: ppl = InStr(CStr(tkv), "I")
        If ppl > 0 Then
            Dim t1 As String: t1 = Left(CStr(tkv), ppl - 1)
            Dim b1 As String: b1 = Mid(CStr(tkv), ppl + 1)
            If b1 = bName And GetCurrencyType(t1) = "TWD" Then
                sortedKeys(ki) = CStr(tkv): ki = ki + 1
            End If
        End If
    End If
End If
Next tkv
```

```
' -----
```

```
Dim sa As Long, sb As Long, tmp1 As String
For sa = twStart To ki - 2
    For sb = sa + 1 To ki - 1
        Dim ppA As Long: ppA = InStr(sortedKeys(sa), "I")
        Dim ppB As Long: ppB = InStr(sortedKeys(sb), "I")
        Dim cA As String: cA = Left(sortedKeys(sa), ppA - 1)
        Dim cB As String: cB = Left(sortedKeys(sb), ppB - 1)
        If IsNumeric(cA) And IsNumeric(cB) Then
            If val(cA) > val(cB) Then
                tmp1 = sortedKeys(sa): sortedKeys(sa) = sortedKeys(sb): sortedKeys(sb) = tmp1
            End If
        End If
    Next sb
Next sa
```

```
Dim usStart As Long: usStart = ki
```

```
' -- broker --
```

```
For Each tkv In positions.keys
    If positions(tkv)(0) > 0 Then
        Dim pp2 As Long: pp2 = InStr(CStr(tkv), "I")
        If pp2 > 0 Then
            Dim t2 As String: t2 = Left(CStr(tkv), pp2 - 1)
            Dim b2 As String: b2 = Mid(CStr(tkv), pp2 + 1)
            If b2 = bName And GetCurrencyType(t2) = "USD" Then
                sortedKeys(ki) = CStr(tkv): ki = ki + 1
            End If
        End If
    End If
End If
Next tkv
```

```
' -----
```

```
Dim tmp2 As String
For sa = usStart To ki - 2
    For sb = sa + 1 To ki - 1
        Dim ppA2 As Long: ppA2 = InStr(sortedKeys(sa), "I")
        Dim ppB2 As Long: ppB2 = InStr(sortedKeys(sb), "I")
        Dim cA2 As String: cA2 = Left(sortedKeys(sa), ppA2 - 1)
        Dim cB2 As String: cB2 = Left(sortedKeys(sb), ppB2 - 1)
        If cA2 > cB2 Then
```

```

        tmp2 = sortedKeys(sa): sortedKeys(sa) = sortedKeys(sb): sortedKeys(sb) = tmp2
    End If
Next sb
Next sa
Next bi

' -----
Dim addedKeys As Object: Set addedKeys = CreateObject("Scripting.Dictionary")
Dim si2 As Long
For si2 = 0 To ki - 1: addedKeys(sortedKeys(si2)) = 1: Next si2
For Each tkv In positions.keys
    If positions(tkv)(0) > 0 Then
        If Not addedKeys.Exists(CStr(tkv)) Then
            sortedKeys(ki) = CStr(tkv): ki = ki + 1
        End If
    End If
Next tkv

' -----
ReDim posData(1 To validCount, 1 To 16)
posCount = 0

Dim si As Long
For si = 0 To validCount - 1
    Dim compositeKey As String: compositeKey = sortedKeys(si)
    Dim pipePos As Long: pipePos = InStr(compositeKey, "|")
    Dim tickerStr As String: tickerStr = Left(compositeKey, pipePos - 1)
    Dim brokerPart As String: brokerPart = Mid(compositeKey, pipePos + 1)

    Dim d As Variant: d = positions(compositeKey)
    Dim netQty As Double: netQty = d(0)
    If netQty <= 0 Then GoTo NextSi

    posCount = posCount + 1
    Dim p As Long: p = posCount

    Dim totCost As Double: totCost = d(2)
    Dim entryDt As Date
    If d(5) > 0 Then entryDt = CDate(d(5)) Else entryDt = Date

    Dim AvgCost As Double
    If netQty > 0 Then AvgCost = totCost / netQty

    Dim curType As String: curType = GetCurrencyType(tickerStr)
    Dim livePrice As Double: livePrice = GetStockPrice(tickerStr)

    Dim mktVal As Double, unrlPnl As Double
    If livePrice > 0 Then
        mktVal = livePrice * netQty
        unrlPnl = mktVal - totCost
    Else
        mktVal = totCost: unrlPnl = 0
    End If

    Dim mktTWD As Double, unrlTWD As Double
    If curType = "USD" Then
        mktTWD = mktVal * exRate: unrlTWD = unrlPnl * exRate
    Else
        mktTWD = mktVal: unrlTWD = unrlPnl
    End If

    totalCost = totalCost + IIf(curType = "USD", totCost * exRate, totCost)
    totalUnrl = totalUnrl + unrlTWD

    Dim chgPct As Double
    If AvgCost > 0 And livePrice > 0 Then chgPct = (livePrice - AvgCost) / AvgCost

    posData(p, 1) = tickerStr
    posData(p, 2) = tickerStr
    posData(p, 3) = GetCompanyName(tickerStr)
    posData(p, 4) = entryDt

```

```

posData(p, 5) = Date - entryDt + 1
posData(p, 6) = brokerPart
posData(p, 7) = CStr(d(6))
posData(p, 8) = mktTWD
posData(p, 9) = netQty
posData(p, 10) = AvgCost
posData(p, 11) = livePrice
posData(p, 12) = chgPct
posData(p, 13) = unr1TWD
posData(p, 14) = d(9)
posData(p, 15) = d(9)
posData(p, 16) = d(7)

```

```
NextSi:
```

```
Next si
```

```
End Sub
```

```
Private Function BuildPositions(wsTr As Worksheet) As Object
```

```
Dim dict As Object
```

```
Set dict = CreateObject("Scripting.Dictionary")
```

```
Dim lastRow As Long
```

```
lastRow = wsTr.Cells(wsTr.Rows.Count, "A").End(xlUp).row
```

```
If lastRow < 2 Then Set BuildPositions = dict: Exit Function
```

```
Dim data As Variant
```

```
data = wsTr.Range("B2:N" & lastRow).Value ' -- M -- N
```

```
Dim i As Long
```

```
For i = 1 To UBound(data, 1)
```

```
Dim ticker As String: ticker = CStr(data(i, 2))
```

```
Dim action As String: action = CStr(data(i, 3))
```

```
Dim shares As Double: shares = val(data(i, 4))
```

```
Dim netAmt As Double: netAmt = val(data(i, 8))
```

```
Dim Sector As String: Sector = CStr(data(i, 9))
```

```
Dim pTgt As Double: pTgt = val(data(i, 10))
```

```
Dim strat As String: strat = CStr(data(i, 11))
```

```
Dim Beta As Double: Beta = val(data(i, 12))
```

```
Dim broker As String
```

```
On Error Resume Next
```

```
broker = Trim(CStr(data(i, 13)))
```

```
On Error GoTo 0
```

```
If broker = "" Then broker = "Default"
```

```
If ticker <> "" Then
```

```
Dim posKey As String: posKey = ticker & "I" & broker
```

```
If Not dict.Exists(posKey) Then
```

```
dict.Add posKey, Array(0#, 0#, 0#, 0#, 0#, 0#, "", 0#, "", 0#, broker)
```

```
End If
```

```
Dim d As Variant: d = dict(posKey)
```

```
If pTgt > 0 Then d(7) = pTgt
```

```
If strat <> "" Then d(8) = strat
```

```
If Beta <> 0 Then d(9) = Beta
```

```
If Sector <> "" Then d(6) = Sector
```

```
d(10) = broker
```

```
Select Case UCase(action)
```

```
Case "BUY"
```

```
d(0) = d(0) + shares
```

```
d(2) = d(2) + netAmt ' -- -- 榑 -- G -- Shares
```

```
If d(5) = 0 Then d(5) = CLng(CDate(data(i, 1)))
```

```
Case "SELL"
```

```
Dim avg As Double
```

```
If d(0) > 0 Then avg = d(2) / d(0) Else avg = 0
```

```
d(0) = d(0) - shares
```

```
d(2) = d(2) - avg * shares
```

```
If d(0) <= 0.0001 Then d(0) = 0: d(2) = 0: d(5) = 0
```

```
Case "ADJUSTCOST"
```

```
d(2) = d(2) + netAmt
```

```
End Select
```

```
dict(posKey) = d
```

```
End If
```

```
Next i
```

```
Set BuildPositions = dict
End Function
```

```
' Portfolio Beta
```

```
Private Function CalcPortBeta(posData() As Variant) As Double
```

```
CalcPortBeta = 0
```

```
If Not IsArray(posData) Then Exit Function
```

```
On Error Resume Next
```

```
Dim n As Long: n = UBound(posData, 1)
```

```
If Err.Number <> 0 Or n < 1 Then Exit Function
```

```
On Error GoTo 0
```

```
Dim totalMkt As Double, totalBeta As Double
```

```
Dim i As Long
```

```
For i = 1 To n
```

```
totalMkt = totalMkt + posData(i, 8)
```

```
totalBeta = totalBeta + posData(i, 8) * posData(i, 14)
```

```
Next i
```

```
If totalMkt > 0 Then CalcPortBeta = totalBeta / totalMkt
```

```
End Function
```

```
Private Function FindNearestPrice(ByRef dict As Object, ByVal targetDate As Date) As Double
```

```
Dim checkDate As Date
```

```
checkDate = targetDate
```

```
Dim daysBack As Integer
```

```
For daysBack = 1 To 10
```

```
checkDate = checkDate - 1
```

```
If dict.Exists(checkDate) Then
```

```
FindNearestPrice = dict(checkDate)
```

```
Exit Function
```

```
End If
```

```
Next daysBack
```

```
FindNearestPrice = 0
```

```
End Function
```

```
' Helpers
```

```
Private Function PnLColor(v As Double) As Long
```

```
If v > 0 Then
```

```
PnLColor = RGB(255, 80, 80)
```

```
ElseIf v < 0 Then
```

```
PnLColor = RGB(0, 210, 100)
```

```
Else
```

```
PnLColor = RGB(200, 200, 200)
```

```
End If
```

```
End Function
```

```
Private Function GetExRate(ws As Worksheet) As Double
```

```
On Error Resume Next
```

```
GetExRate = val(ws.Range("B2").Value)
```

```
On Error GoTo 0
```

```
If GetExRate <= 0 Then GetExRate = 31.6
```

```
End Function
```

```
Private Function GetInceptionDate(ws As Worksheet) As Date
```

```
On Error Resume Next
```

```
GetInceptionDate = CDate(ws.Range("InceptionDate").Value)
```

```
On Error GoTo 0
```

```
If GetInceptionDate <= DateSerial(2000, 1, 1) Then GetInceptionDate = DateSerial(2025, 12, 21)
```

```
End Function
```

```
Private Function GetStartingCapital(ws As Worksheet) As Double
```

```
On Error Resume Next
```

```
GetStartingCapital = val(ws.Range("StartingCapital").Value)
```

```

    On Error GoTo 0
    If GetStartingCapital <= 0 Then GetStartingCapital = 550000
End Function

```

```

' One-time config setup

```

```

Sub SetupPortfolioConfig()
    Dim ws As Worksheet
    Set ws = ThisWorkbook.Sheets(SH_PORT)

    Dim inDate As String
    inDate = InputBox("Inception Date (yyyy/m/d):", "Config", "2025/12/27")
    If Not IsDate(inDate) Then MsgBox "Invalid date format.": Exit Sub

    Dim startCap As String
    startCap = InputBox("Starting Capital (TWD, e.g. 70000):", "Config", "70000")
    If Not IsNumeric(startCap) Then MsgBox "Invalid number.": Exit Sub

    On Error Resume Next
    ThisWorkbook.names("InceptionDate").Delete
    ThisWorkbook.names("StartingCapital").Delete
    On Error GoTo 0

    ws.Range("S1").Value = CDate(inDate)
    ws.Range("S2").Value = CDBl(startCap)
    ThisWorkbook.names.Add "InceptionDate", ws.Range("S1")
    ThisWorkbook.names.Add "StartingCapital", ws.Range("S2")
    ws.Columns("S").Hidden = True

    MsgBox "Config saved!" & vbCr & "Inception: " & inDate & vbCr & "Capital: " & startCap, vbInformation
End Sub

```

```

' Auto-refresh

```

```

Sub StartAutoRefresh()
    g_AutoOn = True
    Call RebuildPortfolioDashboard
    g_NextTime = Now + TimeSerial(0, 1, 0)
    Application.OnTime g_NextTime, "AutoRefreshLoop"
    MsgBox "Auto-refresh started (every 60 sec)", vbInformation
End Sub

```

```

Sub AutoRefreshLoop()
    If Not g_AutoOn Then Exit Sub
    Call RebuildPortfolioDashboard
    g_NextTime = Now + TimeSerial(0, 1, 0)
    Application.OnTime g_NextTime, "AutoRefreshLoop"
End Sub

```

```

Sub StopAutoRefresh()
    g_AutoOn = False
    On Error Resume Next
    Application.OnTime g_NextTime, "AutoRefreshLoop", , False
    On Error GoTo 0
    Application.StatusBar = False
    MsgBox "Auto-refresh stopped.", vbInformation
End Sub

```

```

Private Sub WriteKV(ws As Worksheet, r As Long, label As String, _
    val As Double, fmt As String, colorPnL As Boolean)
    ws.cells(r, 1).Value = label
    ws.cells(r, 1).Font.Color = RGB(150, 150, 150)
    ws.cells(r, 2).Value = val
    ws.cells(r, 2).NumberFormat = fmt
    ws.cells(r, 2).Font.Bold = True
    If colorPnL Then
        ws.cells(r, 2).Font.Color = PnLColor(val)
    Else
        ws.cells(r, 2).Font.Color = RGB(221, 221, 221)
    End If

```

End Sub

```

Sub BuildHoldingsCorrelation()
    Application.ScreenUpdating = False

    Dim wsTr As Worksheet
    Set wsTr = ThisWorkbook.Sheets("Transactions")

    Dim positions As Object
    Set positions = BuildPositions(wsTr)

    ' -----
    Dim uniqTickers As Object: Set uniqTickers = CreateObject("Scripting.Dictionary")
    Dim tkv As Variant
    For Each tkv In positions.keys
        If positions(tkv)(0) > 0 Then
            Dim pkStr As String: pkStr = CStr(tkv)
            Dim barPos As Long: barPos = InStr(pkStr, "|")
            Dim pureT As String
            If barPos > 0 Then pureT = Left(pkStr, barPos - 1) Else pureT = pkStr
            pureT = UCase(Trim(pureT))
            If pureT <> "" And Not uniqTickers.Exists(pureT) Then
                uniqTickers.Add pureT, 1
            End If
        End If
    Next tkv

    Dim tickerCount As Long: tickerCount = uniqTickers.count
    If tickerCount < 2 Then
        MsgBox "Need at least 2 holdings.", vbInformation
        Application.ScreenUpdating = True: Exit Sub
    End If

    ' -----
    Dim rawTickers() As String
    ReDim rawTickers(0 To tickerCount - 1)
    Dim ki As Long: ki = 0
    For Each tkv In uniqTickers.keys
        rawTickers(ki) = CStr(tkv): ki = ki + 1
    Next tkv

    ' -----
    ' -----
    Dim tickers() As String
    ReDim tickers(0 To tickerCount - 1)
    Dim sortIdx() As Long
    ReDim sortIdx(0 To tickerCount - 1)

    ' Step 1: -- x -- index -- J
    Dim twCount As Long: twCount = 0
    Dim i As Long
    For i = 0 To tickerCount - 1
        If GetCurrencyType(rawTickers(i)) = "TWD" Then
            sortIdx(twCount) = i
            twCount = twCount + 1
        End If
    Next i

    ' Step 2: fill US tickers next
    Dim usIdx As Long: usIdx = twCount
    For i = 0 To tickerCount - 1
        If GetCurrencyType(rawTickers(i)) = "USD" Then
            sortIdx(usIdx) = i
            usIdx = usIdx + 1
        End If
    Next i

    ' Step 3: sort TW tickers numerically
    Dim a As Long, b As Long, tmpL As Long
    For a = 0 To twCount - 2

```

```

    For b = a + 1 To twCount - 1
        Dim codeA As String: codeA = Replace(Replace(UCase(rawTickers(sortIdx(a))), ".TWO", ""), ".TW", "")
        Dim codeB As String: codeB = Replace(Replace(UCase(rawTickers(sortIdx(b))), ".TWO", ""), ".TW", "")
        If IsNumeric(codeA) And IsNumeric(codeB) Then
            If val(codeA) > val(codeB) Then
                tmpL = sortIdx(a): sortIdx(a) = sortIdx(b): sortIdx(b) = tmpL
            End If
        End If
    Next b
Next a

' Step 4: sort US tickers alphabetically
For a = twCount To tickerCount - 2
    For b = a + 1 To tickerCount - 1
        If rawTickers(sortIdx(a)) > rawTickers(sortIdx(b)) Then
            tmpL = sortIdx(a): sortIdx(a) = sortIdx(b): sortIdx(b) = tmpL
        End If
    Next b
Next a

' Step 5: apply sorted order
For i = 0 To tickerCount - 1
    tickers(i) = rawTickers(sortIdx(i))
Next i

' -----
Dim priceData() As Object
ReDim priceData(0 To tickerCount - 1)
For i = 0 To tickerCount - 1
    Application.StatusBar = "Fetching " & tickers(i) & " (" & (i + 1) & "/" & tickerCount & ")..."
    Set priceData(i) = GetHistoryPrices(tickers(i))
Next i

' -----
Dim commonDates() As Date
Dim usedCount As Long
Call CorrGetCommonDates(priceData, tickerCount, commonDates, usedCount)

If usedCount < 4 Then
    Dim diagMsg As String
    diagMsg = "Insufficient common trading days (" & usedCount & " day(s))." & vbCrLf & vbCrLf & _
        "History row count per ticker:" & vbCrLf
    Dim dIdx As Long
    For dIdx = 0 To tickerCount - 1
        Dim dcnt As Long: dcnt = 0
        If Not priceData(dIdx) Is Nothing Then dcnt = priceData(dIdx).count
        diagMsg = diagMsg & " " & tickers(dIdx) & " : " & dcnt & " rows"
        If dcnt = 0 Then diagMsg = diagMsg & " <-- API returned nothing"
        diagMsg = diagMsg & vbCrLf
    Next dIdx
    diagMsg = diagMsg & vbCrLf & "Tip: re-import Attach module, save / close / reopen workbook, retry."
    MsgBox diagMsg, vbInformation, "Holdings Correlation"
    Application.ScreenUpdating = True: Exit Sub
End If

' -----
' -----
Dim retCount As Long: retCount = usedCount - 1
Dim returns() As Double
ReDim returns(0 To tickerCount - 1, 0 To retCount - 1)

Dim j As Long
For i = 0 To tickerCount - 1
    For j = 0 To retCount - 1
        Dim px1 As Double: px1 = 0
        Dim px2 As Double: px2 = 0
        On Error Resume Next
        px1 = priceData(i)(commonDates(j))
        px2 = priceData(i)(commonDates(j + 1))
        On Error GoTo 0
        If px1 > 0 And px2 > 0 Then returns(i, j) = Log(px2 / px1)
    
```

```

        Next j
    Next i

' -----
Dim corrMatrix() As Double
ReDim corrMatrix(0 To tickerCount - 1, 0 To tickerCount - 1)
Dim r As Long, c As Long
For r = 0 To tickerCount - 1
    For c = 0 To tickerCount - 1
        corrMatrix(r, c) = CorrPearson(returns, r, c, retCount)
    Next c
Next r

' -----
Call HoldingsCorrRenderSheet(tickers, tickerCount, corrMatrix, usedCount, commonDates)

Application.ScreenUpdating = True
Application.StatusBar = "HoldingsCorr updated: " & Format(Now, "hh:mm:ss")
MsgBox "Holdings correlation matrix updated.", vbInformation
End Sub

' -----
Private Sub HoldingsCorrRenderSheet(tickers() As String, tickerCount As Long, _
    corrMatrix() As Double, usedCount As Long, _
    commonDates() As Date)

Dim wsC As Worksheet
On Error Resume Next: Set wsC = ThisWorkbook.Sheets("HoldingsCorr"): On Error GoTo 0
If wsC Is Nothing Then
    Set wsC = ThisWorkbook.Sheets.Add(After:=ThisWorkbook.Sheets(ThisWorkbook.Sheets.count))
    wsC.Name = "HoldingsCorr"
End If

wsC.Cells.Clear
With wsC.Cells
    .Interior.Color = RGB(0, 0, 0)
    .Font.Color = RGB(200, 200, 200)
    .Font.Name = "Consolas"
    .Font.Size = 9
End With
wsC.Activate
ActiveWindow.DisplayGridlines = False

' -----
Dim retCount As Long: retCount = usedCount - 1
With wsC.Cells(1, 1)
    .Value = "HOLDINGS CORRELATION | " & retCount & "-DAY | " & _
        Format(commonDates(0), "m/d/yy") & " ~ " & _
        Format(commonDates(usedCount - 1), "m/d/yy") & _
        " | " & Format(Now, "hh:mm:ss")
    .Font.Color = RGB(255, 192, 0)
    .Font.Bold = True
    .Font.Size = 11
End With
With wsC.Range(wsC.Cells(2, 1), wsC.Cells(2, tickerCount + 2))
    .Interior.Color = RGB(255, 192, 0)
    .RowHeight = 3
End With

' -----
Dim HDR As Long: HDR = 4
wsC.Columns(1).ColumnWidth = 12
Dim i As Long
For i = 0 To tickerCount - 1
    wsC.Columns(i + 2).ColumnWidth = 7
Next i

' -----
With wsC.Cells(HDR, 1)
    .Value = "Row / Col"
    .Font.Color = RGB(100, 100, 100)
    .Interior.Color = RGB(10, 10, 10)

```

```

        .HorizontalAlignment = xlCenter
End With

```

```

For i = 0 To tickerCount - 1
    With wsC.cells(HDR, i + 2)
        .Value = tickers(i)
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
        .Font.Size = 8
        .Interior.Color = RGB(10, 10, 10)
        .HorizontalAlignment = xlCenter
        .VerticalAlignment = xlBottom
    End With
    wsC.Rows(HDR).RowHeight = 20
Next i

```

```

' -----
Dim j As Long

```

```

For i = 0 To tickerCount - 1
    Dim rn As Long: rn = HDR + 1 + i
    wsC.Rows(rn).RowHeight = 18

```

```

' -----
With wsC.cells(rn, 1)
    .Value = tickers(i)
    .Font.Color = RGB(255, 192, 0)
    .Font.Bold = True
    .Font.Size = 9
    .HorizontalAlignment = xlCenter
    .Interior.Color = RGB(10, 10, 10)
End With

```

```

For j = 0 To tickerCount - 1
    Dim cv As Double: cv = corrMatrix(i, j)
    Dim bgC As Long, fgC As Long

```

```

' -----
If i = j Then

```

```

    bgC = RGB(40, 28, 0)
    fgC = RGB(255, 192, 0)

```

```

ElseIf cv >= 0.2 Then

```

```

    If cv >= 0.7 Then
        bgC = RGB(100, 0, 0): fgC = RGB(255, 130, 130)
    ElseIf cv >= 0.4 Then

```

```

        bgC = RGB(70, 0, 0): fgC = RGB(230, 100, 100)
    Else

```

```

        bgC = RGB(45, 5, 5): fgC = RGB(210, 80, 80)
    End If

```

```

ElseIf cv <= -0.2 Then

```

```

    If cv <= -0.7 Then
        bgC = RGB(0, 70, 20): fgC = RGB(80, 255, 140)
    ElseIf cv <= -0.4 Then

```

```

        bgC = RGB(0, 50, 15): fgC = RGB(60, 220, 110)
    Else

```

```

        bgC = RGB(0, 35, 10): fgC = RGB(40, 200, 90)
    End If

```

```

Else

```

```

    bgC = RGB(18, 18, 18): fgC = RGB(150, 150, 150)

```

```

End If

```

```

With wsC.cells(rn, j + 2)
    .Value = Round(cv, 2)
    .NumberFormat = "0.00"
    .HorizontalAlignment = xlCenter
    .Font.Bold = True
    .Font.Size = 8
    .Interior.Color = bgC

```

```

        .Font.Color = fgC
    ' -----
    With .Borders
        .LineStyle = xlContinuous
        .Color = RGB(0, 0, 0)
        .Weight = xlThin
    End With
End With
Next j
Next i

' -----
Dim LR As Long: LR = HDR + tickerCount + 2
wsC.cells(LR, 1).Value = "LEGEND"
wsC.cells(LR, 1).Font.Color = RGB(255, 192, 0)
wsC.cells(LR, 1).Font.Bold = True

Dim leg As Variant
leg = Array( _
    Array(">= 0.70 High +corr", RGB(100, 0, 0), RGB(255, 130, 130)), _
    Array("0.40~0.69 Mid +corr", RGB(70, 0, 0), RGB(230, 100, 100)), _
    Array("0.20~0.39 Low +corr", RGB(45, 5, 5), RGB(210, 80, 80)), _
    Array("-0.19~0.19 Neutral", RGB(18, 18, 18), RGB(150, 150, 150)), _
    Array("-0.20~-0.39 Low -corr", RGB(0, 35, 10), RGB(40, 200, 90)), _
    Array("-0.40~-0.69 Mid -corr", RGB(0, 50, 15), RGB(60, 220, 110)), _
    Array("<= -0.70 High -corr", RGB(0, 70, 20), RGB(80, 255, 140)))

Dim li As Long
For li = 0 To UBound(leg)
    With wsC.cells(LR + 1 + li, 1)
        .Value = leg(li)(0)
        .Interior.Color = leg(li)(1)
        .Font.Color = leg(li)(2)
        .Font.Bold = True
        .Font.Size = 9
    End With
    wsC.Rows(LR + 1 + li).RowHeight = 16
Next li

' -----

wsC.Activate
End Sub

' -----
Private Sub CorrGetCommonDates(priceData() As Object, tickerCount As Long, _
    ByRef commonDates() As Date, ByRef usedCount As Long)
    Dim allDates As Object: Set allDates = CreateObject("Scripting.Dictionary")
    Dim dkey As Variant

    ' -----
    For Each dkey In priceData(0).keys
        allDates(dkey) = 1
    Next dkey

    ' -----
    Dim i As Long
    For i = 1 To tickerCount - 1
        Dim toRemove() As Variant
        Dim removeCount As Long: removeCount = 0
        ReDim toRemove(0 To allDates.count)
        For Each dkey In allDates.keys
            If Not priceData(i).Exists(dkey) Then
                toRemove(removeCount) = dkey
                removeCount = removeCount + 1
            End If
        Next dkey
        Dim k As Long
        For k = 0 To removeCount - 1
            allDates.Remove toRemove(k)
        Next k
    Next i

```

```

Dim dateCount As Long: dateCount = allDates.count
If dateCount = 0 Then usedCount = 0: Exit Sub

' -----
Dim sortedDates() As Date
ReDim sortedDates(0 To dateCount - 1)
Dim di As Long: di = 0
For Each dkey In allDates.keys
    sortedDates(di) = CDate(dkey): di = di + 1
Next dkey

Dim a As Long, b As Long, tmpD As Date
For a = 0 To dateCount - 2
    For b = a + 1 To dateCount - 1
        If sortedDates(a) > sortedDates(b) Then
            tmpD = sortedDates(a): sortedDates(a) = sortedDates(b): sortedDates(b) = tmpD
        End If
    Next b
Next a

' -----
Dim wantPx As Long: wantPx = 15
Dim startIdx As Long: startIdx = IIf(dateCount > wantPx, dateCount - wantPx, 0)
usedCount = dateCount - startIdx

ReDim commonDates(0 To usedCount - 1)
For i = 0 To usedCount - 1
    commonDates(i) = sortedDates(startIdx + i)
Next i
End Sub

' -----
Private Function CorrPearson(returns() As Double, r1 As Long, r2 As Long, n As Long) As Double
    If n < 2 Then CorrPearson = 0: Exit Function

    Dim sumX As Double, sumY As Double, sumXY As Double
    Dim sumX2 As Double, sumY2 As Double, j As Long

    For j = 0 To n - 1
        Dim x As Double: x = returns(r1, j)
        Dim y As Double: y = returns(r2, j)
        sumX = sumX + x: sumY = sumY + y
        sumXY = sumXY + x * y
        sumX2 = sumX2 + x * x
        sumY2 = sumY2 + y * y
    Next j

    Dim denom As Double
    denom = Sqr((n * sumX2 - sumX ^ 2) * (n * sumY2 - sumY ^ 2))
    If denom = 0 Then CorrPearson = 0 Else CorrPearson = (n * sumXY - sumX * sumY) / denom
End Function

' -----
Private Sub CorrRenderSheet(tickers() As String, tickerCount As Long, _
    corrMatrix() As Double, usedCount As Long, _
    commonDates() As Date)

    Dim wsC As Worksheet
    On Error Resume Next: Set wsC = ThisWorkbook.Sheets("HoldingsCorr"): On Error GoTo 0
    If wsC Is Nothing Then
        Set wsC = ThisWorkbook.Sheets.Add(After:=ThisWorkbook.Sheets(ThisWorkbook.Sheets.count))
        wsC.Name = "HoldingsCorr"
    End If

    wsC.cells.Clear
    With wsC.cells
        .Interior.Color = RGB(0, 0, 0)
        .Font.Color = RGB(221, 221, 221)
        .Font.Name = "Consolas"
        .Font.Size = 10
    End With

```

```

wsC.Activate
ActiveWindow.DisplayGridlines = False

' -----
Dim retCount As Long: retCount = usedCount - 1
With wsC.cells(1, 1)
    .Value = "CORRELATION MATRIX | " & retCount & "-DAY RETURNS | " & _
        Format(commonDates(0), "m/d/yy") & " ~ " & _
        Format(commonDates(usedCount - 1), "m/d/yy") & _
        " | Updated: " & Format(Now, "hh:mm:ss")
    .Font.Color = RGB(255, 192, 0)
    .Font.Bold = True
    .Font.Size = 12
End With
With wsC.Range(wsC.cells(2, 1), wsC.cells(2, tickerCount + 3))
    .Interior.Color = RGB(255, 192, 0)
    .RowHeight = 3
End With

' -----
Dim HDR As Long: HDR = 4
With wsC.cells(HDR, 1)
    .Value = "ROW / COL"
    .Font.Color = RGB(150, 150, 150)
    .Interior.Color = RGB(10, 10, 10)
    .HorizontalAlignment = xlCenter
End With

Dim i As Long, j As Long
For i = 0 To tickerCount - 1
    With wsC.cells(HDR, i + 2)
        .Value = tickers(i)
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
        .Interior.Color = RGB(10, 10, 10)
        .HorizontalAlignment = xlCenter
    End With
Next i

' -----
For i = 0 To tickerCount - 1
    Dim rn As Long: rn = HDR + 1 + i
    wsC.Rows(rn).RowHeight = 20

    With wsC.cells(rn, 1)
        .Value = tickers(i)
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
        .HorizontalAlignment = xlCenter
        .Interior.Color = RGB(10, 10, 10)
    End With

    For j = 0 To tickerCount - 1
        Dim cv As Double: cv = corrMatrix(i, j)
        Dim bgC As Long, fgC As Long

        If i = j Then
            ' -- 鵝?-- u -- G -- 蓄v -- P -- 蓄v = 1.00
            bgC = RGB(40, 30, 0): fgC = RGB(255, 192, 0)
        ElseIf cv >= 0.7 Then
            ' -- 蚰 --
            bgC = RGB(90, 10, 10): fgC = RGB(255, 100, 100)
        ElseIf cv >= 0.4 Then
            ' -- 蚰 --
            bgC = RGB(55, 20, 0): fgC = RGB(230, 140, 70)
        ElseIf cv >= 0.1 Then
            ' -- C -- 蚰 --
            bgC = RGB(22, 22, 22): fgC = RGB(200, 200, 160)
        ElseIf cv >= -0.1 Then
            ' -----
            bgC = RGB(15, 15, 15): fgC = RGB(170, 170, 170)
        ElseIf cv >= -0.4 Then
            ' -- C -- 蚰t --
            bgC = RGB(0, 22, 28): fgC = RGB(80, 200, 200)
        Else
            ' -- 蚰t --
            bgC = RGB(0, 45, 20): fgC = RGB(0, 210, 100)
        End If
    End If
End For

```

```

        With wsC.cells(rn, j + 2)
            .Value = Round(cv, 2)
            .NumberFormat = "0.00"
            .HorizontalAlignment = xlCenter
            .Font.Bold = True
            .Interior.Color = bgC
            .Font.Color = fgC
        End With
    Next j
Next i

' -----
Dim LR As Long: LR = HDR + tickerCount + 3
wsC.cells(LR, 1).Value = "COLOR LEGEND"
wsC.cells(LR, 1).Font.Color = RGB(255, 192, 0)
wsC.cells(LR, 1).Font.Bold = True

Dim leg As Variant
leg = Array( _
    Array("? 0.70    HIGH POSITIVE", RGB(90, 10, 10), RGB(255, 100, 100)), _
    Array("0.40 ~ 0.69    MODERATE POSITIVE", RGB(55, 20, 0), RGB(230, 140, 70)), _
    Array("0.10 ~ 0.39    LOW POSITIVE", RGB(22, 22, 22), RGB(200, 200, 160)), _
    Array("-0.09 ~ 0.09    NEUTRAL", RGB(15, 15, 15), RGB(170, 170, 170)), _
    Array("-0.10 ~ -0.39    LOW NEGATIVE", RGB(0, 22, 28), RGB(80, 200, 200)), _
    Array("? -0.40    HIGH NEGATIVE", RGB(0, 45, 20), RGB(0, 210, 100)) _
)

Dim li As Long
For li = 0 To UBound(leg)
    With wsC.cells(LR + 1 + li, 1)
        .Value = leg(li)(0)
        .Interior.Color = leg(li)(1)
        .Font.Color = leg(li)(2)
        .Font.Bold = True
        .HorizontalAlignment = xlLeft
    End With
    wsC.Rows(LR + 1 + li).RowHeight = 18
Next li

' -----
wsC.Columns(1).ColumnWidth = 20
Dim col As Long
For col = 2 To tickerCount + 1
    wsC.Columns(col).ColumnWidth = IIf(tickerCount > 10, 8, 10)
Next col

wsC.Activate
End Sub

```

Option Explicit

```

' =====
' MARKET SCANNER v2.1
' =====

Private Const W_BIAS20 As Double = 0.25
Private Const W_BIAS60 As Double = 0.25
Private Const W_BIAS5 As Double = 0.1
Private Const W_DISTH As Double = 0.15
Private Const W_CHG180 As Double = 0.15
Private Const W_BIAS120 As Double = 0.1

' =====
' MAIN ENTRY
' =====

Sub RunCompanyResearch(market As String, Sector As String)
    Dim wsRes As Worksheet
    On Error Resume Next
    Set wsRes = ThisWorkbook.Sheets("Company research")
    On Error GoTo 0
    If wsRes Is Nothing Then MsgBox "Sheet 'Company research' not found.", vbExclamation: Exit Sub

    Dim tickerList As Variant
    tickerList = GetSectorTickers(market, Sector)
    If IsEmpty(tickerList) Then
        MsgBox "No tickers found for: " & market & " / " & Sector, vbExclamation
        Exit Sub
    End If

    Application.ScreenUpdating = False

    With wsRes
        .Range("C:S").Clear
        .Range("C:S").Interior.Color = RGB(0, 0, 0)
        .Range("C:S").Font.Color = RGB(200, 200, 200)
        .Range("C:S").Font.Name = "Consolas"
        .Range("C:S").Font.Size = 9
    End With

    With wsRes.cells(1, 3)
        .Value = "MARKET SCANNER — " & market & " | " & Sector & " | " & Format(Now, "yyyy/mm/dd hh:mm")
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
        .Font.Size = 13
    End With
    wsRes.Range(wsRes.cells(1, 3), wsRes.cells(1, 19)).Interior.Color = RGB(10, 10, 10)

    Dim hdrs As Variant
    hdrs = Array("TICKER", "COMPANY", "SECTOR", "PRICE", _
        "HIGH DIST%", "180D CHG%", "DAY CHG%", _
        "BIAS5", "BIAS20", "BIAS60", "BIAS120", "BIAS240", _
        "PE", "FWD PE", "PEG", "TREND", "SCORE")

    Dim ci As Integer
    For ci = 0 To UBound(hdrs)
        With wsRes.cells(2, ci + 3)
            .Value = hdrs(ci)
            .Font.Color = RGB(255, 192, 0)
            .Font.Bold = True
            .Interior.Color = RGB(10, 10, 10)
            .HorizontalAlignment = xlCenter
        End With
    Next ci
    With wsRes.Range(wsRes.cells(2, 3), wsRes.cells(2, 19)).Borders(xlEdgeBottom)
        .LineStyle = xlContinuous
        .Color = RGB(255, 192, 0)
        .Weight = xlThin
    End With

    Dim rowNum As Long: rowNum = 3

```

```
Dim i As Long
```

```
For i = LBound(tickerList) To UBound(tickerList)
```

```
    Dim tkr As String: tkr = CStr(tickerList(i))
```

```
    Application.StatusBar = "Scanning [" & i + 1 & "/" & (UBound(tickerList) + 1) & "]" & tkr
```

```
    DoEvents
```

```
    Dim tech As TechIndicators
```

```
    tech = GetTechData(tkr)
```

```
    Dim nm As String: nm = GetCompanyName(tkr)
```

```
    Dim rowBg As Long: rowBg = IIf((rowNum Mod 2) = 1, RGB(12, 12, 12), RGB(20, 20, 20))
```

```
    With wsRes.Range(wsRes.cells(rowNum, 3), wsRes.cells(rowNum, 19))
```

```
        .Interior.Color = rowBg
```

```
        .HorizontalAlignment = xlCenter
```

```
        .Font.Name = "Consolas"
```

```
        .Font.Size = 9
```

```
    End With
```

```
    wsRes.Rows(rowNum).RowHeight = 16
```

```
    With wsRes.cells(rowNum, 3)
```

```
        .Value = tkr
```

```
        .Font.Color = RGB(255, 192, 0)
```

```
        .Font.Bold = True
```

```
    End With
```

```
    wsRes.cells(rowNum, 4).Value = nm
```

```
    wsRes.cells(rowNum, 4).HorizontalAlignment = xlLeft
```

```
    wsRes.cells(rowNum, 4).Font.Color = RGB(210, 210, 210)
```

```
    wsRes.cells(rowNum, 5).Value = Sector
```

```
    wsRes.cells(rowNum, 5).Font.Color = RGB(150, 150, 150)
```

```
If tech.Success Then
```

```
    wsRes.cells(rowNum, 6).Value = tech.price
```

```
    wsRes.cells(rowNum, 6).NumberFormat = "#,##0.00"
```

```
    WriteColoredPct wsRes.cells(rowNum, 7), tech.DistFromHigh, "0.00%", False
```

```
    WriteColoredPct wsRes.cells(rowNum, 8), tech.Change180D, "0.00%", True
```

```
    WriteColoredPct wsRes.cells(rowNum, 9), tech.ChangePercent, "0.00%", True
```

```
    WriteColoredPct wsRes.cells(rowNum, 10), tech.Bias5, "0.00%", True
```

```
    WriteColoredPct wsRes.cells(rowNum, 11), tech.Bias20, "0.00%", True
```

```
    WriteColoredPct wsRes.cells(rowNum, 12), tech.Bias60, "0.00%", True
```

```
    WriteColoredPct wsRes.cells(rowNum, 13), tech.Bias120, "0.00%", True
```

```
    WriteColoredPct wsRes.cells(rowNum, 14), tech.Bias240, "0.00%", True
```

```
    ' ——— PE / FWD PE / PEG — fetch directly —————
```

```
    Dim pe As Double, fpe As Double, peg As Double
```

```
    Call FetchFundamentals(tkr, pe, fpe, peg)
```

```
If pe > 0 Then
```

```
    wsRes.cells(rowNum, 15).Value = pe
```

```
    wsRes.cells(rowNum, 15).NumberFormat = "0.0"
```

```
    wsRes.cells(rowNum, 15).Font.Color = RGB(180, 180, 255)
```

```
Else
```

```
    wsRes.cells(rowNum, 15).Value = "N/A"
```

```
    wsRes.cells(rowNum, 15).Font.Color = RGB(100, 100, 100)
```

```
End If
```

```
If fpe > 0 Then
```

```
    wsRes.cells(rowNum, 16).Value = fpe
```

```
    wsRes.cells(rowNum, 16).NumberFormat = "0.0"
```

```
    wsRes.cells(rowNum, 16).Font.Color = RGB(180, 180, 255)
```

```
Else
```

```
    wsRes.cells(rowNum, 16).Value = "N/A"
```

```
    wsRes.cells(rowNum, 16).Font.Color = RGB(100, 100, 100)
```

```
End If
```

```
If peg > 0 Then
```

```
    wsRes.cells(rowNum, 17).Value = peg
```

```
    wsRes.cells(rowNum, 17).NumberFormat = "0.00"
```

```
    wsRes.cells(rowNum, 17).Font.Color = IIf(peg < 1.5, RGB(0, 220, 100), RGB(200, 150, 100))
```

```

Else
    wsRes.cells(rowNum, 17).Value = "N/A"
    wsRes.cells(rowNum, 17).Font.Color = RGB(100, 100, 100)
End If

Call ApplyScanTrend(wsRes.cells(rowNum, 18), tech)

Dim score As Double: score = CalcMomentumScore(tech)
With wsRes.cells(rowNum, 19)
    .Value = score
    .NumberFormat = "0"
    .Font.Bold = True
    Call ApplyScoreColor(wsRes.cells(rowNum, 19), score)
End With
Else
    wsRes.cells(rowNum, 6).Value = "ERR"
    wsRes.cells(rowNum, 6).Font.Color = RGB(255, 102, 0)
    wsRes.cells(rowNum, 18).Value = "N/A"
    wsRes.cells(rowNum, 19).Value = 0
End If
    rowNum = rowNum + 1
Next i

' Sort by Score
If rowNum > 4 Then
    wsRes.Range(wsRes.cells(3, 3), wsRes.cells(rowNum - 1, 19)).Sort _
        Key1:=wsRes.cells(3, 19), Order1:=xlDescending, Header:=xlNo
End If

' Summary
Call DrawScanSummary(wsRes, rowNum + 1, market, Sector, tickerList, rowNum - 3)

' Column widths
wsRes.Columns("C:S").AutoFit
wsRes.Columns("C").ColumnWidth = 8    ' Ticker: 1/3 of default (~8 units)
wsRes.Columns("D").ColumnWidth = 28    ' Company name

Application.StatusBar = "Scan complete — " & Format(Now, "hh:mm:ss")
Application.ScreenUpdating = True
MsgBox "Scan complete! " & (rowNum - 3) & " stocks scanned.", vbInformation
End Sub

' =====
' Dedicated fundamentals fetcher — handles both JSON formats
' {"raw": 25.3} AND direct "key": 25.3
' =====
Sub FetchFundamentals(ticker As String, ByRef pe As Double, ByRef fpe As Double, ByRef peg As Double)
    pe = 0: fpe = 0: peg = 0

    ticker = UCase(Trim(ticker))
    If InStr(ticker, ".TW") = 0 And InStr(ticker, ".TWO") = 0 And IsNumeric(ticker) Then
        ticker = ticker & ".TW"
    End If

    Dim http As Object
    On Error Resume Next
    Set http = CreateObject("WinHttp.WinHttpRequest.5.1")
    If http Is Nothing Then Set http = CreateObject("MSXML2.XMLHTTP")
    On Error GoTo 0

    ' Try quoteSummary price module first (most reliable for PE)
    Dim url As String
    url = "https://query1.finance.yahoo.com/v10/finance/quoteSummary/" & ticker & _
        "?modules=summaryDetail,defaultKeyStatistics"

    Dim resp As String
    On Error Resume Next
    With http
        .Open "GET", url, False
        .setRequestHeader "User-Agent", "Mozilla/5.0"
        .send
    End With

```

```

    If .status = 200 Then resp = .responseText
End With
On Error GoTo 0

If Len(resp) < 50 Then Exit Sub

' Clean whitespace
resp = Replace(Replace(Replace(resp, " ", ""), vbCr, ""), vbLf, "")

' Extract with both formats
pe = ExtractNum(resp, "trailingPE")
fpe = ExtractNum(resp, "forwardPE")
peg = ExtractNum(resp, "pegRatio")

' Fallback: try v8 chart endpoint regularMarketPE
If pe = 0 Then
    Dim url2 As String
    url2 = "https://query1.finance.yahoo.com/v8/finance/chart/" & ticker & "?interval=1d&range=1d"
    Dim resp2 As String
    On Error Resume Next
    With http
        .Open "GET", url2, False
        .setRequestHeader "User-Agent", "Mozilla/5.0"
        .send
        If .status = 200 Then resp2 = .responseText
    End With
    On Error GoTo 0
    If Len(resp2) > 50 Then
        resp2 = Replace(Replace(Replace(resp2, " ", ""), vbCr, ""), vbLf, "")
        pe = ExtractNum(resp2, "trailingPE")
        fpe = ExtractNum(resp2, "forwardPE")
    End If
End If
End Sub

```

' Handles both {"raw":25.3} and direct 25.3 formats

```

Private Function ExtractNum(json As String, key As String) As Double
    ExtractNum = 0

```

```

' Format 1: "key":{"raw":25.3

```

```

Dim s1 As String: s1 = "" & key & "" & {"raw":25.3

```

```

Dim pos As Long: pos = InStr(json, s1)

```

```

If pos > 0 Then

```

```

    Dim vs As Long: vs = pos + Len(s1)

```

```

    Dim ve As Long: ve = InStr(vs, json, ",")

```

```

    Dim ve2 As Long: ve2 = InStr(vs, json, "}")

```

```

    If ve = 0 Or (ve2 > 0 And ve2 < ve) Then ve = ve2

```

```

    If ve > vs Then

```

```

        Dim tmp As String: tmp = Mid(json, vs, ve - vs)

```

```

        If IsNumeric(tmp) And CDBl(tmp) > 0 Then ExtractNum = CDBl(tmp): Exit Function
    End If

```

```

End If

```

```

' Format 2: "key":25.3

```

```

Dim s2 As String: s2 = "" & key & ""

```

```

pos = InStr(json, s2)

```

```

Do While pos > 0

```

```

    Dim afterKey As Long: afterKey = pos + Len(s2)

```

```

    ' Skip until colon

```

```

    Dim colonPos As Long: colonPos = InStr(afterKey, json, ":")

```

```

    If colonPos > afterKey + 3 Then Exit Do ' colon too far, skip

```

```

    If colonPos > 0 Then

```

```

        Dim valStart As Long: valStart = colonPos + 1

```

```

        ' Skip opening brace if present (means it's an object, not direct)

```

```

        If Mid(json, valStart, 1) = "{" Then

```

```

            pos = InStr(pos + 1, json, s2)

```

```

            GoTo NextOccurrence
        End If

```

```

        Dim valEnd As Long: valEnd = InStr(valStart, json, ",")

```

```

        Dim valEnd2 As Long: valEnd2 = InStr(valStart, json, "}")

```

```

        If valEnd = 0 Or (valEnd2 > 0 And valEnd2 < valEnd) Then valEnd = valEnd2
    End If

```

```

        If valEnd > valStart Then
            Dim tv As String: tv = Mid(json, valStart, valEnd - valStart)
            tv = Replace(tv, "\"", "")
            If IsNumeric(tv) And CDBl(tv) > 0 Then ExtractNum = CDBl(tv): Exit Function
        End If
    End If
NextOccurrence:
    pos = InStr(pos + 1, json, s2)
Loop
End Function

' =====
' Summary bar
' =====

Private Sub DrawScanSummary(ws As Worksheet, sumRow As Long, _
    market As String, Sector As String, _
    tickers As Variant, count As Long)
    Dim cntBull As Long, cntBear As Long, cntNeut As Long
    Dim r As Long
    For r = 3 To 3 + count - 1
        Dim tv As String: tv = ws.Cells(r, 18).Value
        If InStr(tv, "BULL") > 0 Then cntBull = cntBull + 1
        If InStr(tv, "BEAR") > 0 Then cntBear = cntBear + 1
        If InStr(tv, "NEUT") > 0 Then cntNeut = cntNeut + 1
    Next r

    With ws.Range(ws.Cells(sumRow, 3), ws.Cells(sumRow, 19))
        .Interior.Color = RGB(15, 15, 15)
        .Font.Bold = True
    End With
    ws.Cells(sumRow, 3).Value = "SUMMARY": ws.Cells(sumRow, 3).Font.Color = RGB(255, 192, 0)
    ws.Cells(sumRow, 4).Value = "Total: " & count: ws.Cells(sumRow, 4).Font.Color = RGB(200, 200, 200)
    ws.Cells(sumRow, 5).Value = "BULLISH: " & cntBull: ws.Cells(sumRow, 5).Font.Color = RGB(255, 80, 80)
    ws.Cells(sumRow, 6).Value = "BEARISH: " & cntBear: ws.Cells(sumRow, 6).Font.Color = RGB(0, 210, 100)
    ws.Cells(sumRow, 7).Value = "NEUTRAL: " & cntNeut: ws.Cells(sumRow, 7).Font.Color = RGB(200, 200, 200)
    If count > 0 Then
        Dim bp As Double: bp = cntBull / count
        ws.Cells(sumRow, 8).Value = bp
        ws.Cells(sumRow, 8).NumberFormat = "0%"
        ws.Cells(sumRow, 8).Font.Color = IIf(bp > 0.5, RGB(255, 80, 80), RGB(0, 210, 100))
    End If
End Sub

' =====
' Momentum Score
' =====

Private Function CalcMomentumScore(tech As TechIndicators) As Double
    Dim s As Double
    s = s + NormScore(tech.Bias20, -0.3, 0.3) * 25
    s = s + NormScore(tech.Bias60, -0.3, 0.3) * 25
    s = s + NormScore(tech.Bias5, -0.15, 0.15) * 10
    s = s + NormScore(-tech.DistFromHigh, 0, 0.4) * 15
    s = s + NormScore(tech.Change180D, -0.5, 1) * 15
    s = s + NormScore(tech.Bias120, -0.3, 0.3) * 10
    CalcMomentumScore = WorksheetFunction.Max(0, WorksheetFunction.Min(100, s))
End Function

Private Function NormScore(val As Double, minV As Double, maxV As Double) As Double
    If maxV = minV Then NormScore = 0.5: Exit Function
    NormScore = (val - minV) / (maxV - minV)
    If NormScore < 0 Then NormScore = 0
    If NormScore > 1 Then NormScore = 1
End Function

' =====
' Cell helpers
' =====

Private Sub WriteColoredPct(cell As Range, val As Double, fmt As String, positiveIsGood As Boolean)
    cell.Value = val
    cell.NumberFormat = fmt

```

```

' If the value is exactly 0, color it Gray
If val = 0 Then
    cell.Font.Color = RGB(150, 150, 150)
Exit Sub
End If

' Determine colors based on the positiveIsGood flag
If positiveIsGood Then
    If val > 0 Then
        cell.Font.Color = RGB(0, 210, 100) ' Positive is Good -> Green
    Else
        cell.Font.Color = RGB(255, 80, 80) ' Negative is Bad -> Red
    End If
Else
    ' If positiveIsGood is False, we assume Negative is Good
    If val < 0 Then
        cell.Font.Color = RGB(0, 210, 100) ' Negative is Good -> Green
    Else
        cell.Font.Color = RGB(255, 80, 80) ' Positive is Bad -> Red
    End If
End If

End Sub

Private Sub ApplyScanTrend(cell As Range, tech As TechIndicators)
    cell.Font.Bold = True: cell.HorizontalAlignment = xlCenter
    If tech.Bias5 > 0 And tech.Bias20 > 0 And tech.Bias60 > 0 Then
        cell.Value = "STRONG BULL": cell.Font.Color = RGB(255, 60, 60): cell.Interior.Color = RGB(40, 0, 0)
    ElseIf tech.Bias20 > 0 And tech.Bias60 > 0 Then
        cell.Value = "BULLISH": cell.Font.Color = RGB(255, 120, 120): cell.Interior.Color = RGB(25, 0, 0)
    ElseIf tech.Bias5 < 0 And tech.Bias20 < 0 And tech.Bias60 < 0 Then
        cell.Value = "STRONG BEAR": cell.Font.Color = RGB(0, 255, 100): cell.Interior.Color = RGB(0, 35, 0)
    ElseIf tech.Bias20 < 0 And tech.Bias60 < 0 Then
        cell.Value = "BEARISH": cell.Font.Color = RGB(100, 220, 100): cell.Interior.Color = RGB(0, 20, 0)
    ElseIf tech.Bias20 > 0 Then
        cell.Value = "WEAK BULL": cell.Font.Color = RGB(255, 192, 0)
    ElseIf tech.Bias20 < 0 Then
        cell.Value = "WEAK BEAR": cell.Font.Color = RGB(150, 200, 150)
    Else
        cell.Value = "NEUTRAL": cell.Font.Color = RGB(150, 150, 150)
    End If
End Sub

Private Sub ApplyScoreColor(cell As Range, score As Double)
    Select Case True
        Case score >= 75: cell.Font.Color = RGB(255, 60, 60): cell.Interior.Color = RGB(40, 0, 0)
        Case score >= 55: cell.Font.Color = RGB(255, 160, 80): cell.Interior.Color = RGB(25, 10, 0)
        Case score >= 40: cell.Font.Color = RGB(200, 200, 200): cell.Interior.Color = RGB(20, 20, 20)
        Case score >= 25: cell.Font.Color = RGB(100, 200, 120): cell.Interior.Color = RGB(0, 15, 5)
        Case Else: cell.Font.Color = RGB(0, 210, 100): cell.Interior.Color = RGB(0, 25, 10)
    End Select
End Sub

' =====
' Export helpers
' =====

Sub ExportTopToWatchlist()
    Dim wsRes As Worksheet
    On Error Resume Next: Set wsRes = ThisWorkbook.Sheets("Company research"): On Error GoTo 0
    If wsRes Is Nothing Then MsgBox "Run a scan first.", vbExclamation: Exit Sub

    Dim topN As Long
    topN = CLng(InputBox("Export top N:", "Export", "10"))
    If topN <= 0 Then Exit Sub

    Dim wsWatch As Worksheet
    On Error Resume Next: Set wsWatch = ThisWorkbook.Sheets("Watchlist"): On Error GoTo 0
    If wsWatch Is Nothing Then
        Set wsWatch = ThisWorkbook.Sheets.Add(After:=wsRes)
        wsWatch.Name = "Watchlist"
    End If

```

```

wsWatch.cells.Clear
With wsWatch.cells: .Interior.Color = RGB(0, 0, 0): .Font.Color = RGB(200, 200, 200): .Font.Name = "Consolas": .Font.Size = 10: End With
wsWatch.cells(1, 1).Value = "WATCHLIST — " & Format(Now, "yyyy/mm/dd hh:mm"): wsWatch.cells(1, 1).Font.Color = RGB(255, 192, 0):
wsWatch.cells(1, 1).Font.Bold = True

Dim headers As Variant: headers = Array("TICKER", "COMPANY", "SCORE", "TREND", "BIAS20", "BIAS60", "PE", "FWD PE")
Dim ci As Integer
For ci = 0 To UBound(headers)
    With wsWatch.cells(2, ci + 1): .Value = headers(ci): .Font.Color = RGB(255, 192, 0): .Font.Bold = True: .Interior.Color = RGB(
10, 10, 10): End With
Next ci

Dim LR As Long: LR = wsRes.cells(wsRes.Rows.count, 3).End(xlUp).row
Dim cc As Long: cc = WorksheetFunction.Min(topN, LR - 2)
Dim wr As Long: wr = 3
Dim r As Long
For r = 3 To 3 + cc - 1
    wsWatch.Range(wsWatch.cells(wr, 1), wsWatch.cells(wr, 8)).Interior.Color = IIf((wr Mod 2) = 1, RGB(12, 12, 12), RGB(20, 20, 20
))
    wsWatch.cells(wr, 1).Value = wsRes.cells(r, 3).Value: wsWatch.cells(wr, 1).Font.Color = RGB(255, 192, 0): wsWatch.cells(wr, 1)
.Font.Bold = True
    wsWatch.cells(wr, 2).Value = wsRes.cells(r, 4).Value
    wsWatch.cells(wr, 3).Value = wsRes.cells(r, 19).Value
    wsWatch.cells(wr, 4).Value = wsRes.cells(r, 18).Value
    wsWatch.cells(wr, 5).Value = wsRes.cells(r, 11).Value: wsWatch.cells(wr, 5).NumberFormat = "0.00%"
    wsWatch.cells(wr, 6).Value = wsRes.cells(r, 12).Value: wsWatch.cells(wr, 6).NumberFormat = "0.00%"
    wsWatch.cells(wr, 7).Value = wsRes.cells(r, 15).Value
    wsWatch.cells(wr, 8).Value = wsRes.cells(r, 16).Value
    wr = wr + 1
Next r
wsWatch.Columns("A:H").AutoFit: wsWatch.Columns("B").ColumnWidth = 28
wsWatch.Activate
MsgBox "Top " & cc & " exported!", vbInformation
End Sub

Sub ExportScanCSV()
Dim wsRes As Worksheet
On Error Resume Next: Set wsRes = ThisWorkbook.Sheets("Company research"): On Error GoTo 0
If wsRes Is Nothing Then Exit Sub
Dim path As String: path = ThisWorkbook.path & "\Scan_" & Format(Now, "yyyymmdd_hhmmss") & ".csv"
Dim fNum As Integer: fNum = FreeFile
Open path For Output As #fNum
Dim LR As Long: LR = wsRes.cells(wsRes.Rows.count, 3).End(xlUp).row
Dim r As Long, c As Integer, line As String
For r = 2 To LR
    line = ""
    For c = 3 To 19
        Dim v As String: v = CStr(wsRes.cells(r, c).Value)
        If InStr(v, ",") > 0 Then v = """" & v & """"
        line = line & IIf(c > 3, ",", "") & v
    Next c
    Print #fNum, line
Next r
Close #fNum
MsgBox "Exported: " & path, vbInformation
End Sub

Sub OpenMarketScanner()
frmMarketScanner.Show
End Sub

Sub ColorizeRange(rng As Range)
Dim cell As Range
For Each cell In rng
    ' 外層判斷：確認是否為數字
    If IsNumeric(cell.Value) Then

        ' 內層判斷：改為標準的區塊寫法 (Block If)
        If cell.Value > 0 Then

```

```

        cell.Font.Color = RGB(255, 100, 100)
    ElseIf cell.Value < 0 Then
        cell.Font.Color = RGB(100, 255, 100)
    End If ' <--- 補上內層 If 的結尾

```

```

End If ' <--- 原本外層 IsNumeric 的結尾
Next cell
End Sub

```

```

' =====
' SECTOR TICKERS — EXPANDED
' =====

```

```

Function GetSectorTickers(market As String, Sector As String) As Variant
    Dim arr As Variant: arr = Empty

```

```

    If market = "TW" Then
        Select Case Sector
            Case "01. 半導體 - 晶圓代工 (Foundry)"
                arr = Array("2330", "2303", "5347", "6770", "3443", "4961", "3034", "6669")
            Case "02. 半導體 - IC設計 (IC Design)"
                arr = Array("2454", "2379", "3034", "3035", "4961", "2458", "8016", "3529", "6643", "4966", "3661", "6547", "5274", "3130", "6770", "2385", "3044", "4919", "6770", "3017")
            Case "03. 半導體 - 封測 (Packaging)"
                arr = Array("3711", "2449", "6239", "6271", "8150", "3450", "2412", "6257", "6269", "3189", "6274", "3016")
            Case "04. 半導體 - IP/ASIC (Silicon IP)"
                arr = Array("3661", "3443", "3035", "3529", "6669", "6643", "6533", "4966", "3665", "6643")
            Case "05. 電腦週邊 - AI 伺服器 (AI Server)"
                arr = Array("2382", "3231", "2356", "6669", "2317", "2376", "3017", "3697", "6213", "2399", "3704", "6456", "3023", "6197", "3005")
            Case "06. 電腦週邊 - 品牌與代工 (OEM/ODM)"
                arr = Array("2357", "2324", "2353", "2301", "4938", "3231", "2382", "2356", "6669", "2317")
            Case "07. 電腦週邊 - 工業電腦 (IPC)"
                arr = Array("2395", "6414", "8050", "6245", "3540", "5443", "6185", "3548")
            Case "08. 通信網路 - 網通設備 (Networking)"
                arr = Array("2345", "3704", "5388", "6285", "3596", "2332", "4906", "4943", "3518", "6415", "3526", "2465", "6456")
            Case "09. 光電業 - 面板與光學 (Panel & Optics)"
                arr = Array("2409", "3481", "3008", "3406", "3376", "6706", "3189", "2384", "3044", "6274", "3296", "6443", "4919")
            Case "10. 電子零組件 - PCB (印刷電路板)"
                arr = Array("3037", "2368", "2313", "4958", "6269", "8046", "3044", "3189", "4938", "2382", "6257", "3042", "8011", "6274")
            Case "11. 電子零組件 - 被動元件 (Passive)"
                arr = Array("2327", "2492", "2456", "3026", "2438", "3557", "6269", "2478", "3018", "2485")
            Case "12. 電子零組件 - 連接器 (Connectors)"
                arr = Array("3533", "3605", "3023", "2345", "6598", "3588", "2308", "6289")
            Case "13. 電子零組件 - 電源供應 (Power Supply)"
                arr = Array("2308", "6278", "6412", "2301", "3483", "6449", "3591", "1537", "4977", "3577")
            Case "14. 金融 - 金控 (FHC)"
                arr = Array("2881", "2882", "2891", "2886", "2884", "2892", "2885", "2880", "2883", "2890", "2887", "5880", "2801")
            Case "15. 金融 - 銀行與證券 (Bank/Securities)"
                arr = Array("5880", "2834", "2887", "6005", "2897", "2845", "2838", "2849", "5876", "2849", "6015", "6016")
            Case "16. 航運 (Container Shipping)"
                arr = Array("2603", "2609", "2615", "2617", "5608", "2208")
            Case "17. 傳產 - 散裝與航空 (Bulk & Aviation)"
                arr = Array("2606", "2637", "2618", "2610", "5608", "2634", "2605", "2601", "2608", "2614")
            Case "18. 鋼鐵 (Steel)"
                arr = Array("2002", "2014", "2006", "2027", "2031", "2023", "2059", "2062", "2069", "2038", "2034")
            Case "19. 塑膠 (Plastics)"
                arr = Array("1301", "1303", "1326", "1304", "1308", "1310", "1313", "2915", "1307", "1711")
            Case "20. 紡織 (Textiles)"
                arr = Array("1402", "1476", "1477", "1417", "1409", "1414", "1463", "1464", "1474", "1475")
            Case "21. 水泥 (Cement)"
                arr = Array("1101", "1102", "1103", "1104", "1108", "1109", "1110")
            Case "22. 電機機械"
                arr = Array("1503", "1504", "1513", "1519", "4566", "1514", "1516", "1528", "1560", "2308", "1536", "1590")
            Case "23. 電器電纜"
                arr = Array("1605", "1609", "1603", "1608", "1612", "1614", "1615", "1616", "1618", "1626")
            Case "24. 生技醫療 (Biotech)"
                arr = Array("1795", "4123", "4128", "4743", "6446", "4108", "4144", "4157", "4174", "4116", "1701", "4155", "4174", "6547", "4103")
            Case "25. 化學工業 (Chemicals)"
                arr = Array("1722", "1710", "4770", "1701", "1737", "1773", "1750", "1785", "1730", "1742")

```

```

Case "26. 汽車工業 (Auto)"
    arr = Array("2201", "2204", "2207", "1319", "2206", "2209", "2212", "2213", "2219", "2230", "6191")
Case "27. 建材營造 (Construction)"
    arr = Array("2501", "2542", "2548", "5522", "2538", "2543", "2547", "2560", "2514", "2524", "2530")
Case "28. 食品工業 (Food)"
    arr = Array("1216", "1210", "1227", "1215", "1218", "1229", "1231", "1232", "1233", "1234", "1236", "1256")
Case "29. 觀光餐旅 (Tourism)"
    arr = Array("2707", "2727", "2723", "5706", "2706", "2712", "2718", "2722", "2726", "2731")
Case "30. 綠能環保 (Green Energy)"
    arr = Array("9933", "6869", "8341", "3708", "6414", "1560", "3576", "6690", "3536", "3622", "1563", "3577")
End Select

ElseIf market = "US" Then
    Select Case Sector
    Case "01. Tech - Mega Cap (七巨頭)"
        arr = Array("AAPL", "MSFT", "GOOGL", "GOOG", "AMZN", "NVDA", "META", "TSLA", "AVGO", "ORCL", "ADBE", "NFLX", "AMD", "INTC", "QCOM")
    Case "02. Tech - Software Infrastructure (軟體基建)"
        arr = Array("MSFT", "ORCL", "ADBE", "PLTR", "PANW", "CRWD", "SNOW", "FTNT", "NET", "ZS", "OKTA", "S", "DDOG", "GTLB", "ESTC", "MDB", "CFLT")
    Case "03. Tech - Software Application (應用軟體)"
        arr = Array("CRM", "INTU", "NOW", "UBER", "SAP", "ADP", "WDAY", "CDNS", "ADSK", "DDOG", "ROP", "FICO", "HUBS", "SHOP", "SQ", "BILL", "PCTY")
    Case "04. Tech - Semiconductors (半導體)"
        arr = Array("NVDA", "AVGO", "TSM", "AMD", "QCOM", "TXN", "MU", "ADI", "LRCX", "AMAT", "INTC", "NXPI", "MRVL", "MCHP", "ON", "KLAC", "ASML", "MPWR")
    Case "05. Tech - Consumer Electronics (消費電子)"
        arr = Array("AAPL", "SONY", "HPQ", "DELL", "APH", "TEL", "GLW", "STX", "WDC", "NTAP", "PSTG", "ROKU", "SMCI", "LOGI")
    Case "06. Tech - Info Services (資訊服務)"
        arr = Array("IBM", "ACN", "FI", "IT", "FIS", "CTSH", "BR", "EPAM", "GLOB", "INFY", "LDOS", "SAIC", "BAH", "CACI")
    Case "07. Financial - Banks Diversified (綜合銀行)"
        arr = Array("JPM", "BAC", "WFC", "C", "HSBC", "USB", "PNC", "TFC", "FITB", "RF", "HBAN", "KEY", "CFG", "MTB", "ZION", "CMA", "EWBC")
    Case "08. Financial - Credit Services (信貸服務)"
        arr = Array("V", "MA", "AXP", "PYPL", "COF", "SYF", "DFS", "SQ", "FIS", "FI", "GPN", "AFRM", "SOFI", "UPST")
    Case "09. Financial - Asset Management (資產管理)"
        arr = Array("BLK", "BX", "KKR", "GS", "MS", "APO", "BK", "STT", "AMP", "IVZ", "TROW", "BEN", "VIRT", "LPLA", "MORN")
    Case "10. Financial - Insurance (保險)"
        arr = Array("BRK-B", "PGR", "CB", "MMC", "AON", "TRV", "ALL", "AFL", "MET", "PRU", "AIG", "L", "UNM", "HIG", "RNR", "RE", "WRB", "CINF")
    Case "11. Healthcare - Drug Manufacturers (製藥)"
        arr = Array("LLY", "JNJ", "MRK", "ABBV", "PFE", "NVS", "AZN", "BMY", "RHHBY", "SNY", "GSK", "TEVA", "PRGO", "VTRS", "JAZZ")
    Case "12. Healthcare - Biotechnology (生技)"
        arr = Array("AMGN", "VRTX", "GILD", "REGN", "MRNA", "BIIB", "ALNY", "IONS", "EXEL", "SRPT", "NTLA", "BEAM", "CRSP", "EDIT")
    Case "13. Healthcare - Medical Devices (醫材)"
        arr = Array("ABT", "MDT", "SYK", "ISRG", "BSX", "EW", "BDX", "ZBH", "STE", "HOLX", "TFX", "NVCR", "ATRC", "SWAV", "INSP", "NARI")
    Case "14. Healthcare - Plans (健保)"
        arr = Array("UNH", "ELV", "CI", "CVS", "HUM", "CNC", "MOH", "OSCR", "CLOV")
    Case "15. Cyclical - Auto Manufacturers (汽車)"
        arr = Array("TSLA", "TM", "F", "GM", "HMC", "RACE", "RIVN", "LCID", "NIO", "LI", "XPEV", "STLA", "APTV", "MGA", "BWA", "LEA")
    Case "16. Cyclical - Internet Retail (電商)"
        arr = Array("AMZN", "BABA", "PDD", "JD", "EBAY", "DASH", "BKNG", "ABNB", "EXPE", "W", "CHWY")
    Case "17. Cyclical - Restaurants (餐飲)"
        arr = Array("MCD", "SBUX", "CMG", "YUM", "QSR", "DRI", "TXRH", "SHAK", "WING", "JACK", "WEN", "CAKE", "EAT", "BLMN")
    Case "18. Cyclical - Travel & Leisure (旅遊)"
        arr = Array("MAR", "HLT", "CCL", "RCL", "LVS", "WYNN", "MGM", "CZR", "NCLH", "UAL", "DAL", "LUV", "AAL", "ALK", "JBLU")
    Case "19. Defensive - Discount Stores (零售)"
        arr = Array("WMT", "COST", "TGT", "DG", "DLTR", "TJX", "ROST", "NKE", "LULU", "PVH", "VFC", "RL", "SKX", "CROX", "YETI")
    Case "20. Defensive - Beverages (飲料)"
        arr = Array("KO", "PEP", "MNST", "KDP", "STZ", "BUD", "SAM", "TAP", "ABEV", "CELH", "FIZZ")
    Case "21. Defensive - Household (家用品)"
        arr = Array("PG", "CL", "KMB", "EL", "K", "GIS", "CPB", "SJM", "MKC", "HRL", "CAG", "BG", "INGR", "POST", "LANC")
    Case "22. Comm - Internet Content (網路內容)"
        arr = Array("GOOGL", "META", "NFLX", "DASH", "PINS", "SNAP", "RDDT", "MTCH", "IAC", "ZG", "ANGI", "CDW")
    Case "23. Comm - Telecom (電信)"

```

```

arr = Array("TMUS", "VZ", "T", "CMCSA", "CHTR", "LUMN", "IDT", "OOMA", "BAND")
Case "24. Comm - Entertainment (娛樂)"
arr = Array("DIS", "WBD", "LYV", "EA", "TTWO", "NFLX", "PARA", "FOX", "FOXA", "OMC", "IPG", "DKNG", "PENN")
Case "25. Industrials - Aerospace (航太軍工)"
arr = Array("GE", "RTX", "BA", "LMT", "GD", "NOC", "LHX", "TDG", "HEI", "KTOS", "RKL", "IRDM", "BWXT", "CW", "HII", "
MOOG")
Case "26. Industrials - Logistics (物流)"
arr = Array("UPS", "FDX", "UNP", "CSX", "NSC", "XPO", "SAIA", "JBHT", "ODFL", "CHRW", "LSTR", "GXO", "RXO", "MRTN", "K
NX")
Case "27. Industrials - Farm & Heavy (重工)"
arr = Array("CAT", "DE", "ETN", "ITW", "PCAR", "EMR", "PH", "CMI", "IR", "ROK", "AME", "FTV", "GNRC", "GGG", "GTLS", "
TTC")
Case "28. Energy - Oil & Gas (石油天然氣)"
arr = Array("XOM", "CVX", "SHEL", "TTE", "COP", "SLB", "EOG", "MPC", "PSX", "VLO", "OXY", "DVN", "MRO", "APA", "HAL", "
BKR", "CTRA", "EQT")
Case "29. Basic Materials (原物料)"
arr = Array("LIN", "SHW", "FCX", "SCCO", "NEM", "APD", "ECL", "DD", "PPG", "RPM", "FMC", "MOS", "CF", "HUN", "CTVA", "
AXTA")
Case "30. Real Estate (REITs)"
arr = Array("PLD", "AMT", "EQIX", "PSA", "O", "SPG", "CCI", "DLR", "WELL", "AVB", "EQR", "ESS", "UDR", "MAA", "NNN", "
WPC", "VICI", "GLPI", "IIPR")
Case "31. ETFs - Major Indices (指數與區域)"
arr = Array("SPY", "VOO", "QQQ", "VTI", "VT", "IWM", "VUG", "VTV", "SCHG", "VGT", "XLK", "XLF", "XLV", "XLE", "XLY", "
XLP", "XLI", "XLU", "XLRE", "XLB", "XLC", "SMH", "SOXX", "ARKK", "GDX", "ICLN", "TAN", "JETS", "XBI")
End Select
End If

GetSectorTickers = arr
End Function

```

Option Explicit

```

' =====
' FetchTWSE
' 呼叫 TWSE/TPEX 即時 API，取得最新成交價（盤中）或收盤價
' ex_ch 範例: "tse_2337.tw" 或 "otc_6510.tw"
' =====

Private Function FetchTWSE(http As Object, exch As String) As Double

    Dim url As String
    url = "https://mis.twse.com.tw/stock/api/getStockInfo.jsp?ex_ch=" & _
        exch & "&json=1&delay=0&_=" & CStr(CLng((Now - DateSerial(1970, 1, 1)) * 86400))

    Dim response As String
    On Error Resume Next
    With http
        .Open "GET", url, False
        .setRequestHeader "User-Agent", "Mozilla/5.0"
        .setRequestHeader "Referer", "https://mis.twse.com.tw/"
        .setRequestHeader "Cache-Control", "no-cache"
        .send
        If .status = 200 Then response = .responseText
    End With
    On Error GoTo 0

    If Len(response) < 20 Then FetchTWSE = 0: Exit Function

    ' —— 優先取 "z"（最新成交價，盤中/收盤） ——
    Dim price As Double
    price = TWSEExtract(response, ""z"":""")

    ' —— 若 z 為 "-"（休市/未開盤），改取昨日收盤 "y" ——
    If price = 0 Then price = TWSEExtract(response, ""y"":""")

    FetchTWSE = price
End Function

' =====
' TWSEExtract 從 TWSE JSON 取數字欄位
' 格式: "z":120.50 或 "y":121.00
' =====

Private Function TWSEExtract(json As String, tag As String) As Double
    Dim pos As Long: pos = InStr(json, tag)
    If pos = 0 Then TWSEExtract = 0: Exit Function

    pos = pos + Len(tag)
    Dim ep As Long: ep = pos
    Do While ep <= Len(json)
        Select Case Mid(json, ep, 1)
            Case ":", ",", "}", "]" : Exit Do
        End Select
        ep = ep + 1
    Loop

    Dim s As String: s = Trim(Mid(json, pos, ep - pos))
    If s = "-" Or s = "" Then TWSEExtract = 0: Exit Function
    If IsNumeric(s) Then TWSEExtract = CDbl(s)
End Function

' =====
' FetchYahooClose 使用 interval=1d&range=5d
' 從 close[] 陣列取最後一個有效正數
' =====

Private Function FetchYahooClose(http As Object, ticker As String) As Double

    Dim url As String
    url = "https://query2.finance.yahoo.com/v8/finance/chart/" & ticker & "?interval=1d&range=5d"

    Dim response As String
    On Error Resume Next

```

```

With http
    .Open "GET", url, False
    .setRequestHeader "User-Agent", "Mozilla/5.0 (Windows NT 10.0; Win64; x64)"
    .setRequestHeader "Referer", "https://finance.yahoo.com/"
    .setRequestHeader "Cache-Control", "no-cache"
    .send
    If .status = 200 Then response = .responseText
End With
On Error GoTo 0

If Len(response) < 200 Then FetchYahooClose = 0: Exit Function

' 定位 quote 區塊內的 close[]
Dim qPos As Long: qPos = InStr(response, """"quote"":{")
If qPos = 0 Then qPos = 1

Dim csStart As Long: csStart = InStr(qPos, response, """"close"":[")
If csStart = 0 Then FetchYahooClose = 0: Exit Function
csStart = csStart + 9

Dim csEnd As Long: csEnd = InStr(csStart, response, "]")
If csEnd <= csStart Then FetchYahooClose = 0: Exit Function

Dim closeArr() As String
closeArr = Split(Mid(response, csStart, csEnd - csStart), ",")

Dim i As Long, price As Double
For i = UBound(closeArr) To LBound(closeArr) Step -1
    Dim cv As String: cv = Trim(closeArr(i))
    ' 正確寫法：拆成兩層 If
    If IsNumeric(cv) Then
        If CDBl(cv) > 0 Then
            price = CDBl(cv)
            Exit For
        End If
    End If
End For
Next i

FetchYahooClose = price
End Function

' =====
' DebugPriceTest v6.0
' =====
Sub DebugPriceTest()

    Dim testTicker As String
    testTicker = InputBox("輸入股票代號 (如 2337 或 AAPL) :", "Price Debug v6", "2337")
    If testTicker = "" Then Exit Sub

    testTicker = UCase(Trim(testTicker))
    If IsNumeric(testTicker) Then testTicker = testTicker & ".TW"
    If Not g_PriceCache Is Nothing Then g_PriceCache.RemoveAll

    Dim http As Object
    Set http = CreateObject("WinHttp.WinHttpRequest.5.1")

    Dim isTW As Boolean
    isTW = (InStr(testTicker, ".TW") > 0 Or InStr(testTicker, ".TWO") > 0)

    Dim twseResp As String, yahooResp As String
    Dim twsePrice As Double, yahooPrice As Double, finalPrice As Double
    Dim code As String

    If isTW Then
        code = Replace(Replace(testTicker, ".TWO", ""), ".TW", "")

        ' TWSE 請求
        Dim twseUrl As String
        twseUrl = "https://mis.twse.com.tw/stock/api/getStockInfo.jsp?ex_ch=tse_" & _
            code & ".tw&json=1&delay=0"
    
```

```

With http
    .Open "GET", twseUrl, False
    .setRequestHeader "User-Agent", "Mozilla/5.0"
    .setRequestHeader "Referer", "https://mis.twse.com.tw/"
    .send
    twseResp = .responseText
End With
twsePrice = FetchTWSE(http, "tse_" & code & ".tw")
End If

' Yahoo 請求
Dim yahooUrl As String
yahooUrl = "https://query2.finance.yahoo.com/v8/finance/chart/" & testTicker & "?interval=1d&range=5d"
With http
    .Open "GET", yahooUrl, False
    .setRequestHeader "User-Agent", "Mozilla/5.0"
    .setRequestHeader "Cache-Control", "no-cache"
    .send
    yahooResp = .responseText
End With
yahooPrice = FetchYahooClose(http, testTicker)

finalPrice = GetStockPrice(testTicker)

Dim info As String
If isTW Then
    info = "【TWSE API】" & vbCrLf & _
        "URL      : tse_" & code & ".tw" & vbCrLf & _
        "價格      : " & twsePrice & vbCrLf & _
        "原始      : " & Left(twseResp, 300) & vbCrLf & vbCrLf
End If
info = info & "【Yahoo Finance】" & vbCrLf & _
    "URL      : " & yahooUrl & vbCrLf & _
    "Len      : " & Len(yahooResp) & " chars" & vbCrLf & _
    "價格      : " & yahooPrice & vbCrLf & vbCrLf & _
    "★ GetStockPrice 最終結果: " & finalPrice

MsgBox info, vbInformation, "Price Debug v6: " & testTicker
Set http = Nothing
End Sub

```

Option Explicit

```

' =====
' TICKER INSIGHT v1.0
' Per-ticker FIFO trade history & performance dashboard
' Bloomberg terminal style, US color convention (green +, red -)
'
' ENTRY POINTS:
'   Sub OpenTickerInsight()           - call from Alt+F8 or button
'   Sub RefreshTickerInsight(ticker) - full rebuild for a ticker
'   Sub RefreshTickerProjection       - recompute PROJECTION block only
'   Sub TickerInsight_OnChange(t)     - dispatcher for sheet events
'
' WORKSHEET LAYOUT (auto-created on first run):
'   Row 1: TICKER <GO>: [B1 input]
'   Row 3: 3 section titles  (A=ACTIVE | D=LIFETIME | G=PROJECTION)
'   Row 4: NET EXPOSURE    | REALIZED PNL    | LAST PRICE
'   Row 5: ENTRY (W.AP)    | LIFETIME EFF   | PRICE TARGET [H5 input]
'   Row 6: LAST PRICE      | VELOCITY/DAY  | PROJECTED PNL
'   Row 7: UNREALIZED PNL |              | PROJECTED RETURN %
'   Row 8: RETURN %        |              |
'   Row 10: 4) RECENT TRADE HISTORY
'   Row 11: ENTRY | EXIT | DAYS | SHARES | PNL | RETURN % | BROKER
'   Row 12+: ... newest-first ...
'
' INSTALLATION:
' 1. VBE > File > Import File > TickerInsight.bas (this file)
' 2. Run OpenTickerInsight (Alt+F8) - "Ticker Insight" sheet created
' 3. In VBE left panel, double-click the "Ticker Insight" sheet
'    under "Microsoft Excel Objects". Paste this Worksheet_Change:
'
'   Private Sub Worksheet_Change(ByVal Target As Range)
'       Dim hitB1 As Boolean
'       Dim hitH5 As Boolean
'       hitB1 = Not Intersect(Target, Me.Range("B1")) Is Nothing
'       hitH5 = Not Intersect(Target, Me.Range("H5")) Is Nothing
'       If Not (hitB1 Or hitH5) Then Exit Sub
'       If Target.Count > 1 Then Exit Sub
'       If hitB1 And Trim(Me.Range("B1").Value) = "" Then Exit Sub
'
'       Application.EnableEvents = False
'       On Error GoTo ErrorHandler
'
'       If hitB1 Then
'           Call RefreshTickerInsight(CStr(Me.Range("B1").Value))
'       ElseIf hitH5 Then
'           Call RefreshTickerProjection
'       End If
'
'   ErrorHandler:
'       Application.EnableEvents = True
'       If Err.Number <> 0 Then
'           MsgBox "Ticker Insight ?航炊: " & Err.Description, _
'               vbCritical, "?航炊?IV " & Err.Number
'       End If
'   End Sub
'
' 4. Type a ticker in B1 (e.g. CRDO, 2330) and optionally H5 target.

```

```

Private Const SH_TI As String = "Ticker Insight"
Private Const SH_TR As String = "Transactions"
Private Const SH_REAL As String = "Realized"
Private Const SH_RR4 As String = "RR4"

```

```

' Transaction column letters (from existing schema)

```

```

Private Const COL_DATE As String = "B"
Private Const COL_TICKER As String = "C"
Private Const COL_ACTION As String = "D"
Private Const COL_SHARES As String = "E"
Private Const COL_NETAMT As String = "I"

```

```

Private Const COL_BROKER As String = "N"

' Exchange rate: USD -> TWD. All aggregate PnL / exposure values for
' US tickers are multiplied by this constant before display. Per-share
' prices (Entry, Last, Price Target) stay in NATIVE currency.
Private Const FX_USD_TWD As Double = 31.6

' -----
' One FIFO-paired trade record (one BUY lot matched to one SELL)
' -----

Private Type TIRecord
    EntryDate As Date
    ExitDate As Date
    DaysHeld As Long
    shares As Double
    Cost As Double
    Proceeds As Double
    PnL As Double
    ReturnPct As Double
    broker As String
End Type

' -----
' One residual lot still open after all transactions processed
' -----

Private Type TILot
    lotDate As Date
    shares As Double
    CostPerShare As Double
    broker As String
End Type

' =====
' PUBLIC ENTRY: open the dashboard (auto-creates sheet)
' Event hooking is done via Worksheet_Change pasted on the sheet's
' code page (see install snippet in module header).
' =====

Public Sub OpenTickerInsight()
    Dim ws As Worksheet
    Set ws = EnsureTickerInsightSheet()

    ' One-shot defensive cleanup: nuke any stray shapes/conditional
    ' formats inherited from previous versions of the sheet.
    Application.EnableEvents = False
    Call RemoveSheetShapes(ws)
    On Error Resume Next
    ws.cells.FormatConditions.Delete
    ws.AutoFilterMode = False
    On Error GoTo 0
    Application.EnableEvents = True

    ws.Activate
    ws.Range("B1").Select

    ' If sheet already has a ticker, do a full refresh.
    ' Otherwise just redraw the empty shell so the user always sees structure.
    Dim existing As String: existing = Trim(CStr(ws.Range("B1").Value))
    If existing <> "" Then
        Call RefreshTickerInsight(existing)
    Else
        Application.EnableEvents = False
        Call DrawShellHeaders(ws, vbNullString)
        Application.EnableEvents = True
    End If
End Sub

' =====
' DIAGNOSTIC: dumps the state of the Ticker Insight pipeline for the
' ticker currently in B1 (or a passed-in one). Useful when no values
' appear in the dashboard. Run via Alt+F8 -> DiagnoseTicker.
' =====

```

```

Public Sub DiagnoseTicker(Optional ByVal forcedTicker As String = "")
    Dim msg As String
    Dim lf As String: lf = vbCrLf

    On Error GoTo ErrTrap

    ' --- 1. Sheets ---
    msg = msg & "=== SHEET CHECK ===" & lf
    Dim wsTI As Worksheet, wsTr As Worksheet
    On Error Resume Next
    Set wsTI = ThisWorkbook.Sheets(SH_TI)
    Set wsTr = ThisWorkbook.Sheets(SH_TR)
    On Error GoTo ErrTrap

    msg = msg & "Ticker Insight: " & IIf(wsTI Is Nothing, "MISSING", "OK") & lf
    msg = msg & "Transactions : " & IIf(wsTr Is Nothing, "MISSING", "OK") & lf

    If wsTr Is Nothing Then GoTo ShowMsg

    ' --- 2. Ticker resolution ---
    msg = msg & lf & "=== TICKER ===" & lf
    Dim ticker As String
    If forcedTicker <> "" Then
        ticker = forcedTicker
    ElseIf Not (wsTI Is Nothing) Then
        ticker = CStr(wsTI.Range("B1").Value)
    End If
    ticker = UCase(Trim(ticker))

    If ticker = "" Then
        msg = msg & "(B1 empty - type a ticker then re-run, or DiagnoseTicker " & _
            Chr(34) & "CRDO" & Chr(34) & ") " & lf
        GoTo ShowMsg
    End If

    msg = msg & "Input          : [" & ticker & "]" & lf
    msg = msg & "Normalized       : [" & NormalizeTicker(ticker) & "]" & lf
    msg = msg & "Is TW ticker      : " & IsTWTicker(ticker) & lf
    msg = msg & "FX multiplier    : " & GetFXToTWD(ticker) & lf
    msg = msg & "Live price       : " & GetStockPriceSafe(ticker) & lf

    ' --- 3. Transactions scan ---
    msg = msg & lf & "=== TRANSACTIONS SCAN ===" & lf
    Dim lastRow As Long
    lastRow = wsTr.Cells(wsTr.Rows.Count, "A").End(xlUp).row
    msg = msg & "Total rows       : " & (lastRow - 1) & lf

    Dim normT As String: normT = NormalizeTicker(ticker)
    Dim matchCount As Long, sample As String, samples As Long
    Dim r As Long
    For r = 2 To lastRow
        Dim cellT As String: cellT = CStr(wsTr.Cells(r, COL_TICKER).Value)
        If NormalizeTicker(cellT) = normT Then
            matchCount = matchCount + 1
            If samples < 5 Then
                sample = sample & " R" & r & ": " & _
                    CStr(wsTr.Cells(r, COL_DATE).Value) & " " & _
                    "[" & cellT & "]" & " " & _
                    CStr(wsTr.Cells(r, COL_ACTION).Value) & " " & _
                    "sh=" & CStr(wsTr.Cells(r, COL_SHARES).Value) & " " & _
                    "net=" & CStr(wsTr.Cells(r, COL_NETAMT).Value) & " " & _
                    "brk=" & CStr(wsTr.Cells(r, COL_BROKER).Value) & lf
                samples = samples + 1
            End If
        End If
    Next r

    msg = msg & "Matching rows : " & matchCount & lf
    If matchCount > 0 Then
        msg = msg & lf & "Sample rows (up to 5):" & lf & sample
    Else

```

```

msg = msg & "(none -- check ticker spelling and column C of Transactions)" & lf
msg = msg & "Hint: Transactions C column normalised forms of first 10 tickers:" & lf
Dim cnt As Long: cnt = 0
For r = 2 To lastRow
    If cnt < 10 Then
        Dim t2 As String: t2 = CStr(wsTr.cells(r, COL_TICKER).Value)
        If t2 <> "" Then
            msg = msg & " [" & t2 & "] -> [" & NormalizeTicker(t2) & "]" & lf
            cnt = cnt + 1
        End If
    End If
Next r
End If

' --- 4. FIFO build ---
msg = msg & lf & "=== FIFO RESULT ===" & lf
Dim records() As TIRecord, recCount As Long
Dim lots() As TILot, lotCount As Long
Call BuildFIFOHistory(ticker, records, recCount, lots, lotCount)
msg = msg & "recCount      : " & recCount & " (closed FIFO pairs)" & lf
msg = msg & "lotCount       : " & lotCount & " (open lots = active pos)" & lf

If recCount > 0 Then
    msg = msg & lf & "First record:" & lf
    msg = msg & " Entry=" & records(1).EntryDate & " Exit=" & records(1).ExitDate & lf
    msg = msg & " Days=" & records(1).DaysHeld & " Shares=" & records(1).shares & lf
    msg = msg & " Cost=" & records(1).Cost & " Proceeds=" & records(1).Proceeds & lf
    msg = msg & " PnL=" & records(1).PnL & " Ret%=" & records(1).ReturnPct & lf
    msg = msg & " Broker=" & records(1).broker & lf
End If

If lotCount > 0 Then
    msg = msg & lf & "First open lot:" & lf
    msg = msg & " Date=" & lots(1).lotDate & " Shares=" & lots(1).shares & lf
    msg = msg & " CPS=" & lots(1).CostPerShare & " Broker=" & lots(1).broker & lf
End If

' --- 5. Format probe ---
msg = msg & lf & "=== FORMAT PROBE ===" & lf
msg = msg & "MoneyFmt(NT$) = " & MoneyFmt("NT$") & lf
msg = msg & "PnLFmt(NT$) = " & PnLFmt("NT$") & lf

' --- 6. DrawLifetimeMetrics dry-run ---
If Not (wsTI Is Nothing) Then
    msg = msg & lf & "=== DRAW PROBE ===" & lf
    On Error Resume Next
    Err.Clear
    Call DrawLifetimeMetrics(wsTI, ticker, records, recCount)
    If Err.Number <> 0 Then
        msg = msg & "DrawLifetimeMetrics: ERR " & Err.Number & " - " & Err.Description & lf
    Else
        msg = msg & "DrawLifetimeMetrics: OK (no error)" & lf
    End If
    On Error GoTo ErrTrap
End If

ShowMsg:
MsgBox msg, vbInformation, "Ticker Insight Diagnostic"
Exit Sub

ErrTrap:
msg = msg & lf & lf & "*** DIAGNOSTIC EXCEPTION ***" & lf & _
    "Err " & Err.Number & ": " & Err.Description
MsgBox msg, vbCritical, "Ticker Insight Diagnostic - error"
End Sub

' =====
' DISPATCHER: called from ThisWorkbook.Workbook_SheetChange
' =====

Public Sub TickerInsight_OnChange(ByVal target As Range)
    On Error Resume Next

```

```

If target.CountLarge > 1 Then Exit Sub
Select Case target.Address
    Case "$B$1"
        Call RefreshTickerInsight(CStr(target.Value))
    Case "$H$5"
        Call RefreshTickerProjection
End Select
End Sub

```

```

' FULL REFRESH: rebuild every section for a ticker

```

```

Public Sub RefreshTickerInsight(ByVal ticker As String)
    ticker = UCase(Trim(ticker))
    If ticker = "" Then Exit Sub

    Dim ws As Worksheet: Set ws = EnsureTickerInsightSheet()

```

```

Application.EnableEvents = False
Application.ScreenUpdating = False
On Error GoTo ErrTrap

```

```

' --- Detect ticker change via hidden A2 tracker -----
' A2 stores the PREVIOUS ticker (invisible: black-on-black).
' B1 has already been updated by the user, so we can't use it.
Dim oldTicker As String
oldTicker = UCase(Trim(CStr(ws.Cells(2, 1).Value)))
Dim savedTarget As Variant: savedTarget = ws.Cells(5, 8).Value

```

```

Call ClearTickerData(ws)
Call DrawShellHeaders(ws, ticker)

```

```

' If user is refreshing the SAME ticker, preserve PRICE TARGET.
' If they switched tickers, leave H5 empty (set by shell).
If oldTicker = ticker And IsNumeric(savedTarget) Then
    If CDbl(savedTarget) > 0 Then ws.Cells(5, 8).Value = CDbl(savedTarget)
End If

```

```

Dim records() As TIRecord, recCount As Long
Dim lots() As TILot, lotCount As Long
Call BuildFIFOHistory(ticker, records, recCount, lots, lotCount)

```

```

If recCount = 0 And lotCount = 0 Then
    With ws.Cells(4, 1)
        .Value = "No transactions found for [" & ticker & "]"
        .Font.Color = RGB(255, 80, 80)
        .Font.Bold = True
        .Font.Italic = True
    End With
    GoTo StampTicker
End If

```

```

Call DrawActivePosition(ws, ticker, lots, lotCount)
Call DrawLifetimeMetrics(ws, ticker, records, recCount)
Call DrawProjection(ws, ticker, lots, lotCount)
Call DrawTradeHistory(ws, ticker, records, recCount)

```

```

StampTicker:
' Stamp current ticker into invisible A2 for next refresh's compare
With ws.Cells(2, 1)
    .Value = ticker
    .Font.Color = RGB(0, 0, 0)
    .Interior.Color = RGB(0, 0, 0)
End With

```

```

Done:
Application.EnableEvents = True
Application.ScreenUpdating = True
Exit Sub

```

```

ErrTrap:

```

```

' Surface errors instead of silently swallowing - vital for debug
Dim errN As Long: errN = Err.Number
Dim errD As String: errD = Err.Description
Application.EnableEvents = True
Application.ScreenUpdating = True
MsgBox "RefreshTickerInsight error" & vbCrLf & vbCrLf & _
    "Err " & errN & ": " & errD & vbCrLf & _
    "Ticker: " & ticker, vbCritical, "Ticker Insight"
End Sub

' =====
' LIGHT REFRESH: recompute PROJECTION only (price target changed)
' =====

Public Sub RefreshTickerProjection()
    Dim ws As Worksheet
    On Error Resume Next: Set ws = ThisWorkbook.Sheets(SH_TI): On Error GoTo 0
    If ws Is Nothing Then Exit Sub

    Dim ticker As String: ticker = UCase(Trim(CStr(ws.Range("B1").Value)))
    If ticker = "" Then Exit Sub

    Application.EnableEvents = False
    Application.ScreenUpdating = False
    On Error GoTo ProjErr

    Dim records() As TIRecord, recCount As Long
    Dim lots() As TILot, lotCount As Long
    Call BuildFIFOHistory(ticker, records, recCount, lots, lotCount)
    Call DrawProjection(ws, ticker, lots, lotCount)

```

ProjDone:

```

Application.EnableEvents = True
Application.ScreenUpdating = True
Exit Sub

```

ProjErr:

```

Dim errN As Long: errN = Err.Number
Dim errD As String: errD = Err.Description
Application.EnableEvents = True
Application.ScreenUpdating = True
MsgBox "RefreshTickerProjection error" & vbCrLf & vbCrLf & _
    "Err " & errN & ": " & errD, vbCritical, "Ticker Insight"
End Sub

```

' =====
' SHEET MANAGEMENT
' =====

Private Function EnsureTickerInsightSheet() As Worksheet

```

    Dim ws As Worksheet
    On Error Resume Next
    Set ws = ThisWorkbook.Sheets(SH_TI)
    On Error GoTo 0
    If ws Is Nothing Then
        Set ws = ThisWorkbook.Sheets.Add( _
            After:=ThisWorkbook.Sheets(ThisWorkbook.Sheets.count))
        ws.Name = SH_TI
        Call InitSheetLayout(ws)
    End If
    Set EnsureTickerInsightSheet = ws
End Function

```

Private Sub InitSheetLayout(ws As Worksheet)

```

' Suppress events around shell setup so writing B1 doesn't recursively
' trigger Workbook_SheetChange while the sheet is half-built.
Dim prevEvents As Boolean
prevEvents = Application.EnableEvents
Application.EnableEvents = False

With ws.cells
    .Interior.Color = RGB(0, 0, 0)
    .Font.Color = RGB(221, 221, 221)

```

```

        .Font.Name = "Consolas"
        .Font.Size = 10
    End With

    ws.Activate
    On Error Resume Next
    ActiveWindow.DisplayGridlines = False
    On Error GoTo 0

    ' Column widths: 3 metric groups + history area
    ws.Columns("A").ColumnWidth = 22 ' Label / Entry date
    ws.Columns("B").ColumnWidth = 16 ' Value / Exit date
    ws.Columns("C").ColumnWidth = 8 ' Days
    ws.Columns("D").ColumnWidth = 22 ' Label / Shares
    ws.Columns("E").ColumnWidth = 16 ' Value / PnL
    ws.Columns("F").ColumnWidth = 10 ' Return %
    ws.Columns("G").ColumnWidth = 22 ' Label / Broker
    ws.Columns("H").ColumnWidth = 16 ' Value

    On Error Resume Next
    ws.Tab.Color = RGB(255, 192, 0)
    On Error GoTo 0

    ' Draw the full static shell immediately so the user sees structure
    ' (even before they type a ticker). RefreshTickerInsight will repaint
    ' this same shell + fill in values.
    Call DrawShellHeaders(ws, vbNullString)

    Application.EnableEvents = prevEvents
End Sub

' -----
' Draws every STATIC element of the dashboard (labels, section
' titles, history table header). Called from InitSheetLayout (empty
' ticker) and from RefreshTickerInsight (with current ticker).
' -----
Private Sub DrawShellHeaders(ws As Worksheet, ticker As String)
    ' --- Row 1: TICKER <GO>: input ---
    With ws.cells(1, 1)
        .Value = "TICKER <GO>:"
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
        .Font.Size = 11
    End With
    With ws.cells(1, 2)
        .Value = ticker
        .Interior.Color = RGB(255, 192, 0)
        .Font.Color = RGB(0, 0, 0)
        .Font.Bold = True
        .Font.Size = 11
        .HorizontalAlignment = xlCenter
    End With
    ws.Rows(1).RowHeight = 22

    ' --- Row 3: section titles ---
    With ws.cells(3, 1)
        .Value = "1) ACTIVE POSITION"
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
    End With
    With ws.cells(3, 4)
        .Value = "2) LIFETIME METRICS"
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
    End With
    With ws.cells(3, 7)
        .Value = "3) PROJECTION"
        .Font.Color = RGB(255, 192, 0)
        .Font.Bold = True
    End With
    ws.Rows(3).RowHeight = 20

```

```

' --- Row 4-8: metric labels (cyan) ---
Call WriteLabel(ws, 4, 1, "NET EXPOSURE")
Call WriteLabel(ws, 5, 1, "ENTRY PRICE (W.AP)")
Call WriteLabel(ws, 6, 1, "LAST PRICE")
Call WriteLabel(ws, 7, 1, "UNREALIZED PNL")
Call WriteLabel(ws, 8, 1, "RETURN %")

Call WriteLabel(ws, 4, 4, "REALIZED PNL")
Call WriteLabel(ws, 5, 4, "LIFETIME EFF")
Call WriteLabel(ws, 6, 4, "VELOCITY/DAY")

Call WriteLabel(ws, 4, 7, "LAST PRICE")
Call WriteLabel(ws, 5, 7, "PRICE TARGET")
Call WriteLabel(ws, 6, 7, "PROJECTED PNL")
Call WriteLabel(ws, 7, 7, "PROJECTED RETURN %")

' Row heights for metric block
Dim rh As Long
For rh = 4 To 8: ws.Rows(rh).RowHeight = 18: Next rh

' --- Row 10: history title ---
With ws.cells(10, 1)
    .Value = "4) RECENT TRADE HISTORY"
    .Font.Color = RGB(255, 192, 0)
    .Font.Bold = True
End With
ws.Rows(10).RowHeight = 20

' --- Row 11: history table header ---
Dim hdrs As Variant
hdrs = Array("ENTRY DATE", "EXIT DATE", "DAYS", "SHARES", _
    "REALIZED PNL", "RETURN %", "BROKER")
Dim c As Long
For c = 0 To UBound(hdrs)
    With ws.cells(11, c + 1)
        .Value = hdrs(c)
        .Font.Color = RGB(150, 150, 150)
        .Font.Bold = True
        .HorizontalAlignment = xlLeft
    End With
Next c
With ws.Range(ws.cells(11, 1), ws.cells(11, UBound(hdrs) + 1)).Borders(xlEdgeBottom)
    .LineStyle = xlContinuous
    .Color = RGB(255, 192, 0)
    .Weight = xlThin
End With
ws.Rows(11).RowHeight = 18

' --- Yellow background for H5 (PRICE TARGET input) ---
With ws.cells(5, 8)
    .Interior.Color = RGB(255, 192, 0)
    .Font.Color = RGB(0, 0, 0)
    .Font.Bold = True
    .Font.Size = 11
    .NumberFormat = "0.00"
    .HorizontalAlignment = xlCenter
End With
End Sub

Private Sub ClearTickerData(ws As Worksheet)
' Wipe everything except row 1 (header) and row 2 (gutter)
With ws.Range(ws.cells(3, 1), ws.cells(500, 8))
    .ClearContents
    .Interior.Color = RGB(0, 0, 0)
    .Font.Color = RGB(221, 221, 221)
    .Font.Bold = False
    .Font.Italic = False
    .Font.Size = 10
    .NumberFormat = "General"
    .HorizontalAlignment = xlGeneral

```

```

        .Borders.LineStyle = xlNone
        ' Strip any leftover conditional formatting (icon sets, data bars)
        .FormatConditions.Delete
    End With

    ' Remove ALL shapes (charts, autosshapes, sparklines, smart tags etc.)
    ' that may have been left on the sheet. We own this sheet entirely.
    Call RemoveSheetShapes(ws)

    ' Turn off any AutoFilter that may have been applied
    On Error Resume Next
    ws.AutoFilterMode = False
    On Error GoTo 0
End Sub

'-----
' Deletes every Shape on the worksheet (charts, autosshapes, comment
' callouts, smart tags, sparklines). Loops backwards to avoid index
' shifting as items are removed.
'-----
Private Sub RemoveSheetShapes(ws As Worksheet)
    On Error Resume Next
    Dim i As Long
    For i = ws.Shapes.Count To 1 Step -1
        ws.Shapes(i).Delete
    Next i
    On Error GoTo 0
End Sub

'=====
' FIFO HISTORY BUILDER
' Walks Transactions chronologically per (ticker, broker).
' BUY pushes a lot. SELL pops lots from the front and emits one
' TIRRecord per lot consumed (so one SELL that eats 3 lots = 3 rows).
' ADJUSTCOST distributes per share across all current lots.
'=====
Private Sub BuildFIFOHistory(ByVal ticker As String, _
    ByRef records() As TIRRecord, _
    ByRef recCount As Long, _
    ByRef lots() As TILot, _
    ByRef lotCount As Long)

    recCount = 0
    lotCount = 0
    ReDim records(1 To 256)
    ReDim lots(1 To 64)

    Dim wsTr As Worksheet
    On Error Resume Next: Set wsTr = ThisWorkbook.Sheets(SH_TR): On Error GoTo 0
    If wsTr Is Nothing Then Exit Sub

    Dim lastRow As Long
    lastRow = wsTr.Cells(wsTr.Rows.Count, "A").End(xlUp).row
    If lastRow < 2 Then Exit Sub

    ' Collect rows for this ticker (normalised so 2330 / 2330.TW / 2330.TWO match)
    Dim tickerNorm As String: tickerNorm = NormalizeTicker(ticker)
    Dim tRows() As Long, tCount As Long
    ReDim tRows(1 To lastRow)
    Dim r As Long
    For r = 2 To lastRow
        If NormalizeTicker(CStr(wsTr.Cells(r, COL_TICKER).Value)) = tickerNorm Then
            tCount = tCount + 1
            tRows(tCount) = r
        End If
    Next r
    If tCount = 0 Then Exit Sub

    ' Sort by date ascending (insertion sort - tCount usually small)
    Dim i As Long, j As Long, tmp As Long
    For i = 1 To tCount - 1
        For j = i + 1 To tCount

```

```

        If SafeDate(wsTr.cells(tRows(i), COL_DATE).Value) > _
            SafeDate(wsTr.cells(tRows(j), COL_DATE).Value) Then
            tmp = tRows(i): tRows(i) = tRows(j): tRows(j) = tmp
        End If
    Next j
Next i

' Per-broker FIFO lot queue
Dim queues As Object: Set queues = CreateObject("Scripting.Dictionary")

Dim k As Long
For k = 1 To tCount
    r = tRows(k)
    Dim tDate As Date: tDate = SafeDate(wsTr.cells(r, COL_DATE).Value)
    Dim action As String: action = UCase(Trim(CStr(wsTr.cells(r, COL_ACTION).Value)))
    Dim shares As Double: shares = SafeNum(wsTr.cells(r, COL_SHARES).Value)
    Dim netAmt As Double: netAmt = SafeNum(wsTr.cells(r, COL_NETAMT).Value)
    Dim broker As String: broker = Trim(CStr(wsTr.cells(r, COL_BROKER).Value))
    If broker = "" Then broker = "Default"

    If shares <= 0 And action <> "ADJUSTCOST" Then GoTo NextTx

    If Not queues.Exists(broker) Then queues.Add broker, New Collection
    Dim queue As Collection: Set queue = queues(broker)

    Select Case action
        Case "BUY"
            Dim cps As Double
            If shares > 0 Then cps = netAmt / shares Else cps = 0
            queue.Add Array(tDate, shares, cps)

        Case "SELL"
            Dim sellPx As Double
            If shares > 0 Then sellPx = netAmt / shares Else sellPx = 0
            Dim remaining As Double: remaining = shares
            Do While remaining > 0.0000001 And queue.count > 0
                Dim frontLot As Variant: frontLot = queue(1)
                Dim lotDate As Date: lotDate = CDate(frontLot(0))
                Dim lotShares As Double: lotShares = CDb1(frontLot(1))
                Dim lotCps As Double: lotCps = CDb1(frontLot(2))

                Dim consumed As Double
                If lotShares <= remaining + 0.0000001 Then
                    consumed = lotShares
                Else
                    consumed = remaining
                End If

                ' Emit a FIFO trade record
                recCount = recCount + 1
                If recCount > UBound(records) Then _
                    ReDim Preserve records(1 To UBound(records) * 2)
                With records(recCount)
                    .EntryDate = lotDate
                    .ExitDate = tDate
                    .DaysHeld = CLng(tDate - lotDate)
                    .shares = consumed
                    .Cost = consumed * lotCps
                    .Proceeds = consumed * sellPx
                    .PnL = .Proceeds - .Cost
                    If .Cost <> 0 Then
                        .ReturnPct = .PnL / .Cost
                    Else
                        .ReturnPct = 0
                    End If
                    .broker = broker
                End With

                ' Update / pop the lot
                If consumed >= lotShares - 0.0000001 Then
                    queue.Remove 1
                End If
            Loop
        End Select
Next k

```

```

        Else
            queue.Add Array(lotDate, lotShares - consumed, lotCps), Before:=1
            queue.Remove 2
        End If
        remaining = remaining - consumed
    Loop

    Case "ADJUSTCOST"
        Dim totSh As Double: totSh = 0
        Dim li As Long
        For li = 1 To queue.count
            totSh = totSh + Cdbl(queue(li)(1))
        Next li
        If totSh > 0 Then
            Dim adjPerSh As Double: adjPerSh = netAmt / totSh
            For li = 1 To queue.count
                Dim adjLot As Variant: adjLot = queue(li)
                Dim newCps As Double: newCps = Cdbl(adjLot(2)) + adjPerSh
                queue.Remove li
                If li > queue.count Then
                    queue.Add Array(adjLot(0), adjLot(1), newCps)
                Else
                    queue.Add Array(adjLot(0), adjLot(1), newCps), Before:=li
                End If
            Next li
        End If
    End Select
NextTx:
Next k

If recCount > 0 Then ReDim Preserve records(1 To recCount)

' Flatten residual lots (active position)
Dim brokerKey As Variant
For Each brokerKey In queues.keys
    Dim bq As Collection: Set bq = queues(brokerKey)
    Dim li2 As Long
    For li2 = 1 To bq.count
        lotCount = lotCount + 1
        If lotCount > UBound(lots) Then _
            ReDim Preserve lots(1 To UBound(lots) * 2)
        With lots(lotCount)
            .lotDate = CDate(bq(li2)(0))
            .shares = Cdbl(bq(li2)(1))
            .CostPerShare = Cdbl(bq(li2)(2))
            .broker = CStr(brokerKey)
        End With
    Next li2
Next brokerKey

If lotCount > 0 Then ReDim Preserve lots(1 To lotCount)
End Sub

' =====
' SECTION 1: ACTIVE POSITION (columns A-B)
' =====

Private Sub DrawActivePosition(ws As Worksheet, ticker As String, _
    lots() As TILot, lotCount As Long)
    Dim totalShares As Double, totalCost As Double
    Dim i As Long
    For i = 1 To lotCount
        totalShares = totalShares + lots(i).shares
        totalCost = totalCost + lots(i).shares * lots(i).CostPerShare
    Next i

    Dim fx As Double: fx = GetFXTToTWD(ticker) ' 31.6 for US, 1 for TW
    Dim lastPx As Double: lastPx = GetStockPriceSafe(ticker)
    Dim wap As Double, unrl As Double, retPct As Double, netExp As Double

    If totalShares > 0 Then
        wap = totalCost / totalShares
    End If

```

```

    netExp = lastPx * totalShares * fx          ' aggregate -> TWD
    unrl = (lastPx - wap) * totalShares * fx    ' aggregate -> TWD
    If totalCost > 0 Then retPct = (lastPx - wap) * totalShares / totalCost
End If

' NOTE: labels are drawn by DrawShellHeaders. We only fill values here.
If totalShares > 0 Then
    Call WriteValueMoney(ws, 4, 2, netExp, "NT$")
    Call WriteValuePrice(ws, 5, 2, wap)        ' per-share native (no fx)
Else
    With ws.cells(4, 2)
        .Value = "NO ACTIVE POSITION"
        .Font.Color = RGB(150, 150, 150)
        .Font.Italic = True
        .Interior.Color = RGB(0, 0, 0)
        .Borders.LineStyle = xlNone
    End With
    Call WriteDash(ws, 5, 2)
End If

If totalShares > 0 And lastPx > 0 Then
    Call WriteValuePrice(ws, 6, 2, lastPx)      ' per-share native (no fx)
    Call WriteValuePnL(ws, 7, 2, unrl, "NT$")
    Call WriteValuePct(ws, 8, 2, retPct)
Else
    Call WriteDash(ws, 6, 2)
    Call WriteDash(ws, 7, 2)
    Call WriteDash(ws, 8, 2)
End If
End Sub

```

```

' =====
' SECTION 2: LIFETIME METRICS (columns D-E)
' LIFETIME EFF = Profit Factor (gross win / gross loss)
' VELOCITY/DAY = Realized PnL / sum(holding days), min 1 per trade
' =====

```

```

Private Sub DrawLifetimeMetrics(ws As Worksheet, ticker As String, _
                                records() As TIRecord, recCount As Long)
    Dim totalPnL As Double, grossWin As Double, grossLoss As Double
    Dim totalDays As Long
    Dim i As Long
    For i = 1 To recCount
        totalPnL = totalPnL + records(i).PnL
        If records(i).PnL > 0 Then grossWin = grossWin + records(i).PnL
        If records(i).PnL < 0 Then grossLoss = grossLoss + Abs(records(i).PnL)
        Dim d As Long: d = records(i).DaysHeld
        If d < 1 Then d = 1
        totalDays = totalDays + d
    Next i

```

```

    Dim fx As Double: fx = GetFXTToTWD(ticker)    ' 31.6 for US, 1 for TW

```

```

' --- Row 4: REALIZED PNL (TWD-converted) ---

```

```

Call WriteValuePnL(ws, 4, 5, totalPnL * fx, "NT$")

```

```

' --- Row 5: LIFETIME EFF (Profit Factor) ---

```

```

' Defensive: clear borders/background, set NumberFormat BEFORE Value,
' and Round the value so even if format application fails the cell
' still displays 2 decimals max.

```

```

With ws.cells(5, 5)

```

```

    .Borders.LineStyle = xlNone

```

```

    .Interior.Color = RGB(0, 0, 0)

```

```

    .HorizontalAlignment = xlLeft

```

```

    If grossLoss > 0 Then

```

```

        Dim pf As Double: pf = grossWin / grossLoss

```

```

        .NumberFormat = "0.00"

```

```

        .Value = Round(pf, 2)

```

```

        If pf >= 1 Then

```

```

            .Font.Color = RGB(0, 210, 100)

```

```

        Else

```

```

            .Font.Color = RGB(255, 80, 80)

```

```

        End If
        .Font.Bold = True
    ElseIf grossWin > 0 Then
        .NumberFormat = "@"
        .Value = "infinite"      ' only wins, no losses
        .Font.Color = RGB(0, 210, 100)
        .Font.Bold = True
    Else
        .NumberFormat = "@"
        .Value = ChrW(&H2014)    ' em dash (consistent with WriteDash)
        .Font.Color = RGB(150, 150, 150)
    End If
End With

' --- Row 6: VELOCITY/DAY (TWD-converted, PnL per holding day) ---
Dim velocity As Double
If totalDays > 0 Then velocity = totalPnL / totalDays
Call WriteValuePnL(ws, 6, 5, velocity * fx, "NT$")
End Sub

' =====
' SECTION 3: PROJECTION (columns G-H)
' H5 is the user-editable PRICE TARGET (yellow background)
' =====

Private Sub DrawProjection(ws As Worksheet, ticker As String, _
    lots() As TILot, lotCount As Long)
    Dim totalShares As Double
    Dim i As Long
    For i = 1 To lotCount: totalShares = totalShares + lots(i).shares: Next i

    Dim lastPx As Double: lastPx = GetStockPriceSafe(ticker)
    Dim fx As Double: fx = GetFXTToTWD(ticker)

    ' NOTE: labels drawn by DrawShellHeaders. Values only here.
    ' Row 4: LAST PRICE (per-share native)
    If lastPx > 0 Then
        Call WriteValuePrice(ws, 4, 8, lastPx)
    Else
        Call WriteDash(ws, 4, 8)
    End If

    ' Row 5: PRICE TARGET (per-share native input) - preserve existing value
    With ws.cells(5, 8)
        If Not IsNumeric(.Value) Then .Value = vbNullString
    End With

    Dim target As Double
    If IsNumeric(ws.cells(5, 8).Value) Then target = Cdbl(ws.cells(5, 8).Value)

    ' Row 6: PROJECTED PNL (TWD-converted)
    If totalShares > 0 And target > 0 And lastPx > 0 Then
        Dim projPnL As Double: projPnL = (target - lastPx) * totalShares * fx
        Call WriteValuePnL(ws, 6, 8, projPnL, "NT$")
    Else
        Call WriteDash(ws, 6, 8)
    End If

    ' Row 7: PROJECTED RETURN %
    If target > 0 And lastPx > 0 Then
        Dim projRet As Double: projRet = (target / lastPx) - 1
        Call WriteValuePct(ws, 7, 8, projRet)
    Else
        Call WriteDash(ws, 7, 8)
    End If
End Sub

' =====
' SECTION 4: RECENT TRADE HISTORY (table, newest first)
' =====

Private Sub DrawTradeHistory(ws As Worksheet, ticker As String, _
    records() As TIRecord, recCount As Long)

```

```

' NOTE: Title (Row 10) and headers (Row 11) are drawn by DrawShellHeaders.
' We only fill data rows starting Row 12.
If recCount = 0 Then Exit Sub

' Sort by ExitDate descending (newest first)
Dim i As Long, j As Long, tmp As TIRRecord
For i = 1 To recCount - 1
    For j = i + 1 To recCount
        If records(j).ExitDate > records(i).ExitDate Then
            tmp = records(i): records(i) = records(j): records(j) = tmp
        End If
    Next j
Next i

Dim fx As Double: fx = GetFXTToTWD(ticker)
Dim pnlFmtStr As String: pnlFmtStr = PnLFmt("NT$") ' quoted-literal NT$ format

Dim r As Long: r = 12
For i = 1 To recCount
    With ws.cells(r, 1)
        .NumberFormat = "yyyy-mm-dd"
        .Value = records(i).EntryDate
        .Font.Color = RGB(221, 221, 221)
        .HorizontalAlignment = xlLeft
        .Borders.LineStyle = xlNone
    End With
    With ws.cells(r, 2)
        .NumberFormat = "yyyy-mm-dd"
        .Value = records(i).ExitDate
        .Font.Color = RGB(221, 221, 221)
        .HorizontalAlignment = xlLeft
        .Borders.LineStyle = xlNone
    End With
    With ws.cells(r, 3)
        .Value = records(i).DaysHeld
        .Font.Color = RGB(221, 221, 221)
        .HorizontalAlignment = xlCenter
        .Borders.LineStyle = xlNone
    End With
    With ws.cells(r, 4)
        .NumberFormat = "#,##0.##"
        .Value = records(i).shares
        .Font.Color = RGB(221, 221, 221)
        .HorizontalAlignment = xlRight
        .Borders.LineStyle = xlNone
    End With
    ' --- PnL converted to TWD ---
    Dim pnlTWD As Double: pnlTWD = records(i).PnL * fx
    With ws.cells(r, 5)
        .NumberFormat = pnlFmtStr
        .Value = pnlTWD
        .Font.Color = USPnLColor(pnlTWD)
        .Font.Bold = True
        .HorizontalAlignment = xlRight
        .Borders.LineStyle = xlNone
    End With
    With ws.cells(r, 6)
        .NumberFormat = "0.00%"
        .Value = Round(records(i).ReturnPct, 4)
        .Font.Color = USPnLColor(records(i).PnL)
        .HorizontalAlignment = xlRight
        .Borders.LineStyle = xlNone
    End With
    With ws.cells(r, 7)
        .Value = records(i).broker
        .Font.Color = RGB(150, 150, 150)
        .Font.Size = 9
        .HorizontalAlignment = xlLeft
        .Borders.LineStyle = xlNone
    End With
    ws.Rows(r).RowHeight = 16

```

```

        r = r + 1
    Next i
End Sub

' =====
' CELL HELPERS
' =====

Private Sub WriteLabel(ws As Worksheet, r As Long, c As Long, txt As String)
    With ws.cells(r, c)
        .Value = txt
        .Font.Color = RGB(0, 200, 255)
        .Font.Bold = False
        .HorizontalAlignment = xlLeft
    End With
End Sub

' All value-writers below set NumberFormat BEFORE Value and explicitly
' clear borders/background to prevent stale formatting from prior renders.
'
' Currency symbol "NT$" must be quoted in the format string. Bare
' "$" in Excel format codes is treated as a locale-aware currency
' specifier (in zh-TW it expands to "NT$"), so an unquoted "NT$..."
' was being mis-parsed and silently rejected on some Excel versions.
' Using Chr(34) (double quote) around the symbol forces literal text.
Private Sub WriteValueMoney(ws As Worksheet, r As Long, c As Long, _
    v As Double, sym As String)
    With ws.cells(r, c)
        .Borders.LineStyle = xlNone
        .Interior.Color = RGB(0, 0, 0)
        .NumberFormat = MoneyFmt(sym)
        .Value = Round(v, 0)
        .Font.Color = RGB(221, 221, 221)
        .Font.Bold = True
        .HorizontalAlignment = xlLeft
    End With
End Sub

Private Sub WriteValuePrice(ws As Worksheet, r As Long, c As Long, v As Double)
    With ws.cells(r, c)
        .Borders.LineStyle = xlNone
        .Interior.Color = RGB(0, 0, 0)
        .NumberFormat = "#,##0.00"
        .Value = Round(v, 2)
        .Font.Color = RGB(221, 221, 221)
        .Font.Bold = True
        .HorizontalAlignment = xlLeft
    End With
End Sub

Private Sub WriteValuePnL(ws As Worksheet, r As Long, c As Long, _
    v As Double, sym As String)
    With ws.cells(r, c)
        .Borders.LineStyle = xlNone
        .Interior.Color = RGB(0, 0, 0)
        .NumberFormat = PnLFmt(sym)
        .Value = Round(v, 0)
        .Font.Color = USPnLColor(v)
        .Font.Bold = True
        .HorizontalAlignment = xlLeft
    End With
End Sub

' Quoted-literal currency format builders.
' For sym="NT$", returns: "NT$"#,##0
' That outer quote pair is part of the Excel format code itself.
Private Function MoneyFmt(sym As String) As String
    Dim q As String: q = Chr(34) & sym & Chr(34)
    MoneyFmt = q & ",##0"
End Function

Private Function PnLFmt(sym As String) As String

```

```

Dim q As String: q = Chr(34) & sym & Chr(34)
PnLFmt = q & "#,##0;- " & q & "#,##0"
End Function

Private Sub WriteValuePct(ws As Worksheet, r As Long, c As Long, v As Double)
    With ws.cells(r, c)
        .Borders.LineStyle = xlNone
        .Interior.Color = RGB(0, 0, 0)
        .NumberFormat = "0.00%"
        .Value = Round(v, 4)
        .Font.Color = USPnLColor(v)
        .Font.Bold = True
        .HorizontalAlignment = xlLeft
    End With
End Sub

Private Sub WriteDash(ws As Worksheet, r As Long, c As Long)
    With ws.cells(r, c)
        .Borders.LineStyle = xlNone
        .Interior.Color = RGB(0, 0, 0)
        .NumberFormat = "@"
        .Value = ChrW(&H2014) ' em dash
        .Font.Color = RGB(150, 150, 150)
        .Font.Bold = False
        .HorizontalAlignment = xlLeft
    End With
End Sub

' US/Bloomberg color convention: green positive, red negative
Private Function USPnLColor(v As Double) As Long
    If v > 0 Then
        USPnLColor = RGB(0, 210, 100)
    ElseIf v < 0 Then
        USPnLColor = RGB(255, 80, 80)
    Else
        USPnLColor = RGB(200, 200, 200)
    End If
End Function

' =====
' TICKER / CURRENCY HELPERS
' =====

' Strip .TW / .TWO suffix and uppercase so "2330", "2330.TW", "2330.TWO"
' all normalise to the same key. Used by FIFO matching and ticker classification.
Private Function NormalizeTicker(t As String) As String
    Dim n As String: n = UCase(Trim(CStr(t)))
    If Len(n) >= 4 Then
        If Right(n, 4) = ".TWO" Then n = Left(n, Len(n) - 4)
    End If
    If Len(n) >= 3 Then
        If Right(n, 3) = ".TW" Then n = Left(n, Len(n) - 3)
    End If
    NormalizeTicker = n
End Function

' Suffix-based classification:
' .TW or .TWO suffix -> TW (no FX conversion)
' Everything else -> US (FX x 31.6 applied to aggregates)
'
' This is a STRICT rule. Special codes like "00981A" without a suffix
' are treated as US. To classify a TW security correctly, the ticker
' MUST be typed with its full suffix (e.g. "00981A.TWO", "2330.TW").
Private Function IsTWTicker(ticker As String) As Boolean
    Dim raw As String: raw = UCase(Trim(CStr(ticker)))
    IsTWTicker = (InStr(raw, ".TW") > 0)
End Function

' Per-share PRICE currency symbol (native; never FX-converted)
' TW -> "NT$" US -> "$"
Private Function GetPriceSymbol(ticker As String) As String

```

```

    If IsTWTicker(ticker) Then GetPriceSymbol = "NT$" Else GetPriceSymbol = "$"
End Function

' Multiplier for aggregate PnL / exposure -> TWD
'   TW ticker -> 1
'   US ticker -> FX_USD_TWD (31.6)
Private Function GetFXToTWD(ticker As String) As Double
    If IsTWTicker(ticker) Then GetFXToTWD = 1 Else GetFXToTWD = FX_USD_TWD
End Function

Private Function GetStockPriceSafe(ticker As String) As Double
    GetStockPriceSafe = 0
    On Error Resume Next
    GetStockPriceSafe = GetStockPrice(ticker)    ' relies on Attach module
    On Error GoTo 0
End Function

' Safer date / number conversion (returns 0 if not parseable)
Private Function SafeDate(v As Variant) As Date
    On Error Resume Next
    If IsDate(v) Then
        SafeDate = CDate(v)
    ElseIf IsNumeric(v) Then
        SafeDate = CDate(CDbl(v))
    End If
    On Error GoTo 0
End Function

Private Function SafeNum(v As Variant) As Double
    On Error Resume Next
    If IsNumeric(v) Then SafeNum = CDbl(v)
    On Error GoTo 0
End Function

```

Option Explicit

```
' My-TW-Coverage GitHub Parser (完整合併版 + 全 Bug 修正)
' Repo : Timeverse/My-TW-Coverage
```

```
' 功能 A (單筆) : 在「設定」B3 輸入代號 → 執行 FetchSingleReport
' 功能 B (批次) : 執行 ParsePilotReports → 寫入四張工作表
```

```
Private Const RAW_BASE As String = "https://raw.githubusercontent.com/Timeverse/My-TW-Coverage/master/"
Private Const API_BASE As String = "https://api.github.com/repos/Timeverse/My-TW-Coverage/"
```

```
' —— 工作表名稱 ——
```

```
Private Const SETTINGS_SHEET As String = "Settings"
Private Const SH_REPORT As String = "Company Report" ' 單筆報告頁 (不含非法字元)
Private Const SH_OVERVIEW As String = "公司總覽"
Private Const SH_BIZ As String = "業務與供應鏈"
Private Const SH_ANNUAL As String = "年度財務"
Private Const SH_QTR As String = "季度財務"
```

```
' 共用輔助：安全偵測 String() 陣列是否已初始化
' 原始寫法 (Not paths) = -1 對陣列做 bitwise NOT → Type Mismatch 崩潰
```

```
Private Function IsArrayInitialized(ByRef arr() As String) As Boolean
    On Error Resume Next
    Dim lb As Long
    lb = LBound(arr)
    IsArrayInitialized = (Err.Number = 0)
    On Error GoTo 0
End Function
```

```
Sub FetchSingleReport()
    Dim token As String: token = GetToken()
    Dim ticker As String
    Dim wsReport As Worksheet

    ' 優先從「公司報告」工作表的 B1 取得代號
    EnsureSheet SH_REPORT
    Set wsReport = ThisWorkbook.Sheets(SH_REPORT)
    ticker = Trim(wsReport.Range("B1").Value)

    ' 如果報告頁 B1 是空的，則回頭去找「設定」工作表的 B3
    If ticker = "" Then
        ticker = Trim(ThisWorkbook.Sheets(SETTINGS_SHEET).Range("B3").Value)
    End If

    If ticker = "" Then
        MsgBox "錯誤：請在「公司報告」B1 或「設定」B3 輸入股票代號。", vbCritical
        Exit Sub
    End If

    Application.StatusBar = "讀取 GitHub 檔案清單..."
    Dim paths() As String
    paths = GetAllMDPaths(token)

    If Not IsArrayInitialized(paths) Then
        MsgBox "錯誤：無法取得目錄，請檢查 Token。", vbCritical
        Application.StatusBar = False
        Exit Sub
    End If

    ' 搜尋路徑... (其餘邏輯不變)
    Dim targetPath As String: targetPath = ""
    Dim i As Long
    For i = 0 To UBound(paths)
```

```

        If InStr(paths(i), ticker) > 0 Then
            targetPath = paths(i)
            Exit For
        End If
    Next i

    If targetPath = "" Then
        MsgBox "搜尋結束：找不到包含「" & ticker & "」的檔案。", vbInformation
        Exit Sub
    End If

    Dim mdContent As String
    mdContent = FetchRaw(targetPath, token)
    If Len(mdContent) > 0 Then
        FormatReportSheet mdContent, targetPath
    End If
    Application.StatusBar = False
End Sub

```

```

' 功能 C：產業搜尋
' 在「設定」B4 輸入產業關鍵字（如 Semiconductors）→ 執行 SearchBySector
' 結果寫入「產業搜尋」工作表，深色主題

```

```

Sub SearchBySector()
    Const SH_SECTOR As String = "產業搜尋"
    Dim token As String: token = GetToken()
    Dim keyword As String
    keyword = Trim(ThisWorkbook.Sheets(SETTINGS_SHEET).Range("B4").Value)

    If keyword = "" Then
        MsgBox "請在「設定」B4 輸入產業關鍵字（如 Semiconductors）", vbExclamation
        Exit Sub
    End If

    Application.ScreenUpdating = False
    Application.StatusBar = "讀取清單..."

    Dim paths() As String: paths = GetAllMDPaths(token)
    If Not IsArrayInitialized(paths) Then
        MsgBox "無法取得清單，請確認 Token。", vbCritical
        GoTo CleanupSector
    End If

    ' 收集符合路徑（不分大小寫比對產業資料夾名稱）
    Dim hits() As String: ReDim hits(0 To UBound(paths))
    Dim hitCount As Long
    Dim i As Long
    For i = 0 To UBound(paths)
        If InStr(LCase(paths(i)), LCase(keyword)) > 0 Then
            hits(hitCount) = paths(i): hitCount = hitCount + 1
        End If
    Next i

    If hitCount = 0 Then
        MsgBox "找不到包含「" & keyword & "」的資料夾。", vbInformation
        GoTo CleanupSector
    End If

    ' 建立 / 清空工作表
    EnsureSheet SH_SECTOR
    Dim ws As Worksheet: Set ws = ThisWorkbook.Sheets(SH_SECTOR)
    ws.Cells.Clear
    ApplySheetDarkBg ws, hitCount + 10

    ' —— 頁首統計列 ——
    With ws.Cells(1, 1)
        .Value = "SECTOR SCAN | " & UCase(keyword) & " | " & hitCount & " COMPANIES | " & Now()
        .Font.Bold = True: .Font.Size = 11
        .Font.Color = RGB(13, 13, 13)
    End With

```

```

        .Interior.Color = RGB(212, 175, 55)
    End With
    ws.Range(ws.cells(1, 1), ws.cells(1, 10)).Merge
    ws.Rows(1).RowHeight = 24

```

```

' ----- 欄位標題 -----
Dim hd As Variant
hd = Array("TICKER", "COMPANY", "SECTOR", "INDUSTRY", "MKT CAP", "EV", "P/E TTM", "P/B", "EV/EBITDA", "PATH")
Dim ci As Long
For ci = 0 To UBound(hd)
    With ws.cells(2, ci + 1)
        .Value = hd(ci)
        .Font.Bold = True
        .Font.Color = RGB(212, 175, 55)
        .Interior.Color = RGB(30, 30, 30)
        .HorizontalAlignment = xlCenter
    End With
Next ci
ws.Rows(2).RowHeight = 20

```

```

' ----- 逐一抓取並解析 -----
Dim r As Long: r = 3
Dim j As Long
For j = 0 To hitCount - 1
    Application.StatusBar = "抓取 " & (j + 1) & " / " & hitCount & ": " & hits(j)
    Dim mc As String: mc = FetchRaw(hits(j), token)
    If Len(mc) = 0 Then GoTo NextHit

    Dim fp() As String: fp = Split(hits(j), "/")
    Dim fnb As String: fnb = Replace(fp(UBound(fp)), ".md", "")
    Dim tk As String, co As String
    If InStr(fnb, "_") > 0 Then
        tk = Split(fnb, "_")(0): co = Split(fnb, "_")(1)
    Else
        tk = fnb: co = fnb
    End If

```

```

' 估值指標
Dim fSec As String: fSec = ExtractSectionLoose(mc, "財務概況")
Dim vSec As String: vSec = ExtractSubSectionLoose(fSec, "估值指標")
Dim pe As String, pb As String, evebitda As String
pe = ParseValuationField(vSec, 0)
pb = ParseValuationField(vSec, 3)
evebitda = ParseValuationField(vSec, 4)

```

```

' 寫入每一欄
Dim rowVals(0 To 9) As Variant
rowVals(0) = tk
rowVals(1) = co
rowVals(2) = RegexFirst(mc, "\\\*板塊[::]\\*\\s*(.+)")
rowVals(3) = RegexFirst(mc, "\\\*產業[::]\\*\\s*(.+)")
rowVals(4) = RegexFirst(mc, "\\\*市值[::]\\*\\s*(.+)")
rowVals(5) = RegexFirst(mc, "\\\*企業價值[::]\\*\\s*(.+)")
rowVals(6) = pe
rowVals(7) = pb
rowVals(8) = evebitda
rowVals(9) = hits(j)

```

```

Dim k As Long
For k = 0 To 9
    With ws.cells(r, k + 1)
        .Font.Color = RGB(210, 210, 210)
        Select Case k
            Case 0 ' Ticker
                .Value = rowVals(k)
                .Font.Bold = True
                .Font.Color = RGB(212, 175, 55)
            Case 6, 7, 8 ' 估值數字
                If IsNumeric(rowVals(k)) Then
                    .Value = CDBl(rowVals(k))
                    .NumberFormat = "0.00"
                End If
            Case Else
                .Value = rowVals(k)
        End Select
    End With
Next k

```

```

        .Font.Color = RGB(24, 144, 255)
    Else
        .Value = rowVals(k)
        .Font.Color = RGB(130, 130, 130)
    End If
Case Else
    .Value = rowVals(k)
End Select
' 偶數列微亮底色
If (r Mod 2) = 0 Then
    .Interior.Color = RGB(22, 22, 22)
End If
End With
Next k
r = r + 1
NextHit:
    If j Mod 10 = 0 Then DoEvents
Next j

' 欄寬
ws.Columns("A").ColumnWidth = 10
ws.Columns("B").ColumnWidth = 20
ws.Columns("C").ColumnWidth = 22
ws.Columns("D").ColumnWidth = 22
ws.Columns("E").ColumnWidth = 16
ws.Columns("F").ColumnWidth = 16
ws.Columns("G:I").ColumnWidth = 12
ws.Columns("J").ColumnWidth = 50

ws.Activate
ws.cells(1, 1).Select
ActiveWindow.FreezePanels = False
ws.cells(3, 1).Select
ActiveWindow.FreezePanels = True

CleanupSector:
    Application.ScreenUpdating = True
    Application.StatusBar = False
    If hitCount > 0 Then
        MsgBox "完成！共找到 " & hitCount & " 家 " & keyword & " 公司。", vbInformation
    End If
End Sub

' —— 從估值表格取出指定欄（資料列第一列 = 標題列下面那列） ——
Private Function ParseValuationField(vSec As String, colIdx As Long) As String
    If vSec = "" Then Exit Function
    Dim normalized As String
    normalized = Replace(Replace(vSec, vbCrLf, vbCrLf), vbCr, vbCrLf)
    Dim lines() As String: lines = Split(normalized, vbCrLf)
    Dim rowCount As Long
    Dim ln As Variant
    For Each ln In lines
        Dim ls As String: ls = Trim(CStr(ln))
        If Left(ls, 1) = "I" And InStr(ls, "---") = 0 And Len(ls) > 1 Then
            rowCount = rowCount + 1
            If rowCount = 2 Then ' 第 2 個有效列 = 數值列
                Dim cA() As String: cA = SplitCells(ls)
                If colIdx <= UBound(cA) Then ParseValuationField = Trim(cA(colIdx))
                Exit Function
            End If
        End If
    Next ln
End Function

' —— 基礎背景渲染：全域純黑（無死角版） ——
Private Sub ApplySheetDarkBg(ws As Worksheet, lastRow As Long)
    ' 1. 將整張工作表的所有儲存格均設為純黑底色與無邊框
    With ws.cells
        .Interior.Color = RGB(0, 0, 0)
        .Borders.LineStyle = xlNone
    End With

```

```

' 2. 僅針對會有資料出現的範圍設定預設字型，避免消耗多餘記憶體
With ws.Range(ws.cells(1, 1), ws.cells(lastRow + 30, 22))
    .Font.Color = RGB(210, 210, 210)
    .Font.Name = "Consolas"
    .Font.Size = 10
End With

ws.Tab.Color = RGB(21, 96, 130)
ws.Activate
ActiveWindow.DisplayGridlines = False
End Sub
' —— 單筆報告排版主引擎 ——

Private Sub FormatReportSheet(content As String, mdPath As String)
    EnsureSheet SH_REPORT
    Dim ws As Worksheet: Set ws = ThisWorkbook.Sheets(SH_REPORT)

    ' 從路徑解析 Ticker 與公司名稱
    Dim parts() As String: parts = Split(mdPath, "/")
    Dim filename As String: filename = Replace(parts(UBound(parts)), ".md", "")
    Dim ticker As String, company As String
    If InStr(filename, "_") > 0 Then
        ticker = Split(filename, "_")(0)
        company = Split(filename, "_")(1)
    Else
        ticker = filename: company = ""
    End If

    ' 1. 清空並初始化純黑背景
    ws.cells.Clear
    ApplySheetDarkBg ws, 120

    ' 2. 建立頂部輸入控制區
    With ws.cells(1, 1)
        .Value = "查詢代號:"
        .Font.Color = RGB(130, 130, 130)
        .Font.Bold = True: .Font.Size = 9
    End With

    With ws.cells(1, 2)
        .Value = ticker
        .HorizontalAlignment = xlCenter
        .Font.Color = RGB(255, 255, 255)
        .Font.Bold = True
        .Interior.Color = RGB(0, 0, 0) ' 純黑背景
        .Borders.LineStyle = xlContinuous
        .Borders.Color = RGB(212, 175, 55)
    End With

    With ws.cells(1, 3)
        .Value = " ← 輸入後直接執行"
        .Font.Color = RGB(100, 100, 100): .Font.Size = 8
    End With
    ws.Rows(1).RowHeight = 25

    Dim nr As Long
    ' Row 2: 公司名稱大標題
    With ws.cells(2, 1)
        .Value = company & " " & ticker
        .Font.Size = 16: .Font.Bold = True
        .Font.Color = RGB(212, 175, 55)
        .Font.Name = "Consolas"
        .Interior.Color = RGB(0, 0, 0)
    End With
    ws.Rows(2).RowHeight = 35

    nr = 3

    ' —— Row 3: 基本資訊 stat bar ——
    Dim statLabels As Variant, statVals As Variant
    statLabels = Array("板塊", "產業", "市值", "企業價值")

```

```

statVals = Array( _
    RegexFirst(content, "\\\*\\*板塊[::]\\*\\*\\s*(.+)", _
    RegexFirst(content, "\\\*\\*產業[::]\\*\\*\\s*(.+)", _
    RegexFirst(content, "\\\*\\*市值[::]\\*\\*\\s*(.+)", _
    RegexFirst(content, "\\\*\\*企業價值[::]\\*\\*\\s*(.+)" _
)
Dim si As Long
For si = 0 To 3
    Dim col As Long: col = si * 3 + 1
    With ws.cells(nr, col)
        .Value = UCase(statLabels(si))
        .Font.Color = RGB(130, 130, 130)
        .Font.Size = 8: .Font.Bold = True
    End With
    With ws.cells(nr, col + 1)
        .Value = statVals(si)
        .Font.Color = RGB(212, 175, 55)
        .Font.Bold = True: .Font.Size = 10
    End With
Next si
ws.Rows(nr).RowHeight = 18
nr = nr + 1

Dim sepRange As Range: Set sepRange = ws.Range(ws.cells(nr, 1), ws.cells(nr, 18))
sepRange.Interior.Color = RGB(212, 175, 55)
ws.Rows(nr).RowHeight = 2
nr = nr + 1

' —— 業務簡介 ——
DarkSectionHeader ws, nr, " BUSINESS OVERVIEW", RGB(0, 0, 0), RGB(212, 175, 55)
nr = nr + 1
Dim biz As String
biz = CleanWikilinks(StripMetaLines(ExtractSectionLoose(content, "業務")))
With ws.cells(nr, 1)
    .Value = biz
    .WrapText = True
    .Font.Color = RGB(190, 190, 190)
    .Font.Size = 10
End With
ws.Rows(nr).RowHeight = 80
ws.Range(ws.cells(nr, 1), ws.cells(nr, 18)).Merge
nr = nr + 1

' —— 供應鏈 (已優化 Regex 邏輯) ——
DarkSectionHeader ws, nr, " SUPPLY CHAIN", RGB(0, 0, 0), RGB(24, 144, 255)
nr = nr + 1
Dim scSection As String: scSection = ExtractSectionLoose(content, "供應鏈")
Dim scData(0 To 2) As Variant
' 使用高度寬容的正則匹配：無視星號位置，抓取冒號後所有文字
scData(0) = Array("UPSTREAM", CleanWikilinks(RegexFirst(scSection, "上游.*?[::]\\*\\s*([^\r\n]+)")))
scData(1) = Array("MIDSTREAM", CleanWikilinks(RegexFirst(scSection, "中游.*?[::]\\*\\s*([^\r\n]+)")))
scData(2) = Array("DOWNSTREAM", CleanWikilinks(RegexFirst(scSection, "下游.*?[::]\\*\\s*([^\r\n]+)")))
Dim sci As Long
For sci = 0 To 2
    With ws.cells(nr, 1)
        .Value = scData(sci)(0)
        .Font.Color = RGB(130, 130, 130): .Font.Bold = True: .Font.Size = 8
    End With
    With ws.cells(nr, 2)
        .Value = scData(sci)(1)
        .Font.Color = RGB(210, 210, 210)
        .WrapText = True
    End With
    ws.Range(ws.cells(nr, 2), ws.cells(nr, 12)).Merge
    ws.Rows(nr).RowHeight = 18
    nr = nr + 1
Next sci

' —— 主要客戶 / 供應商 ——
DarkSectionHeader ws, nr, " CUSTOMERS & SUPPLIERS", RGB(0, 0, 0), RGB(24, 144, 255)
nr = nr + 1

```

```

Dim csSection As String: csSection = ExtractSectionLoose(content, "主要客戶")
Dim csData(0 To 1) As Variant
csData(0) = Array("CUSTOMERS", CleanWikilinks(ExtractListItems(ExtractSubSectionLoose(csSection, "主要客戶"))))
csData(1) = Array("SUPPLIERS", CleanWikilinks(ExtractListItems(ExtractSubSectionLoose(csSection, "主要供應商"))))
Dim csi As Long
For csi = 0 To 1
    With ws.cells(nr, 1)
        .Value = csData(csi)(0)
        .Font.Color = RGB(130, 130, 130): .Font.Bold = True: .Font.Size = 8
    End With
    With ws.cells(nr, 2)
        .Value = csData(csi)(1)
        .Font.Color = RGB(210, 210, 210)
        .WrapText = True
    End With
    ws.Range(ws.cells(nr, 2), ws.cells(nr, 18)).Merge
    ws.Rows(nr).RowHeight = 18
    nr = nr + 1
Next csi

```

' —— 財務概況 ——

```
Dim finSection As String: finSection = ExtractSectionLoose(content, "財務")
```

```
If Len(finSection) > 0 Then
```

' —— 估值指標 (矩陣轉置直式排列) ——

```
Dim vSec As String: vSec = ExtractSubSectionLoose(finSection, "估值指標")
```

```
If Len(vSec) > 0 Then
```

' 【新增邏輯】提取括號內的股價基準日期

```
Dim priceDateStr As String
```

' 尋找 "as of 2026-03-26" 這種格式

```
priceDateStr = RegexFirst(content, "估值指標\s*\([^)]*as of\s*([0-9\-]+)[^)]*\)")
```

```
Dim valHeader As String
```

```
If priceDateStr <> "" Then
```

```
    valHeader = " VALUATION METRICS (As of " & priceDateStr & ")"
```

```
Else
```

```
    valHeader = " VALUATION METRICS"
```

```
End If
```

```
nr = nr + 1
```

' 將抓取到的日期融合進紅色區塊標題

```
DarkSectionHeader ws, nr, valHeader, RGB(0, 0, 0), RGB(255, 100, 100)
```

```
nr = nr + 1
```

```
Dim vLines() As String: vLines = Split(Replace(vSec, vbCr, ""), vbLf)
```

```
Dim vHead() As String, vData() As String
```

```
Dim validRowCount As Long: validRowCount = 0
```

```
Dim lineIt As Variant
```

' 拆解 Markdown 表格為兩組陣列

```
For Each lineIt In vLines
```

```
    Dim lineStr As String: lineStr = Trim(CStr(lineIt))
```

```
    If Left(lineStr, 1) = "I" And InStr(lineStr, "---") = 0 Then
```

```
        validRowCount = validRowCount + 1
```

```
        If validRowCount = 1 Then
```

```
            vHead = SplitCells(lineStr)
```

```
        ElseIf validRowCount = 2 Then
```

```
            vData = SplitCells(lineStr)
```

```
        Exit For
```

```
    End If
```

```
End If
```

```
Next lineIt
```

' 垂直印出陣列

```
If validRowCount = 2 Then
```

```
    Dim vi As Long
```

```
    For vi = 0 To UBound(vHead)
```

```
        If Trim(vHead(vi)) <> "" Then ' 過濾掉潛在的空欄位
```

```
            With ws.cells(nr, 1) ' 欄位名稱放在A欄
```

```
                .Value = Trim(vHead(vi))
```

```

        .Font.Color = RGB(130, 130, 130)
        .Font.Bold = True
        .HorizontalAlignment = xlRight
    End With
    With ws.cells(nr, 2) ' 數值放在B欄
        .Value = Trim(SafeCell(vData, vi))
        .Font.Color = RGB(255, 100, 100)
        .Font.Bold = True
        .HorizontalAlignment = xlLeft
    End With
    ws.Range(ws.cells(nr, 2), ws.cells(nr, 4)).Merge
    nr = nr + 1
End If
Next vi
End If
End If

' 年度財務
Dim aSec As String: aSec = ExtractSubSectionLoose(finSection, "年度")
If Len(aSec) > 0 Then
    nr = nr + 1
    DarkSectionHeader ws, nr, " ANNUAL FINANCIALS", RGB(0, 0, 0), RGB(82, 196, 26)
    nr = nr + 1
    nr = DarkRenderMdTable(ws, aSec, nr, RGB(0, 0, 0), RGB(82, 196, 26))
End If

' 季度財務
Dim qSec As String: qSec = ExtractSubSectionLoose(finSection, "季度")
If Len(qSec) > 0 Then
    nr = nr + 1
    DarkSectionHeader ws, nr, " QUARTERLY FINANCIALS", RGB(0, 0, 0), RGB(250, 173, 20)
    nr = nr + 1
    nr = DarkRenderMdTable(ws, qSec, nr, RGB(0, 0, 0), RGB(250, 173, 20))
End If
End If

' 欄寬重設
ws.Columns("A").ColumnWidth = 26
Dim c As Long
For c = 2 To 8
    ws.Columns(c).ColumnWidth = 14
Next c

ws.Activate
ActiveWindow.FreezePanels = False
ws.cells(4, 1).Select
ActiveWindow.FreezePanels = True
ws.Range("B1").Select
End Sub

' —— 深色區塊標題 ——
Private Sub DarkSectionHeader(ws As Worksheet, r As Long, _
    title As String, bgColor As Long, accentColor As Long)
    Dim rng As Range: Set rng = ws.Range(ws.cells(r, 1), ws.cells(r, 18))
    rng.Interior.Color = bgColor
    ws.cells(r, 1).Interior.Color = accentColor
    ws.cells(r, 1).Value = "" ' accent strip
    With ws.cells(r, 2)
        .Value = title
        .Font.Bold = True
        .Font.Color = accentColor
        .Font.Size = 10
        .Font.Name = "Consolas"
        .Interior.Color = bgColor
    End With
    ws.Range(ws.cells(r, 2), ws.cells(r, 18)).Merge
    ws.Rows(r).RowHeight = 22
End Sub

' —— Markdown 表格渲染器 (純黑版) ——
Private Function DarkRenderMdTable(ws As Worksheet, mdBlock As String, _
    startRow As Long, _

```

```

                headerBg As Long, accentColor As Long) As Long

Dim normalized As String
normalized = Replace(Replace(mdBlock, vbCrLf, vbLf), vbCr, vbLf)
Dim lines() As String: lines = Split(normalized, vbLf)

Dim r As Long: r = startRow
Dim isHeader As Boolean: isHeader = True
Dim rowNum As Long: rowNum = 0
Dim ln As Variant

For Each ln In lines
    Dim ls As String: ls = Trim(CStr(ln))
    If Left(ls, 1) = "!" And InStr(ls, "---") = 0 And Len(ls) > 1 Then
        Dim cA() As String: cA = SplitCells(ls)
        Dim c As Long
        For c = 0 To UBound(cA)
            Dim cv As String: cv = CleanWikilinks(Trim(cA(c)))
            With ws.cells(r, c + 1)
                ' 移除交錯底色，全部設定為純黑
                .Interior.Color = IIf(isHeader, headerBg, RGB(0, 0, 0))

                If isHeader Then
                    .Value = cv
                    .Font.Bold = True
                    .Font.Color = accentColor
                    .HorizontalAlignment = xlCenter
                ElseIf c = 0 Then
                    .Value = cv
                    .Font.Color = RGB(170, 170, 170)
                    .Font.Bold = True
                ElseIf IsNumeric(cv) Then
                    .Value = CDbl(cv)
                    .NumberFormat = "0.00"
                    Dim labelCell As String
                    labelCell = LCase(Trim(cA(0)))
                    If InStr(labelCell, "margin") > 0 Or InStr(labelCell, "income") > 0 Or _
                        InStr(labelCell, "profit") > 0 Then
                        If CDbl(cv) >= 0 Then
                            .Font.Color = RGB(82, 196, 26)
                        Else
                            .Font.Color = RGB(220, 53, 69)
                        End If
                    Else
                        .Font.Color = RGB(210, 210, 210)
                    End If
                    .HorizontalAlignment = xlRight
                Else
                    .Value = cv
                    .Font.Color = RGB(130, 130, 130)
                    .HorizontalAlignment = xlCenter
                End If
            End With
        Next c
        If Not isHeader Then rowNum = rowNum + 1
        isHeader = False
        r = r + 1
    End If
Next ln
DarkRenderMdTable = r
End Function

' —— 寬鬆版 ExtractSection (heading 後可接任意字元) ——
Private Function ExtractSectionLoose(content As String, heading As String) As String
    Static re As Object
    If re Is Nothing Then Set re = CreateObject("VBScript.RegExp")
    re.Global = False: re.MultiLine = False
    re.pattern = "## " & heading & "[^\r\n]*\r?\n([\s\S]+?)(?=\r?\n## |$)"
    Dim m As Object: Set m = re.Execute(content)
    If m.count > 0 Then ExtractSectionLoose = m(0).SubMatches(0) Else ExtractSectionLoose = ""
End Function

' —— 寬鬆版 ExtractSubSection (### heading 後可接任意字元) ——

```

```

Private Function ExtractSubSectionLoose(content As String, heading As String) As String
    Static re As Object
    If re Is Nothing Then Set re = CreateObject("VBScript.RegExp")
    re.Global = False: re.MultiLine = False
    re.pattern = "### " & heading & "[^\r\n]*\r?\n([\s\S]+?)(?=\r?\n###|$)"
    Dim m As Object: Set m = re.Execute(content)
    If m.count > 0 Then ExtractSubSectionLoose = m(0).SubMatches(0) Else ExtractSubSectionLoose = ""
End Function

```

功能 B：批次解析所有公司

```

Sub ParsePilotReports()
    Dim token As String: token = GetToken()

    If token = "" Then
        MsgBox "請先在「" & SETTINGS_SHEET & "」工作表 B1 填入 GitHub Token。", vbExclamation
        Exit Sub
    End If

    Application.ScreenUpdating = False
    Application.Calculation = xlCalculationManual

    SetupSheets

    Application.StatusBar = "讀取檔案清單..."
    Dim paths() As String
    paths = GetAllMDPaths(token)

    If Not IsArrayInitialized(paths) Then
        MsgBox "無法取得檔案清單，請確認 Token 是否有效或網路狀態。", vbCritical
        GoTo Cleanup
    End If

    Dim total As Long: total = UBound(paths) + 1
    Dim i As Long, written As Long
    Dim nextRows(1 To 4) As Long
    Dim n As Long
    For n = 1 To 4: nextRows(n) = 2: Next n

    For i = 0 To UBound(paths)
        Application.StatusBar = "解析 " & (i + 1) & " / " & total & " (" & paths(i) & ")"
        Dim mdContent As String
        mdContent = FetchRaw(paths(i), token)
        If Len(mdContent) > 0 Then
            WriteAllSheets mdContent, paths(i), nextRows
            written = written + 1
        End If
        If (i + 1) Mod 50 = 0 Then DoEvents
    Next i

    AutoFitSheets

Cleanup:
    Application.ScreenUpdating = True
    Application.Calculation = xlCalculationAutomatic
    Application.StatusBar = False
    MsgBox "完成！共解析 " & written & " 家公司", vbInformation
End Sub

```

工作表初始化（批次用）

```

Private Sub SetupSheets()
    EnsureSheet SETTINGS_SHEET
    With ThisWorkbook.Sheets(SETTINGS_SHEET)
        If .Range("A1").Value = "" Then

```

```

        .Range("A1").Value = "GitHub Token"
        .Range("A1").Font.Bold = True
        .Range("B1").Value = " (請貼上 ghp_xxxx Token) "
        .Range("B1").Font.Color = RGB(150, 150, 150)
    End If
    If .Range("A3").Value = "" Then
        .Range("A3").Value = "單筆查詢代號"
        .Range("A3").Font.Bold = True
    End If
End With

EnsureSheet SH_OVERVIEW
EnsureSheet SH_BIZ
EnsureSheet SH_ANNUAL
EnsureSheet SH_QTR

ClearDataSheet SH_OVERVIEW
ClearDataSheet SH_BIZ
ClearDataSheet SH_ANNUAL
ClearDataSheet SH_QTR

WriteHeaders
End Sub

Private Sub EnsureSheet(ByVal shName As String)
    Dim safeName As String: safeName = CleanSheetName(shName)
    If Len(safeName) = 0 Then Exit Sub

    On Error Resume Next
    Dim ws As Worksheet: Set ws = ThisWorkbook.Sheets(safeName)
    On Error GoTo 0

    If ws Is Nothing Then
        Set ws = ThisWorkbook.Sheets.Add(After:=ThisWorkbook.Sheets(ThisWorkbook.Sheets.count))
        On Error Resume Next
        ws.Name = safeName
        On Error GoTo 0
    End If
End Sub

Private Function CleanSheetName(ByVal txt As String) As String
    Dim illegal As Variant: illegal = Array("\", "/", "?", "*", "[", "]", ":")
    Dim i As Integer
    txt = Replace(txt, vbCr, ""): txt = Replace(txt, vbLf, "")
    For i = LBound(illegal) To UBound(illegal)
        txt = Replace(txt, illegal(i), "_")
    Next i
    If Len(txt) > 31 Then txt = Left(txt, 31)
    CleanSheetName = Trim(txt)
End Function

Private Sub ClearDataSheet(ByVal shName As String)
    With ThisWorkbook.Sheets(shName)
        If .UsedRange.Rows.count > 1 Then
            .Rows("2:" & .UsedRange.Rows.count + 1).ClearContents
        End If
    End With
End Sub

Private Sub WriteHeaders()
    WriteHeaderRow SH_OVERVIEW, Array("Ticker", "公司名稱", "產業資料夾", "板塊", "產業", "市值 (百萬台幣)", "企業價值 (百萬台幣)", RGB(31, 78, 121))
    WriteHeaderRow SH_BIZ, Array("Ticker", "公司名稱", "產業資料夾", "業務簡介", "上游", "中游", "下游", "主要客戶", "主要供應商", RGB(55, 86, 35))
    WriteHeaderRow SH_ANNUAL, Array("Ticker", "公司名稱", "產業資料夾", "年度營收1", "年度營收2", "年度營收3", "毛利率1", "毛利率2", "毛利率3", "營業利益率1", "營業利益率2", "營業利益率3", "EPS1", "EPS2", "EPS3"), RGB(123, 44, 44)
    WriteHeaderRow SH_QTR, Array("Ticker", "公司名稱", "產業資料夾", "Q1營收", "Q2營收", "Q3營收", "Q4營收", "Q1 EPS", "Q2 EPS", "Q3 EPS", "Q4 EPS"), RGB(123, 82, 19)
End Sub

Private Sub WriteHeaderRow(shName As String, headers As Variant, bgColor As Long)

```

```

Dim ws As Worksheet: Set ws = ThisWorkbook.Sheets(shName)
Dim i As Long
For i = 0 To UBound(headers)
    With ws.cells(1, i + 1)
        .Value = headers(i)
        .Font.Bold = True
        .Font.Color = RGB(255, 255, 255)
        .Font.Name = "Arial"
        .Interior.Color = bgColor
        .HorizontalAlignment = xlCenter
    End With
Next i
ws.Rows(1).RowHeight = 24
ws.Activate
ActiveWindow.FreezePanels = False
ws.cells(2, 4).Select
ActiveWindow.FreezePanels = True
End Sub

```

' API 請求

```

Private Function GetAllMDPaths(token As String) As String()
    Dim url As String: url = API_BASE & "git/trees/master?recursive=1"
    Dim json As String: json = HttpGet(url, token)
    If json = "" Then Exit Function

    Static re As Object
    If re Is Nothing Then Set re = CreateObject("VBScript.RegExp")
    re.Global = True
    re.pattern = """"path"":""(Pilot_Reports/[^\s]+/[^\s]+\md)""""

    Dim matches As Object: Set matches = re.Execute(json)
    If matches.count = 0 Then Exit Function

    Dim result() As String
    ReDim result(0 To matches.count - 1)
    Dim i As Long
    For i = 0 To matches.count - 1
        result(i) = matches(i).SubMatches(0)
    Next i
    GetAllMDPaths = result
End Function

```

' ——— FetchRaw: 改用 Contents API + raw accept header ———

' 原本走 raw.githubusercontent.com, 對中文檔名 URL 編碼敏感, 容易靜默失敗。

' Contents API 讓 GitHub 伺服器端解析路徑, Accept: v3.raw 直接回傳純文字,

' 兩者合用可完全繞過中文編碼問題, 且失敗時能看到真實 HTTP 狀態碼。

```

Private Function FetchRaw(mdPath As String, token As String) As String
    ' 使用 Contents API (不是 raw 域名)
    Dim url As String: url = API_BASE & "contents/" & UriEncodePath(mdPath)

    On Error GoTo ErrHandler
    Dim http As Object: Set http = CreateObject("WinHttp.WinHttpRequest.5.1")
    http.Open "GET", url, False
    http.setRequestHeader "User-Agent", "Excel-VBA-TW-Coverage"
    If token <> "" Then http.setRequestHeader "Authorization", "token " & token
    ' 關鍵: 告訴 GitHub API 直接回傳原始文字, 而非 JSON 包裝
    http.setRequestHeader "Accept", "application/vnd.github.v3.raw"
    http.send

    If http.status = 200 Then
        FetchRaw = http.responseText
    Else
        ' 顯示真實錯誤碼, 方便診斷 (404=找不到路徑, 401=Token 無效, 403=權限不足)
        MsgBox "下載失敗!" & vbCrLf & _
            "HTTP " & http.status & ": " & http.StatusText & vbCrLf & vbCrLf & _
            "路徑: " & mdPath & vbCrLf & _
            "請求網址: " & url, vbCritical, "FetchRaw 診斷"
    End If

```

```

    FetchRaw = ""
End If
Exit Function
ErrorHandler:
    MsgBox "網路連線失敗：" & Err.Description, vbCritical, "FetchRaw 網路錯誤"
    FetchRaw = ""
End Function

' HttpGet 僅供 GetAllMDPaths (抓目錄清單) 使用，維持 JSON accept header
Private Function HttpGet(url As String, token As String) As String
    On Error GoTo ErrorHandler
    Dim http As Object: Set http = CreateObject("WinHttp.WinHttpRequest.5.1")
    http.Open "GET", url, False
    http.setRequestHeader "User-Agent", "Excel-VBA-TW-Coverage"
    If token <> "" Then http.setRequestHeader "Authorization", "token " & token
    http.setRequestHeader "Accept", "application/vnd.github.v3+json"
    http.send
    If http.status = 200 Then HttpGet = http.responseText Else HttpGet = ""
    Exit Function
ErrorHandler:
    HttpGet = ""
End Function

'
' 批次資料解析與寫入
'
Private Sub WriteAllSheets(content As String, mdPath As String, nextRows() As Long)
    Dim parts() As String: parts = Split(mdPath, "/")
    If UBound(parts) < 2 Then Exit Sub

    Dim sectorFolder As String: sectorFolder = parts(1)
    Dim filename As String: filename = parts(2)
    Dim fnBase As String: fnBase = Left(filename, Len(filename) - 3)

    Dim ticker As String, company As String
    Dim underPos As Long: underPos = InStr(fnBase, "_")
    If underPos > 0 Then
        ticker = Left(fnBase, underPos - 1)
        company = Mid(fnBase, underPos + 1)
    Else
        ticker = fnBase: company = fnBase
    End If

    Dim 板塊 As String, 產業 As String, 市值 As String, EV As String, 業務 As String
    Dim 上游 As String, 中游 As String, 下游 As String, 客戶 As String, 供應商 As String
    Dim ann(1 To 12) As String
    Dim qtr(1 To 8) As String

    板塊 = RegexFirst(content, "\\\*板塊[::]\\\\*\\\\s*(.+)")
    產業 = RegexFirst(content, "\\\*產業[::]\\\\*\\\\s*(.+)")
    市值 = Replace(RegexFirst(content, "\\\*市值[::]\\\\*\\\\s*([\\d,\\s]+)"), ",", "", "")
    EV = Replace(RegexFirst(content, "\\\*企業價值[::]\\\\*\\\\s*([\\d,\\s]+)"), ",", "", "")

    業務 = CleanWikilinks(StripMetaLines(ExtractSection(content, "業務簡介")))
    If Len(業務) > 400 Then 業務 = Left(業務, 400) & "..."

    Dim scSection As String: scSection = ExtractSection(content, "供應鏈位置")
    上游 = CleanWikilinks(RegexFirst(scSection, "\\\*上游[::]\\\\*\\\\s*(.+)"))
    中游 = CleanWikilinks(RegexFirst(scSection, "\\\*中游[::]\\\\*\\\\s*(.+)"))
    下游 = CleanWikilinks(RegexFirst(scSection, "\\\*下游[::]\\\\*\\\\s*(.+)"))

    Dim csSection As String: csSection = ExtractSection(content, "主要客戶及供應商")
    客戶 = CleanWikilinks(ExtractListItems(ExtractSubSection(csSection, "主要客戶")))
    供應商 = CleanWikilinks(ExtractListItems(ExtractSubSection(csSection, "主要供應商")))

    ParseFinancials ExtractSection(content, "財務概況"), ann, qtr

    Dim ws As Worksheet, c As Long

```

```
Set ws = ThisWorkbook.Sheets(SH_OVERVIEW)
```

```
ws.cells(nextRows(1), 1).Resize(1, 7).Value = Array(ticker, company, sectorFolder, 板塊, 產業, 市值, EV)
```

```
Set ws = ThisWorkbook.Sheets(SH_BIZ)
```

```
ws.cells(nextRows(2), 1).Resize(1, 9).Value = Array(ticker, company, sectorFolder, 業務, 上游, 中游, 下游, 客戶, 供應商)
```

```
Set ws = ThisWorkbook.Sheets(SH_ANNUAL)
```

```
ws.cells(nextRows(3), 1).Resize(1, 3).Value = Array(ticker, company, sectorFolder)
```

```
For c = 1 To 12: ws.cells(nextRows(3), 3 + c) = ann(c): Next c
```

```
Set ws = ThisWorkbook.Sheets(SH_QTR)
```

```
ws.cells(nextRows(4), 1).Resize(1, 3).Value = Array(ticker, company, sectorFolder)
```

```
For c = 1 To 8: ws.cells(nextRows(4), 3 + c) = qtr(c): Next c
```

```
Dim n As Long
```

```
For n = 1 To 4: nextRows(n) = nextRows(n) + 1: Next n
```

```
End Sub
```

```
Private Sub ParseFinancials(finSection As String, ann() As String, qtr() As String)
```

```
    If finSection = "" Then Exit Sub
```

```
    Static re As Object
```

```
    If re Is Nothing Then Set re = CreateObject("VBScript.RegExp")
```

```
    re.Global = True
```

```
    re.MultiLine = True
```

```
    re.pattern = "(\\l.+\\l(\\r?\\n\\l\\r)(\\l *[-:| ]+ *\\l(\\r?\\n\\l\\r))?(\\l.+\\l(\\r?\\n\\l\\r))+)"
```

```
    Dim tables As Object: Set tables = re.Execute(finSection)
```

```
    Dim t As Object
```

```
    For Each t In tables
```

```
        Dim lines() As String: lines = Split(t.Value, vbCrLf)
```

```
        Dim rowLines As New Collection
```

```
        Dim ln As Variant
```

```
        For Each ln In lines
```

```
            Dim lineStr As String
```

```
            lineStr = Trim(Replace(CStr(ln), vbCrLf, ""))
```

```
            If Left(lineStr, 1) = "I" And InStr(lineStr, "---") = 0 Then
```

```
                rowLines.Add SplitCells(lineStr)
```

```
            End If
```

```
        Next ln
```

```
    Dim row As Variant
```

```
    For Each row In rowLines
```

```
        Dim cells() As String: cells = row
```

```
        If UBound(cells) < 1 Then GoTo NextRow
```

```
        Dim label As String: label = Trim(cells(0))
```

```
        Dim v1 As String: v1 = SafeCell(cells, 1)
```

```
        Dim v2 As String: v2 = SafeCell(cells, 2)
```

```
        Dim v3 As String: v3 = SafeCell(cells, 3)
```

```
        Dim v4 As String: v4 = SafeCell(cells, 4)
```

```
        If InStr(label, "營收") > 0 Or InStr(label, "Revenue") > 0 Then
```

```
            If v4 <> "" Then
```

```
                qtr(1) = v1: qtr(2) = v2: qtr(3) = v3: qtr(4) = v4
```

```
            Else
```

```
                ann(1) = v1: ann(2) = v2: ann(3) = v3
```

```
            End If
```

```
        ElseIf InStr(label, "毛利率") > 0 Or InStr(label, "Gross") > 0 Then
```

```
            ann(4) = v1: ann(5) = v2: ann(6) = v3
```

```
        ElseIf InStr(label, "營業利益率") > 0 Or InStr(label, "Operating") > 0 Then
```

```
            ann(7) = v1: ann(8) = v2: ann(9) = v3
```

```
        ElseIf InStr(label, "EPS") > 0 Then
```

```
            If v4 <> "" Then
```

```
                qtr(5) = v1: qtr(6) = v2: qtr(7) = v3: qtr(8) = v4
```

```
            Else
```

```
                ann(10) = v1: ann(11) = v2: ann(12) = v3
```

```
            End If
```

```
        End If
```

```
NextRow:
```

```
    Next row
```

```
Next t
```

End Sub

' Markdown 解析工具

' [BUG FIX] \Z 在 VBScript RegExp 不支援，最後一個段落永遠抓不到。

' MultiLine=False 時，\$ 等同整個字串結尾，行為正確。

Private Function ExtractSection(content As String, heading As String) As String

Static re As Object

If re Is Nothing Then Set re = CreateObject("VBScript.RegExp")

re.Global = False

re.MultiLine = False

re.pattern = "## " & heading & "\r?\n([\s\S]+?)(?=\r?\n## !\$)"

Dim m As Object: Set m = re.Execute(content)

If m.count > 0 Then ExtractSection = m(0).SubMatches(0) Else ExtractSection = ""

End Function

' [BUG FIX] 同上

Private Function ExtractSubSection(content As String, heading As String) As String

Static re As Object

If re Is Nothing Then Set re = CreateObject("VBScript.RegExp")

re.Global = False

re.MultiLine = False

re.pattern = "### " & heading & "\r?\n([\s\S]+?)(?=\r?\n###!\$)"

Dim m As Object: Set m = re.Execute(content)

If m.count > 0 Then ExtractSubSection = m(0).SubMatches(0) Else ExtractSubSection = ""

End Function

Private Function StripMetaLines(text As String) As String

Static re As Object

If re Is Nothing Then Set re = CreateObject("VBScript.RegExp")

re.Global = True: re.MultiLine = True

re.pattern = "^**\\.+?[:]**\\.+\\r?\\n?"

StripMetaLines = Trim(re.Replace(text, ""))

End Function

Private Function CleanWikilinks(text As String) As String

If text = "" Then Exit Function

Static re As Object

If re Is Nothing Then Set re = CreateObject("VBScript.RegExp")

re.Global = True

re.pattern = "\\[(.+)\\]"

CleanWikilinks = re.Replace(text, "\$1")

End Function

Private Function ExtractListItems(text As String) As String

If text = "" Then Exit Function

Static re As Object

If re Is Nothing Then Set re = CreateObject("VBScript.RegExp")

re.Global = True: re.MultiLine = True

re.pattern = "^[-*]\\s*(.+) "

Dim matches As Object: Set matches = re.Execute(text)

If matches.count = 0 Then Exit Function

Dim results() As String: ReDim results(0 To matches.count - 1)

Dim i As Long

For i = 0 To matches.count - 1

results(i) = Trim(matches(i).SubMatches(0))

Next i

ExtractListItems = Join(results, "、")

End Function

Private Function RegexFirst(text As String, pattern As String) As String

Static re As Object

If re Is Nothing Then Set re = CreateObject("VBScript.RegExp")

re.Global = False: re.MultiLine = True

re.pattern = pattern

Dim m As Object: Set m = re.Execute(text)

If m.count > 0 Then RegexFirst = Trim(m(0).SubMatches(0)) Else RegexFirst = ""

End Function

Private Function SafeCell(cells() As String, idx As Long) As String

 If idx <= UBound(cells) Then SafeCell = Trim(Replace(cells(idx), ",", "")) Else SafeCell = ""

End Function

Private Function SplitCells(line As String) As String()

 Dim inner As String: inner = line

 If Left(inner, 1) = "|" Then inner = Mid(inner, 2)

 If Right(inner, 1) = "|" Then inner = Left(inner, Len(inner) - 1)

 Dim parts() As String: parts = Split(inner, "|")

 Dim i As Long

 For i = 0 To UBound(parts): parts(i) = Trim(parts(i)): Next i

 SplitCells = parts

End Function

' URL 編碼

' [BUG FIX] Hex(code) 對 code<16 只回傳 1 位，輸出 %A 而非 %0A。

' Right("0" & Hex(code), 2) 強制補零至兩位。

Private Function UrlEncodePath(path As String) As String

 Dim result As String, c As String

 Dim i As Long, code As Long

 For i = 1 To Len(path)

 c = Mid(path, i, 1)

 code = AscW(c)

 ' [BUG FIX] AscW 回傳有號 16-bit, U+8000~U+FFFF 的字元 (如「電」U+96FB=38651)

 ' 超過 32767 後 AscW 回傳負數 (-26885)，負數 < &H800 為 True，

 ' 誤用 2-byte 公式產生錯誤位元組序列 → GitHub 回 404。

 ' 加 65536 將負數還原為正確的 Unicode code point。

 If code < 0 Then code = code + 65536

 If (code >= 65 And code <= 90) Or (code >= 97 And code <= 122) Or _

 (code >= 48 And code <= 57) Or c = "/" Or c = "_" Or c = "-" Or c = "." Then

 result = result & c

 ElseIf code < 128 Then

 result = result & "%" & Right("0" & Hex(code), 2)

 Else

 result = result & EncodeUTF8Char(c)

 End If

 Next i

 UrlEncodePath = result

End Function

Private Function EncodeUTF8Char(c As String) As String

 Dim code As Long: code = AscW(c)

 If code < 0 Then code = code + 65536 ' [BUG FIX] AscW 對 U+8000+ 回傳負數，加 65536 還原正確 code point

 Dim result As String

 If code < &H800 Then

 result = "%" & Right("0" & Hex(&HCO Or (code \ 64)), 2) & _

 %" & Right("0" & Hex(&H80 Or (code Mod 64)), 2)

 ElseIf code < &H10000 Then

 result = "%" & Right("0" & Hex(&HE0 Or (code \ 4096)), 2) & _

 %" & Right("0" & Hex(&H80 Or ((code \ 64) Mod 64)), 2) & _

 %" & Right("0" & Hex(&H80 Or (code Mod 64)), 2)

 End If

 EncodeUTF8Char = UCase(result)

End Function

Private Sub AutoFitSheets()

 Dim shNames As Variant: shNames = Array(SH_OVERVIEW, SH_BIZ, SH_ANNUAL, SH_QTR)

 Dim s As Variant

 For Each s In shNames

 Dim ws As Worksheet: Set ws = ThisWorkbook.Sheets(CStr(s))

 ws.Columns.AutoFit

 If CStr(s) = SH_BIZ Then

 If ws.Columns(4).ColumnWidth > 60 Then ws.Columns(4).ColumnWidth = 60

 ws.Columns(4).WrapText = True

```
End If
Next s
End Sub
```

```
' Token 管理
```

```
Private Function GetToken() As String
    On Error Resume Next
    Dim ws As Worksheet: Set ws = ThisWorkbook.Sheets(SETTINGS_SHEET)
    If ws Is Nothing Then Exit Function
    Dim val As String: val = Trim(ws.Range("B1").Value)
    If val <> "" And InStr(val, "請貼上") = 0 Then GetToken = val
    On Error GoTo 0
End Function

Sub SetToken()
    Dim token As String
    token = InputBox("請貼上 GitHub Personal Access Token : ", "設定 Token")
    If token = "" Then Exit Sub
    EnsureSheet SETTINGS_SHEET
    With ThisWorkbook.Sheets(SETTINGS_SHEET)
        .Range("A1").Value = "GitHub Token"
        .Range("A1").Font.Bold = True
        .Range("B1").Value = token
        .Range("B1").Font.Color = RGB(100, 100, 100)
        .Range("B1").NumberFormat = "@"
    End With
    MsgBox "Token 已儲存到「" & SETTINGS_SHEET & "」B1。", vbInformation
End Sub
```